Engineering Chemistry · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry

DRAFT — UE26CY151A · Unit_01 — Molecular spectroscopy, phase equilibria, and computational chemistry · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · UE26CY151A · Unit_01: Molecular spectroscopy, phase equilibria, and computational chemistry
2. 2 · Interaction of radiation with matter
3. 3 · Electromagnetic spectrum
4. 4 · Rotational spectroscopy
5. 5 · Vibrational spectroscopy(harmonic oscillator)
6. 6 · Vibrational spectroscopy(anharmonic oscillator)
7. 7 · Electronic spectroscopy - Franck-Condon principle (Part II)
8. 8 · Electronic spectroscopy(Progression)
9. 9 · Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential
10. 10 · Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part II
11. 11 · Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I
12. 12 · Reduced phase rule, phase diagram of two-component system - Pb-Ag system
13. 13 · Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part I
14. 14 · Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part III
15. 15 · Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part II
16. 16 · Source slide deck: Born Oppenheimer approximation
17. 17 · Source slide deck: Experiential learning Based on Phase Diagram
18. 18 · Historical question-source pages · unit mapping candidates
19. Appendix A · Symbol glossary
20. Appendix B · Sources, coverage and gaps
21. Appendix C · Revision and retrieval practice

M1

Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry

1 UE26CY151A · Unit_01: Molecular spectroscopy, phase equilibria, and computational chemistry

This unit master organizes source material for Engineering Chemistry, Unit_01, officially titled Molecular spectroscopy, phase equilibria, and computational chemistry. It includes 14 video transcripts and 12 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26CY151A, Unit_01 (Molecular spectroscopy, phase equilibria, and computational chemistry). Bucket folder present: yes. Videos: 14; slide decks: 12. EV0000001

2 Interaction of radiation with matter

This lesson is indexed from the source video titled Interaction of radiation with matter. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000002 · EV0000003

Graphs and diagrams

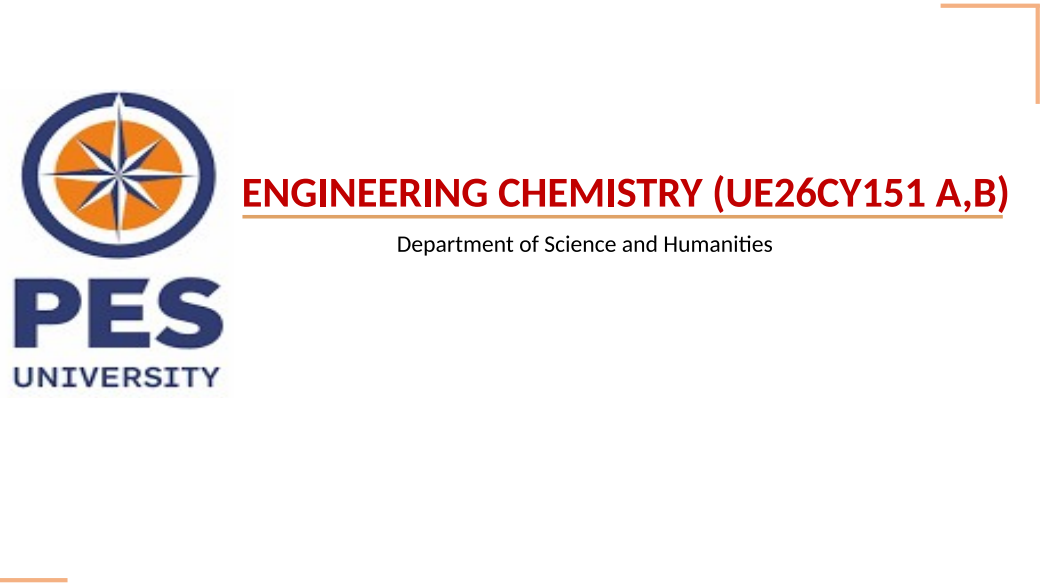

Figure G000001. Interaction of electromagnetic radiation with matter · source slide page 1.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

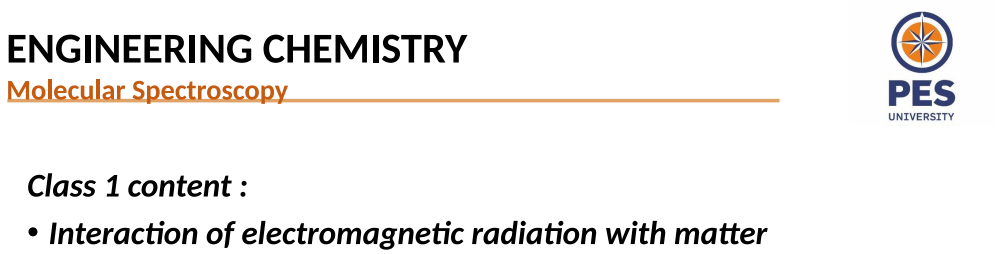

Figure G000002. Interaction of electromagnetic radiation with matter · source slide page 2.

EV0000262

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

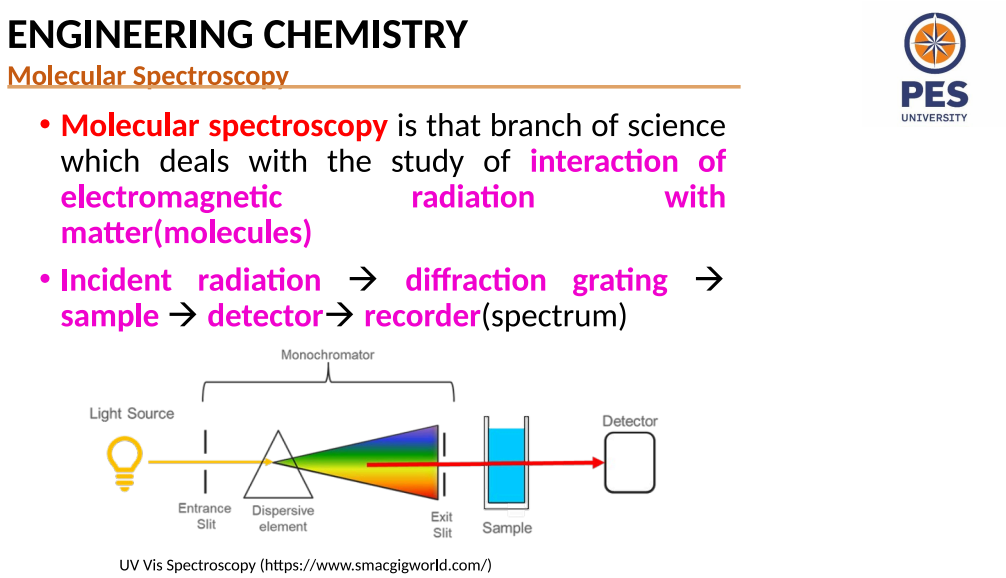

Figure G000003. Interaction of electromagnetic radiation with matter · source slide page 3.

EV0000265

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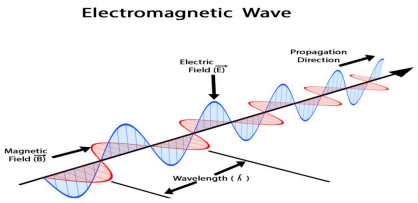
ENGINEERING CHEMISTRY

Molecular Spectroscopy

Some important features of **electromagnetic radiations** are:

- They have **dual character** i.e. particle character as well as wave character, for example, a beam of light is a stream of particles called photons moving through the space in the form of waves
- These waves are associated with **electric and magnetic fields** oscillating perpendicular to each other and also perpendicular to the direction of propagation
- All electromagnetic radiations travel with the **velocity of light**



Source:https://byjus.com/physics/characteristics-of-em-waves/

Figure G000004. Interaction of electromagnetic radiation with matter · source slide page 4.

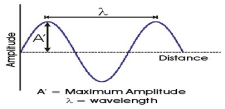
EV0000268

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Molecular Spectroscopy

Waves can be characterised by the following **properties** :



http://www.chemistry.wustl.edu/~coursedev/Online%20tutorials/Waves.htm

- **Wavelength** : The distance between two successive crests and troughs of the wave ; **λ** ; m , nm or Å
- **Frequency** : The number of cycles completed in a certain amount of time ; **ν** ;Hz or s^{-1}
- **Wave number** : Number of complete waves or cycles contained in unit distance ; $\bar{\nu}$, cm^{-1}
- **Energy** of electromagnetic radiation is given by **E= hν** , where E is energy, h is Planck's constant,(h= 6.625 x 10^{-34} Js), and ν is frequency
- Wavelength is related to frequency by **ν = c/λ** , where c is the speed of light, λ is wavelength, and ν is frequency
- Wave number is related to wavelength by $\bar{\nu} = \frac{1}{\lambda}$

Figure G000005. Interaction of electromagnetic radiation with matter · source slide page 5.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

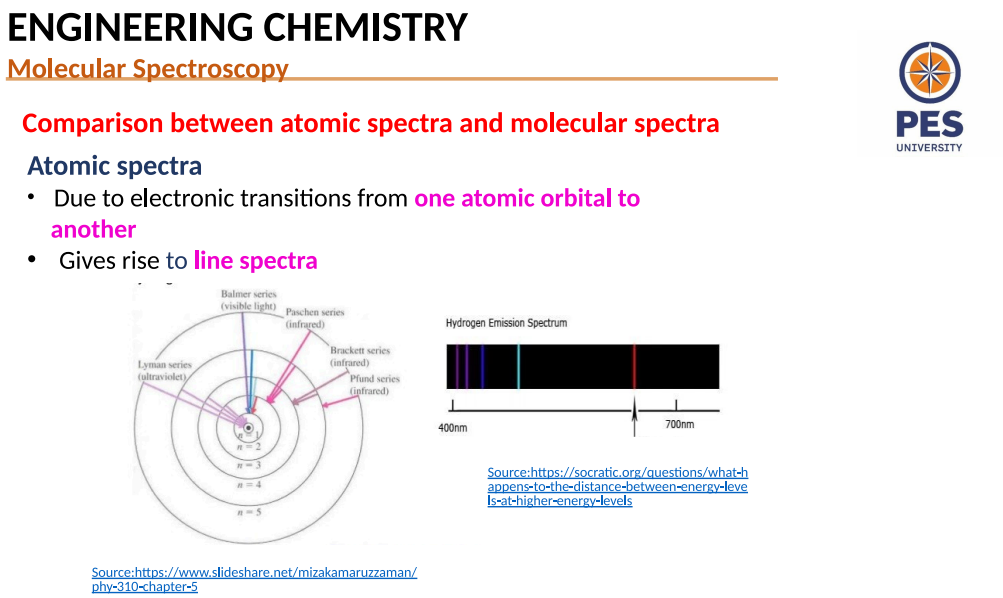

Figure G000006. Interaction of electromagnetic radiation with matter · source slide page 6.

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> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

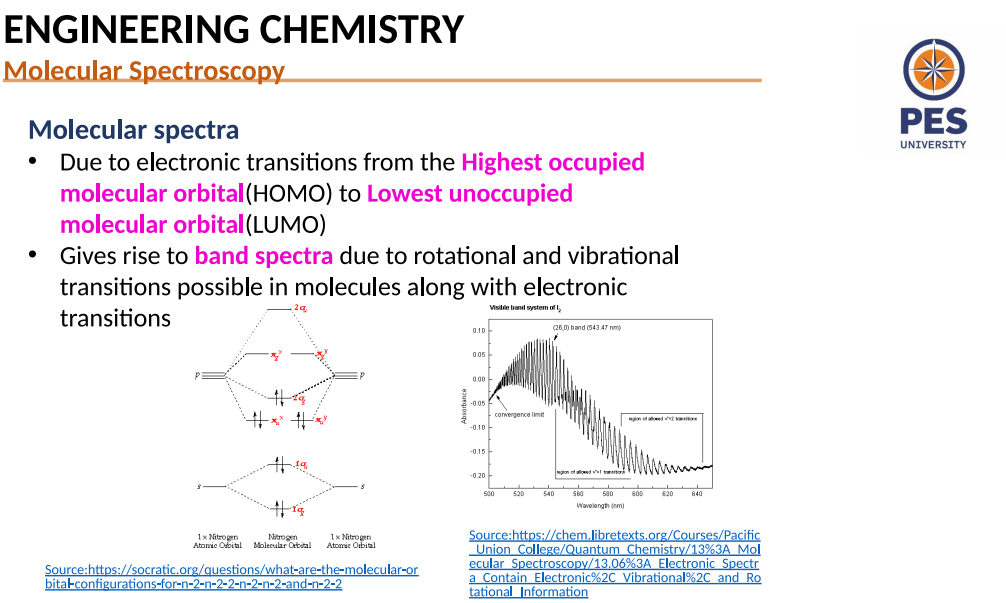

Figure G000007. Interaction of electromagnetic radiation with matter · source slide page 7.

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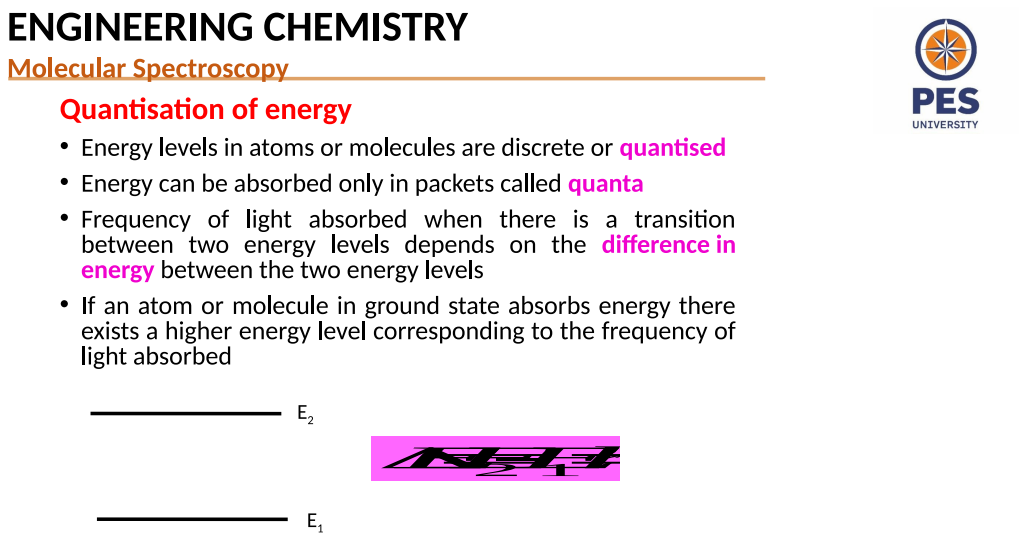

Figure G000008. Interaction of electromagnetic radiation with matter · source slide page 8.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

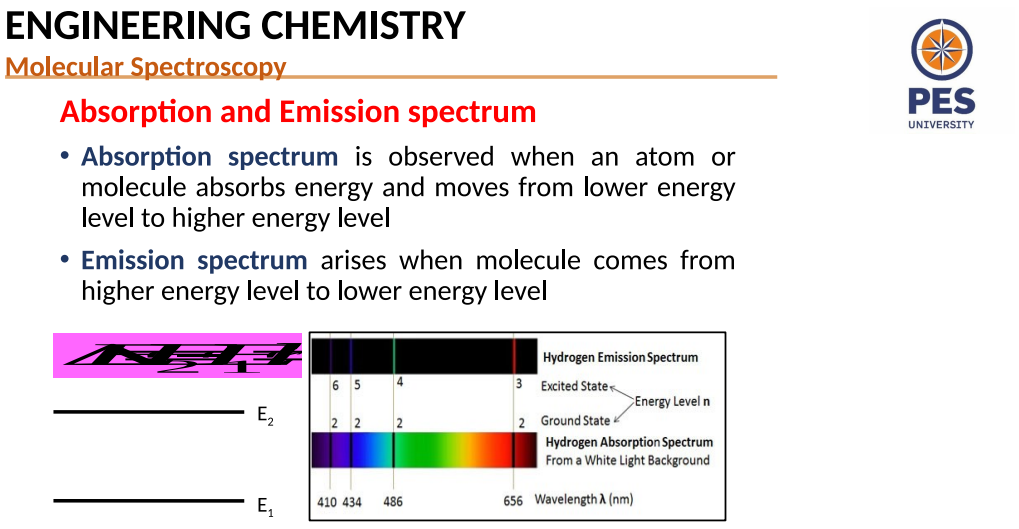

Figure G000009. Interaction of electromagnetic radiation with matter · source slide page 9.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

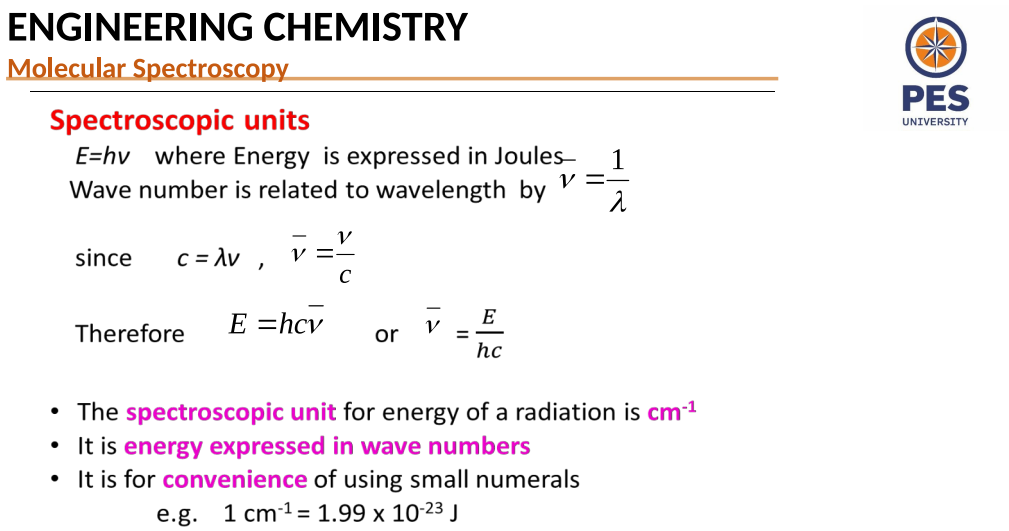

Figure G000010. Interaction of electromagnetic radiation with matter · source slide page 10.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000011. Interaction of electromagnetic radiation with matter · source slide page 11.

EV0000289

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000258 · EV0000259 · EV0000260
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Class 1 content : Interaction of electromagnetic radiation with matter EV0000261 · EV0000262 · EV0000263
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Molecular spectroscopy is that branch of science which deals with the study of interaction of electromagnetic radiation with matter(molecules) Incident radiation ☒ diffraction grating ☒ sample ☒ detector☒ recorder(spectrum) [Image text, OCR] Monochromator Light Source Detector Entrance Dispersive Exit Slit element Slit Sample UV Vis Spectroscopy (https://www.smacgigworld.com/) EV0000264 · EV0000265 · EV0000266
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Some important features of electromagnetic radiations are: They have dual character i.e. particle character as well as wave character, for example, a beam of light is a stream of particles called photons moving through the space in the form of waves These waves are associated with electric and magnetic fields oscillating perpendicular to each other and also perpendicular to the direction of propagation All electromagnetic radiations travel with the velocity of light [Image text, OCR] Electromagnetic Wave Propagation Electric Direction Field (E) Magnetic Field (B) Wavelength (K) Source:https://byjus.com/physics/characteristics-of-em-waves/ EV0000267 · EV0000268 · EV0000269
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Waves can be characterised by the following properties : Wavelength : The distance between two successive crests and troughs of the wave ; λ ; m , nm or Å Frequency : The number of cycles completed in a certain amount of time ; ν ;Hz or s-1 Wave number : Number of complete waves or cycles contained in unit distance ; , cm-1 Energy of electromagnetic radiation is given by E= hν , where E is energy, h is Planck's constant,(h= 6.625 x 10-34 Js), and ν is frequency Wavelength is related to frequency by ν = c/λ , where c is the speed of light, λ is wavelength, and ν is frequency Wave number is related to wavelength by http://www.chemistry.wustl.edu/~coursedev/Online%20tutorials/Waves.htm EV0000270 · EV0000271 · EV0000272
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Comparison between atomic spectra and molecular spectra Atomic spectra Due to electronic transitions from one atomic orbital to another Gives rise to line spectra [Image text, OCR] Figure7shows“permitted”orbitsofanelectron intheBohr modelofahydrogenatom. Balmer series (visible light) Paschen series (infrared) Brackett series Lyman series (infrared) (ultraviolet) Pfund series (infrared) n=5 n=6 Figure7:nottoscale DR.ATAR @ UTM.NS PHY310 Spectral Lines of Hydrogen 34 Source:https://socratic.org/questions/what-happens-to-the-distance-between-energy-levels-at-higher-energy-levels Source:https://www.slideshare.net/mizakamaruzzaman/phy-310-chapter-5 EV0000273 · EV0000274 · EV0000275

7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Molecular spectra Due to electronic transitions from the Highest occupied molecular orbital(HOMO) to Lowest unoccupied molecular orbital(LUMO) Gives rise to band spectra due to rotational and vibrational transitions possible in molecules along with electronic transitions [Image text, OCR] 24 1 x Nitrogen Nitrogen 1 × Nitrogen Atorric Orbital Molecular Orbital Atorric Orbital [Image text, OCR] Visible band system of L 0.10 (26.0) band (543.47 nm) 0.05 0.00 0.05 convergence limit regicn of allowed v=2 transitions -0.10 -0.15 -0.20 regicn of allowed v=1 transitions 500 520 540 560 580 600 620 640 Wavelength (nm) Source:https://chem.libretexts.org/Courses/Pacific_Union_College/Quantum_Chemistry/13%3A_Molecular_Spectroscopy/13.06%3A_Electronic_Spectra_Contain_Electronic%2C_Vibrational%2C_and_Rotational_Information Source:https://socratic.org/questions/what-are-the-molecular-orbital-configurations-for-n-2-n-2-2-n-2-n-2-and-n-2-2 EV0000276 · EV0000277 · EV0000278
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Quantisation of energy Energy levels in atoms or molecules are discrete or quantised Energy can be absorbed only in packets called quanta Frequency of light absorbed when there is a transition between two energy levels depends on the difference in energy between the two energy levels If an atom or molecule in ground state absorbs energy there exists a higher energy level corresponding to the frequency of light absorbed E2 E1 EV0000279 · EV0000280 · EV0000281
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Absorption and Emission spectrum Absorption spectrum is observed when an atom or molecule absorbs energy and moves from lower energy level to higher energy level Emission spectrum arises when molecule comes from higher energy level to lower energy level [Image text, OCR] AtomicSpectralLines WhiteLightEmissionSpectrum ContinuousBlackBodyRadiation HydrogenEmissionSpectrum 6 5 4 3 ExcitedState EnergyLeveln 2 2 2 2 GroundState HydrogenAbsorptionSpectrum FromaWhiteLightBackground 410434 486 656 Wavelength入(nm) E2 E1 EV0000282 · EV0000283 · EV0000284
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy EV0000285 · EV0000286 · EV0000287
11. Source slide page 11 (extracted text): THANK YOU Department of Science and Humanities EV0000288 · EV0000289 · EV0000290

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.316] Hello, everyone. I'm Dr. Lata Pasukuleti and I'll be teaching you this course on engineering chemistry. [00:00:11.236] So the module one talks about molecular spectroscopy. The content is like this. First we are going to look at interaction of electromagnetic radiation with matter, then electromagnetic spectrum, then we discuss the Born Oppenheimer approximation. [00:00:28.676] Further we study BL Ambert's law and then we look at different types of spectroscopy. First we'll study microwave spectroscopy, then we go to IR spectroscopy and finally to electronic spectroscopy. [00:00:43.684] In today's class, we will look at the definition of molecular spectroscopy, what we understand by interaction of electromagnetic radiation with matter, comparison between atomic and molecular spectrum. [00:00:56.260] The quantization of energy concept, absorption and emission spectrum, and finally about spectroscopic units. So first let us define what is molecular spectroscopy. [00:01:11.108] The

word spectroscopy has the in it spectrum. So spectrum when we say we always left refer to light. Now here light is being used as a probe. [00:01:26.148] to study molecules. So this kind of study is called molecular spectroscopy. [00:01:34.084] So how exactly do we use light as a probe to understand the nature of the molecule? This happens because whenever we have a sample, say like this, a sample. [00:01:49.648] And then we have electromagnetic radiation falling on this sample. The electromagnetic radiation has energy and this radiation is going to interact with the matter here. [00:02:05.348] Now because of this interaction, there are bound to be changes in the electromagnetic radiation. Some of the energy could be absorbed, some of it could be transmitted, and some of it could get scattered. [00:02:20.836] So if we can study the radiation which is coming out here, we will be able to understand what is the nature of the molecules. Nature of the matter. Here we are talking of molecular spectroscopy, so we are going to study how it is affecting the molecules. [00:02:38.608] This interaction of electromagnetic radiation with matter is the concept which we are going to use in molecular spectroscopy. [00:02:50.884] So now let us see if I want to design a spectrophotometer. So basically I want to see what is the kind of interaction that is happening between the radiation and the molecule and what information can I get from that. [00:03:06.116] Now molecular spectroscopy is a very strong tool, it's a very good probe to understand matter and in this case molecules. So what would the spectrophotometer look like? [00:03:20.132] Now what do I need? I need incident radiation. So basically I should have a source here which is going to give me radiation. So this is say a source from where radiation is coming out. These [00:03:35.728] radiation and this radiation is going to fall on the matter here, the molecules here, and there is going to be interaction. So basically I need incident radiation which is going to [00:03:46.848] fall on the sample and there is bound to be interaction. Now if I want to study the interaction then I should see the radiation which is coming out. For that I need a detector or an analyzer we call it. Now this detector now is going [00:04:04.528] Going to send this radiation to a diffraction grading. Why? Because it's going to break the radiation into the different components of different wavelengths. And then we need a recorder which is going to record these different wavelengths, and what we get is [00:04:22.096] Known as a spectrum. So that means if radiation falls on a sample and we are able to analyze the radiation which is coming out, then we get what we call as a spectrum. So the molecule is going to [00:04:42.244] dictate what kind of spectrum. So if we if we have different types of molecules we expect to get different types of spectrum. So if we can study the spectrum and interpret it, we will get to know what is the nature of the molecules here.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:01.988] Now, some important features of electromagnetic radiations we need to understand. Now we know that these radiations have dual nature, meaning there is a particle nature and also there is a wave nature. So we can say whenever we have a beam of light. [00:05:21.476] It is a stream of particles which are called [00:05:24.912] Photons and these are moving through space in the form of waves. So we will have particle nature as well as wave nature. The radiation is going to have particle, we will see particle nature as well as wave nature in the form of we can call it as a stream of particles. [00:05:42.576] particles called photons or we can look at it as a form of a wave. Now another important characteristic of these waves is that there is an electric and a magnetic field associated with these [00:05:59.472] Radiations and they are oscillating perpendicular to each other and also perpendicular to the direction of propagation. So here we have. [00:06:07.904] uh electric field

oscillating and we also have the magnetic field which is perpendicular and these are going to be perpendicular to the direction of propagation we can see in the picture over here and all electromagnetic radiations always travel with the velocity of light. So these [00:06:25.584] These are important features which we should remember when we are studying molecular spectroscopy. Now let us [00:06:32.608] C, what are the characteristics of the wave or how is a wave characterized? What are the various properties that a wave possesses? Now you can see the wave is shown in this form. Now always every wave is associated with wavelength. So what is this wavelength? [00:06:50.320] distance between two successive crests and troughs of the wave and it is it is denoted by lambda and it could be in meters, nanometers or angstrom depending on uh what is the [00:07:05.252] distance between two successive casts or traps. Okay. Then we also have frequency. [00:07:12.032] What is frequency the number of cycles completed in a certain amount of time and that is given by nu, and it could be uh in hertz or the second inverse. We also have wave number [00:07:26.832] for every wave and that is number of complete waves or cycles that is contained in a unit distance and that is given by new bar and that is centimeter inverse. So basically wave number wave number is one by lambda. [00:07:44.464] You can make out from the unit itself. So lambda is saying meters or centimeters, this is centimeter inverse. Okay. Now energy of electromagnetic radiation is given by E is equal to H nu, where E is the energy, H is the Planck constant given by 6.625. [00:08:02.032] minus thirty four Joues second and nu is the frequency. [00:08:06.596] And wavelength is related to frequency by C is equal to nu lambda, c is the speed of light, lambda is the wavelength, and nu is the frequency. So these are a few equations which we should always keep in mind when we are going to uh [00:08:23.396] study about the different uh parameters which we are going to find out for a particular molecule. And there is also wave number which is related to [00:08:35.648] Wavelength by nu bar is equal to 1 by lambda. That's exactly what I said. So wave number is related to wavelength by this formula. So all these are very important when we are trying to analyze the spectrum. All these properties, wave properties we should understand. [00:08:56.100] Now, let us compare atomic spectr and molecular spectrum. Atomic spectrum is something which you have already studied. [00:09:07.616] See, we are going to learn about molecular spectrum, but if you recollect what you studied in atomic spectrum, then understanding molecular spectr would be that much more simple. So let us look at atomic spectrum. Now we know that in an atom we have the nucleus in the center. [00:09:25.360] And these are the atomic orbitals which are present around the atom. Now this is a hydrogen atom. This has just one electron in it, and we know that when radiation is going to fall on [00:09:41.380] This hydrogen atom, then these elect this electron is going to be excited to different levels. Which level it goes to depends on what is the amount of energy it absorbs. So it could go from one to two, it could go [00:09:59.152] From one to three, one to four.

EV0000006 · EV0000007 · EV0000008 · EV0000009

Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.600] In to any of these higher energy orbitals, the electron transits when when it absorbs radiation. So that is the kind of interaction what we mean by electromagnetic interaction of radiation with matter. In this case, it is atom. So what we [00:10:19.312] observe is this these levels now higher levels get occupied by the electron but since that is unstable at a higher energy level every system tries to come back to a lower energy so this energy is going to be emitted back and that [00:10:36.880] If we can record, we will get a spectrum like this. Now here we have some radiations which are coming from the uh highest level to say [00:10:53.380] One, two, three, third level. Some will come to the first level.

Some will come back to the second level. And the difference in energy for each of this. [00:11:05.700] is different. So if it comes from this level to this level, the energy is different. When it comes from here to here energy is different. If energy is different, the frequency and the corresponding wavelength are also different. So [00:11:20.560] Based on this difference in energy, we will get different lines in the spectrum. So this is what your hydrogen spectrum looks like, and you can see these lines, these are the emission lines, the lines which are coming due to the [00:11:39.012] Electron coming back from the higher energy level to the lower energy level. One thing we can understand here that if we see a line in the spectrum. [00:11:53.152] This is something you're physically observing. So if we see a line in the spectrum, it means that corresponding to that frequency, there is an energy gap which is present in the [00:12:07.440] In the atom, and that energy gap is between atomic orbitals because we are talking of atoms, it is between the atomic orbitals. So there must be two levels of energy which [00:12:22.052] The gap is equivalent to the energy which is getting emitted. In this case, this is an emission spectrum. So the energy is getting empty. So if you see a line in the spectrum, it means that there exist that two energy levels of that energy gap. [00:12:40.740] That is about atomic spectrum and this spectrum is always is known as the line spectrum because you will see different lines corresponding to the energy emission which is happening here. [00:12:53.860] Now coming to molecular spectrum. Just like you have atomic orbitals, you have studied molecular orbitals also. [00:13:03.780] So in a molecule, atomic orbitals you know combine to give you molecular orbit. So when these combine like this, you get these molecular orbitals, and if you remember they are refer to as sigma, sigma star pi pi star so on. So these [00:13:21.312] Molecular orbitals exist inside the molecule just like in an atom there are atomic orbitals. So these molecular orbitals now which are present have the electrons which are [00:13:33.360] uh being filled up uh following the the same rules which we use for atomic orbitals. Now in this particular case we find that this is the highest occupied molecular orbital. [00:13:46.916] Right? That is known as the Homo, highest occupied molecular orbital. Now when radiation falls on such a molecule, what is going to happen? The electron now is going to transit from the highest or occupied molecular orbital to [00:14:04.784] lowest unoccupied molecular orbital. And when such a transition happens, then we will see a line in the spectrum. That is because [00:14:17.408] Energy gap is there, that much energy is absorbed, say we are looking at absorption, then we'll find that that particular line in the energy is uh the line corresponding to that particular energy is missing in the spectrum. So where did it go? The molecule above [00:14:35.056] Absorb that energy and transit it to the next level, and that is known as the lowest unoccupied molecular orbital. So in a molecule, in a molecule, in the molecule, there is a transition from the Homo to the lum. So we have Homo. [00:14:53.860] And this transition is to the lowest unoccupied molecular orbital. This is what we are going to see in our molecule.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:02.756] So what do we expect? We should see a line in the spectrum, right? But we find that in the case of molecules, we don't see a line. In fact, we see a band like this. You can see this is another example where you're actually seeing a band. [00:15:20.068] If you look closely, if you have good resolution in your spectral photometer, you will find that these this band is actually made of several lines inside. [00:15:33.604] Now that is referred to as the vibrational core structure, and if you have very good resolution within these vibration lines also you will find very fine lines which are referred to as the rotational fine lines. What is this band spectrum in a molecule? [00:15:52.932] We understand that in a

molecule apart from electronic transition there could be other types of transitions possible. Why? Because a molecule can undergo vibration. [00:16:07.184] It can also undergo rotational rotation. So, therefore, apart from the electronic transition, there could be other transitions to different levels, like there could be change in vibrational levels, there could be change in rotation levels. Now such levels do not exist in [00:16:24.912] An atom you have only atomic orbital, so you will see only electronic transition. There's only one line which you will see you will see only lines in the atomic spectrum, but in a molecule, you are going to see a band, and if you have good resolution, you will see finer lines inside this band. So those. [00:16:42.564] As I mentioned, if you see a line in the spectrum, that means there are corresponding energy levels which are equal to the energy gap. [00:16:54.852] The frequency of the absorption or the emission which you see in the spectrum. Therefore, what do we understand so far? That atomic spectrum will give you a line spectrum, whereas molecular spectrum will give you a [00:17:08.992] Uh molecular spectrum will be known as a band spectrum. You will see a band with finer lines. [00:17:16.772] Now let us understand the concept of quantization. We know that energy levels in an atom or a molecule are not continuous, instead they are discrete or quantized. What do peep what do we mean by that? [00:17:32.196] that we have energy levels at certain values only. [00:17:38.180] It's not a continuous energy that is available inside the atom or a molecule. So it is like a a simple uh analogy would be like a ladder. Like in a ladder you have rungs, different rungs. Now [00:17:54.080] I you can put your hand uh leg on one rung or the next rung and the next rung. You cannot stand if you put your leg in between these rung. Exactly like that in an atom, you have these [00:18:06.368] orbitals energy levels in a molecule also energy levels which are at discrete energy value. So such uh [00:18:14.864] And then such an energy level is known as a quantized energy level. It is not continuous, it is quantized. Now because it is quantized, that means it will there will be a level here, there will be a level here. There cannot be any levels in between. Now this energy there. [00:18:32.496] Therefore, can be absorbed only in the form of packets which are known as quanta. [00:18:40.196] This energy gap is will be suppose there is an energy level E1 and there's an energy level E2, the energy gap delta E is equal to E2 minus E1. This difference in energy will be what we say quantize and therefore that is given. [00:18:57.968] By H nu. H is the Planck's constant and mu is the frequency of the radiation which will get absorbed or emitted. So that means energy is quantized means this h nu can be [00:19:14.064] Either as only H new or it could be two H new, three H new, four H new, but you cannot have a quarter H nu or a th three by fourth H new. No, that is what we mean by energy is quantities means the energy gap is always equal to some integral multiple [00:19:31.728] Of H nu. So and the frequency of light absorbed when there is a transition between these two energy levels depends on the energy gap between the two levels. So if the molecule is in this particular energy level, it will move up. [00:19:48.096] to this energy level and this difference in radiation is H nu. So if an atom or a molecule is in the ground state, as I mentioned just now, this is known as the ground state and this is the uh higher le higher energy level then

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:04.580] Corresponding to the energy gap, the frequency of light is going to be absorbed. So this is what we mean by quantization of energy. [00:20:15.492] Now there are two types of spectrum which we can record. One is absorption spectrum, the other one is emission spectrum. What would be an absorption spectrum when light is absorbed. [00:20:29.584] This would give you an absorption spectrum. So when an atom or a molecule moves from [00:20:36.880] The lower energy level to higher energy level, thereby

absorbing a particular uh radiation. Absorbing a particular radiation would be like you will see some lines if you have used white light, then you would see some lines missing. So if you see a line here. [00:20:54.544] That means there is a corresponding energy, uh two energy levels with that corresponding energy gap. That is known as an absorption spectrum, you can see here. And if [00:21:06.212] the light uh has already been absorbed and now it is it's coming the atom or molecule is coming from the higher energy level to the lower energy level then what we see is known as the emission spectrum. So it were a hydrogen atom again you can see [00:21:23.984] the emission spectrum here. So basically what it means that you can up it depends on the detector whether you're [00:21:31.872] studying the absorption spectrum or you're studying the emission spectrum. So they will be absolutely complementary to each other. So whatever light is missing here, radiation is missing here in the em in the absorption spectrum would appear in the emission spectrum. So basically why what does it mean [00:21:49.520] Again, it tells us that based on the energy gap, light is going to be absorbed or emitted. It depends on what we are what is the detector, how we are analyzing or what kind of spectrum we want to study. And most of our study here, we will be looking at absorption spectrum. [00:22:09.924] Now let us study what do we mean by spectroscopic units. Now we know that energy is given by E is equal to H2. Is expressed in joules. [00:22:21.120] Now, wave number is related to wavelength, nu bar is equal to 1 by lambda. We know this relation also. Now, since c is equal to nu lambda, so nu bar can be expressed as nu by c, and therefore energy can be expressed in terms of nu bar. [00:22:38.864] as HC nu bar. Now if I want to express energy not in joules but in terms of nu bar, then I will have to divide the energy by HC. [00:22:56.356] Why do we do that? Because new bar is treated as the spectroscopic unit of energy and basically it just [00:23:06.464] tells us it it corresponds to energy of a radiation and it is in terms of centimeter inverse. Now why should we do that? If we express why should we express in centimeter inverse and not in joules? [00:23:21.104] It is just a matter of convenience. Why? Because if you see you can try to do this conversion, one centimeter inverse is equal to 1.99 into 10 to the power of 23 joules. [00:23:38.148] Minus 23 joules. So therefore 1 centimeter inverse is 1.99 tet power minus 23 joules. So that would mean we are avoiding using these large numbers. Instead, this [00:23:52.676] Amount of energy. Basically, we don't want to use such uh exponential uh terms, large exponential terms. Therefore, we can express in a simple way, we just say one centimeter inverse, five centimeter inverse, it's just a matter of convenience. What it means is this energy in joules. [00:24:10.448] Has been expressed in terms of centimeter inverse. Though we know centimeter inverse is not an SI unit, but still it is for convenience that we will express energy in terms of centimeter inverse, and therefore it is referred to as [00:24:28.016] as the spectroscopic unit for energy. [00:24:34.596] Thank you. And before that, let me tell you that in the next class we are going to study about electromagnetic spectrum. Thank you.

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Full transcript: 3508 normalized words, 103 timestamped segments, 1486.677333 seconds duration. Low-confidence segments flagged: 0.

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3 Electromagnetic spectrum

This lesson is indexed from the source video titled Electromagnetic spectrum. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000022 · EV0000023

Graphs and diagrams

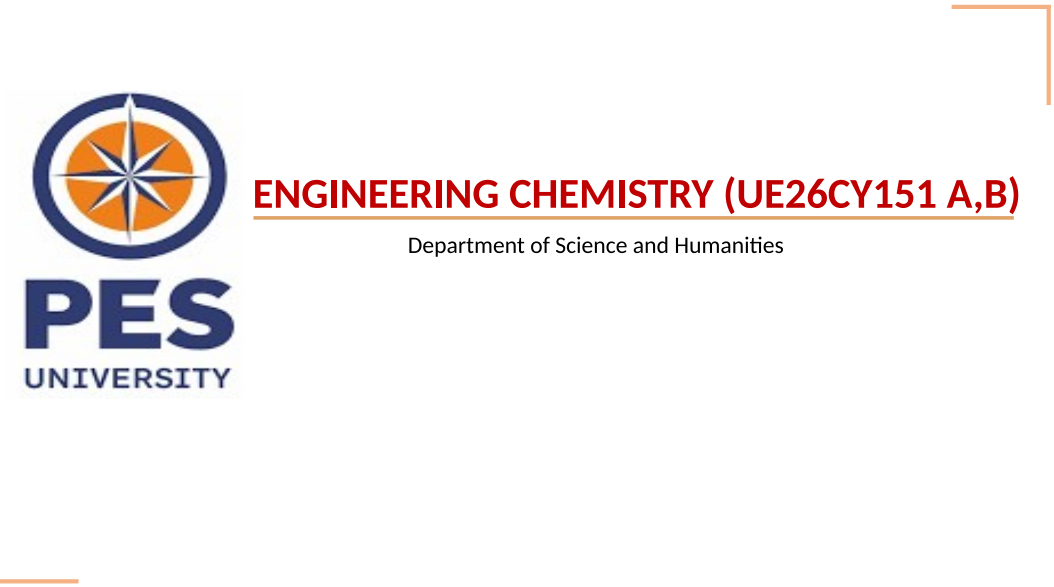

Figure G000012. Electromagnetic spectrum · source slide page 1.

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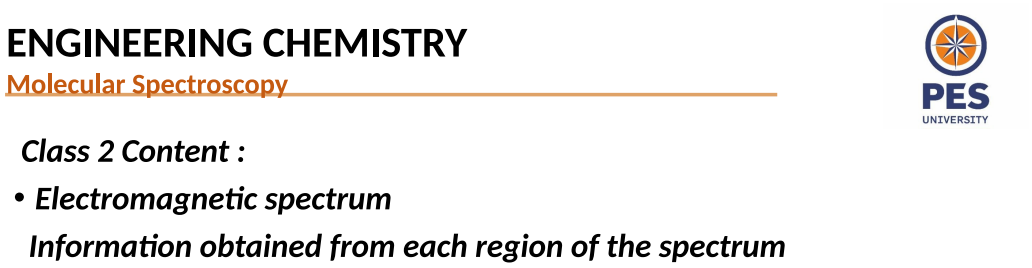

Figure G000013. Electromagnetic spectrum · source slide page 2.

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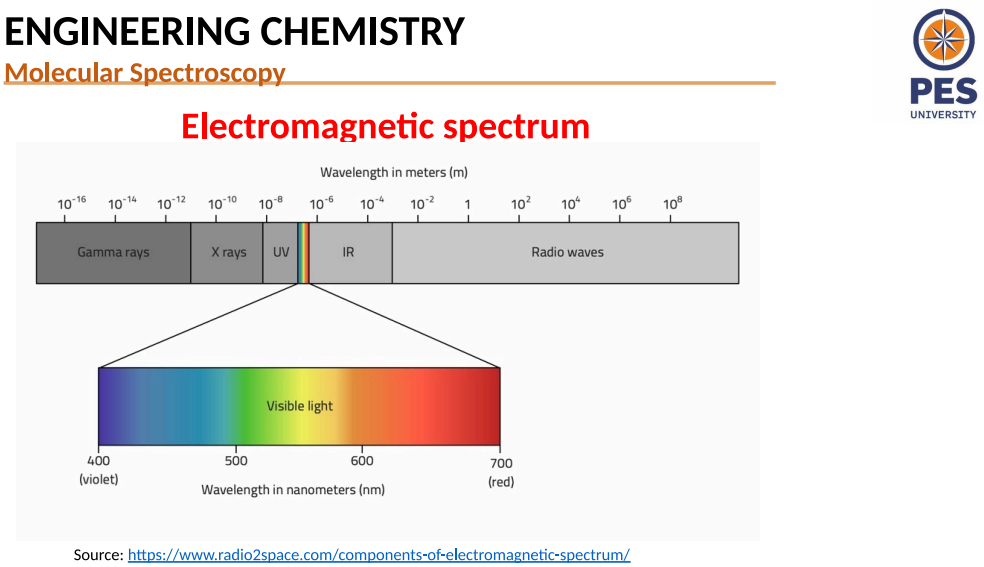

Source: https://www.radio2space.com/components-of-electromagnetic-spectrum/

Figure G000014. Electromagnetic spectrum · source slide page 3.

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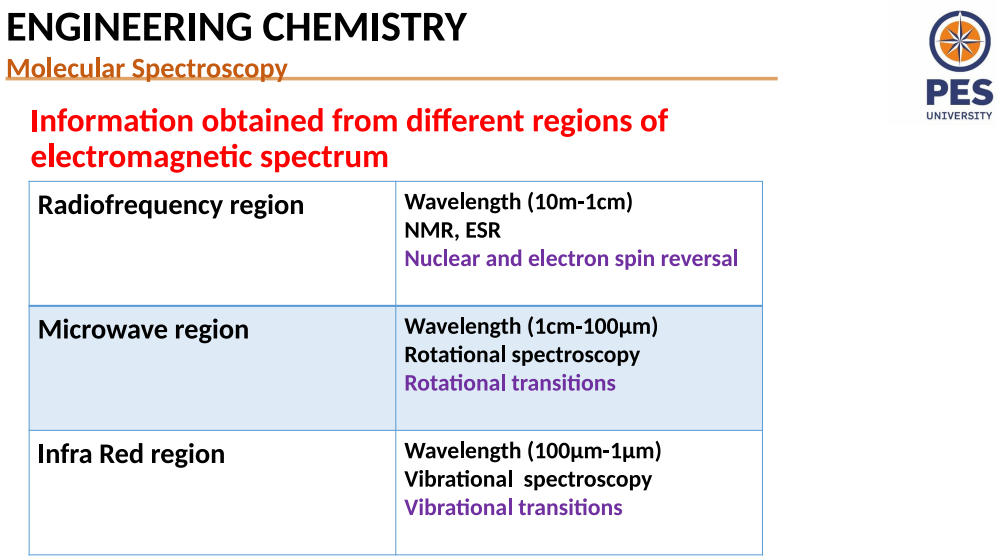

Figure G000015. Electromagnetic spectrum · source slide page 4.

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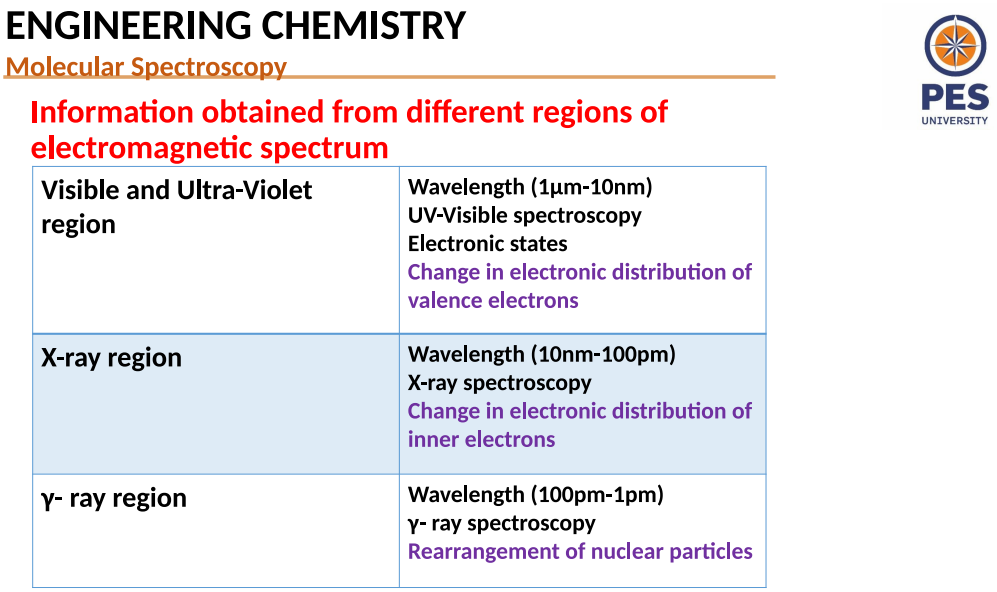

Figure G000016. Electromagnetic spectrum · source slide page 5.

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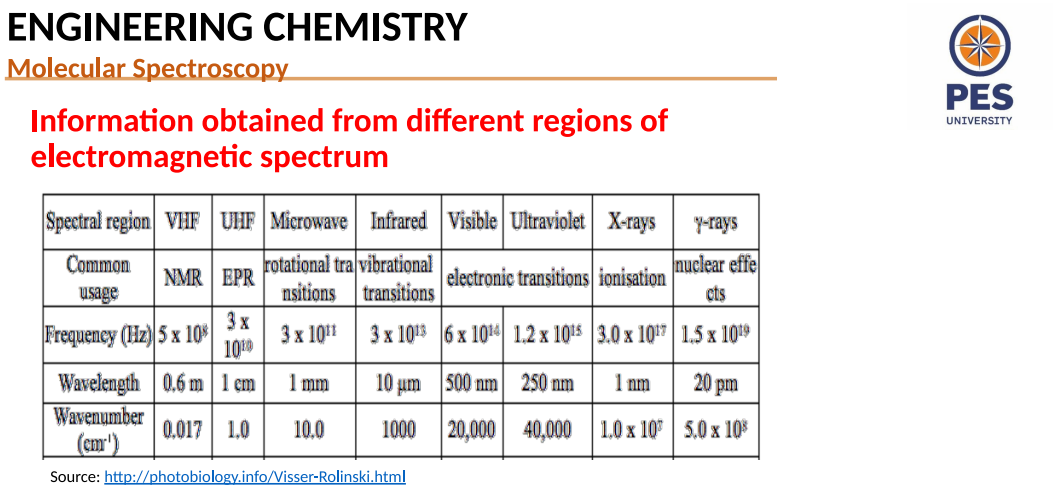

Figure G000017. Electromagnetic spectrum · source slide page 6.

EV0000307

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Figure G000018. Electromagnetic spectrum · source slide page 7.

EV0000310

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000291 · EV0000292 · EV0000293
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Class 2 Content : Electromagnetic spectrum Information obtained from each region of the spectrum EV0000294 · EV0000295 · EV0000296
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Electromagnetic spectrum [Image text, OCR] Wavelengthinmeters(m) 10-16 10-14 10-12 10-10 10-8 10-6 10-4 10-2 1 10² 104 106 108 Gammarays Xrays UV IR Radiowaves Visiblelight 400 500 600 700 (violet) (red) Wavelengthin nanometers(nm) Source: https://www.radio2space.com/components-of-electromagnetic-spectrum/ EV0000297 · EV0000298 · EV0000299
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Information obtained from different regions of electromagnetic spectrum Radiofrequency region | Wavelength (10m-1cm) NMR, ESR Nuclear and electron spin reversal Microwave region | Wavelength (1cm-100μm) Rotational spectroscopy Rotational transitions Infra Red region | Wavelength (100μm-1μm) Vibrational spectroscopy Vibrational transitions EV0000300 · EV0000301 · EV0000302
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Information obtained from different regions of electromagnetic spectrum Visible and Ultra-Violet region | Wavelength (1μm-10nm) UV-Visible spectroscopy Electronic states Change in electronic distribution of valence electrons X-ray region | Wavelength (10nm-100pm) X-ray spectroscopy Change in electronic distribution of inner electrons γ- ray region | Wavelength (100pm-1pm) γ- ray spectroscopy Rearrangement of nuclear particles EV0000303 · EV0000304 · EV0000305
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Information obtained from different regions of electromagnetic spectrum [Image text, OCR] Spectral region VHF UHF Microwave Infrared Visible Ultraviolet X-rays Y-rays Common NMR EPR rotationaltra vibrational electronictransitions ionisation nucleareffe usage nsitions transitions cts Frequency (Hz)5x 108 3x 1010 3 x 1011 3 x 1013 6 x1014 1.2 x1015 3.0 x 1017 1.5 x 1019 Wavelength 0.6 m 1 cm 1 mm 10 μm 500 nm 250 nm 1 nm 20 pm Wavenumber (cm-') 0.017 1.0 10.0 1000 20,000 40,000 1.0 x 107 5.0 x 108 Singlephoton 2.07 x 1.24 x energy (eV) 106 104 1.24 x 103 1.24 x 10-1 2.5 5.0 1.24 x 103 6.2 x 10 Photonenergy 2.03 x 1.20 x 1.20 x 10-1 12.0 239 479 (kJ mol') 104 10 2 1.2 x 10s 6 x 106 Source: http://photobiology.info/Visser-Rolinski.html EV0000306 · EV0000307 · EV0000308
7. Source slide page 7 (extracted text): THANK YOU Department of Science and Humanities EV0000309 · EV0000310 · EV0000311

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.412] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module one on molecular spectroscopy and in today's class we are going to look at electromagnetic spectrum and the information obtained from each region of the spectrum. [00:00:20.804] Light is a form of electromagnetic radiation, which is divided into different regions based on the frequency of the

radiation. Though these demarcations are made, the boundaries are not very precise. [00:00:38.532] Now when the entire range of these energies which comprise the electromagnetic radiation are presented as we can see in the picture here, this is referred to as the electromagnetic spectrum. [00:00:58.404] So we can see there are these low energy beam, that is the radio waves. Then we have infrared radiation, then we have the visible region, then we go on to ultraviolet region. [00:01:12.804] Then more energetic X rays and finally the gamma rays which are high energetic. [00:01:21.156] Now since we are talking of interaction of radiation with matter, each of these regions is going to interact with matter in a different manner, and we will get a lot of information from each of these regions. [00:01:42.372] Let us see what is the information we obtain from the different regions of the electromagnetic spectrum. First, the lowest energy beam, that is the radio frequency region. [00:01:57.124] The wavelength is ten meters to one centimeter. That clearly conveys that the energy of this radiation is very low. Now what is the kind of study we do in this region? [00:02:14.080] It is known as NMR and ESR. NMR stands for nuclear magnetic resonance and ESR stands for electron spin resonance. Electron spin resonance is also referred to as electron paramagnetic resonance sometimes. [00:02:31.424] So NMR and ESR spectroscopy is what we are going to study using the radio frequency uh beam of light. [00:02:43.876] What exactly do uh happens when radio frequency region light falls onto the sample. Now there is going to be a nuclear or spin electron spin reversal. [00:03:00.160] In the case of NMR that is called that is nuclear magnetic resonance there's going to be nuclear spin reversal, whereas in the case of ESR there's going to be electron spin reversal. Now let us briefly understand what do we mean by the spin reversal. [00:03:18.532] We will start with electron spin because that is something you already understand that we can have two spins that is plus half and minus half. And both of them occupy the same orbital. [00:03:37.380] So that means they are degenerate. [00:03:40.996] Both these spins, plus and minus half, are degenerate and they occupy the the same orbital. But in the presence of magnetic field, what happens is these will split into [00:03:58.352] two levels one will be the lower level that is say minus half and the other one will be plus half. What is the difference? Say in one case it is the spin is clockwise and in the other case the spin is anti-clockwise. So [00:04:18.020] When you have a magnetic field, when these electrons are placed in a magnetic field, we find that while the the energy levels now split into one with lower energy and one with higher energy, it is basically [00:04:34.464] How did this happen? Because the electron is spinning and it interacts with the magnetic field. One could be aligned with the magnetic field, the other could be uh uh opposite in direction to the magnetic field. Therefore, we get two energy levels now. [00:04:53.892] Now if a radiation falls on this atom, say we are talking about atom.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:01.444] When a radiation falls on this atom, then we will find that if the energy gap is equivalent to the frequency of the radiation which is falling on this atom. [00:05:16.608] then there will be a transition of the atom from one energy level to the next energy level. So we will find that there is this kind of transition which happens. [00:05:29.472] So what will happen to the atom now? It is going to absorb this radiation and move from one level to the ne to from the lower level to the [00:05:40.576] uh higher energy level. And what has happened because of this movement? There is a spin reversal. So that's exactly what we mean by electron spin reversal. And why are we calling it as a resonance phenomenon? Because whenever [00:05:56.832] The energy gap. The energy gap means the frequency matches with the frequency of the radiation we know resonance will happen and the atom now picks up this

energy and goes from the lower energy level to the higher energy level. So this is [00:06:13.616] what is known as the electron spin resonance. [00:06:18.724] Now the same thing happens for NMR also. So here the nuclei also have spins. Now the nuclei are spinning and whenever there is a charge spinning [00:06:35.076] spinning charge then we know that a magnetic field is created. [00:06:40.356] Now these spins say plus half and minus half again in the case of nuclei again are degenerate. But when there is a magnetic field present, then there is splitting, just like it happened in the electron. [00:06:55.392] For the electron, the same phenomena happens for nuclei also. The spins are now you have one where it is aligned with the magnetic field and the other one where it is uh opposite to the uh magnetic field. [00:07:11.168] So direction of the magnetic field. And now we have two energy levels. When radiation is made to fall on these nuclei, then they are going whenever the energy gap matches with the incident. [00:07:25.888] uh light frequency, incident radiation frequency, the atom now the nuclei are going to absorb that energy. There will be a resonance phenomenon. The nuclei will absorb energy and go to the next level. So this [00:07:44.356] is known as nuclear magnetic resonance. Again what happened? What we observe here is nuclear spin reversal. In the case of nuclei this would be plus my plus half and this would be minus half. Okay. [00:08:00.176] So now that is what we understand by nuclear magnetic resonance and electron spin resonance and the radiation required for such a spin reversal happens to be falling in the radio frequency region. So what does it tell us? [00:08:16.432] that the energy gap is not very large. Therefore, a low frequency beam like the radio frequency region [00:08:26.436] is sufficient to cause cause the spin reversal of either the nuclei or the electron based on uh what is the kind of spectroscopy we are uh studying. [00:08:40.004] Now let us come to the next region that is the microwave region. Now here the wavelength is 1 centimeter to 100 micrometers means it is slightly more energetic beam and the spectroscopy studied in this region is rotational spectroscopy and we talk of [00:08:57.540] rotational levels. We come to know of rotational levels. Now what is this study based on? Now let us see we have an HCl molecule. When the molecule rotates [00:09:12.484] There is a change in the dipole moment. We know that XL is a polar molecule. There is a change in the dipole moment. So as the molecule rotates, there is bound to be the fluctuation in the electric field because of the change in dipole moment. So there's a certain frequency. [00:09:30.256] with which the electric field is uh fluctuating. Now the radiation which we are going to impinge onto the molecule. [00:09:42.116] That also has a fluctuating electric field because it's an electromagnetic radiation, there's a fluctuating electric field and a fluctuating magnetic field. So the electric field is also fluctuating in the radiation. Now when the frequency of this [00:09:57.828] Fluctuating electric field of the molecule due to rotation matches with the fluctuating electric field of the radiation falling on the molecule, resonance phenomena will happen, and the molecule now will move from

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:12.400] one rotational level to the next rotational level. Now why are we calling it as rotational levels? Because it is by virtue of the rotation of the molecule. So this is what we are going to study in rotational spectroscopy. [00:10:29.348] If we have homonuclear molecules like say hydrogen, then we find that we cannot study rotational spectroscopy. Why? Because even when the molecule rotates, there is no rotating electric field, and if you drop radiation. [00:10:48.224] the incident radiation onto this molecule, we will find that there will be no interaction as there is no fluctuating electric field. So homonuclear diatomic molecules are not going to be active in the microwave region. Now let's talk about

infrared region. [00:11:05.200] Now in the infrared region, the wavelength is now becoming slightly uh lesser. So hundred micrometers to one micrometer. And the study we do here is known as vibrational spectroscopy, and now we will learn about vibrational levels. So what happens in this case? [00:11:24.580] When a molecule undergoes vibration, say a diatomic molecule, simple diatomic molecule, when it undergoes vibration, again, say it is stretching. Vibration is like stretching motion or bending motion, we find that there is going to be [00:11:43.568] a change in the dipole moment and again due to this vibrational change there is going to be a [00:11:50.436] fluctuating electric field and just like we studied in rotational spectroscopy in vibrational spectroscopy also there is a change in dipole moment there's a change in the electric field fluctuating electric field when the frequency of radiation impinging onto the sample matches [00:12:08.208] The frequency of the elect fluctuating electric field due to vibration of the molecule, the molecule is going to absorb this radiation, there will be a resonance like phenomena. Molecule will absorb radiation and it will move to the next level. Now, which level will this [00:12:25.776] We call it called as the vibrational levels. Why? Because it is due to the vibrational motion of the molecule. So that is what we study in vibrational spectroscopy. One thing we can see very clearly here that rotational transition requires the [00:12:45.488] Radiation to be in the microwave region and whereas for vibrational transitions the we require infrared region which is much more energetic. So it also tells that the energy levels in the vibration between the energy gap in the [00:13:03.216] vibration levels is much higher. Therefore, it requires infrared region, whereas in the microwave region they are closer as it the transition requires only the uh less energetic beam as compared to infrared. [00:13:18.736] So if you have similarly if you have a uh hydrogen molecule, uh homonuclear diatomic molecule, again vibration is uh like stretching is not going to give us uh any kind of uh any kind of uh [00:13:35.248] change in dichole moment. So when it's a diatomic molecule obviously we have only stretching there cannot be any bending. It's only when you come to triatomic molecules you can notice the bending motion also. So H2 now in this case when it is stretching there is no change in [00:13:52.912] The uh dipole moment, hence no change in electric field. So therefore no fluctuating electric field and there will be no interaction with the uh incident uh radiation. And therefore, we say that the homonuclear diatomic molecules again are not active in the [00:14:13.092] infrared region. Let us get to the next beam of [00:14:21.488] next region that is the visible and the UV region. We know visible comes first and then the UV region, but nevertheless the spectrum with the spectroscopy is referred to as UV visible spectroscopy and the wavelength now has become one micrometer to ten nanometers. [00:14:39.264] So what are we going to study here in UV visible spectroscopy? Here now we are going to look at electronic states. Now in the visible region we know that we can see colours [00:14:54.372] Why? Because we understand that the molecule or the atom is going to absorb energy and

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:02.560] the m atom will go to the uh the uh from the the electron now transits from one atomic level to the next atomic level. Similarly in a molecule it goes from one molecular orbital to the next molecular orbital. I referred to it in the previous class where we talked about Homo to lumo [00:15:20.240] transition so in the case of molecules we have the transition electronic transition happen happening from home to the pneumonia similarly in atomic orbitals also we find that the radiation is absorbed and it goes uh it absorbs radiation goes to the next [00:15:38.916] level and uh energy level and w w whatever is transmitted that is the color you see. So we know that in the visible region [00:15:49.104] There are

electronic transitions happening and that will happen between the different electronic states. So basically we can say that visible and UV region, there is going to be electronic distribution of the valence electrons. [00:16:07.140] We must understand it's only from the Homo to the lumon, that is the highest occupied molecular orbitals in in the case of molecules. That will be the valence electrons, the [00:16:18.576] uh the far uh the higher higher energy electr electrons because they require the minimum energy t the the uh energy is [00:16:28.932] required to move them is much less as compared to the uh one the an electrons which are closer to the nuclei as in the case of atoms and in this case in the molecule also it will move from Homo to lumo. Therefore there is change in electronic disposal [00:16:46.704] Distribution of the valence electrons. So, in case of UV visible region, there is electronic transition happening of the valence electrons, that is from the highest occupied molecular orbitals, or in the case of atom, we call them as the valence electrons. And if [00:17:06.464] U if it happens in the visible region we can see the color. If it's in the UV region, obviously we cannot see the colour. Now coming to X ray region. [00:17:16.320] These are very energetic beams and now the wavelength has come down to picometers, so 10 nanometers to 100 picometers, and this study is known as X-ray spectroscopy. Now here again we will see electronic transitions. [00:17:33.508] But these transitions now will be involving the inner electrons. So what is the difference here? In the case of UV visible, it is the valence electrons which undergo transition and that requires much less energy. Whereas if you want to [00:17:50.416] uh remove the inner electrons then it requires more energy and therefore x-ray region is [00:17:59.300] is what is required the light in the X ray region, the frequency should be high, energy should be more to cause this kind of electronic distribution which involves the inner electrons. Now coming to the last [00:18:16.416] of these regions in the electromagnetic spectrum, the gamma ray region. Now the wavelength here is 100 picometer to one picometer and the spectroscopy is known as the gamma ray spectroscopy. [00:18:31.908] Okay. We are aware that gamma rays are produced whenever there is nuclear reactions happening. We know that nuclei consists of protons and neutrons which are very tightly held. [00:18:47.460] So if there is going to be any kind of rearrangement in the nuclear in the nuclear that is with the protons and neutrons, a lot of energy will be required for such a transformation to take place. So what we study in gamma ray spectroscopy. [00:19:06.944] we have to use very highly energetic beams in order to cause any kind of nuclear uh rearrangement. We know that whenever the radioactive nuclei [00:19:19.088] changes into another nuclei, gamma rays are given out. So here we study the emission spectroscopy. In all the other cases we are concentrating more on the absorption spectroscopy, but here we will study more it's uh the [00:19:36.132] uh emission spectroscopy and based on the gamma rate which is being given out, we can get information about what kind of [00:19:45.808] in nuclear rearrangement happened and what were the nuclei which are getting formed or what is the kind of nuclear reaction which is happening. So these are the various regions of the electromagnet spectrum and we notice that each of these regions

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:03.440] is giving us different kind of information about the atom or the molecule which is being impinged upon by the radiation. [00:20:19.556] One more type of spectroscopy which is very very important is Raman spectroscopy. But it is different from all these like rotational or vibrational spectroscopy. [00:20:34.928] or electronic spectroscopy. Why? Because there we talk about absorption of light, there or emission of light. But here Raman spectroscopy talks about scattering of light. [00:20:48.368] So that is the

major difference between the rotational vibrational spectroscopy and raman spectroscopy. Now what happens when a monochromatic radiation is passed through a transparent medium? [00:21:05.476] Now we know that either light could get absorbed [00:21:10.468] or it could get transmitted. So if we have any kind of uh molecule here, say this is our uh molecule, when we have this incident radiation, we find that either absorption of radiation will happen or [00:21:28.612] Radiation will get transmitted. Most of the radiation we expect apart from what is absorbed is going to be transmitted. So say this is the incident radiation and this is the radiation. [00:21:40.464] transmitted radiation, but some amount of radiation there will be a change in direction and this is referred to as the scattered radiation. [00:21:51.268] Normally in transparent medium we expect very little scattering, but nevertheless there is a possibility that scattering will happen. Now most of the scattering scattered radiation [00:22:05.988] is going to have the wavelength same as whatever is the incident radiation. Whatever incident radiation is there, the same wavelength will be seen most of the time for the scattered radiation and such a scattering is referred to as [00:22:22.704] Ra uh r rarely scattering. But Rama noticed that some of the scattered radiation consists of radiation of a different wavelength from that of the incident wavelength. [00:22:37.216] So this was referred to as this is referred to as the Raman scattering. So you can see here, suppose we have a multi atomic molecule like this, this is the wavelength of the in uh which is incident, and we find that [00:22:51.600] Most of the times the scattering which happens is going to be rally scattering where the incident wavelength and the scattered uh the incident light wavelength and the scattered light wavelength has the uh has is same. But [00:23:06.896] There will be some radiations which we find that the wavelength is more. [00:23:14.176] of the scattered radiation which we refer to as the uh Stokes Raman scattering or they uh when you see in the spectrum they are called as Stokes lines and and some of the radiation we will find will have a smaller wavelength and those are referred to as the anti scale [00:23:31.856] Stokes, Raman, scattering, and in the spectrum they are referred to as the anti-stokes lines. [00:23:38.596] Let us see what are these Stokes and Antistokes like now we can e we can explain this quantum mechanically if you consider light to be made up of photons. And you know that these photons carry [00:23:53.124] E is equal to H nu. We know that E is equal to H nu. So they carry this energy in the form of packets. So if H nu naught is the incident radiation, then [00:24:09.476] these fo these uh photons now what happens is the molecule is going to absorb this radiation go to what we call as a virtual state [00:24:21.444] And again it cannot remain there because it's an unstable state from the ground state. From the ground state it goes to a virtual state and then it [00:24:30.816] comes back with the same frequency of radiation which is referred to as elastic scattering. So it absorbed the radiation and then again it came back with the frequency. [00:24:42.080] Uh same kind the same uh frequency of radiation and this is referred to as valley scattering or this is elastic in nature, elastic scattering. But there could be what Raman uh observed, but there were some radiations which were inelastic scatter were inelastic. [00:24:59.728] scattering was happening. Now in this case what happens from the ground state it goes to the virtual state but when it comes back it is coming back to a higher level it doesn't go back to the

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Transcript block 6 (source interval 001500-001799 seconds): [00:25:12.416] uh initial ground state it ends up at a higher level and therefore now the frequency is much less. [00:25:23.524] So the wavelength is going to be higher. So this will be referred to as stoke scattering. So where is stoke scattering and the lines which you see are referred to as stokes line. So here it has absorbed energy,

but it has come back to the [00:25:40.868] See uh to a higher level, therefore the energy of this beam is much less. It is now you have to subtract this much of energy. [00:25:50.148] Now uh again in what happened there could be another possibility where it absorbs energy goes to uh a different level. Now here what you observe that it is starting from a higher level. It is not at the ground level, it is starting at a higher level, it goes up [00:26:07.632] But when it comes back, then there the it comes back to the ground state. That means now the energy is more. You can see the energy gap, it is much more and therefore now [00:26:21.188] the beam has more energy or smaller wavelength, and this is known as anti stokes scattering. And we find that [00:26:31.408] Whenever this kind of scattering happens, inelastic scattering, there could be you will see stokes lines as well as anti-stokes lines. Of course, it is found that the stokes lines are much more intense as compared to the anti-stokes lines. That is because the population of the [00:26:49.168] ground state will be any time more than that of the higher level. So anti-stokes lines will appear only if there are molecules in this higher level they go absorb energy and then come back to the initial ground state and therefore their energy they are much more [00:27:06.736] So basically what is happening in this, there is exchange of energy between the molecules and radiation while the scattering is taking place. [00:27:18.896] And one advantage of Raman spectroscopy is that homonuclear diatomic molecules I mentioned just now will not show any uh absorption in the microwave or the [00:27:34.016] IR region. So they are microwave and IR inactive, but all these molecules can be Raman active. Now uh if we can record the rotational Raman spectrum or the vibrational Raman spectrum. [00:27:50.928] all the information which we couldn't get from the pure rotational or pure vibrational spectroscopy we can obtain from Raman spectroscopy. [00:28:01.668] So this table summarizes all the different regions and the kind of transitions which are taking place. And of course we can see as we go from one region to the other, the frequency is increasing, the wave number is decreasing. [00:28:19.440] Length is decreasing and the wave number is increasing. So this is about [00:28:27.264] the electromagnetic spectrum and the different regions the information which we get from different regions. So in the next class we are going to talk about Born Oppenheimer approximation and then we will also move to rotational spectroscopy. [00:28:47.620] Thank you.

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Full transcript: 3843 normalized words, 118 timestamped segments, 1728.682667 seconds duration. Low-confidence segments flagged: 0.

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4 Rotational spectroscopy

This lesson is indexed from the source video titled Rotational spectroscopy. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000046 · EV0000047

Graphs and diagrams

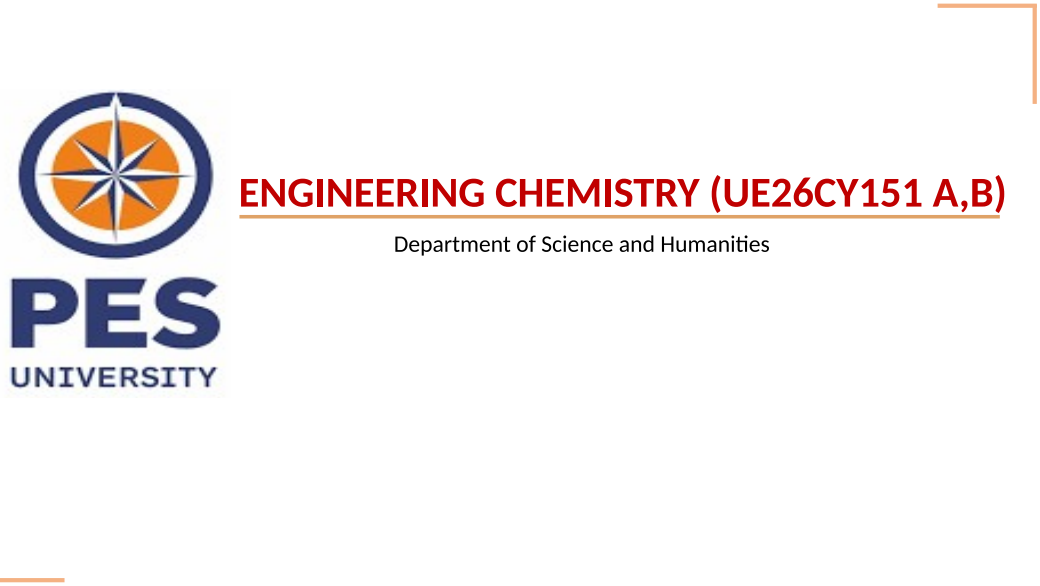

Figure G000025. Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum · source slide page 1.

EV0000331

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000026. Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum · source slide page 2.

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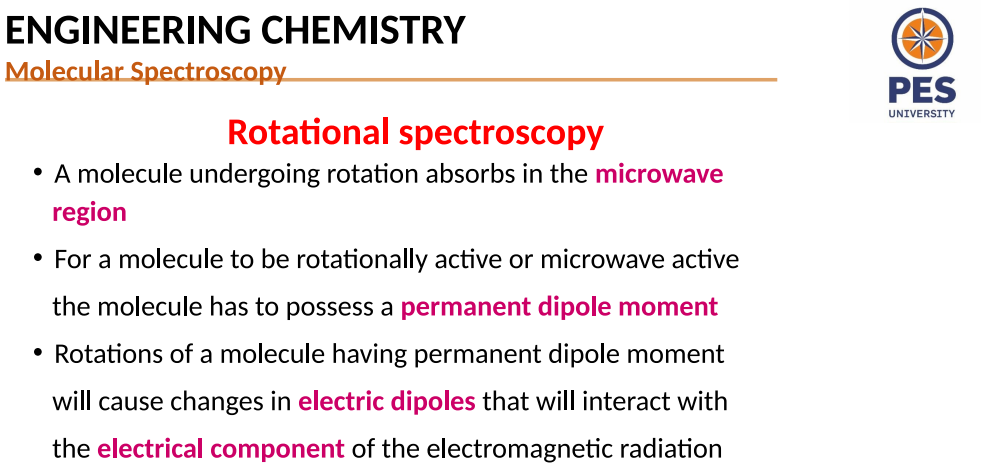

Figure G000027. Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum · source slide page 3.

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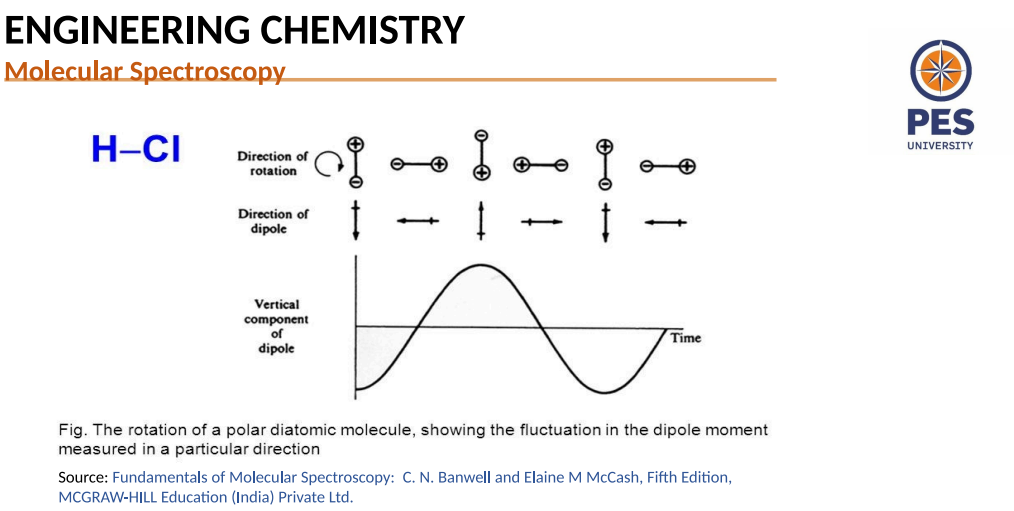

Figure G000028. Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum · source slide page 4.

EV0000340

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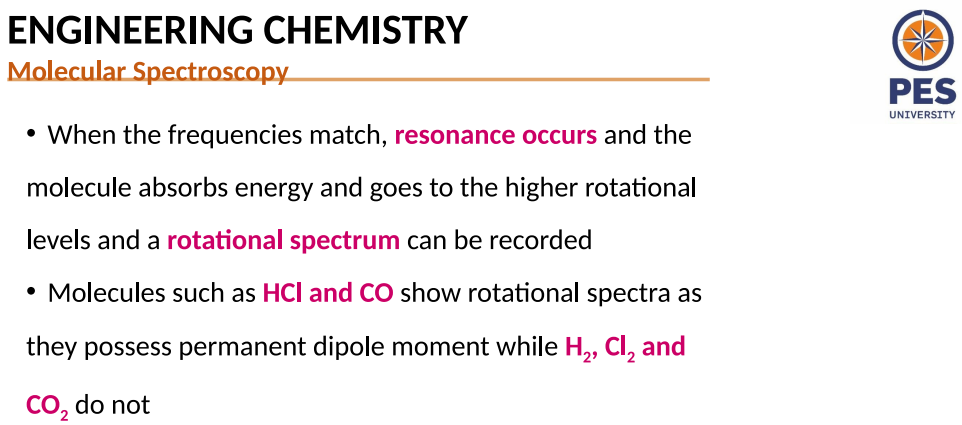

Figure G000029. Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum · source slide page 5.

EV0000343

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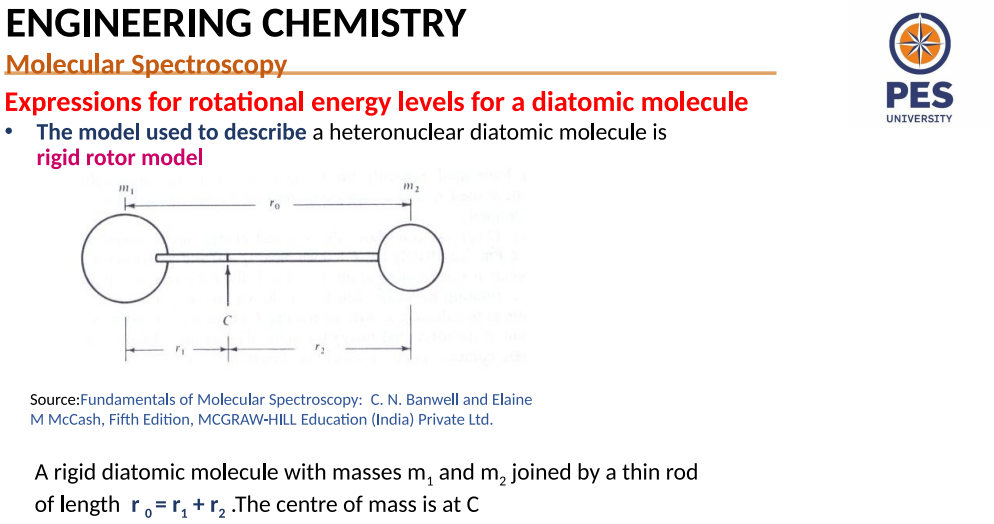

Figure G000030. Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum · source slide page 6.

EV0000346

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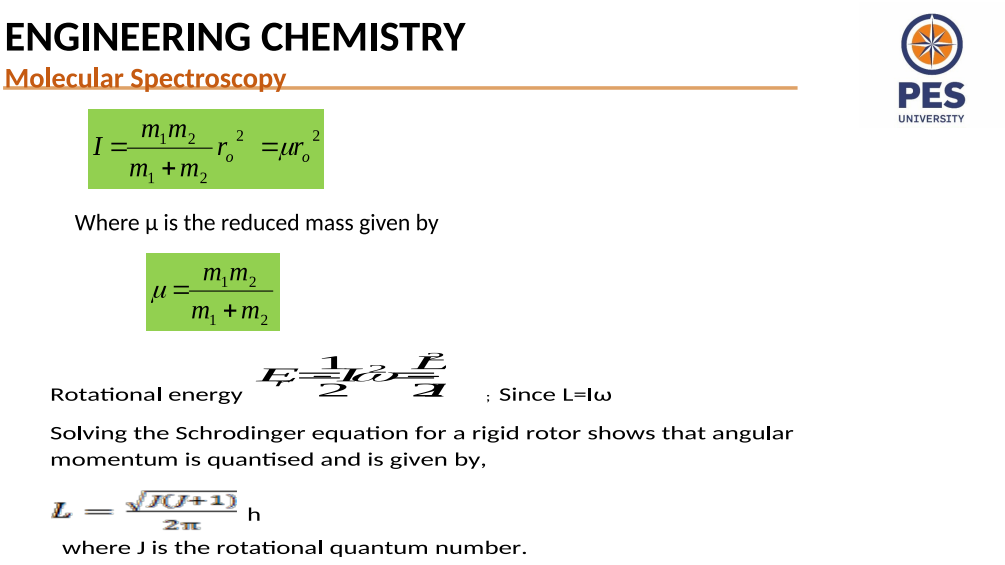

Figure G000031. Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum · source slide page 7.

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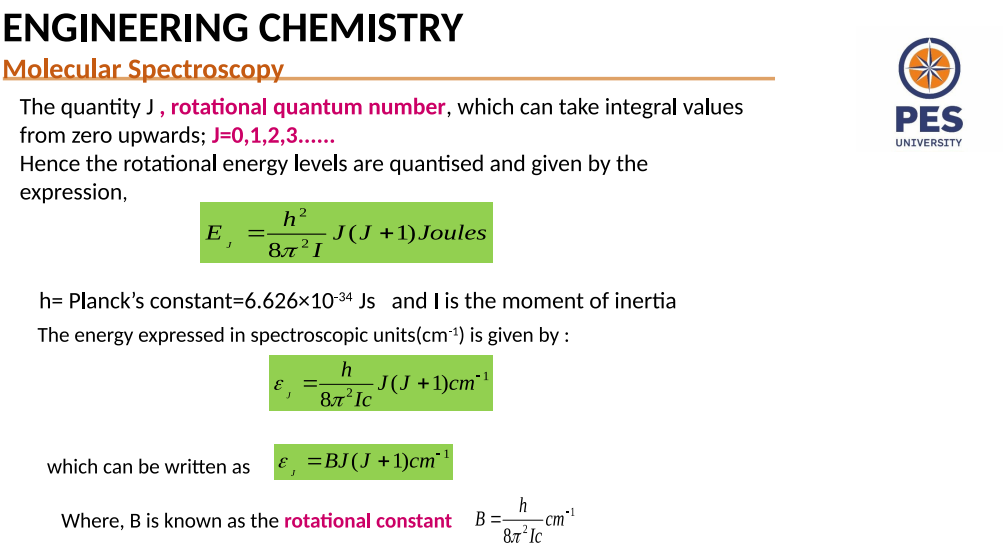

Figure G000032. Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum · source slide page 8.

EV0000352

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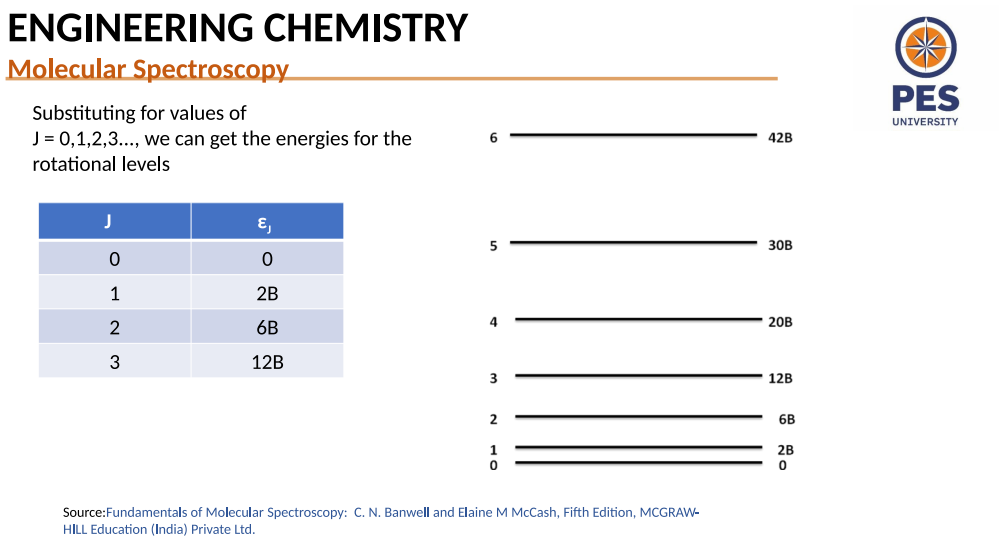

Figure G000033. Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum · source slide page 9.

EV0000355

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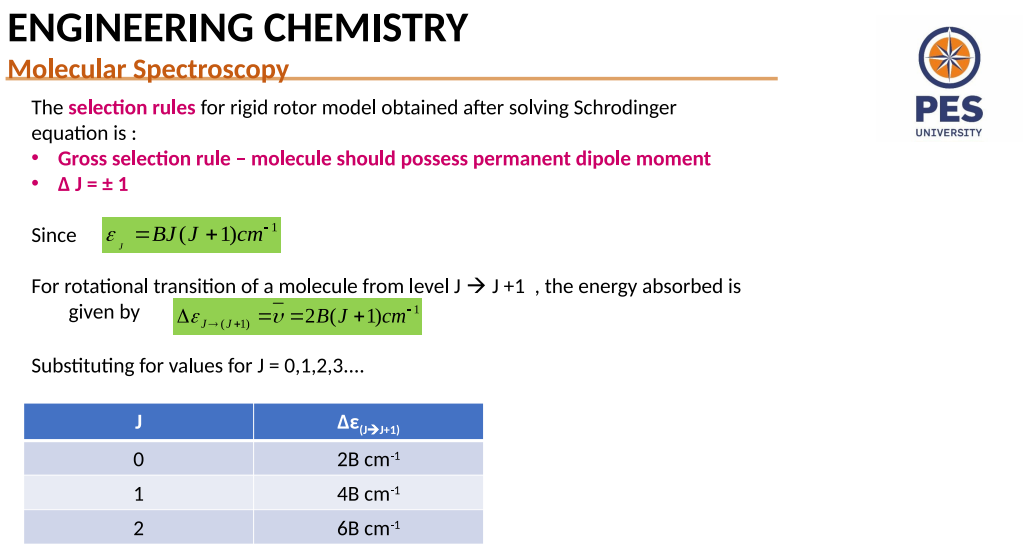

Figure G000034. Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum · source slide page 10.

EV0000358

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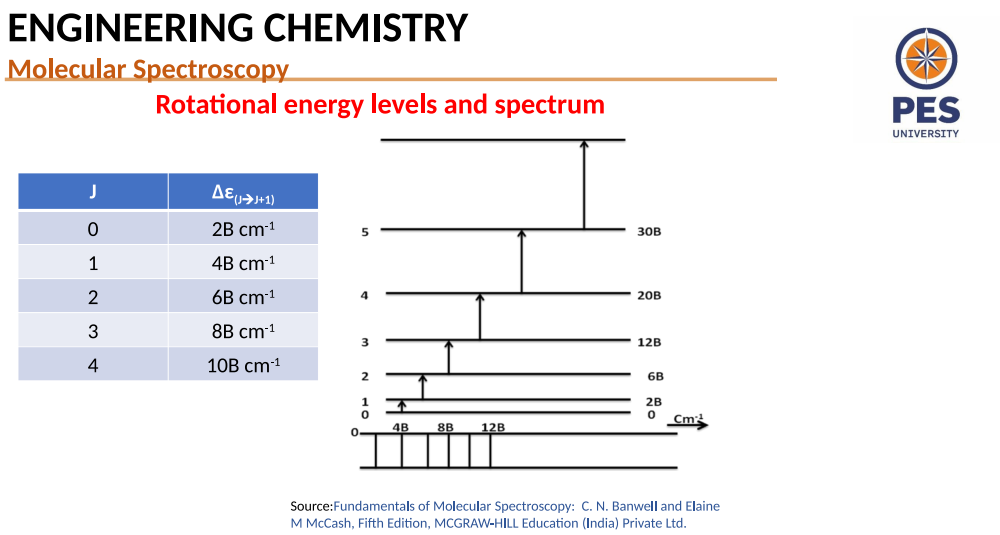

Figure G000035. Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum · source slide page 11.

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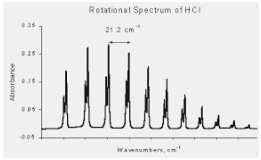
ENGINEERING CHEMISTRY

Molecular Spectroscopy

Information obtained from the rotational spectrum



Source:http://www.physics.dcu.ie/~be/Ps415/Rotational1.pdf

- The **first line in the spectrum** appears at 2B cm^{-1} and the **distance between any two consecutive lines** is constant and is equal to 2B cm^{-1} . We can get value of 'B' from the spectrum and calculate ,I, the moment of inertia using the expression $B = \frac{h}{8\pi^2 Ic} cm^{-1}$

  Since $I = \mu r_o^2$, r_o can be determined ; $r_e = \sqrt{\frac{I}{\mu}}$

  r_o is the **bond length** of the molecule

- The spectrum also reveals that **some higher rotational levels are also populated at room temperature**

Figure G000036. Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum · source slide page 12.

EV0000364

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000037. Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum · source slide page 13.

EV0000367

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000330 · EV0000331 · EV0000332
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Class 4 content: Microwave Spectroscopy – Diatomic rigid rotar model and rotational spectrum EV0000333 · EV0000334 · EV0000335
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Rotational spectroscopy A molecule undergoing rotation absorbs in the microwave region For a molecule to be rotationally active or microwave active the molecule has to possess a permanent dipole moment Rotations of a molecule having permanent dipole moment will cause changes in electric dipoles that will interact with the electrical component of the electromagnetic radiation EV0000336 · EV0000337 · EV0000338

4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy [Image text, OCR] Absorption of ElectromagneticRadiation-The Coupling Mechanism Anelectromagneticwaveis anoscillating electricalfield and interactsonlywithmolecules that can undergoa change in dipole moment The oscillating dipole can beprovided by the rotationof apermanent dipole like for example HCl.Thistypeofinteractionleadstomicrowavespectra H-CI Directionof rotation Directionof dipole Vertical component of Time dipole Fig.Therotationofapolardiatomicmolecule,showing thefluctuationinthedipolemoment measuredinaparticulardirection Source: Fundamentals of Molecular Spectroscopy: C. N. Banwell and Elaine M McCash, Fifth Edition, MCGRAW-HILL Education (India) Private Ltd. EV0000339 · EV0000340 · EV0000341

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy When the frequencies match, resonance occurs and the molecule absorbs energy and goes to the higher rotational levels and a rotational spectrum can be recorded Molecules such as HCl and CO show rotational spectra as they possess permanent dipole moment while H2, Cl2 and CO2 do not EV0000342 · EV0000343 · EV0000344

6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Expressions for rotational energy levels for a diatomic molecule The model used to describe a heteronuclear diatomic molecule is rigid rotor model [Image text, OCR] /111 12 ro Source:Fundamentals of Molecular Spectroscopy: C. N. Banwell and Elaine M McCash, Fifth Edition, MCGRAW-HILL Education (India) Private Ltd. A rigid diatomic molecule with masses m1 and m2 joined by a thin rod of length r 0 = r1 + r2 .The centre of mass is at C EV0000345 · EV0000346 · EV0000347

7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Where μ is the reduced mass given by EV0000348 · EV0000349 · EV0000350

8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy The quantity J , rotational quantum number, which can take integral values from zero upwards; J=0,1,2,3..... Hence the rotational energy levels are quantised and given by the expression, h= Planck's constant=6.626×10-34 Js and I is the moment of inertia The energy expressed in spectroscopic units(cm-1) is given by : which can be written as Where, B is known as the rotational constant EV0000351 · EV0000352 · EV0000353

9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Substituting for values of J = 0,1,2,3..., we can get the energies for the rotational levels J | εJ 0 | 0 1 | 2B 2 | 6B 3 | 12B Source:Fundamentals of Molecular Spectroscopy: C. N. Banwell and Elaine M McCash, Fifth Edition, MCGRAW-HILL Education (India) Private Ltd. EV0000354 · EV0000355 · EV0000356

10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy The selection rules for rigid rotor model obtained after solving Schrodinger equation is : Gross selection rule – molecule should possess permanent dipole moment Δ J = ± 1 Since For rotational transition of a molecule from level J ☒ J +1 , the energy absorbed is given by Substituting for values for J = 0,1,2,3.... J | Δε(J☒J+1) 0 | 2B cm-1 1 | 4B cm-1 2 | 6B cm-1 EV0000357 · EV0000358 · EV0000359

11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Rotational energy levels and spectrum J | Δε(J⊠J+1) 0 | 2B cm-1 1 | 4B cm-1 2 | 6B cm-1 3 | 8B cm-1 4 | 10B cm-1 Source:Fundamentals of Molecular Spectroscopy: C. N. Banwell and Elaine M McCash, Fifth Edition, MCGRAW-HILL Education (India) Private Ltd. EV0000360 · EV0000361 · EV0000362
12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Information obtained from the rotational spectrum Source:http://www.physics.dcu.ie/~be/Ps415/Rotational1.pdf The first line in the spectrum appears at 2B cm-1 and the distance between any two consecutive lines is constant and is equal to 2B cm-1 . We can get value of 'B' from the spectrum and calculate ,I, the moment of inertia using the expression Since I = μ ro2 , ro can be determined ; ro is the bond length of the molecule The spectrum also reveals that some higher rotational levels are also populated at room temperature EV0000363 · EV0000364 · EV0000365
13. Source slide page 13 (extracted text): THANK YOU Department of Science and Humanities EV0000366 · EV0000367 · EV0000368

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.508] Hello everyone, welcome back to this course on engineering chemistry. We continue with module one on molecular spectroscopy and in today's class we are going to derive the expression for rotational energy levels of a diatomic molecule. [00:00:19.428] We will study the selection rules and finally we will arrive at the rotational spectrum. [00:00:29.380] So let us get to the expressions of rotational energy levels for a diatomic molecule, because that is the simplest molecule. In the previous class, we talked about [00:00:42.564] rotational spectroscopy and how we have different energy levels in the molecule when there is a change in dipole moment as the molecule rotates. [00:00:58.788] So hetero nuclear diatomic molecules we say will be microwave active as they are going to transit from one rotation level to another rotation level when there is radiation falling on the molecule. [00:01:19.492] So let us take the case of a diatomic molecule and heteronuclear diatomic molecule of course because homonuclear diatomic molecules will not be microwave active. [00:01:33.060] We need a model in order to understand how the molecule is going to behave, and we can see here what we are going to use is a rigid rotor molecule. So what is a rigid rotor model? We have [00:01:47.780] two atoms heteronuclear so the masses are different and they are held together by a rigid rod and that is having a center of gravity at the point C [00:02:01.060] So if R0 is the distance between the centers, then this distance is R1 and this distance is R2. So this is the model where we have digit rotor model. So here we have two atoms. [00:02:17.840] which are held together by a rigid rod, meaning that whenever there is rotation, this rod will not change its form. So firstly let us get to the moment of inertia for such a molecule. Let us see how the derivation goes. [00:02:36.740] Now we have masses M1 and M2 which are joined by a thin rod, so therefore R0 is going to be R1 plus R2. Now the molecule is rotating N over N and about the center of gravity. [00:02:56.036] Therefore, we can write the balancing equation m1r1 is equal to m2r2. [00:03:03.780] Now, since this equation is h holds true, the moment of inertia which is given by I is equal to m1 r1 squared plus m2 r2 squared can be written as [00:03:18.852] M2 R2 R1 plus M1 R1 R2. All we have done is replaced M1 R1 by M2 R2 and here M2 R2 by M1 R1. [00:03:29.828] And what we get is I is equal to R1 R2 M1 plus M two. Now why are we interested in movement of inertia?

Because there is a uh rotation which is happening. So we will be interested in angular terms, angularity and mass is uh [00:03:47.200] moment of inertia when we talk of uh when we refer to in angular terms. Now since m1 r one is equal to m2 r2 we can also write this as m2 r naught minus r1 because we said that r naught is equal to r one plus r2. [00:04:04.640] So when you substitute here, we can rearrange this and get to the expression for R1, that is M2R0 by M1 plus M2. And why do we require this? Because we want to substitute R1 here. [00:04:19.392] So this is the expression, and similarly we want for R2 also we get M1 R1 is equal to M2 R2. So we substitute and we get M1 R0 minus R2. So therefore R2 is equal to M1 R0 by M1 plus M2. When we substitute for R1 and [00:04:37.104] and R2 in this expression for moment of inertia we get this expression i is equal to m1 m2 divided by m1 plus m2 r naught square which can be written as mu r naught square. [00:04:53.216] Now what is mu this is known as the reduced mass. Reduced mass and that is given by m1 m2 by m1 plus m2. So this is the expression we are looking for.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:09.456] So what have we got now? I is equal to mu r naught square. And what is r naught? It is the distance between the two atoms or the two masses which we have considered here. [00:05:28.356] Now coming to energy. Now rotational energy will be given by half of I omega squared. We are still talking in terms of classical mechanics. So [00:05:43.012] Energy is half I omega squared. Omega is the angular velocity. And [00:05:49.252] This can be written as L squared by 2i. Why? Because L is the angular momentum. So angular momentum is given by moment of inertia multiplied by angular velocity. So here we have i omega squared by 2 that can be written as L squared by [00:06:07.024] 2i. So energy expression in terms of angular momentum is n squared by 2i. Now we have to invoke the quantum mechanical treatment. [00:06:20.900] because we are talking of a single molecule and here classical mechanics or classical laws are not applicable to molecules [00:06:33.520] which are so small. So therefore we will use the Schrödinger equation for the rigid rotor, and then we find when we solve the Schrödinger equation that angular momentum is now given by this expression [00:06:51.236] root of j j plus 1 by 2 pi h or we can say root of j j plus 1 h cross which can also be written as h by 2 pi. Now what is this j here? j is known as the rotational quantum number. [00:07:08.068] What does this expression show us that the angular momentums are all quantized. They are not [00:07:18.400] continuous. In the classical case we said L squared by 2i. L can have any value, but here L cannot have all the values. It can take up only those values which J has because [00:07:33.072] Here do j plus 1 by 2 pi h. So therefore the quantity j is known as the rotational quantum number and it can take up integral values 0, 1, 2, 3. So we will get different values of [00:07:49.520] quantum uh the angular momentum based on the value of j which we substitute here and therefore we can see that angular momentum is quantized. [00:08:01.124] So we can also say based on this that the energy levels are context. How we can substitute for L in this expression for energy, and what we obtain is EJ. [00:08:13.904] is equal to h squared by 8 pi squared i j g plus 1 joules. This energy is expressed in terms of joules and what is h here it is the Planck's constant 6.626 into 10 power minus 34 joules second [00:08:31.504] And I is the moment of inertia of the [00:08:35.908] Molecule. So we get this expression for energy, and we can see very well that the energy is also quantum. Which energy are we talking about? The rotational energy levels inside the molecules is R [00:08:53.312] context. Why? Because they cannot have any value but those which are decided by the value of the rotational quantum number that is J. And we know that J can take up only values of 0, 1, 2, 3. [00:09:10.532] Let us see what these energy levels look like. So we got

an expression in terms of joues. Now we know that in spectroscopy we use only spectroscopic units, that is in terms of centimeter inverse. If I want to convert the energy into [00:09:28.228] wave number we need to divide by H C. So when you do that, this is the expression. That was H squared by 8 pi squared I, so it becomes H by 8 pi squared I C, J times J plus 1 centimeter inverse. Now this is a constant [00:09:45.476] for a particular molecule because H pi and C are constant C is the [00:09:51.648] uh speed of light and uh that is 3 into 10 to the power of 8 meters per second and i is the moment of inertia of the molecule which is again constant. Therefore this can be written in the form of a constant b. So what happens to your expression now it can be written as b times j plus

EV0000050 · EV0000051 · EV0000052 · EV0000053

Transcript block 3 (source interval 000600-000899 seconds): [00:10:09.296] 1 centimeter inverse. And what is the value of B is H over A pi squared I C centimeter inverse and B is known as the rotational constant. Very important. So rotational quantum number is J and B is [00:10:26.864] Is known as the rotational constant and it's expressed in centimeter inverse, it will be h by a pi squared i c. [00:10:35.044] Now that we have an expression for energy levels in centimeter inverse, let us substitute various values of J and see where these energy levels will come up. And these are rotational energy levels because we consider that the molecule is rotating. [00:10:54.240] And that's how we arrived at the expression for the rotational energy levels. So we have when you substitute J is equal to 0 here, we get energy as 0, and we can mark it in this energy level diagram. This is the energy scale. [00:11:11.492] And we have 0 is uh at g is equal to 0, we have energy as 0. When we substitute j is equal to 1, we get 2b. And that's what is shown here. And when you get go to j is equal to 2, we get 6b. Now [00:11:27.088] We can see this is 0, 2b, 6b and so on. We can go to the next level that is 12b. So what do we observe here as we go to higher energy levels, the energy gap, the spacing between the energy levels is going [00:11:44.752] on increasing. So this is how we get [00:11:49.040] to know where are these energy levels inside the molecules. If you want to find out for any energy level, all you have to know is the value of B. So we substitute the value of B and we'll know what is the energy of that particular level. So we got [00:12:06.704] an expression for the energy levels, rotational energy levels, which tell us that the energy levels are quantized. When we substitute the values for J 0, 1 to 3 we get the different energy levels. So we got the rotational levels inside the molecule. [00:12:21.252] But just knowing the rotational level is not going to help us get the spectrum because what we want finally or what we get when we do this spectroscopy is the spectrum. Now how do we get to the [00:12:36.624] Spectrum. We have to follow certain rules which are known as the selection rules. If we understand the selection rules, then we will be able to arrive at the spectrum. The selection rule for rigid rotor model. [00:12:51.044] Is obtained after solving the Schrödinger equation. Now there's a cross selection rule which is applicable to any kind of molecule that a molecule will show rotational spectroscopy only if it possesses permanent dipole moment, that is known as the gross selection. [00:13:08.816] But there is another specific selection rule which is [00:13:13.188] employed and which is applicable only for a rigid rotor heteronuclear diatomic molecule and that says delta J is equal to plus minus one. So what does it mean that the rotational quantum number [00:13:31.056] can change only by one. When it is plus one we refer to absorption spectrum. When it is minus one we refer to the uh emission spectrum. So EJ [00:13:45.632] We know is equal to the rotational levels where are they? Bj plus 1 centimeter inverse. So what does it mean? We have these energy levels inside the molecule. Just now we saw. So what does the selection rule mean to us? Delta J is plus minus 1 means that [00:14:03.344] And

we can have a transition zero to one, we can have a transition one to two, we can have a transition two to three, we can have a transition two to three to four, but we cannot have zero to two, two to four. [00:14:19.332] All those transitions are forbidden. Only transitions allowed are between one level and the next level present in the molecule. So delta j is equal to plus minus one. [00:14:31.984] So what is the energy gap? Why am I interested in the energy gap? Because we know that the radiation will be absorbed based on the energy gap. Rotational levels, of course, we know where they are, but more important for us is to know what is the energy gap. So if we want to calculate using this expression, [00:14:50.916] What is the energy gap between J to J plus 1? So we substitute here J plus 1 first minus for so J. So E j plus 1 minus Ej will give us this expression that is delta E and that is nu bar is the

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:08.752] wave number and that will be the expression becomes 2bj plus 1 centimeter inverse. So energy is at B J J plus 1 centimeter inverse the energy gap is 2 B J plus 1 centimeter inverse. Now we substitute the different values of J is equal to [00:15:26.384] to zero one to three and we will know what is the energy gap. So when it goes from zero to one, j is equal to zero, then we find that [00:15:37.024] It is going so J is equal to zero. That's what we mean. So zero to j plus one. That is one. So we see that the energy gap is two b. When we go to from uh one to two, it will be four b. When we go from two to three, it is going to be six b and so on so forth. We can calculate the energy. [00:15:54.672] the energy gap between the energy levels and we see very clearly as we go to higher energy levels the energy gap is going on increasing. [00:16:04.388] So we can see here very clearly these are the energy levels which we have. And if you have a transit and only delta j is equal to plus minus 1 is what is allowed. And here we can see plus 1 is what we are doing here. So 0 to 1 to be [00:16:21.520] Zero uh then one to two, then it's going to be four b and then six b and eight b and ten b so on so forth. So we can see that the energy gap. [00:16:31.968] the energy gap itself is going on increasing. So here it is two B, here it is four B, here it is six B, here it is eight B and ten B, so on. Now we know that [00:16:44.400] what we see in the spectrum. Now if a source of radiation in the radio frequency region is going to fall on the molecule, we will find that [00:16:54.756] Energy will be absorbed. And what energy will be absorbed? That will be corresponding to the energy gap. So that means when radiation falls on this molecule, the first line will appear at [00:17:08.976] 2b because that is the lowest energy gap, and remember this is in centimeter inverse this way. That means wave numbers energy is increasing. So which requires the minimum energy? You can see from the energy level diagram here that this requires minimum energy. [00:17:26.704] Energy and that will correspond to 2B. How? 0, 2, 1. Then we have 1, 2, 2, that would be 4b, and then we have 6b here, and then we have 8b, 10 P and 12B. So what do we observe here that the energy the lines are appearing at [00:17:45.520] 2b, 4b, 6b, 8b, 10b, and so on, so on so forth for the rotational spectrum. So in the spectrum, you're going to see these lines, and what is the observation here? The first line is appearing at 2b, but [00:18:03.044] The difference between these lines is also very clear. It is going to be 2b. So 2b to 4b, then 4b to 6b. So this difference is constant and that is 2b. Let us see the spectrum. [00:18:22.308] of H C. Now it appears like this. So you can see the lines are equidistant. So you will have the first line appearing at 2b. [00:18:33.316] And the distance between any two lines in the spectrum is going to also be equal to 2b, and that remains constant. So what if we record this spectrum? [00:18:47.552] And from here we can calculate what is to be. So if we know what is to be, either we measure the entire

or the first line, the wavelength or the frequency or the wave number of the first line or the distance between any two consecutive lines in the spectrum, then we can determine what [00:19:05.200] is 2b and from 2b we can find out what is b and using the formula for b that is h by 8 pi square i c we can determine what is i [00:19:17.348] and I is given by mu R0 square. Mu is the reduced mass which is fixed for any uh any molecule which is mu is equal to we already know m1 m2 [00:19:31.428] by m1 plus m2. So that will be equal to uh mu is equal mu is known to us. So i is also you have calculated using the [00:19:43.392] uh B value which you got from the spectrum and we can calculate R0 using this expression which is equal to R0 is equal to root of I by mu. And what is R0 here? What was our model? [00:19:57.744] rigid rotor. So we had a large atom and a small atom and R0 is the distance here. So and what is this distance? This is R0 and what is that? Will be for any molecule it is going to be the bond length. So that means

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:15.520] Using I the microwave spectroscopy or rotational spectroscopy, we can determine the bond length of the molecule. So all you have to do is record the rotational spectrum, get the value of [00:20:31.712] to B, then from there we find out B and then finally we get the value of R0 and that is the bond length of the molecule. Now this spectrum also reveals [00:20:47.044] You can see that the intensity of the lines is increasing and then [00:20:53.412] Intensity depends on the population of the energy levels and therefore we can see from this spectrum that it is not just the ground level which is the most populated. Why we are saying that because this refers to zero to one transition, but this is not [00:21:11.184] The most populated level at room temperature, even some of the higher levels are also populated. It is not just the lowest level or the ground state which is the most populated in the case of rotational levels, even some of the higher levels are also populated, as you can see from the [00:21:31.120] intensity of the lines. [00:21:34.352] So this is how we get the rotational spectrum, and the information we get from rotational spectrum is we can calculate the bond length of the molecule. So in the next class we are going to study another type of spectroscopy, which is the vibrational spectroscopy. [00:21:54.084] Thank you.

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Full transcript: 3173 normalized words, 90 timestamped segments, 1315.733333 seconds duration. Low-confidence segments flagged: 0.

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5 Vibrational spectroscopy(harmonic oscillator)

This lesson is indexed from the source video titled Vibrational spectroscopy(harmonic oscillator). Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000066 · EV0000067

Graphs and diagrams

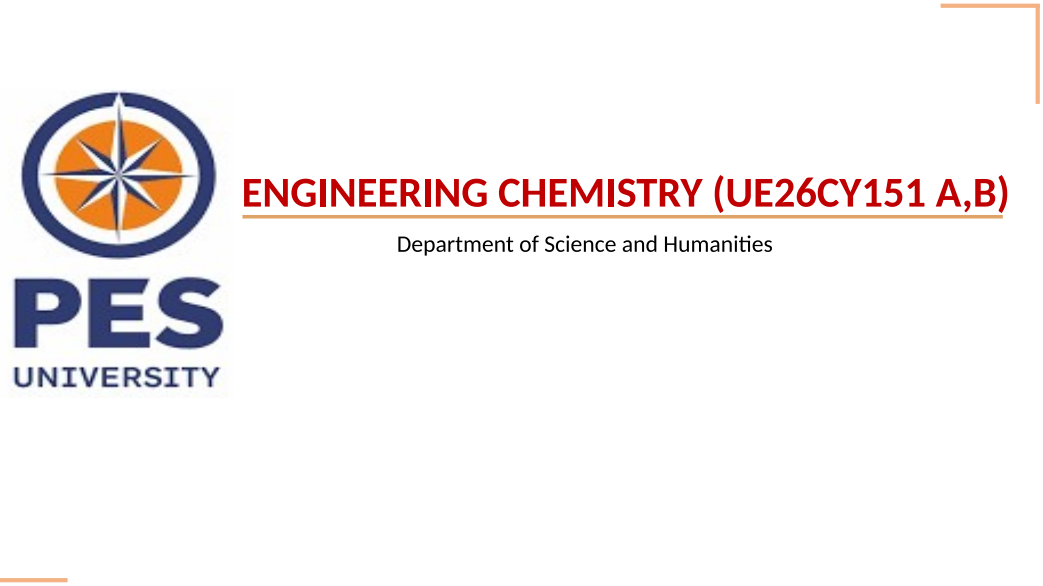

Figure G000038. IR spectroscopy diatomic harmonic oscillator model · source slide page 1.

EV0000370

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Figure G000039. IR spectroscopy diatomic harmonic oscillator model · source slide page 2.

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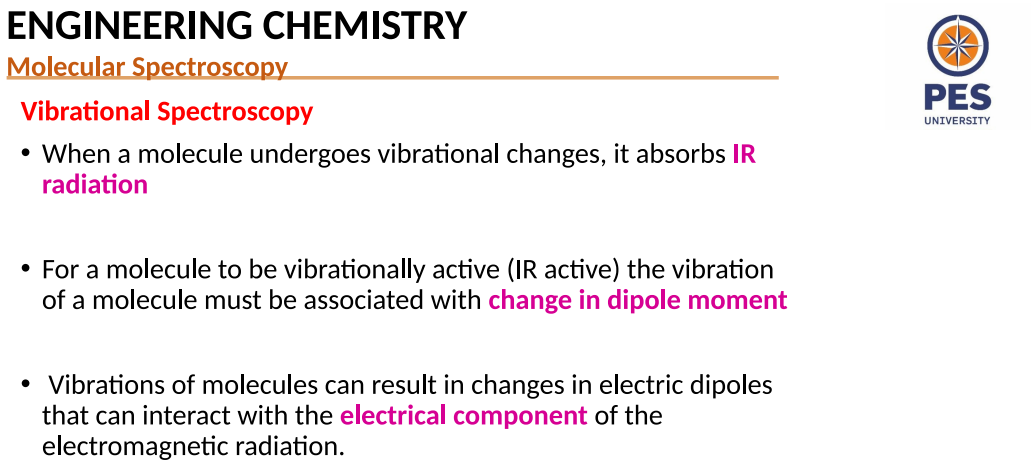

Figure G000040. IR spectroscopy diatomic harmonic oscillator model · source slide page 3.

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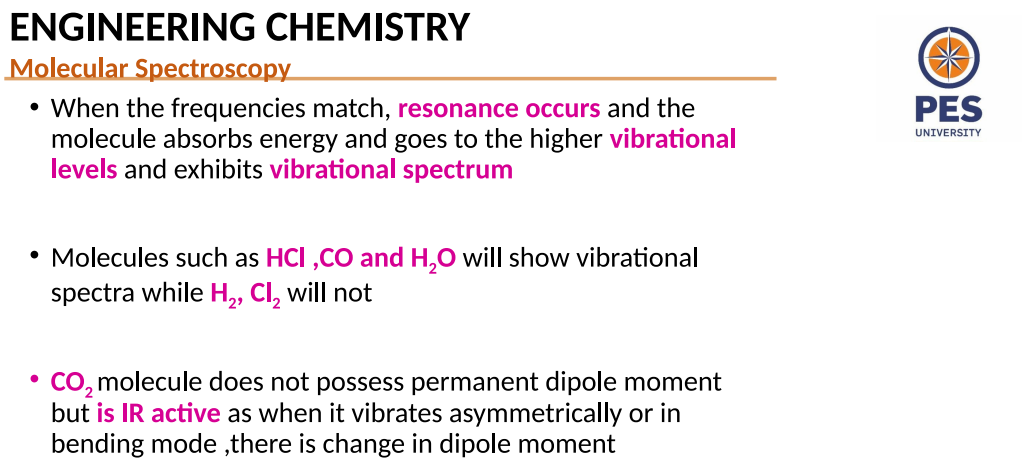

Figure G000041. IR spectroscopy diatomic harmonic oscillator model · source slide page 4.

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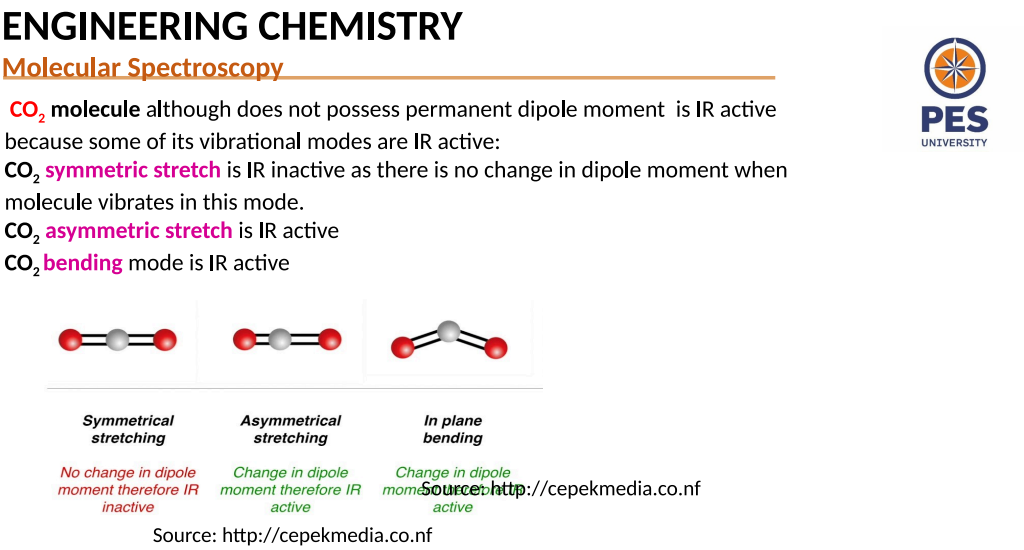

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Figure G000042. IR spectroscopy diatomic harmonic oscillator model · source slide page 5.

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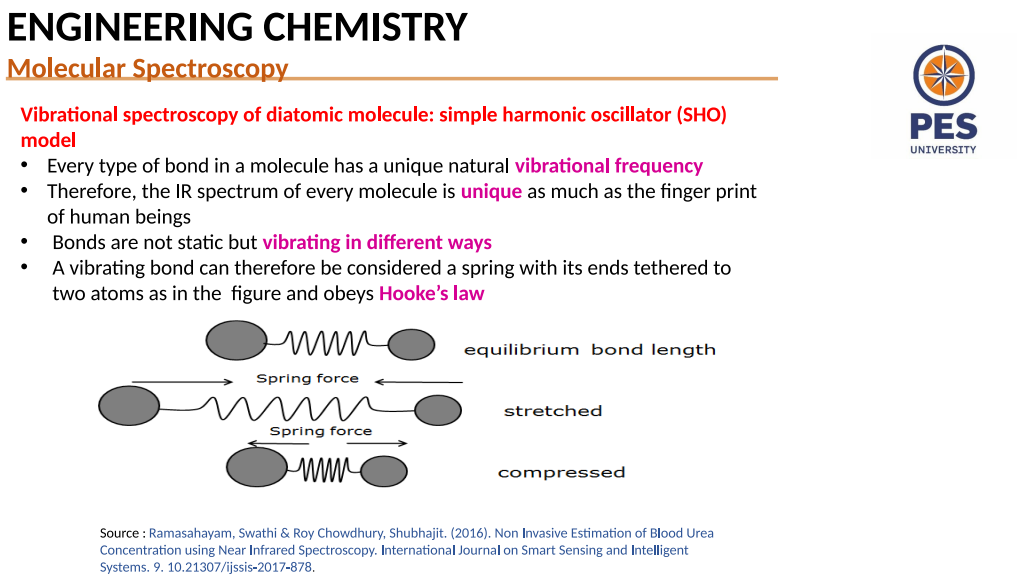

Figure G000043. IR spectroscopy diatomic harmonic oscillator model · source slide page 6.

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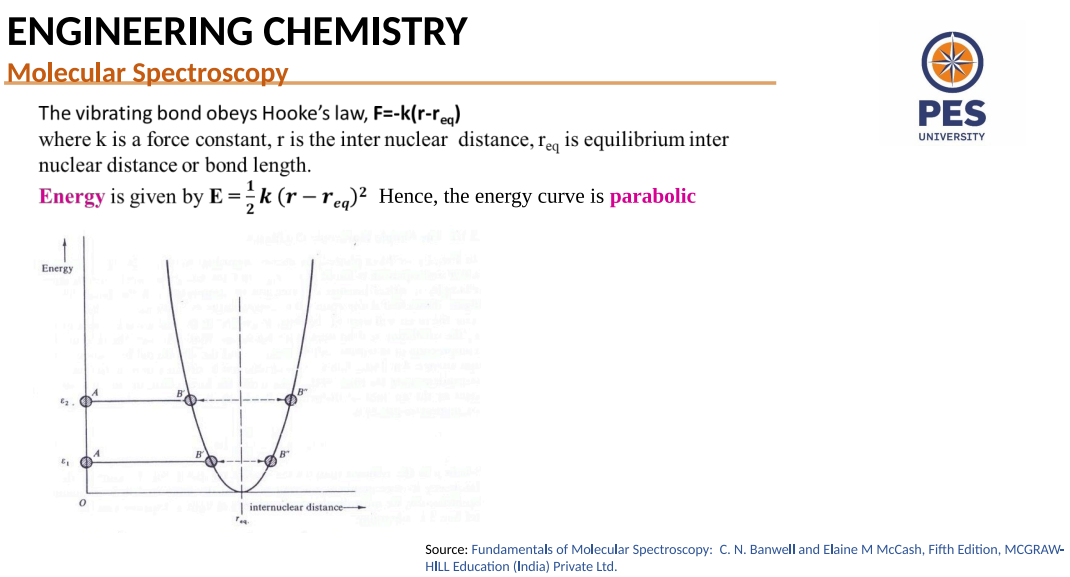

Figure G000044. IR spectroscopy diatomic harmonic oscillator model · source slide page 7.

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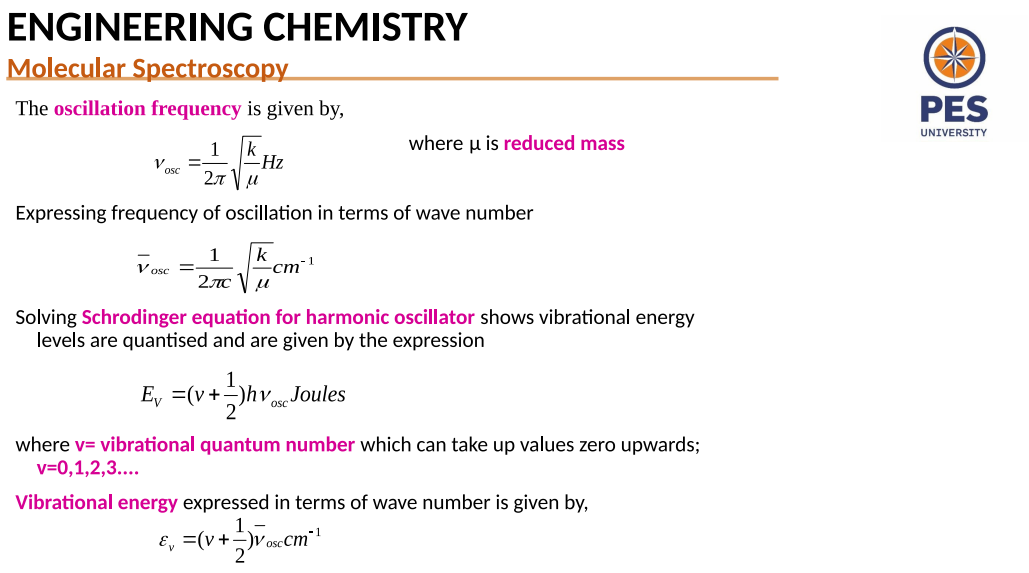

Figure G000045. IR spectroscopy diatomic harmonic oscillator model · source slide page 8.

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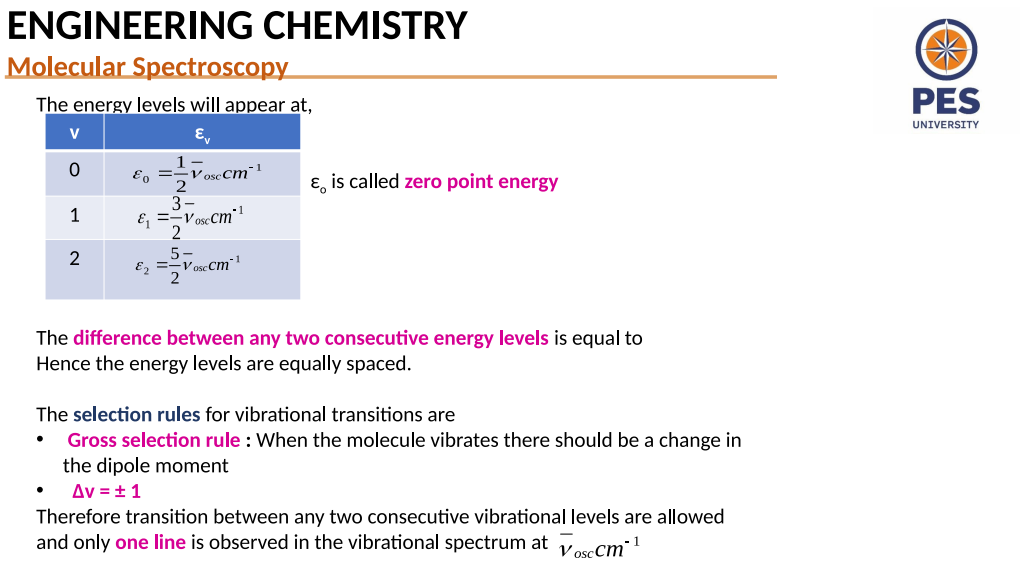

Figure G000046. IR spectroscopy diatomic harmonic oscillator model · source slide page 9.

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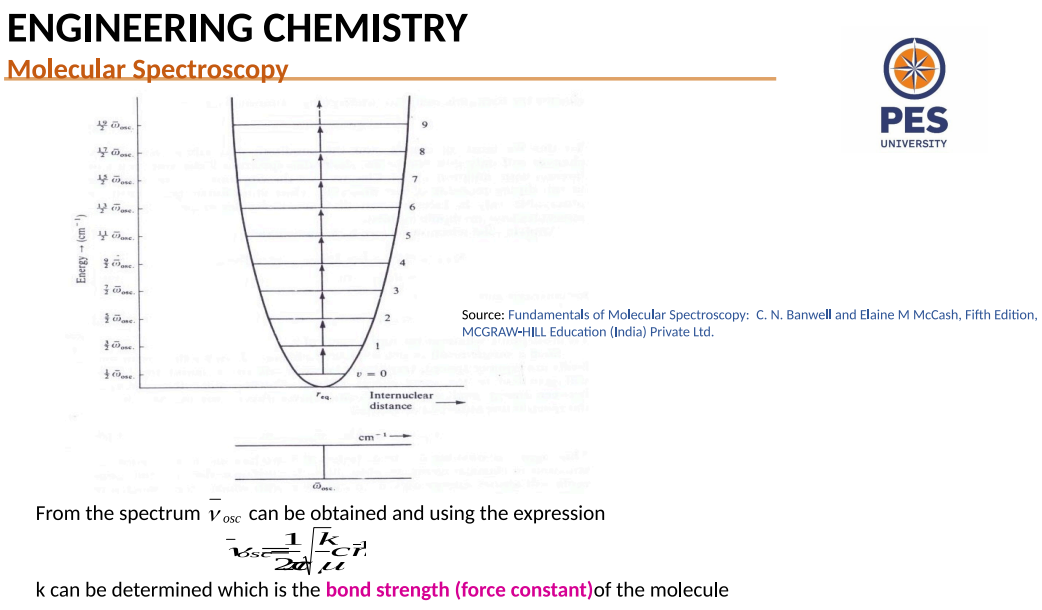

Figure G000047. IR spectroscopy diatomic harmonic oscillator model · source slide page 10.

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Figure G000048. IR spectroscopy diatomic harmonic oscillator model · source slide page 11.

EV0000400

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000369 · EV0000370 · EV0000371
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Class 5 Content: IR spectroscopy (Diatomic Harmonic oscillator model) EV0000372 · EV0000373 · EV0000374
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Vibrational Spectroscopy When a molecule undergoes vibrational changes, it absorbs IR radiation For a molecule to be vibrationally active (IR active) the vibration of a molecule must be associated with change in dipole moment Vibrations of molecules can result in changes in electric dipoles that can interact with the electrical component of the electromagnetic radiation. EV0000375 · EV0000376 · EV0000377
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy When the frequencies match, resonance occurs and the molecule absorbs energy and goes to the higher vibrational levels and exhibits vibrational spectrum Molecules such as HCl ,CO and H2O will show vibrational spectra while H2, Cl2 will not CO2 molecule does not possess permanent dipole moment but is IR active as when it vibrates asymmetrically or in bending mode ,there is change in dipole moment EV0000378 · EV0000379 · EV0000380
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy CO2 molecule although does not possess permanent dipole moment is IR active because some of its vibrational modes are IR active: CO2 symmetric stretch is IR inactive as there is no change in dipole moment when molecule vibrates in this mode. CO2 asymmetric stretch is IR active CO2 bending mode is IR active [Image text, OCR] Spectroscopy Symmetrical Asymmetrical In plane Out of plane stretching stretching bending bending Nochangeindipole Changeindipole Changeindipole Change in dipole momentthereforeIR momentthereforeIR momentthereforeIR momentth inactive active active Change in No changein No change in polarisabilitythereforepolarisabilitythereforepolarisabilitytherefore Ramanactive. Ramaninactive. Ramaninactive. Source: http://cepekmedia.co.nf Source: http://cepekmedia.co.nf EV0000381 · EV0000382 · EV0000383
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Vibrational spectroscopy of diatomic molecule: simple harmonic oscillator (SHO) model Every type of bond in a molecule has a unique natural vibrational frequency Therefore, the IR spectrum of every molecule is unique as much as the finger print of human beings Bonds are not static but vibrating in different ways A vibrating bond can therefore be considered a spring with its ends tethered to two atoms as in the figure and obeys Hooke's law [Image text, OCR] WWO equilibrium bond length Spring force stretched Spring force WW compressed Source : Ramasahayam, Swathi & Roy Chowdhury, Shubhajit. (2016). Non Invasive Estimation of Blood Urea Concentration using Near Infrared Spectroscopy. International Journal on Smart Sensing and Intelligent Systems. 9. 10.21307/ijssis-2017-878. EV0000384 · EV0000385 · EV0000386

7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Hence, the energy curve is parabolic [Image text, OCR] Energy internucleardistance- Figure3.1Parabolic curve ofenergyplotted against the extension or compression of a spring obeyingHooke'slaw. Source: Fundamentals of Molecular Spectroscopy: C. N. Banwell and Elaine M McCash, Fifth Edition, MCGRAW-HILL Education (India) Private Ltd. EV0000387 · EV0000388 · EV0000389
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy The oscillation frequency is given by, where μ is reduced mass Expressing frequency of oscillation in terms of wave number Solving Schrodinger equation for harmonic oscillator shows vibrational energy levels are quantised and are given by the expression where v= vibrational quantum number which can take up values zero upwards; v=0,1,2,3.... Vibrational energy expressed in terms of wave number is given by, EV0000390 · EV0000391 · EV0000392
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy The energy levels will appear at, εo is called zero point energy The difference between any two consecutive energy levels is equal to Hence the energy levels are equally spaced. The selection rules for vibrational transitions are Gross selection rule : When the molecule vibrates there should be a change in the dipole moment $\Delta v = \pm 1$ Therefore transition between any two consecutive vibrational levels are allowed and only one line is observed in the vibrational spectrum at v | εv 0 | 1 | 2 | EV0000393 · EV0000394 · EV0000395
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy [Image text, OCR] 8 6 osc. ↑ @ose. ose. oe ose. Internuclear distance cm @ose. Source: Fundamentals of Molecular Spectroscopy: C. N. Banwell and Elaine M McCash, Fifth Edition, MCGRAW-HILL Education (India) Private Ltd. From the spectrum can be obtained and using the expression k can be determined which is the bond strength (force constant)of the molecule EV0000396 · EV0000397 · EV0000398
11. Source slide page 11 (extracted text): THANK YOU Department of Science and Humanities EV0000399 · EV0000400 · EV0000401

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.932] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module one on molecular spectroscopy. And in today's class we are going to study about vibrational spectroscopy. [00:00:17.296] We will tie try to derive an expression for the vibrational energy levels that is for a harmonic oscillator and also we will arrive at the vibrational spectrum for a harmonic oscillator. [00:00:34.564] So what is vibrational spectroscopy? Now when a molecule undergoes vibration, there is going to be a change in the dipole moment. [00:00:48.292] Like I have explained this before when we were talking about the electromagnetic spectrum. Suppose we have an H Cl molecule and it is going to undergo [00:00:59.300] Vibration. It is vibrating. So when it is vibrating, since it's a diatomic molecule, the only kind of vibration is the stretching type of vibration. So we will see that the molecule stretches. [00:01:13.380] And then compresses stretches and compresses that would result in the a vibrational change in the uh molecule. Now when there is a vibration in the molecule, then the dipole moment is also changing because we know HCl is a [00:01:31.920] polar molecule you have a slight positive charge and a slight negative charge. So when the molecule is stretching there is change in the dipole moment. The direction remains same but same but the

magnitude is changing. Now when there is a change in this dipole moment then the electrical [00:01:49.552] Field is also changing. The electric field now is there is a frequency with which it is vibrating, and therefore the electric field will also fluctuate with a particular frequency. Now if radiation falls on such a molecule. [00:02:08.644] We know that electromagnetic radiation it has an electric field as well as a magnetic field and based on the energy of the radiation, the electric component of the electromagnetic radiation is also going to be fluctuating. [00:02:24.836] Now when this frequency of the radiation, the electric field of the radiation, fluctuation of the radiation for is matching with the molecule, the the electric field of the [00:02:39.232] uh molecule which is changing due to the change in dipole moment if these frequencies match then we know that there is going to be a resonance which is going to take place and therefore the molecule is going to absorb this energy from the radiation and it goes from [00:02:56.880] One vibrational level to the next vibration level. So what is important for us to understand that in vibrational spectroscopy, we are going to use IR radiation. [00:03:12.544] Rotational spectroscopy we know that we use the microwave energy. But here we are going to use the IR radiation. That means [00:03:21.888] that the vibrational energy levels, the energy gap of these vibrational energy levels matches with the energy gap of the IR radiation. Because only when these frequencies there is going to be a resonance phenomena and the molecule [00:03:39.588] goes from the lower vibrational level to the next vibrational level. So what we see as a result of this transition, because there is a transition which is happening [00:03:55.520] from the of the molecule from lower vibration level to the higher vibration levels, we will see that a particular energy is being absorbed from that whole IR region beam. So therefore that would if we can record [00:04:13.168] Is that energy that is getting absorbed? Then we can get what we call as the vibrational spectrum. And once we have the vibrational spectrum with us, we will be able to understand about the nature of the molecule and even [00:04:30.736] Then we can identify what is the molecule. Now each bond has a particular [00:04:40.676] vibration. Therefore, this vibrational spectroscopy is kind of like a fingerprint for the molecule. So we will be able to identify the molecule if we can get the vibrational spectrum of the molecule. [00:04:57.024] Now which type of molecules will show this vibrational spectr, or we can record the vibrational spectra that is for molecules like H Cl, C O and H two O. These are examples. Why these examples? Because

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:13.248] They there when the vibration takes place there has to be a change in the dipole moment. So therefore these molecules we know are polar in nature and therefore we will find that they will vibr when they vibrate there is going to be a change in the dipole moment. But when we [00:05:30.896] Come to homonuclear diatomic molecules like H2 and Cl2, we observe that these will not be active in the IR region. Why? Because they do not [00:05:43.744] possess a dipole moment. So even if the molecule is stretching and compressing, there is no change in the dipole moment, so no change in the electric field. So even if I radiation is falling on such a molecule, there will be no uh phenomena of resonance and hence [00:06:01.392] They are found to be I inactive. So [00:06:05.700] Can we say that if a molecule possesses permanent dichole moment, then it will be active in the IR region, just like we said about rotational spectroscopy? No. Here there is a small difference that CO2 molecule, we know that. [00:06:21.604] It is microwave inactive because it does not possess a permanent dipole moment. But here in this case, in the IR region, we find that CO2 molecule is actually IR active. Why? The reason is [00:06:37.956] Here, the criteria for observing a transition between the

vibration levels is that [00:06:44.432] When the molecule is vibrating, there should be a change in the dipole moment, and that happens in the case of the CO2 molecule. Though it doesn't possess a permanent dipole moment, we know that the CO2 molecule, each of these bonds [00:07:01.680] possess a dipole moment which actually cancel against each other, but when it vibrates we will find that in certain modes of vibration the CO2 molecule we can actually see a change in the dipole moment. So let us see how this is happening. [00:07:21.028] Uh in the case of symmetric stretch, now suppose the CO2 is stretching symmetrically about the [00:07:30.276] center then we about its equilibrium position we will find that there is no change in the dipole moment at all. So we will find that this mode is said to be IR inactive. [00:07:45.444] But if the molecule is undergoing asymmetrical stretching, that is you can see one bond is becoming small, one bond is stretching, and then it goes in the reverse direction. So this is known as asymmetric stretching. If this happens, then we will find that there is definitely a change in the [00:08:03.216] type moment because these two moments are not getting cancelled. So therefore we find that in this mode it becomes IR active. [00:08:12.740] Because there is a change in the diple moment. And then if it is bending, say in the plane it is bending like this, again we see there is a change in the diple moment, these movements are not getting cancelled now, and therefore in the bending mode also we find that it is I active. [00:08:30.528] So overall we can say that the carbon dioxide molecule is IR active, though it does not possess a permanent dipole moment. So and we understand the criteria is that the molecule should on vibration there should be a change in the dipole moment. [00:08:50.980] Now, let us understand the vibrational spectroscopy in more detail, and for that, we will consider the simplest molecule, and that would be a diatomic molecule. And we will arrive at the energy expressions and see. [00:09:10.368] why is the molecule able to show the vibrational spectrum. So diatomic molecule of course we should understand that we have to use a heteronuclear diatomic molecule. And again as we did in rotational spectroscopy we need a model [00:09:26.080] so that we can uh apply that model and get to the energy expression. So the model here we choose for a vibrating molecule, vibrating diatomic molecule is that of a simple harmonic oscillator. So what would be [00:09:43.748] a simple harmonic oscillator. We are taking a heteronuclear diatomic molecule, so one mass will be larger than the other. So M1 and M two. And now [00:09:54.656] we have a spring which is attaching the two atoms or holding the two atoms. So therefore when the molecule stretches the spring is getting stretched, when it is compressing the spring is getting compressed, and in the equilibrium position it remains like this.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:12.432] So we can have the stretching and the compressing for a diatomic molecule uh undergoing vibration. So this model would fit our vitronuclear diatomic molecule very well. And since [00:10:27.300] we it is a simple harmonic motion that we observe and it's a simple harmonic oscillator we can say that the more this vibration is going to obey hook's law. So basically the atoms, this uh bonds they are never at rest. These molecules are never at rest [00:10:45.104] These bonds are always vibrating in different ways. So a har simple harmonic oscillator model for a diatomic model molecule would be more suitable as a model for us. Now we know that every bond in a molecule has a [00:11:02.064] natural vibrational frequency which is very very unique to that bone. Therefore, if we can get the IR spectrum, then we will get a fingerprint of the molecules. And if that [00:11:17.796] will be very unique to the molecule. So let us see we have chosen this model of simple harmonic oscillator and let us try to get to the energy expression. Now since it is a vibrating bond [00:11:32.464] It obeys

Hook's law, simple harmonic oscillator, f is equal to minus of kx. Any displacement is now related to the force by minus of kx. R minus r equilibrium will be the displacement. Now [00:11:49.092] This means that K would basically be F. [00:11:54.820] By X, ignoring the negative negative sign because it's just because it's a restoring force. So what does it tell us this expression that if K is very high, K is known as the spring constant. So if the spring constant or the force constant is very high, that means even for a small distance. [00:12:12.592] We need to apply a large amount of force. So Hook's law is being obeyed over here and the energy would be given by E is equal to half K R minus R equilibrium squared. [00:12:27.440] And looking at this expression we can understand that the energy curve is going to be parabolic. Now let us understand how we got this parabolic curve and [00:12:38.532] Uh what is the uh reason why we get uh energy profile like this? Now, since let us say we are talking of the H Cl molecule which is undergoing vibration, so [00:12:51.056] As a rule, we understand that the heavier atoms we are going to fix at a particular position and it is the lighter atom which is going to be moving when the molecule stretches. So let us say that the Cl is [00:13:06.336] fixed because being the heavier and since it it is at equilibrium so at this is the equilibrium distance or the internuclear distance at Cl that is in the equilibrium position. So we have H over here and this is the uh Cl. Now if we [00:13:23.456] supply energy to the molecule. That means we are trying to expand it. So basically we are trying to give it more energy than the energy of the [00:13:33.104] uh energy of this molecule rises to this position. So what has happened? H now we are stretching. So instead of being at the equilibrium, H has now moved to uh uh a different position and if you compress it then again it is coming to uh [00:13:50.848] another position but here we see in both the cases the energy is higher than at the equilibrium position. Similarly now if we want to stretch the molecule further then this h will the the bond length now has increased. So [00:14:08.036] Earlier this was the bond length, you are providing more energy, you are stretching it more, so the bond length increased and it will reach this position, and when you compress it, it comes back to this position again, both are of higher energy. So this way we can go on supplying more and more and more energy, or rather we are stretching the [00:14:25.808] bond more and more and we get this energy in this in this curve which can be represented by this parabola. So for a harmonic oscillator the energy profile is in the form of a parabola. Now [00:14:43.748] Every one has a [00:14:46.624] Oscillation frequency. So here the molecule will have an oscillation frequency which is given by 1 by 2 pi root of k by mu in in terms of hertz and mu is what is the reduced mass. So we have already defined mu as m1 m2 by

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:05.476] M1 plus M2. Now this frequency is given in wave in Hertz. So if we want to express in terms of wave number because we are interested in spectroscopic units, so nu bar oscillation would be 1 by 2 pi. [00:15:21.860] uh C root of K by mu. So when we want to convert uh frequency into wave number we have to divide by the C which is the velocity of light. [00:15:33.060] 1 by 2 pi C root of K by mu centimeter inverse. So now we know that the mu bar is given by this or this is in terms of spectroscopic units. Mu is a known quantity for a particular molecule. And [00:15:52.132] K, what is K? It is known as a force constant, but in the case of a molecule, what would be K? K would represent the bond strength. So if I can get to the spectrum. [00:16:08.420] Then somehow, if I can find out what is the new bar oscillation for this harmonic oscillator, I will be able to get the value of K using this expression. So let us find out if this is possible. Now what we do just like we did in the [00:16:26.820] uh case of rotational spectroscopy we need to solve the Schrödinger equation because so far we are talking of

classical mechanics but now we know we are talking of a molecule we cannot apply classical mechanic laws mechanics laws therefore we need to you go give it a [00:16:44.592] quantum treatment. So we will solve the Schrödinger equation for a harmonic oscillator, and when we do that we find that the energies now of these levels is given by V plus half H nu. [00:16:59.280] That is in Joule. So what is V now? We got a term V which is known as which is the vibrational quantum number and can take up values zero upwards. That is V is equal to 0, 1, 2, 3, and so on so forth. So [00:17:16.128] That means what this energy now expression because of the V has become quantized. So in classical mechanics we saw that all those energies represented by the parabola was possible. [00:17:30.400] But here now the energies are have become discrete or they have become quantized. So vibration energy now if we want to express in terms of wave numbers spectroscopic units, all we have to do is divide this expression by C. And when we do it in wave numbers we know that [00:17:48.048] that it's represented by this small e and this is equal to v plus half nu bar centimeter inverse. So V plus half nu bar. So here it was V plus half h nu but because we are converting it dividing by H C, so this becomes nu bar centimeter. [00:18:05.616] inverse so this is an important expression and this is going to give us the various energy levels when we start substituting v is equal to 0123. Now [00:18:18.564] Let us see where these energy levels are appearing. So we got that expression when we substitute V is equal to 0 because it was V plus half nu bar. [00:18:30.064] So V is equal to 0 is going to give me half nu bar centimeter inverse. Now this energy is referred to as the zero point energy. [00:18:40.400] If we substitute V is equal to 1, then we get 3 by 2 mu bar centimeter inverse. If you substitute 01 to V, I can get the vibrational levels in the molecule. Now let us draw a rough diagram here and see. [00:18:57.472] What it looks like. So we have the energy here and here we have the R. So it is represented by a harmonic oscillator. So therefore [00:19:10.416] The energy levels like this. In the classical mechanics, all these energies are allowed, but now we see that the energy of the quotient. Now when B is equal to zero [00:19:23.040] What is the energy? Hal new bar. So that means it is not here, but it has a value that is half new bar, and therefore this energy is known as zero point energy. What does it mean that [00:19:38.948] The lowest energy level the lowest energy level is not at an ease equal to zero. We are not getting ease equal to zero, rather we are getting a value for this and that is half new bar. [00:19:54.564] This is referred to as zero point energy. Basically means that a molecule is never addressed, but it is always vibrating with energy which is known as zero energy and given by now let us start two B is equal to one. That is three by one.

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:12.336] two energy scale so let us go to the next one and that will be three by two new bar so now we have the next energy level now when we so this is zero this is one now let us say we have two here again this is [00:20:29.856] five by two new bar and we have a four year save that will be three [00:20:37.764] And then it would be seven by two nubar. So we can go on like this. We get half nubar, three by two nubar, five by two nubar, and seven by two nubar. So you know where the energy levels are. [00:20:55.536] thing is we are interested in the energy gap because that will actually tell us the kind of absorption that is going to take place. Now if you look at the uh suppose we have [00:21:11.408] Between zero and one. What is the difference in energy? That would be B is equal to one. [00:21:19.936] Minus V is equal to zero. V is equal to one is substitute energy expression. And then V is equal to zero is substrate the energy expression and V get the energy gathered. And that would be three by two minus half. [00:21:38.052] new bar and that is equal to new bar. So therefore between these two

energy levels 0 to 1, the energy gap is new bar. Let us go from 3 by 2 to 5 by 2. That is one to two. [00:21:55.984] Again you are five by two minus three by two are so and again when you go from two to three again the energy differences so what do we understand here that energy levels [00:22:08.960] The quantity of energy levels difference is equal to nu bar oscillation centimeter inverse. So the energy levels are equally spaced when your molecule, the diatomic molecule, is showing simple harmonic. [00:22:28.576] Oscillations. So the energy gap. Now you are the energy level. You know where the energy levels are. You also know what is the energy gap. Now knowing this. [00:22:38.704] is still not sufficient for me to arrive at the spectrum. What is important, I should know what are the selection rules. Now coming to selection rules, the gross selection rule [00:22:49.264] When it remains the same that it is same for all molecules. That when a molecule vibrates there should be a change in the diagonal moment. And what is the specific selection rule for the simple harmonic oscillator diatomic molecule? It says that Dell B is equal to plus minus [00:23:06.896] What does it mean? That the only transition allowed will be zero to one or one to two or two to three. You cannot have zero to two or one to three and or any of them any other transition two to four, etc. So the select [00:23:24.464] rule strictly says that the change in the vibrational quantum number for any transition can be only plus minus one or the other transitions are forward. Therefore we see that now zero to one where would the line appear [00:23:43.600] What do you mean by where would the line appear in the spectrum? That it is going to absorb that much energy equal to the energy gap that we have. Now so for from zero to one we see that the line should appear at nubar, the molecule should absorb energy nubar. [00:24:01.296] From V is equal to one, V is equal to two, again the entity are the same. So therefore, we will see that the molecule again absorbs at [00:24:10.096] new and similarly for any of these transitions since uh where selection rule is zero to one one to two two to three only those uh uh delta v is equal to plus minus one so plus one obviously stand for absorption and minus one stand for emission because [00:24:27.728] this is the selection rule and also we know that the energy gap is constant we find that in the spectrum only one line will be observed so for any of these transitions we will find that we all absorb at the same value [00:24:45.296] It is equal to number centimeter in now we can see from the spectrum here [00:24:54.064] Uh from the uh rotation uh the vibrational levels you can see that the energy gap is constant. So here we have studied a lot of energy. And this is

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Transcript block 6 (source interval 001500-001799 seconds): [00:25:11.840] being restributed by L V is equal to plus minus one. Therefore when you see the absorption you can only go to the next constituent level. Therefore the spectrum is going to look like this. [00:25:24.880] Just one line is observed in the spectrum and that will be at new bar centimeter in. [00:25:34.352] Now if I record this vibration spectrum, I'm going to see just one in this line and that line would be equal to nu bar, and I already have the expression with me mu bar is equal to one by two by c root of k by mu. [00:25:49.600] Now from the spectrum and I get a mu bar and then I know what is mu that is the reduced mass of the molecule which I can easily calculate. The other terms are known to me. So when I substitute here, I can get the value of K. [00:26:07.748] from the spectrum. And this is going to [00:26:12.512] Help us find out what is the bond strength of the molecules. So basically we just find out K using this expression just by recording the spectrum and getting the new bar. So we get K and that K is the bond strength of the molecules. [00:26:30.160] So vibrational spectroscopy is going to give us the bone strength of the molecule. Rotational spectroscopy saw that it gives us the bone length and vibrational spectroscopy is going to give us the bone strength. [00:26:46.756]

So this is the harmonic oscillator model for a diatomic diatomic molecule. So in the next class we are talking to talk about how this harmonic oscillator model is not suitable for molecules, and we will take up [00:27:04.528] [ASR low confidence; check audio] Vibrational spectroscopy by changing the model to an end-harmonic oscillator. Thank you.

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Full transcript: 3910 normalized words, 109 timestamped segments, 1631.978667 seconds duration. Low-confidence segments flagged: 1. Review locators: 00:27:04.528

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6 Vibrational spectroscopy(anharmonic oscillator)

This lesson is indexed from the source video titled Vibrational spectroscopy(anharmonic oscillator). Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000090 · EV0000091

Graphs and diagrams

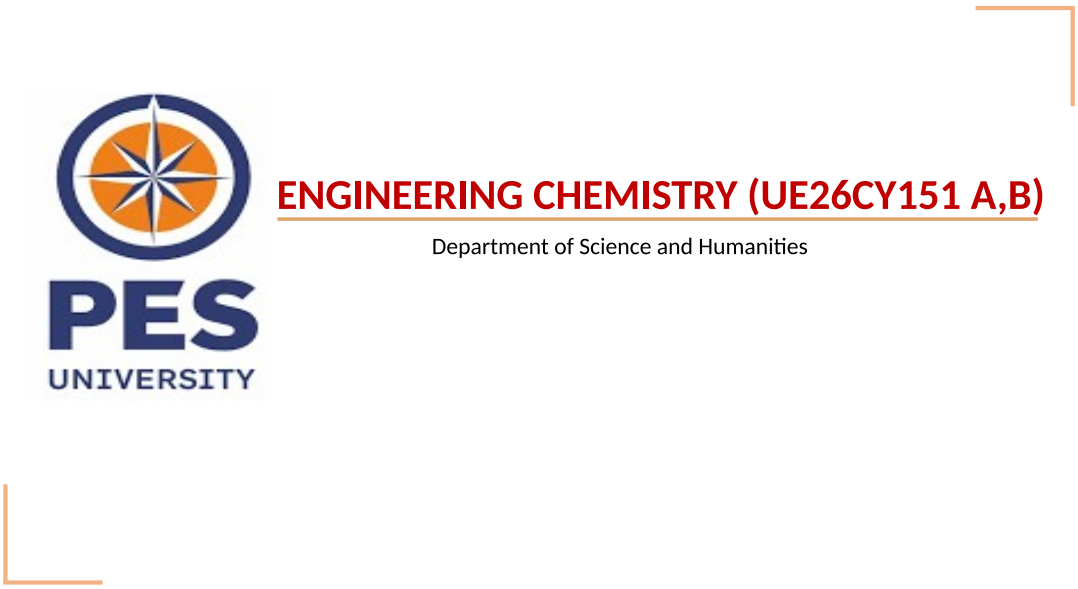

Figure G000049. IR spectroscopy anharmonic oscillator model · source slide page 1.

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Figure G000050. IR spectroscopy anharmonic oscillator model · source slide page 2.

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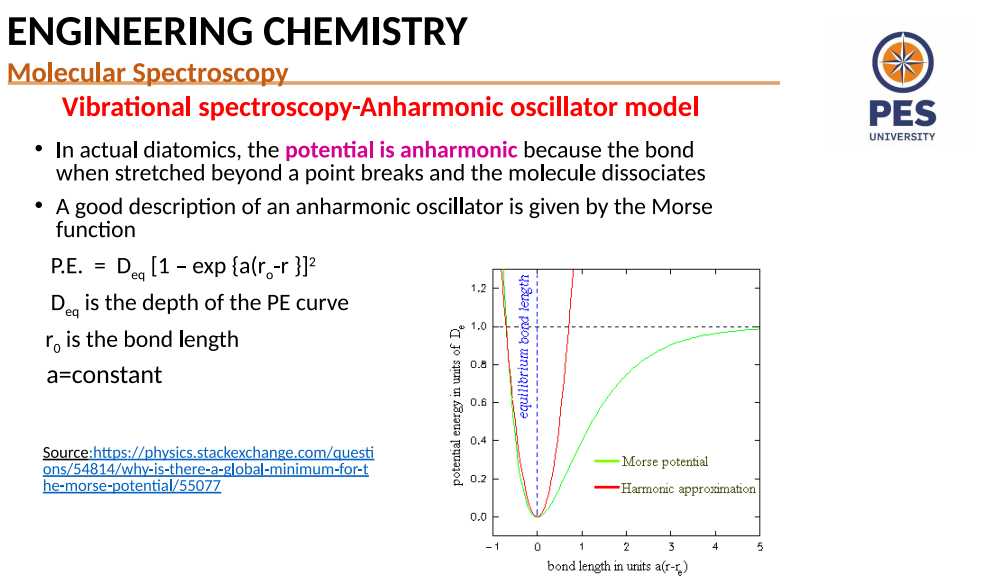

Figure G000051. IR spectroscopy anharmonic oscillator model · source slide page 3.

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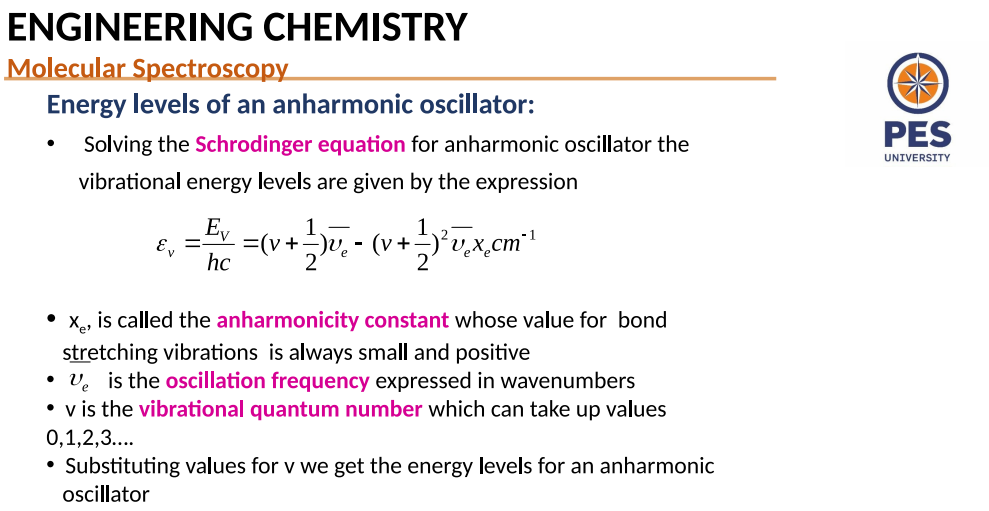

Figure G000052. IR spectroscopy anharmonic oscillator model · source slide page 4.

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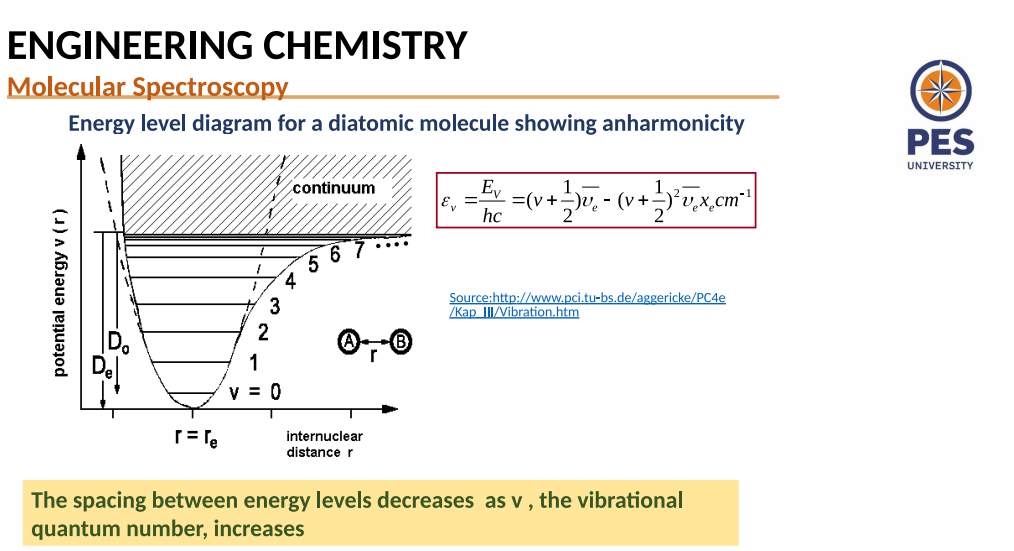

Figure G000053. IR spectroscopy anharmonic oscillator model · source slide page 5.

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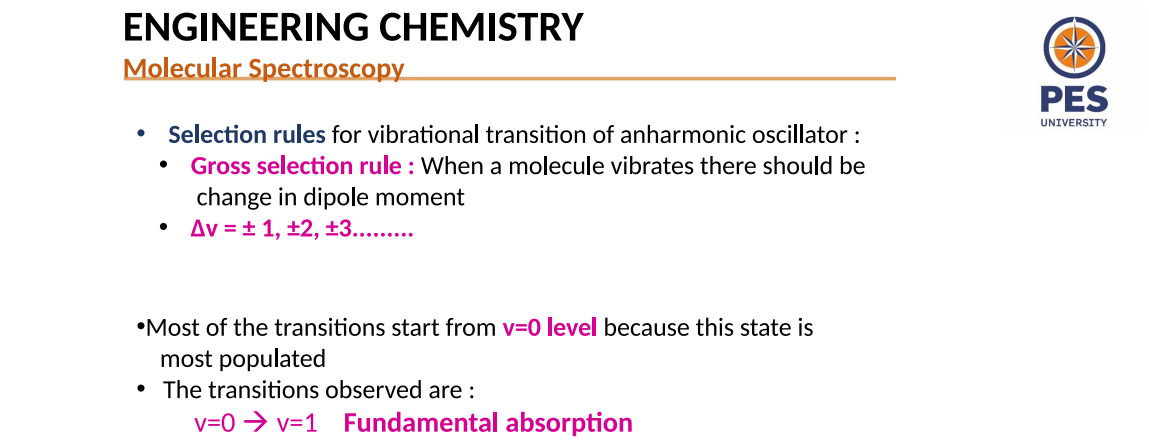

Figure G000054. IR spectroscopy anharmonic oscillator model · source slide page 6.

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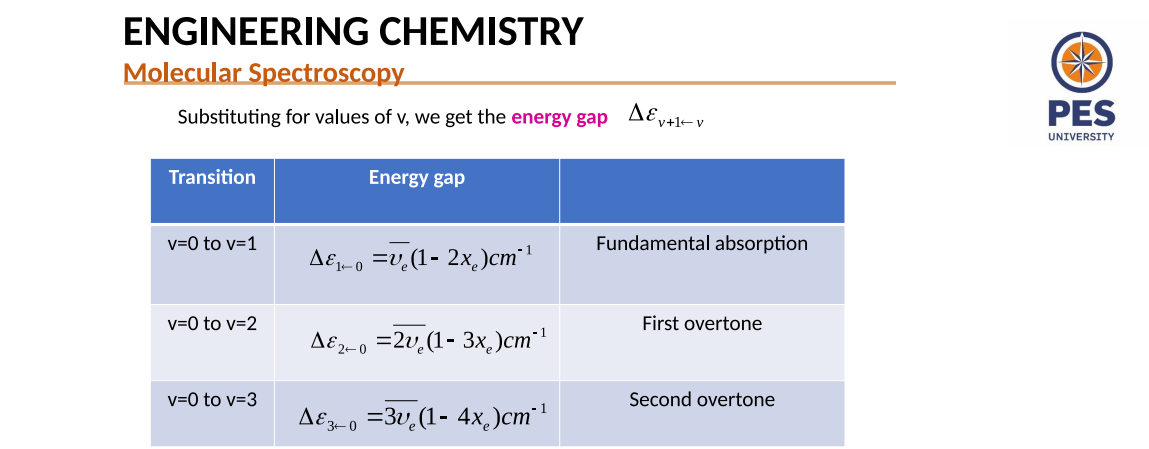

Figure G000055. IR spectroscopy anharmonic oscillator model · source slide page 7.

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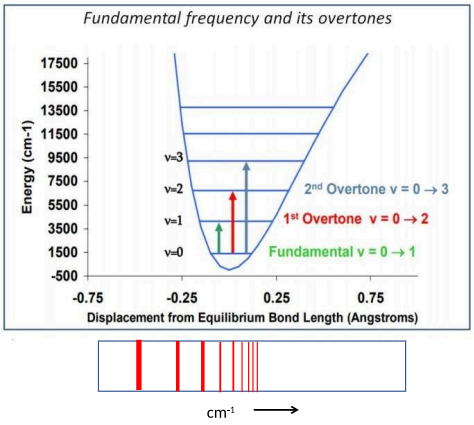
ENGINEERING CHEMISTRY

Molecular Spectroscopy

Vibrational transitions for diatomic molecule undergoing anharmonic oscillations



Source:https://www.researchgate.net/figure/Fundamental-frequency-and-overtones-see-lecture-N-Antaris_fig2_331976119

The spacing between the lines in the spectrum decreases as we go to higher values of v

Figure G000056. IR spectroscopy anharmonic oscillator model · source slide page 8.

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ENGINEERING CHEMISTRY

Molecular Spectroscopy

At **higher temperatures** additional lines are visible in the spectrum as higher vibrational levels may also get populated and these lines originating from the higher vibrational levels are called **hot bands**
e.g., Energy absorbed for the transition from **v=1 to v=2** is given by,

$$\Delta\varepsilon_{2\leftarrow 1} = \overline{\upsilon_e}(1 - 4x_e)cm^{-1}$$

The expression shows that this line will appear to the left of the fundamental absorption in the spectrum

Figure G000057. IR spectroscopy anharmonic oscillator model · source slide page 9.

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Figure G000058. IR spectroscopy anharmonic oscillator model · source slide page 10.

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000402 · EV0000403 · EV0000404
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Class 6 Content: IR Spectroscopy (Anharmonic Oscillator model) EV0000405 · EV0000406 · EV0000407
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy In actual diatomics, the potential is anharmonic because the bond when stretched beyond a point breaks and the molecule dissociates A good description of an anharmonic oscillator is given by the Morse function P.E. = Deq [1 – exp {a(ro-r }]2 Deq is the depth of the PE curve r0 is the bond length a=constant Vibrational spectroscopy-Anharmonic oscillator model [Image text, OCR] 1.2 ⊟ 1.0 s 0.8 0.6 0.4 Morse potential 0.2 Harmonic approximation 0.0 0 1 2 M1 4 5 bond length in units a(r-r) @O.S.Smart, 1995 Source:https://physics.stackexchange.com/questions/54814/why-is-there-a-global-minimum-for-the-morse-potential/55077 EV0000408 · EV0000409 · EV0000410
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Energy levels of an anharmonic oscillator: Solving the Schrodinger equation for anharmonic oscillator the vibrational energy levels are given by the expression xe, is called the anharmonicity constant whose value for bond stretching vibrations is always small and positive is the oscillation frequency expressed in wavenumbers v is the vibrational quantum number which can take up values 0,1,2,3.... Substituting values for v we get the energy levels for an anharmonic oscillator EV0000411 · EV0000412 · EV0000413

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Energy level diagram for a diatomic molecule showing anharmonicity [Image text, OCR] continuum ( ) a Abaua eualod 5 4 3 2 D 1 V 0 [=le internuclear distancer Source:http://www.pci.tu-bs.de/aggericke/PC4e/Kap_III/Vibration.htm The spacing between energy levels decreases as v , the vibrational quantum number, increases EV0000414 · EV0000415 · EV0000416
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Selection rules for vibrational transition of anharmonic oscillator : Gross selection rule : When a molecule vibrates there should be change in dipole moment $\Delta v = \pm 1, \pm 2, \pm 3$..... Most of the transitions start from v=0 level because this state is most populated The transitions observed are : v=0 ⊠ v=1 Fundamental absorption v=0⊠ v= 2 First overtone v=0 ⊠ v=3 Second overtone EV0000417 · EV0000418 · EV0000419
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Substituting for values of v, we get the energy gap Transition | Energy gap | v=0 to v=1 | | Fundamental absorption v=0 to v=2 | | First overtone v=0 to v=3 | | Second overtone EV0000420 · EV0000421 · EV0000422
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Vibrational transitions for diatomic molecule undergoing anharmonic oscillations [Image text, OCR] Fundamentalfrequencyanditsovertones 17500 15500 13500 11500 9500 v=3 7500 v=2 2ndOvertonev=0→3 5500 V=1 1stOvertonev=0→2 3500 1500 v=0 Fundamentalv=0→1 -500 -0.75 -0.25 0.25 0.75 DisplacementfromEquilibrium BondLength(Angstroms) Source:https://www.researchgate.net/figure/Fundamental-frequency-and-overtones-see-lecture-N-Antaris_fig2_331976119 The spacing between the lines in the spectrum decreases as we go to higher values of v cm-1 EV0000423 · EV0000424 · EV0000425
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy At higher temperatures additional lines are visible in the spectrum as higher vibrational levels may also get populated and these lines originating from the higher vibrational levels are called hot bands e.g., Energy absorbed for the transition from v=1 to v=2 is given by, The expression shows that this line will appear to the left of the fundamental absorption in the spectrum EV0000426 · EV0000427 · EV0000428
10. Source slide page 10 (extracted text): THANK YOU Department of Science and Humanities EV0000429 · EV0000430 · EV0000431

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.772] Hello everyone, welcome back to this course on engineering chemistry. We continue with module one on molecular spectroscopy, and in today's class we will look at the anharmonic oscillator model for vibrational spectroscopy. [00:00:16.420] Then we consider and derive the expression for vibrational energy levels for an anharmonic oscillator, and finally we will try to get the vibrational spectrum of an anharmonic oscillator. [00:00:30.980] In the previous class we considered the molecule to behave like a harmonic oscillator and therefore the energy was given in the form of a parabola. That's because we considered E is equal to [00:00:47.952] half k x squared that is for a harmonic oscillator and that would result in the energy profile being that of a parabola. [00:00:57.316] Now, a real molecule is not going to behave the same way. The reason being that when you consider it as a harmonic oscillator, we are assuming that we can go on stretching the molecule and it will remain intact.

Like we can extend the [00:01:17.060] bond length as much as we want, but still the molecule remains intact is what we see in this curve. But the real picture is something different. We know that if we stretch a molecule beyond a point, it is going to dissociate. So therefore we need [00:01:34.832] to change the model for real molecules from simple harmonic oscillator to that of the anharmonic oscillator. So therefore [00:01:44.544] When we say the molecule behaves unharmonically or vibrates in an anharmonic fashion, then the best way to represent the energy profile is given by this curve which is shown in green. [00:02:00.036] Now this is known as the Morse curve and it seems to fit the model very well. [00:02:07.024] Why? Because suppose we have two atoms and we are trying to form a molecule. So we are trying to bring one atom is stationary, the other atom we are trying to bring from infinity. As we come closer, we are reducing the bond length because [00:02:24.656] What is the bond length? This is the bond length. That is the equilibrium bond length. So when we are bringing the molecule uh atom closer, we know that we are reducing the bond length and as a result the energy is decreasing. [00:02:40.196] But when we come to this point where we have reached the equilibrium bond length of the molecule, then this is the most stable and therefore represented by minimum in the curve, minimum in the energy curve. Now if we go on bringing the molecule. [00:02:57.232] Atoms even further closer, then we are compressing the molecule, and again the energy is going to increase. So we see that the molecule actually is following this energy profile, and if we [00:03:13.840] extend the molecule beyond a particular limit, then it is going to dissociate. So this Morse curve or Morse potential represents the real molecule more closer than the [00:03:29.120] uh than the m ideal molecule which would be represented by the harmonic oscillator. So we are going to now study the anharmonic oscillator and try to get to the energy levels in an anharmonic oscillator. [00:03:44.932] uh model using the anharmonic oscillator model. Now one important fact we can notice here [00:03:51.760] that at very low stretching when we are not expanding the molecule too much then the harmonic oscillator and the anharmonic oscillator almost behave in a similar fashion but when you stretch the molecule uh further then the behavior completely [00:04:09.424] changes. So that is what we understand from this graph and for now we are going to use the anharmonic oscillator model or in other words the energy profile is going to be represented by the Morse function. Now what is MOS function? It says potential [00:04:26.992] energy is equal to d equilibrium 1 minus exponential a times r naught minus r the whole square. Now d equilibrium is the depth of the potential energy curve. So you can see here that it is [00:04:40.912] Till there is zero, and you can very well make out that this would represent the dissociation energy of the molecule. [00:04:50.532] So let us get the energy levels of an anharmonic oscillator. So we are assuming that the molecule is vibrating anharmonically. So now since the model is changed, the energy level expressions are also going to change.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:07.760] And how do we get the energy level expression? Just like we did in rotational uh spectroscopy or when we were talking of harmonic oscillator, we have to solve the Schrodinger equation for the harmonic oscillator and the vibrational energy levels now will be given by this [00:05:25.392] uh expression uh E that is we are now talking in centimeter inverse so that means the energy in joules divided by H. So basically energy expressed in spectroscopic units that is in centimeter inverse. So energy will be divided by H C and what we get [00:05:42.960] is the expression V plus half nu bar. Now if you remember for the harmonic oscillator what did we have? V plus half nu bar was the expression for a harmonic oscillator but here we have a second term because of the anharmonicity and that is given by

[00:06:00.528] by minus of V plus half squared nu e bar X E. So nu E bar because we are representing in terms of centimeter inverse. So here we have [00:06:13.476] And what is V here? V is the vibrational quantum number. V is the vibrational quantum number which is going to take up the value 0, 1, 2, 3. And therefore by substituting these values of V, V can [00:06:31.248] can get the different energy levels which are present in the anharmonic oscillator. But only thing here we have some new term here. What is this new term? XE. Now this is known as the anharmonicity constant. [00:06:48.816] and whose value for bond stretching vibrations is always small and [00:06:55.872] positive. We have to remember this. It is small and uh positive. So that is the anharmonicity constant and nu e of course is the oscillation frequency and frequency we are saying but it is expressed in wave numbers. So basically it is mu E. [00:07:12.944] But so the oscillation frequency expressed in wave number. So this is now our expression for the energy levels of the anharmonic oscillator. [00:07:26.212] What do these energy levels look like? So we got this expression now by solving the Schrodinger equation. All we have to do is substitute values for V is equal to 0, 1, 2, 3. So when you substitute the value of 0, then we get this energy level at V is equal to 0. [00:07:43.984] And you can notice here that we have used the anharmonic oscillator model. So basically the energy profile is that of an anharmonic oscillator. Now we go on substituting v is equal to 0, 1, 2, 3 into this expression. Now when we substitute v is equal to [00:08:01.552] To zero, then we are going to uh get an energy level here. When we substitute V is equal to one, obviously we got this energy level here. Now if you compare with the [00:08:16.676] harmonic oscillator the energy level for V is equal to zero is going to be at the zero point energy of the anharmonic oscillator not at that point but at a point which is slightly [00:08:33.152] lower so you can see we have subtracted a small positive value from here therefore the value now v is equal to zero the zero point energy how do you get zero point energy when we substitute b is equal to zero then we are getting the zero point energy [00:08:52.176] So the same way we will get for an harmonic oscillator also, but nevertheless we can understand zero point energy is going to be at a slightly lower level as compared to that of the harmonic oscillator. Similarly, when we go to V is equal to 1, we get this energy level again. [00:09:09.808] Again slightly lower than that of the harmonic oscillator. Now when we come to V is equal to 2, now we notice something. [00:09:18.020] happening here. When we substitute V is equal to 2, there is again a decrease in energy because of the second term, but remember that this term is getting squared. So that means each time we are subtracting a larger [00:09:35.792] number what does it mean V is equal to 2 level will be much more closer to V is equal to 1 as compared to in the uh harmonic oscillator model and similarly 3 4 5 6 as we go on substituting the [00:09:53.360] Energy levels are going to come closer and closer and we see that as we go to higher levels the vibrational energy levels are actually coming closer.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:06.564] What did we see in the case of rotational energy levels? That the energy levels are going farther and farther apart. [00:10:13.600] When we came to harmonic oscillator, we saw that the energy gap remains constant. You go to higher levels of V also, energy gap remains constant. That's why we saw only one line in the spectrum. But here, what do we observe? Because of this second term getting subtraction. [00:10:31.280] And larger and larger term is getting subtracted each time, we find that the energy levels are coming closer together. Now [00:10:40.868] Here we can see that D equilibrium, that is the dissociation energy. And there's one more D naught over here. What would this be? This is the dissociation energy required by the molecule. What do we mean? Because we know that in the case

of vibration, [00:10:58.640] A molecule is always going to possess the zero point energy. So whatever energy it requires to dissociate, it already has that V is equal to zero. So the dissociation energy required to break down the molecule is going to be this value D naught. [00:11:19.012] Now, just getting the energy levels are not sufficient. Now we need to have the selection rules. [00:11:27.364] Now in the case of harmonic oscillator or in rotational spectroscopy, we know that the energy levels uh where do they exist, but finally coming to the spectrum will require us [00:11:40.640] the uh require that we know the selection rule. So again we have the gross selection rule which remains the same. When a molecule vibrates there should be a change in the dipole moment. And what is the specific selection rule now? [00:11:54.752] Delta V is equal to plus minus one, plus minus two, plus minus three, so on. That means [00:12:01.056] All kind of transitions are going to be allowed. In the case of harmonic oscillator, what was the selection rule? Only plus minus one. And all other transitions were forbidden. But here we have plus minus one, plus minus two, plus minus three, meaning that all the transitions are going to be permitted. [00:12:18.704] If you remember, let us draw this and see what it looks like. E versus R. And now your energy profile is like this. So what it means, delta, and we know that the energy levels are going to be. [00:12:37.380] Coming closer like this. Okay. Just a rough diagram. And what do we mean plus minus one is allowed? So plus one, let us say we are talking of uh absorption. Minus will stand for emission. Plus minus one is allowed. Then plus minus [00:12:55.940] Two is also allowed. And again plus minus three is also allowed. So that means you will have a whole lot of transitions that are possible. And you can have 0 to 1, 0 to 2, 0 to 3, you can have 1 to 2, 1 to 3, 1 to 4. [00:13:13.040] Two to one two to three, two to four, all these transitions are going to be allowed. So what will happen to your spectrum now? It is going to be very, very complicated and it'll be difficult for us to interpret the spectrum. [00:13:26.948] but we have something coming to help us. And what is that? Most of the transitions now start from V is equal to zero level. [00:13:38.208] Why? Because this is the state which is the most populated state. So most of the molecules are going to be at V is equal to zero level at room temperature. So therefore [00:13:50.096] All of the transition, most of the transitions are going to start from V is equal to 0. So you will hardly see any transitions happening from V is equal to 1, V is equal to 2, etc. Though the selection rule actually allows it. But now the spectrum is going to be much much simpler. [00:14:07.760] Why? Because all transitions will start from these equal to zero level at room temperature when you're recording the spectrum. So what are the most important transitions which will which you will see as intense transitions in the spectrum? [00:14:24.960] So first one will happen from V is equal to 0 to 1 and that is known as the fundamental absorption. Then second one will be 0 to 2, which is known as the first overtone, and then we have v is equal to 0 to 3, that is known as the second overtone. So you see all transitions are starting from [00:14:42.608] From V is equal to zero. So you will see in the spectrum fundamental absorption, first overturn, second overturn. You can have other [00:14:52.048] uh transitions also two is equal to four or five, but then those will be very weak in intensity. In fact fundamental absorption will be the one which will be most intense and then we have first overtone and second overtone. After that the intensity of the transitions become very very weak because more energy will be required.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:13.284] So if we substitute the value of V is equal to 0 and V is equal to 1, meaning that we want to see the transition now. Fundamental absorption would correspond to V is equal to 0 to V is equal to 1. So if we take the

energy expression and substitute for V is equal to [00:15:31.888] To 1 and minus V is equal to 0, the expression we get is this mu E bar 1 minus 2 XC centimeter inverse. Remember that we are expressing energies in terms of wave numbers. Though I have written energy gap here, what do we mean that? [00:15:49.456] we are expressing energy in terms of uh centimeter inverse and that's what I told you in the first class that we use spectroscopic units it is just for convenience okay now let us calculate for first overtone so [00:16:02.528] You all should substitute for yourself and see whether you are getting this expression. So V is equal to 0 to V is equal to 2. Go to the energy expression, substitute V is equal to 2 minus V is equal to 0, and this is the expression you get for the first overtone. And similarly, second overtone, you [00:16:20.176] Going to get an expression like this. So, what do we understand from this energy gap? We know that the line in the spectrum you're going to see. [00:16:32.496] is based on the energy gap because that is the amount of energy the molecule is going to absorb to go from one level to the next level. So here it is new e bar 1 minus 2 XE. If it was [00:16:49.732] A harmonic oscillator and then we would have seen the line at nu bar, but now there is a 1 minus 2 XC clearly indicating that the energy gap is now less. Similarly, we will have we would have had something closer to 2 nu bar provided the selection. [00:17:07.504] allowed us a plus two and similarly three mue bar if it allowed us a plus three. So we these are the energy gaps which you see in the which you will see in the and harmonic oscillator and now corresponding to these energy gaps [00:17:25.072] We will see lines in the spectrum. [00:17:28.512] Let us observe what happens. So this is a diagram where you see that for a diatomic molecule which is undergoing anharmonic oscillations, what are the transitions you see and what is the spectrum you get. So as I said, fundamental absorption. [00:17:46.096] will be the first line because the I told you we represent in centimeter inverse energy is increasing this way and you know that this will require the minimum energy. Nevertheless the intensity of this line is going to be [00:18:03.632] very high. So you can see here a thick line over here because most of the molecules intensity always depends on the population of the state. And since requires minimum energy, though it is a short line it would kind of give you an impression that this [00:18:20.544] Will be the weakest line, but no here the energy gap. It's short only because the energy gap is less. Therefore, and therefore it appears as the first line. And with low energy, and the second line is your first overtone, which is 0 to 2, and that line is appearing here. [00:18:38.224] Have 0 to 3 and you have the turn line after this, it actually the intensity goes on falling down, and but one more observation which you have to make here that because these energy levels are coming closer and closer, we will find that the energy [00:18:55.792] Energy gap also decreases as we go further and further in the the spacing between the energy lines is also becoming lesser and lesser. So you can see that the spectrum [00:19:11.588] like this where in initially the gap is more but you can see that the lines are coming uh closer and closer. So the lines are basically converging and you can say that if you get a spectrum like this then [00:19:28.912] the molecule is showing anharmonic behavior. If you see the lines converging like this, it means that the molecule is showing anharmonic behavior. Now [00:19:40.836] This is what we see at room temperature. Now, suppose we do a spectrum at higher temperature, we find that additional lines are visible. Now, where do these additional lines come from? So what has happened now? Now apart from [00:19:57.092] V is equal to zero, the V is equal to one level also got populated because of the high temperatures. Some molecules gained energy and now they are at V is equal to one level. So when they go there and if you are recording the spectrum.

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:15.056] And light infrared radiation is going to fall on this such molecule, then we will find that there will be some transitions happening from V is equal to one also. So since there is no selection rule, I mean in the sense that selection rule is delta V is equal to plus minus one plus minus two plus [00:20:32.720] Minus 3, we will find lines originating from V is equal to 1 as well. Nevertheless, the intensity will still be not so high, it will still be less, and therefore [00:20:45.572] We will find that such lines will appear in the spectrum if we do it at a higher temperature, and the reason being the higher vibration levels also are getting populated, and rightly so they are called as hot bands because you see them at higher temperature. [00:21:03.344] Temperature. Now, what would the be the frequency of such an absorption? Now, if we say let us take an example, we it is at V is equal to 1 and it is going to V is equal to 2. When you substitute in the energy expression where it changes from goes from 1 to 2, [00:21:20.912] we get this expression nu e bar 1 minus 4 x centimeter inverse. So where would this line appear in the spectrum? If you remember fundamental absorption was nu E bar 1 minus 2 XE. [00:21:37.956] centimeter inverse. This was for the fundamental absorption. So if you compare, you can see that this now [00:21:48.400] Is going to this energy gap is further less. So, therefore, where will it appear in the spectrum? It will appear to the left of the fundamental absorption in the spectrum. So you have the most intense line as the fundamental absorption, but to the left of this, you will see a [00:22:06.160] faint line and that is referred to as the hot band and provided you are doing it at a high temperature. So this is about vibrational spectroscopy and about the anharmonic oscillator. [00:22:20.516] In the next class, we are going to study about another type of spectroscopy, which is the electronic spectroscopy. Thank you.

EV0000106 · EV0000107 · EV0000108 · EV0000109

Full transcript: 3400 normalized words, 89 timestamped segments, 1351.509333 seconds duration. Low-confidence segments flagged: 0.

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7 Electronic spectroscopy - Franck-Condon principle (Part II)

This lesson is indexed from the source video titled Electronic spectroscopy - Franck-Condon principle (Part II). Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000110 · EV0000111

Graphs and diagrams

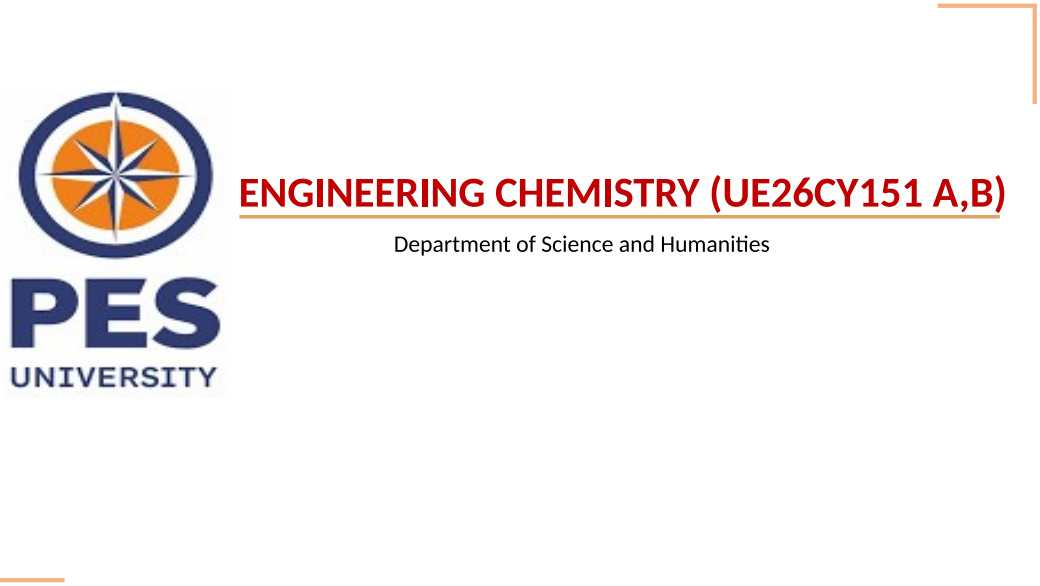

Figure G000059. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 1.

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Figure G000060. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 2.

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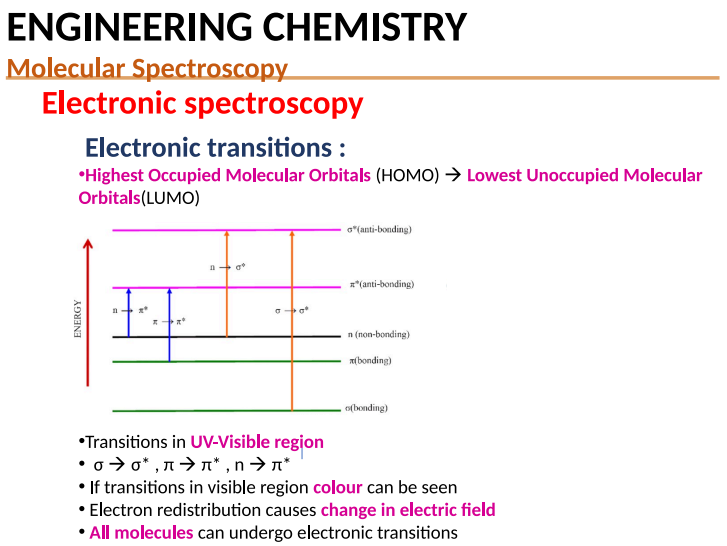

Figure G000061. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 3.

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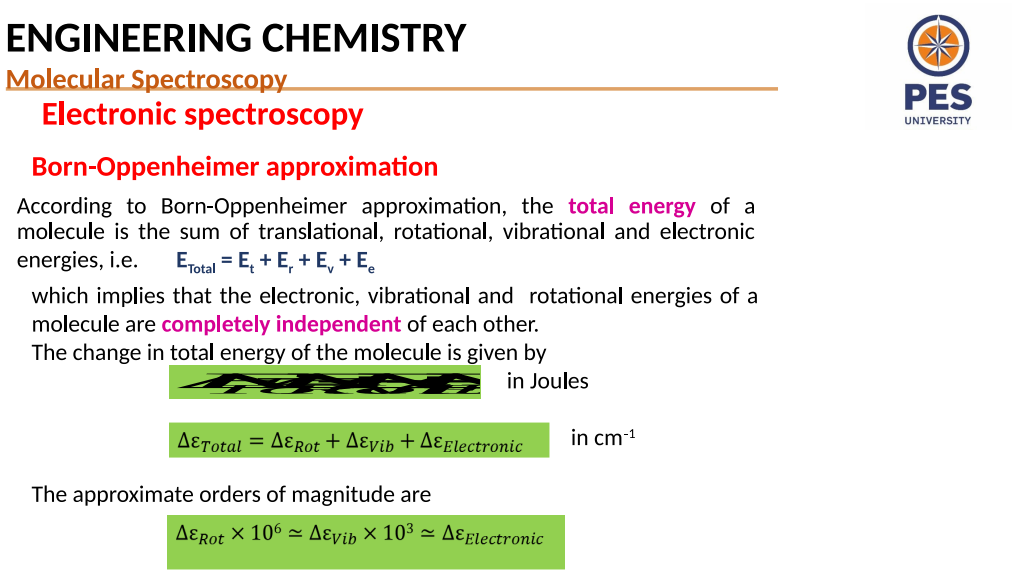

Figure G000062. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 4.

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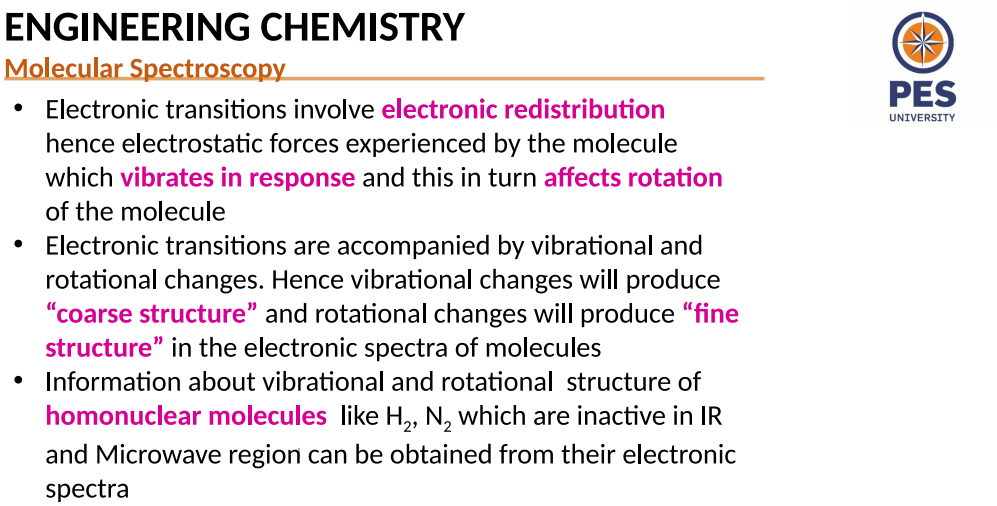

Figure G000063. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 5.

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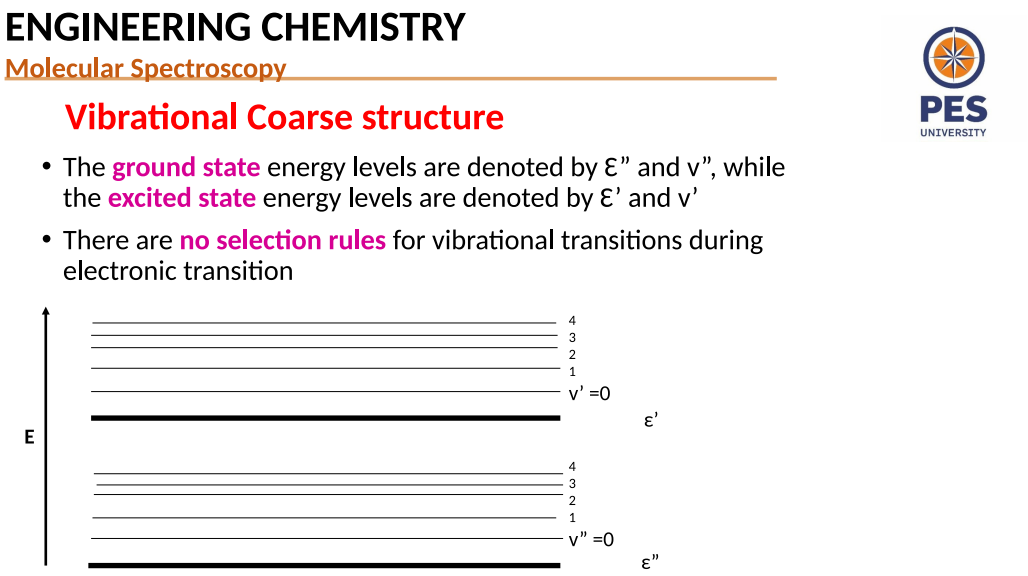

Figure G000064. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 6.

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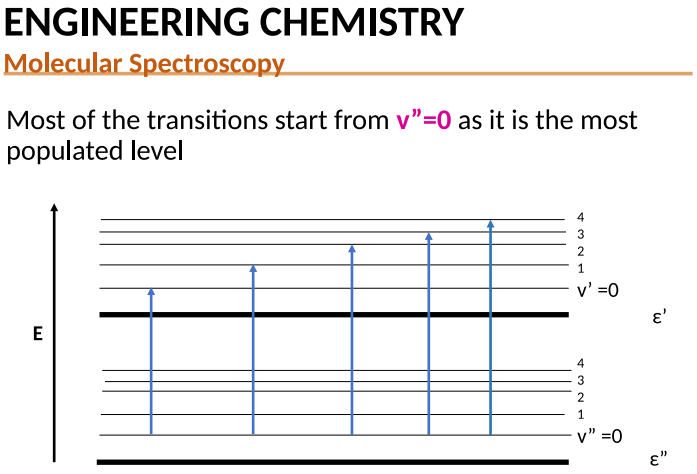

Figure G000065. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 7.

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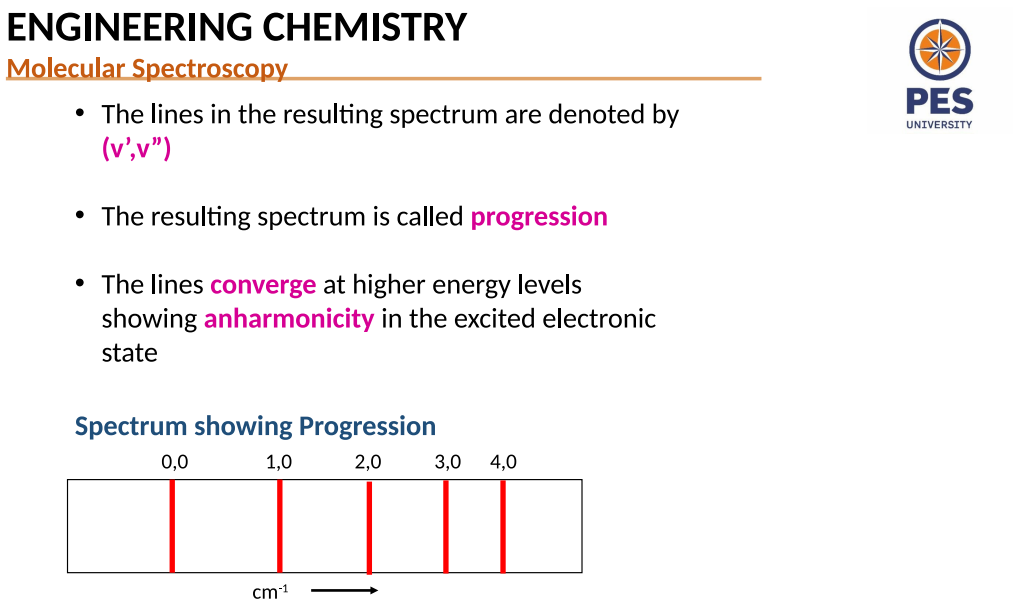

Figure G000066. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 8.

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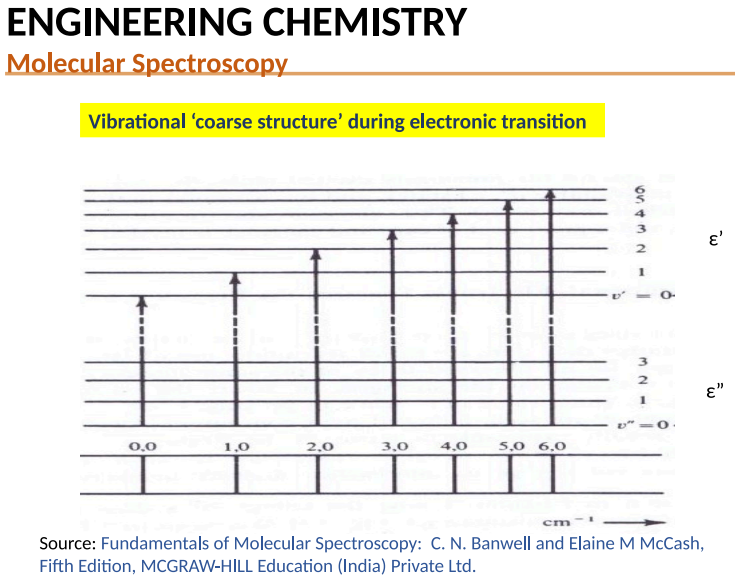

Figure G000067. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 9.

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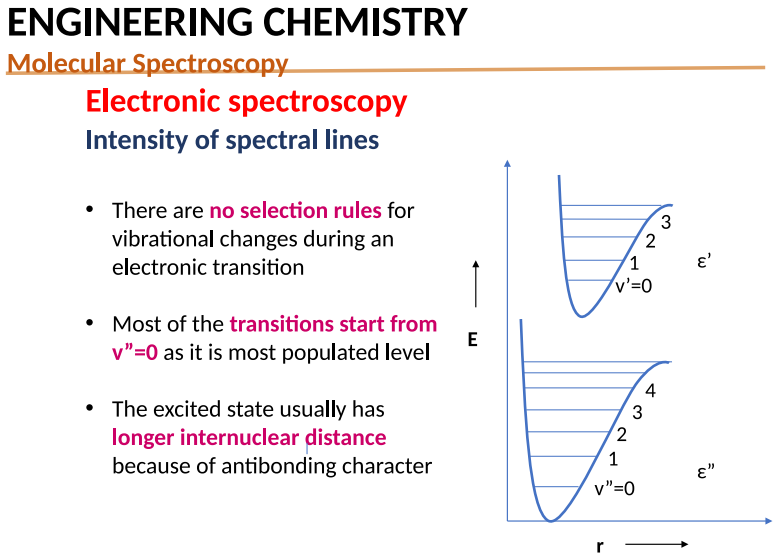

Figure G000068. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 10.

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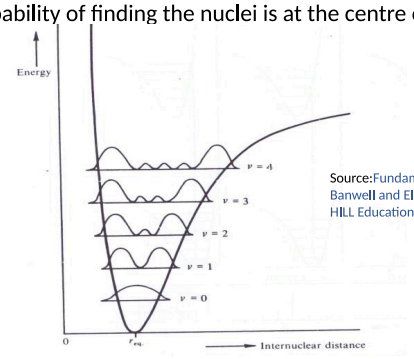
ENGINEERING CHEMISTRY

Molecular Spectroscopy

The transitions arise from the centre of v"=0 because the maximum probability of finding the nuclei is at the centre of v"=0



Source:Fundamentals of Molecular Spectroscopy: C. N. Banwell and Elaine M McCash, Fifth Edition, MCGRAW-HILL Education (India) Private Ltd.

Probability distribution for a diatomic molecule according to quantum theory

Figure G000069. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 11.

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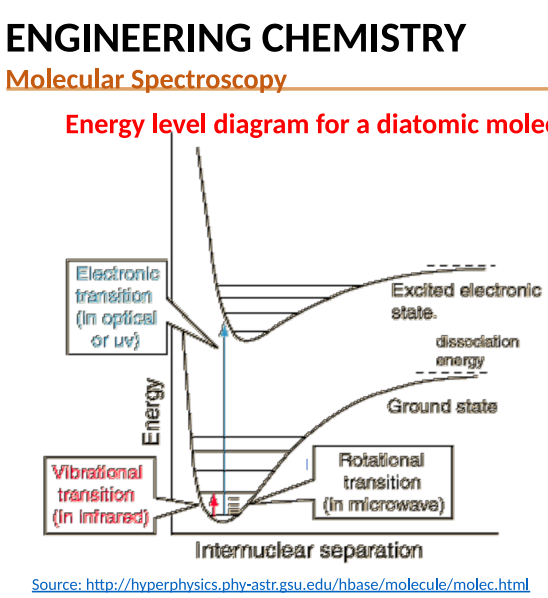

Figure G000070. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 12.

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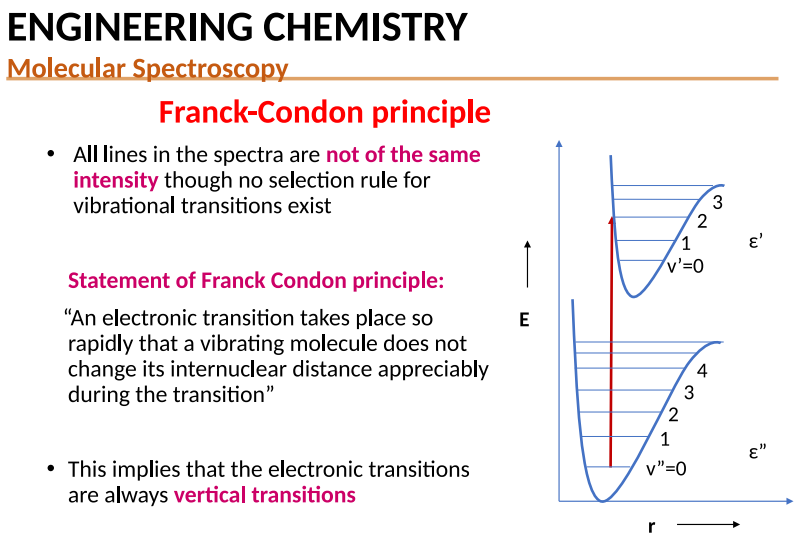

Figure G000071. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 13.

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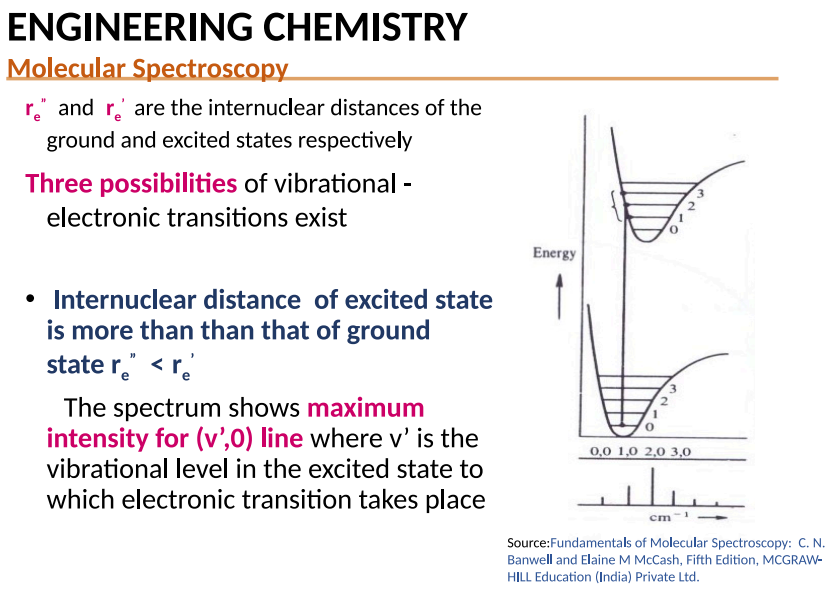

Figure G000072. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 14.

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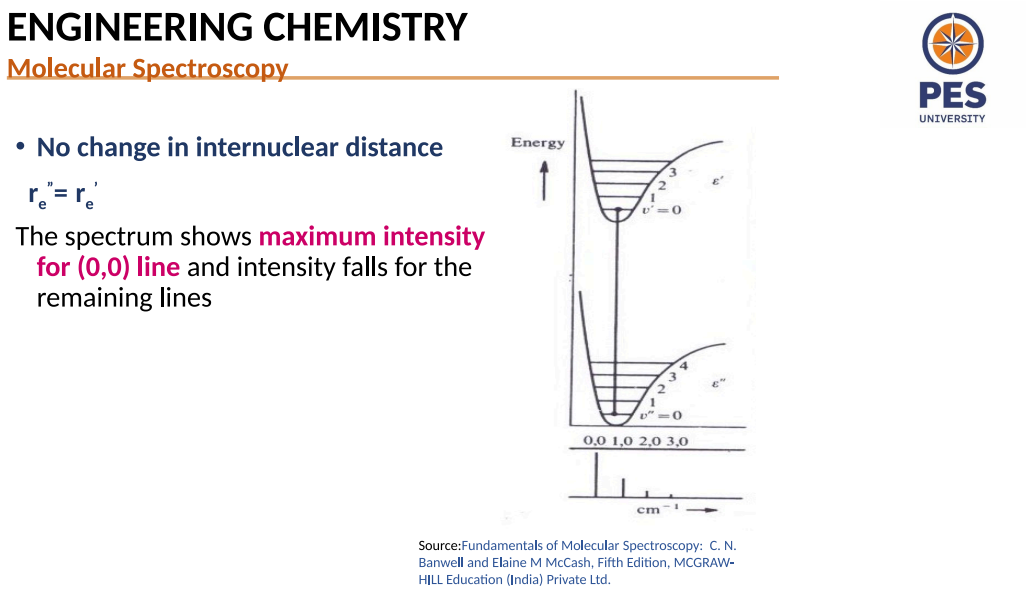

Figure G000073. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 15.

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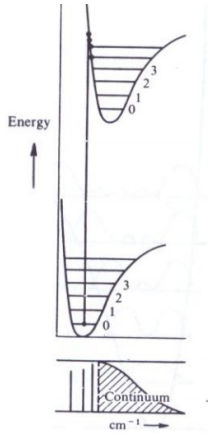
ENGINEERING CHEMISTRY

Molecular Spectroscopy

- **Internuclear distance of excited state is considerably greater than than that of ground state $r_e'' << r_e'$**

The spectrum shows **continuum** since the transition results in the molecule going beyond dissociation energy. The molecule dissociates into atoms which gain kinetic energy. Since this energy is not quantised a continuum is seen in the spectrum



Source:Fundamentals of Molecular Spectroscopy: C. N. Banwell and Elaine M McCash, Fifth Edition, MCGRAW-HILL Education (India) Private Ltd.

Figure G000074. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 16.

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Figure G000075. Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle · source slide page 17.

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000432 · EV0000433 · EV0000434
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Class 7 content: Electronic spectroscopy - Vibrational course structure, Frank Condon principle EV0000435 · EV0000436 · EV0000437
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Electronic spectroscopy Electronic transitions : Highest Occupied Molecular Orbitals (HOMO) ☒ Lowest Unoccupied Molecular Orbitals(LUMO) Transitions in UV-Visible region σ ☒ σ* , π ☒ π* , n ☒ π* If transitions in visible region colour can be seen Electron redistribution causes change in electric field All molecules can undergo electronic transitions [Image text, OCR] *(anti-bonding) π* (anti-bonding) π* ππ* n(non-bonding) π(bonding) o(bonding) EV0000438 · EV0000439 · EV0000440
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Electronic spectroscopy Born-Oppenheimer approximation According to Born-Oppenheimer approximation, the total energy of a molecule is the sum of translational, rotational, vibrational and electronic energies, i.e. ETotal = Et + Er + Ev + Ee which implies that the electronic, vibrational and rotational energies of a molecule are completely independent of each other. The change in total energy of the molecule is given by in Joules in cm-1 The approximate orders of magnitude are EV0000441 · EV0000442 · EV0000443

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Electronic transitions involve electronic redistribution hence electrostatic forces experienced by the molecule which vibrates in response and this in turn affects rotation of the molecule Electronic transitions are accompanied by vibrational and rotational changes. Hence vibrational changes will produce “coarse structure” and rotational changes will produce “fine structure” in the electronic spectra of molecules Information about vibrational and rotational structure of homonuclear molecules like H2, N2 which are inactive in IR and Microwave region can be obtained from their electronic spectra EV0000444 · EV0000445 · EV0000446
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Vibrational Coarse structure The ground state energy levels are denoted by Ɛ” and v”, while the excited state energy levels are denoted by Ɛ’ and v’ There are no selection rules for vibrational transitions during electronic transition ε” ε’ 4 3 2 1 v” =0 4 3 2 1 v’ =0 E EV0000447 · EV0000448 · EV0000449
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Most of the transitions start from v”=0 as it is the most populated level ε” ε’ 4 3 2 1 v” =0 4 3 2 1 v’ =0 E EV0000450 · EV0000451 · EV0000452
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy The lines in the resulting spectrum are denoted by (v’,v”) The resulting spectrum is called progression The lines converge at higher energy levels showing anharmonicity in the excited electronic state Spectrum showing Progression 0,0 1,0 2,0 3,0 4,0 cm-1 EV0000453 · EV0000454 · EV0000455
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Vibrational ‘coarse structure’ during electronic transition [Image text, OCR] 65 4 2 u"=0 0.0 1,0 2,0 3,0 4,0 5,06.0 cm-1 Figure 6.1Thevibrational‘coarse'structureof theband formed duringelectronicabsorption from theground(u"=O)statetoahigherstate. ε’ ε” Source: Fundamentals of Molecular Spectroscopy: C. N. Banwell and Elaine M McCash, Fifth Edition, MCGRAW-HILL Education (India) Private Ltd. EV0000456 · EV0000457 · EV0000458
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Electronic spectroscopy Intensity of spectral lines There are no selection rules for vibrational changes during an electronic transition Most of the transitions start from v”=0 as it is most populated level The excited state usually has longer internuclear distance because of antibonding character E r ε” ε’ v”=0 v’=0 1 2 3 4 1 2 3 EV0000459 · EV0000460 · EV0000461
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy The transitions arise from the centre of v”=0 because the maximum probability of finding the nuclei is at the centre of v”=0 [Image text, OCR] Energy Internucleardistance Figure 6.2Theprobability distributionfor a diatomicmolecule accordingto thequantum theory.The nuclei are most likely to be found at distances apart given by the maxima of the curveforeachvibrationalstate. Source:Fundamentals of Molecular Spectroscopy: C. N. Banwell and Elaine M McCash, Fifth Edition, MCGRAW-HILL Education (India) Private Ltd. Probability distribution for a diatomic molecule according to quantum theory EV0000462 · EV0000463 · EV0000464

12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Energy level diagram for a diatomic molecule [Image text, OCR] Electronic transition Excitedelectronic (in optical state. or uv) dissoclation energy Groundstate Vibrational Rotational transition transition (in infrared) (inmicrowave) Internuclearseparation Source: http://hyperphysics.phy-astr.gsu.edu/hbase/molecule/molec.html EV0000465 · EV0000466 · EV0000467
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Franck-Condon principle All lines in the spectra are not of the same intensity though no selection rule for vibrational transitions exist Statement of Franck Condon principle: “An electronic transition takes place so rapidly that a vibrating molecule does not change its internuclear distance appreciably during the transition” This implies that the electronic transitions are always vertical transitions E r ε” ε’ v”=0 v’=0 1 2 3 4 1 2 3 EV0000468 · EV0000469 · EV0000470
14. Source slide page 14 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy [Image text, OCR] Energy 0.01.02.03.0 0.01,02,03.0 tinuum cm cm cm (a) (b) (c) Figure 6.3 The operation of theFranck-Condon principle for(a) internuclear distances equal inupper and lower states,(b)upper-state internuclear distance a littlegreater than thatin the lower state,and(c)upper-state distance considerably greater. re” and re’ are the internuclear distances of the ground and excited states respectively Three possibilities of vibrational - electronic transitions exist Internuclear distance of excited state is more than than that of ground state re” < re’ The spectrum shows maximum intensity for (v’,0) line where v’ is the vibrational level in the excited state to which electronic transition takes place Source:Fundamentals of Molecular Spectroscopy: C. N. Banwell and Elaine M McCash, Fifth Edition, MCGRAW-HILL Education (India) Private Ltd. EV0000471 · EV0000472 · EV0000473
15. Source slide page 15 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy No change in internuclear distance re”= re’ The spectrum shows maximum intensity for (0,0) line and intensity falls for the remaining lines [Image text, OCR] Energy 0.01.02.03.0 0.01,02,03.0 tinuum cm cm cm (a) (b) (c) Figure 6.3 The operation of theFranck-Condon principle for(a) internuclear distances equal inupper and lower states,(b)upper-state internuclear distance a littlegreater than thatin the lower state,and(c)upper-state distance considerably greater. Source:Fundamentals of Molecular Spectroscopy: C. N. Banwell and Elaine M McCash, Fifth Edition, MCGRAW-HILL Education (India) Private Ltd. EV0000474 · EV0000475 · EV0000476
16. Source slide page 16 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Internuclear distance of excited state is considerably greater than than that of ground state re” << re’ The spectrum shows continuum since the transition results in the molecule going beyond dissociation energy. The molecule dissociates into atoms which gain kinetic energy. Since this energy is not quantised a continuum is seen in the spectrum Source:Fundamentals of Molecular Spectroscopy: C. N. Banwell and Elaine M McCash, Fifth Edition, MCGRAW-HILL Education (India) Private Ltd. EV0000477 · EV0000478 · EV0000479
17. Source slide page 17 (extracted text): THANK YOU Department of Science and Humanities EV0000480 · EV0000481 · EV0000482

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.644] Hello everyone. Welcome back to this course on engineering chemistry. We will continue with module one on molecular

spectroscopy. This will be the last class of this module. [00:00:14.468] Today's class we are going to discuss about the intensity of spectral lines and the Frank Condon principle. We are studying electronic spectroscopy. [00:00:26.848] Which means that when we allow UV visible light to fall on the molecule, the electron can undergo transition from the ground state to the excited state. Now the ground state is always represented by a double prime and the excited state is represented by [00:00:44.496] single prime. So E double prime stands for the ground electronic state and E prime stands for the ground excited state. And within the electronic state we know that there are vibrational level and these vibration levels you can see are [00:01:02.692] Coming closer and closer, the energy gap between the levels is coming, uh is becoming less, which shows that the molecule is an harmonic in nature. Similarly, in the excited state also we have these vibrational levels and they are coming closer. Also, the energy profile is a molecular. [00:01:20.464] curve that shows us that the molecule is undergoing anharmonic oscillations and V double prime equal to 01234 is representing the vibrational levels in the ground state and v prime equal to 0123 represents the [00:01:39.168] vibration levels in the excited state. Now in order to understand the intensity of spectral lines, we need to know certain rules and certain facts about the [00:01:51.536] Transitions. Now there are no selection rules for vibrational changes during an electronic transition. It means that when an electron undergoes transition from the ground state to the excited state, [00:02:06.256] it can end up in any of these vibrational levels. There is no rule [00:02:13.136] forbidding any kind of vibrational change. So if we have electron coming from V double prime equal to 0, it could go to V prime equal to 0, 1, 2, 3, or if it is starting from V double prime equal to 1. Similarly it could end up in any of these [00:02:30.800] Vibrational levels of the excited state. So that is what the rule says that there is no forbidden transition as far as vibrational changes are considered. So that means the spectrum is going to be very complicated. [00:02:49.060] but we have a rule which helps us simplify the spectrum [00:02:54.628] that most of the transitions start from V double prime equal to zero as it is the most populated level. So most of the molecules now will be at V double prime equal to zero, most of the electrons will be residing in V double prime equal to zero with at room temperature. [00:03:12.400] And that becomes the most populated level. And therefore, most of the transitions will start from V double prime equal to zero. So when you have UV light falling on the molecule, the electron now moves from V double prime equal to zero to any of the [00:03:29.968] These uh vibration levels in the excited state, that's because of the rule one. [00:03:35.812] Now the third rule says the excited state usually has longer internuclear distance because of the anti-bonding character. Now let us see the internuclear distance in the case of the ground state, which is this. And if you look at the internuclear distance of the excited state, we see [00:03:53.584] that it is definitely larger. Now this could be due to the antibonding character of the excited state. Now how do we explain that? [00:04:05.124] We know that bond order from molecular orbital theory we know that bond order is given by number of bonding electrons minus the number of antibonding electrons divided by two. [00:04:18.148] So if the antibonding character is more, the number of antibonding electrons are more and therefore we get a small bond order. And we know that bond order and bond length are inversely proportional. So if the bond order is small, that means the bond length is going to be more. [00:04:37.924] The next rule says the transitions arise from the center of V double prime equal to 0 because the maximum probability of finding the nuclei is at the center of V double prime equal to 0. Now these represent the vibrational levels and here [00:04:57.296] They should be actually V double prime equal to zero, V double prime equal to one, two, three, four. But because we have shown only one uh electron state, we

have written it as V is equal to zero, one, two, three. But since we are talking of electronic transitions

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:14.928] we should denote the ground state as V double prime equal to zero. [00:05:19.748] Now what is the probability of finding the nuclear? Now psi is the wave function. The psi squared becomes the probability of finding the nuclear. For V is equal to 0, we find that this is the shape. [00:05:36.868] And here we see that in the center there is maximum probability of finding the nuclei. [00:05:44.736] As we go to V is equal to 1, we find that it becomes like this. And then as we go to higher and higher values of V, we find that the probability of finding the nuclei is more at the uh extremity as for V [00:06:01.808] is equal to zero, the probability of finding the electrons in is more in the center. And therefore that is going to help us further [00:06:14.112] understand the pattern. So what is the rule? V double prime equal to zero is the one which is most populated level at room temperature and [00:06:25.552] Here from the probability graph we can understand that the center is where we are going to have the electrons. It is not that there are no electrons here or the nuclei do not exist in these positions at any point of time, but more probably [00:06:43.248] of finding the nuclei is at the center for V double prime equal to zero. Now we are in a position to understand the energy level diagram for a diatomic molecule. [00:06:58.016] If you remember in the initial classes I had given this diagram, but we did not clearly understand why we are drawing the diagram in this way for a diatomic molecule. Now we we have understood. [00:07:14.724] Rotational spectroscopy, vibrational spectroscopy, as well as electronic spectroscopy and the rules which are governing the energy profile and the placement of these. [00:07:26.864] uh s uh energy levels. So let us now clearly understand what an energy level diagram for a diatomic molecule would look like. Now here we have the [00:07:36.432] ground state, electronic state, and here we have the excited electronic state. So one thing we observe is the Morse curve. So for both ground as well as the excited state and therefore we know that it is [00:07:50.052] And harmonic in nature, and also the vibrational levels seem to be coming closer, and that also is an indication that we are talking of anharmonic uh anharmonically oscillating molecule. So this is the uh ground state and this is the excited state, and within the excited state. [00:08:07.824] The ground state we have these vibrational levels, and within the excited state also we have these vibrational lepths. [00:08:17.604] Within these vibrational levels we have the rotational levels. Now these rotational levels you can see that as you increase the [00:08:27.856] J value, then we find that these rotational levels become further and further apart. And that is what we have understood. So we have the [00:08:38.692] ground electronic state, ground excited state. Within each of these electronic levels we have the vibrational levels, and within the vibrational levels we have this uh rotational levels. So [00:08:52.880] If I want an electronic transition to take place, or if I want the molecule to move from the ground electronic state to the higelectronic state, it requires a lot of energy and that energy is corresponding to UV visible light. [00:09:09.040] Whereas if I want just a vibrational change in the molecule, then this small amount of energy is what is required and that happens to be in the infrared region. [00:09:22.628] Similarly, if I want just a rotational change, then it's this very small amount of energy which needs to be provided, and that comes from the microwave region. So now we can understand. [00:09:36.624] why we draw the energy level diagram this way and also we know that because of the anti-bonding character there is a slight displacement in the excited state. Let us now look at Frank Cordon principle, which is going to explain the intensity of the spectrum.

[00:09:56.068] Now we find that all lines in the spectra are not of the same intensity, though there is no selection rule for vibrational changes to take place.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:06.948] So when an electron undergoes transition from the ground state to the excited state, there is no restriction on the vibrational changes. So that means all lines should be of equal intensity, but it is not found to be so. [00:10:24.676] That is explained with the help of Frank Contin principle, which states that an electronic transition takes place so rapidly that a vibrating molecule does not change its internuclear distance appreciably during the transition. [00:10:41.220] the electron being very light undergoes transition very quickly and the nuclei being bulky they take time to adjust to this transition and therefore for the electron it is [00:10:57.072] as if the nuclei are stationary. So we find that the electronic transition happens to be vertical transitions. [00:11:05.776] So the electron moves as if there is no change in internuclear distance, and as a result, what will happen? That in this case, what you see here that the electron has moved from V double prime equal to 0 to V prime equal to 2. And therefore, since [00:11:23.504] V double prime equal to zero is the state which is most populated and at the same time we know that transitions will happen from the center of V double prime equal to zero. So most of the transitions are happening from happening from here and therefore two zero line will be most intense in this. [00:11:41.136] particular case. Let us study this in little more detail. RE double prime and r e prime are the internuclear distances of the ground and excited states respectively. [00:11:53.796] So this represents E double prime, that is the ground state, electronic state, and this represents the excited electronic state. And RE double prime would correspond to the internuclear distance for the [00:12:08.336] ground state whereas R prime would stand for the internuclear distance of the excited state. Now when there is an electronic transition taking place, there are three possibilities. [00:12:23.556] Let us see what they are. The first one states that the internuclear distance of the excited state is more than that of the ground state. So here we find that RE double prime that is RE double prime is less than RE prime. [00:12:41.188] Now in this case if we follow the flank francon principle, we know that is recollect the rules, what is it that we [00:12:49.808] Double prime equal to zero is the most populated state and transitions most of the transitions will happen from the center because that is the where we have the maximum probability of finding the nuclear. So the electron transition happens from here and it's always a vertical transition. [00:13:07.568] So we see that here the electron has landed in V prime equal to 2. So therefore in this case we find that [00:13:17.712] Two zero is the line with maximum intensity, but that doesn't mean we don't have other lines, there are other lines also, but the probability of finding the nucleide electron is much less. [00:13:31.168] in these areas and therefore we find that there are uh lines corresponding to 1000 and also three zero four zero [00:13:42.692] etc but though those intensities are much less as compared to two zero and here we see that two zero line is here and this is centimeter inverse meaning that energy is increasing in this way one zero definitely would be to the left of two zero [00:14:00.464] Why? Because the position represents the energy gap. [00:14:06.176] Now if the molecule is moving from the electronic transition is happening from V double prime equal to zero to v double prime equal to one, definitely the energy is much less as compared to when there is a two zero transition. And therefore we find that one zero is less than [00:14:23.856] in energy so we will show the line here but nevertheless the intensity is less because the intensity is always more in the center of e double prime equal to 0 and similarly 0 0 will be to the left 30 though the intensity is less it is appearing

to the right of 2 0. [00:14:41.636] Another case would be when there is no change in the internuclear distance. So you can see here that RE double prime is equal to RE prime. In that case, again there is going to be a vertical transition and we find that. [00:14:56.708] It happens from the center and ends up in the center of V prime equal to 0. And therefore the maximum intensity intensity now will be for the 00 line. And there are other possibilities that the probability of finding the new

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:14.480] Does exist on either side and therefore we find one zero two zero three zero lines are also seen in the spectrum. Nevertheless, one zero will be to the right of zero zero because zero zero requires the minimum energy, one zero will require little more energy. [00:15:32.048] And since the spectrum showing the energy scale in this direction, therefore we find that one zero will be here. So in this case where there is the inter nuclear distance of our uh of the excited state and the ground state are equal. [00:15:48.064] then in that case zero zero nine is that it has the maximum intensity. [00:15:54.564] The third case would be where the internuclear distance of the excited state is considerably greater than that of the ground state. So that means RE double prime is much less than RE prime. [00:16:10.336] You can see that it is very much displaced to the right. So now when an electronic transition happens, the electron ends up at uh beyond the dissociation energy of the excited state. Now if you look here, this would represent the dissociation energy of the ground state, and this would represent the [00:16:30.192] Dissociation energy of the excited state. So when an electronic transition happens, it has gone beyond the dissociation energy. So the molecule is going to break into atoms. These atoms gain kinetic energy, and now the energy is not quantized anymore. This kinetic energy. [00:16:47.952] Energy is not quantized, and when energy is not quantized, we cannot get discrete lines, and what we see is a continuum. You can see in the spectrum here, we see a continuum in the spectrum. Nevertheless, there are some lines to the left of the continuum. [00:17:05.520] And those are of lesser intensity because there is probability of finding the nuclei around the center also. But nevertheless, the maximum intensity is from the [00:17:20.580] the la for the line which is originating from the center. But since gone beyond the dissociation energy, what we see is a continuum in the spectrum. So this way Frank Condon's principle is helping us understand [00:17:36.432] why the intensity of spectrines is different for each of these cases, though there is no particular restriction for vibrational changes. [00:17:50.788] That brings us to the end of module one. In the next class, we are going to talk about nanomaterials. Thank you.

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Full transcript: 2590 normalized words, 75 timestamped segments, 1080.576 seconds duration. Low-confidence segments flagged: 0.

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8 Electronic spectroscopy(Progression)

This lesson is indexed from the source video titled Electronic spectroscopy(Progression). Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text

is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000126 · EV0000127

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.996] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module one on molecular spectroscopy and we are going to study electronic spectroscopy in this class. We will revisit the Born-Oppenheimer approximation. [00:00:18.992] And then finally we will study the vibrational core structure, which is also called progression. In electronic spectroscopy, we will find out where the electronic levels in the molecule. [00:00:36.804] Like in rotational spectroscopy, we talked about rotational levels. In vibrational spectroscopy, we talked about the vibration levels. In electronic spectroscopy, we will talk about the electronic levels which are there inside the molecules. [00:00:51.716] and now when a UV visible light is made to fall on the sample, the energy is sufficient to cause electronic [00:01:01.920] transitions and these transitions will happen from the highest occupied molecular orbital to the lowest unoccupied molecular orbital. So what do we understand? For these electronic transitions to take place, UV visible light [00:01:19.600] has the correct frequency which matches with the frequency or the energy gap between these electronic levels. So what are the different types of electronic transitions that can take place? [00:01:36.292] In molecular orbital theory we studied that we have sigma molecular orbitals, pi orbitals, then we have n and then we have the pi star and then the sigma star. [00:01:49.220] Now the there can be any of these transitions uh which are possible due to UV visible light. So it can be sigma to sigma star. [00:02:01.156] Now we understand that sigma and pi are known as the bonding orbitals, whereas pi star and sigma star are known as the antibonding orbitals. And n is known as a non-bonding orbital. So if a molecule has no pair of electrons, that is going to be placed in [00:02:18.928] in the non-bonding orbital. So we can have sigma to sigma star, then you can have n to sigma star [00:02:28.772] You can also have pi to pi star and you can have n to pi star. So all these kind of transitions are possible in the UV visible region. Now if the energy is [00:02:46.656] Very high. If the energy gap is very high, then we it will fall in the UV region and therefore we cannot see the color. But if these transitions happen to fall in the visible region or the energy gap is same as that of the visible region, then we will be able to see the uh color. [00:03:04.336] of the particular molecule. Now when these electronic transitions happen, [00:03:10.960] There is electron redistribution, meaning if the electron goes in one of the molecular orbitals, which is the highest molecular or highest occupied molecular orbital, and there is a transition which happens to the lowest unoccupied molecular orbital, then this there is a electron redistribution. [00:03:28.592] which is going to change the electric field. And therefore [00:03:34.320] we find that all molecules can undergo electronic transitions. If you remember rotational and vibrational uh spectra were not possible and we could not record those spectra because uh for the homonuclear diatomic molecules but [00:03:51.952] Electronic transitions are possible for all kind of molecules. Let us revisit the Born Oppenheimer approximation. [00:04:02.532] Because now we understand what do we mean by rotational levels, what we mean by vibrational levels, and what is a electronic level. So what did Born Oppenheimer what does Born Oppenheimer approximation say that the total energy of a molecule is given by [00:04:20.304] the sum of its rotational energy, the vibrational energy, and the electronic energy. So what does it mean that [00:04:30.208] Each of these energies can be treated

independently. You can calculate what is the rotational energy of the molecule, what is the vibrational energy of the molecule, and what is the electronic energy of the molecule, sum them up and that will give us the total energy of the molecule. [00:04:48.388] Now, if we are talking about change in total energy, because what we are interested in is the energy gap. So if you are talking of change in the total energy of the molecule, that would be the sum of whatever is the rotational change in the molecule, whatever is the

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:06.160] And the electronic [00:05:10.112] uh change in energy of the molecule if this is given in joules. But most of the times in we express energy in spectroscopic units, therefore we write it as small uh E, so del E total is equal to the del E rotational, del E vibrational plus the [00:05:27.792] Del E electronic and remember these are now expressed in wave numbers. So why is it that this approximation holds? [00:05:37.988] That is because the energy gap between the electronic charge electronic levels is much much higher than that of the vibrational levels. So here we can see what is that value. So del E electronic will be approximately del E1. [00:05:55.760] vibration into ten to r uh power of three. So that means the energy gap [00:06:01.776] in an electron in the like between electronic levels is almost thousand times that b what is there between the vibrational levels and similarly for rotational levels it is e del e rotational multiplied by ten to the power of six. So this is the reason [00:06:18.624] why the Born Oppenheimer approximation holds because what are we what is the approximation here that each of these energies are totally independent of each other because the [00:06:31.648] difference in these energy levels is very, very high. So therefore we can use this approximation that the total energy of the molecule can be given as a sum independent sum of the rotational vibration and electronic energies and [00:06:48.528] assuming that there is no electro there is no coupling between any of these energies. So that brings us to a very uh important point that if we have a molecule in a particular electronic state. [00:07:04.132] Now we know that within this electronic state we have the vibrational levels. Within the vibrational levels we have the rotation levels. You can see very clearly the difference in energy is very high. So in an electronic state we are going to have different vibrational levels. [00:07:18.320] or and within the vibration levels and rotation levels. So if I say there are uh two electrons in the same electronic state, but their energies could be different. That is true. [00:07:31.104] Why because they could be in the same electronic state, but they could be in different vibrational levels within the same electronic state. And also they could be in different rotational levels within the same vibrational state. So therefore the total energy would be the sum of [00:07:48.784] Which electronic state it is in and what is the vibrational state and what is its rotational state state. So the two electrons could have absolutely different energies when you sum up those energies. So basically saying that the bond open hybrid approximation. [00:08:06.352] is saying that an electron can ha can be in the same electronic state, but the total energy could be different. So for each electron now the treatment has to be different. [00:08:20.164] Now a little more about the electronic transitions and what is the information we are going to get from this electronic spectroscopy. Now electronic transition as I mentioned just now it involves electronic [00:08:35.616] redistribution means the electron is now it was at one particular level and it went to another level. So there is an electron redistribution. Therefore, there will be a distortion in the electrostatic forces or in distortion in the electric field which is felt by the molecule. [00:08:55.076] Now, since there is a change in the electric field felt by the molecule, how does the molecule respond? It is going to vibrate. Now, when a molecule

vibrates, it is going to affect its rotation also. So meaning [00:09:12.704] Whenever there is an electronic transition, there is a vibrational change also happening, and there is a rotational change also happening. [00:09:23.492] Now, if we can record the vibrational changes during an electronic transition, if we can record the vibrational changes which are happening, we will get what is known as the core structure of the molecule. [00:09:40.192] And if we have very good resolution, we can even [00:09:44.932] see the rotational changes which are happening which is referred to as the fine structure in the electronic spectr of the molecule. So what are we seeing here that we are using UV visible light, so there is going to be electronic transition

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.596] But along with the electronic transition, there could be some vibrational changes. If we can somehow record those vibrational changes, we can get the [00:10:13.536] core structure that's called the vibrational core structure and if the resolution of our instrument is very high we can even see the fine structure. Remember right in the first class I said that molecular spectra are [00:10:25.904] band spectrum so now you can understand that the band is the electronic transition within that when you see these lines these fine lines they are due to the uh core structure and further if you see finer lines then that is referred to as the uh [00:10:43.536] So that is now is very clear it is due to the vibrational and rotational changes which are happening along with the electronic transition. So that means [00:10:57.508] For homonuclear diatomic molecules like hydrogen and nitrogen, which were otherwise inactive in the IR and microwave region, we can [00:11:08.976] record the electronic spectrum and if we can get the vibrational and the rotational structure from this electronic spectrum. [00:11:18.608] Then we will be able to get all the parameters, the rotational parameters, the vibrational parameters for these homonuclear diatomic molecules. So we can know about the bond length, the bond strength, all these which could be determined using. [00:11:37.056] uh pure rotational and pure vibrational uh spectroscopy but unfortunately the hydrogen and nitrogen molecules which are uh homonuclear diatomic molecules they are not active because they don't possess dipole moment and there is no change in dipole moment when the molecule either [00:11:54.704] rotates or when it vibrates. But all these molecules they have these molecular orbitals filled with electrons. So electronic transition are possible and therefore using the electronic spectrum we can get to the vibrational core structure and the fine structure and that is [00:12:12.272] How we can determine all the rotational and vibrational parameters for these molecules. Now let us understand the vibrational course structure. [00:12:25.060] How are we going to use it to determine the vibrational parameters of the molecule, though? [00:12:33.728] We are using UV visible light, that means it is going to result in electronic transitions. Now let us look at the diatomic molecule. This is the structure where we have this lower electronic state that is known as the ground state, and this will be the excited state. [00:12:51.376] State. So we have two electronic states here, the ground state and the excited state. Now within the ground state, we know there are going to be vibrational levels. So we can see there is a V double prime equal to 0, 1, 2, 3, 4. And what we notice here [00:13:09.632] that the lines are becoming closer and closer. What are these lines representing? They are representing the energy levels. So which levels? Vibrational energy levels within the electronic state. So these vibration levels are getting closer and closer as we go to higher values of V. What does it mean that [00:13:27.424] These are this is representing an anharmonic oscillator and we notice that V double prime equal to zero happens to be slightly away from the [00:13:39.216] uh e is equal to zero because we know that always molecules will possess

this zero point energy. So we should not start V double prime equal to zero from here, we start it slightly slightly above. Now okay coming to this nomenclature. [00:13:53.696] The always the ground state is given the double prime means E double prime stands for the ground electronic state and e prime always stands for the excited state. Similarly, we will label the vibrational levels also by V with V double prime when we say V double. [00:14:11.408] prime we are referring to the ground state and V prime refers to the excited electronic state. [00:14:18.112] So, therefore, we have V double prime equal to 01234, and here we have V prime equal to 01234. And just like we have vibrational levels in the ground state, we will also have vibrational levels in the excited state. And therefore, it is very important that we distinguish in the [00:14:36.160] the nomenclature. So when I say V prime equal to three then I'm referring to the V prime equal to three this energy level of the excited state. And when I say V double prime equal to one then it is referring to the ground state. So the ground state energy levels are given [00:14:53.808] double prime like e double prime and b double prime and also excited state are given single prime that is e prime and b prime and one important fact which we have to remember that there are no selection rules

EV0000134 · EV0000135 · EV0000136 · EV0000137

Transcript block 4 (source interval 000900-001199 seconds): [00:15:09.700] For vibrational transitions during an electronic transition. Now, what does this statement mean? That [00:15:17.088] If whenever there is a molecule which is going from ground state to the excited state, we must remember that we are using UV visible light, that means the transition has to happen from here to here and not within the vibrational uh level, say or of the same. [00:15:34.768] electronic state no we are using UV visible light so therefore the transition has to happen to the next electronic state now it can [00:15:44.112] go from zero to zero it can go from zero to one zero to two and also it can go from V V double prime equal to one and go to two three four of the [00:15:57.024] uh higher levels. What does it mean? There are no selection rules to be followed. So it can uh you can have a zero zero transition, you can have a zero one transition, zero two transition. So there's no restriction on delta V [00:16:12.224] When when we we are talking about an electronic transition, during an electronic transition, there is no selection rule for the vibrational transition. [00:16:22.084] Let us come to the next rule. Now that would mean again that the spectrum is going to be very, very complicated. But again, this fact helps us to simplify the spectrum. [00:16:36.464] Why? Because we know at room temperature it is V double prime equal to zero. Now we should not write V is equal to zero, we should say V double prime equal to zero because we are talking of two different electronic states. So V double prime equal to zero is the most populated state [00:16:54.000] uh populated uh level and therefore all transition are going to start from v double prime equal to zero. Well at least most of the transitions will start from v double prime equal to zero as that is the most populated level. So what happens to your transitions now? [00:17:11.664] you can have V double prime equal to 0 to V prime equal to 0. [00:17:16.580] So here remember we have a 00 transition also, and then it can go from 0 to 1, 0 to 2, 0 to 3, 0 to 4. This is how the energy gap is increasing. So the lowest energy level would rest correspond to V double prime equal to 0 to V [00:17:34.352] prime equal to zero and slowly as it goes to higher levels. All these are permitted because there is no selection rule. And also now all the transitions are starting from V double prime equal to zero. So that has helped us to simplify the spectrum. [00:17:53.316] Now, the lines which are coming into the resulting spectrum. We just now saw that there will be transitions and how do we label these lines? It's very important to know how we are labeling them. So it is V prime. [00:18:07.632] comma V double prime what does it mean that the

lines are going to be labeled according to the level where it goes that is this is the higher vibrational level [00:18:20.356] vibration level of the high uh excited electronic state and this is the vibration level where it is originating. So therefore if there is a transition from zero to zero then we will get a line here that is zero zero. And if [00:18:36.768] the transition is from zero to one. So it is one zero because we always call uh label the lines like this the higher [00:18:47.780] Electronic state vibration level is named first, is it given first and then the lower. So we see the the second one number will always be the origin from where it is originating. So we will get different lines in the spectrum based on the different types of transition. [00:19:05.552] And again, we can see centimeter inverse is the scale here. That means the energy is going this way. I have to put the 00 line here first and not the 1, 0 or [00:19:17.456] The reverse way. Why? Because the energy gap is going on increasing as we move from uh left to right. So we know that zero zero requires minimum energy, then we have one zero, two zero, three zero, and four zero. So such a spectrum is known as [00:19:35.088] Progression. Why? Because the numbers 0, 1, 2, 3, 4, they are changing in such a manner to uh where it is increasing by one for every line here. [00:19:50.528] Understand all these lines are originating from zero. So therefore zero zero, one zero, two zero, three zero and four zero would rebuild the lines you will see in the spectrum and such a spectrum is known as a progression. Now in an interesting

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:07.908] fact which you see here as we go to these other uh hights where the energy is more we find that the lines are coming closer together. [00:20:20.036] So that is because there is an harmonicity in the excited electronic state. These lines [00:20:29.600] If it was harmonic, then all these lines would be equidistant. But since there is an harmonicity in the upper electronic state, we find that these lines are actually converging. They will start coming closer and closer as we go to [00:20:47.312] the uh higher values of energy. So let us look at what the vibrational core structure looks like. [00:21:00.388] We have understood this very clearly. So V double prime equal to 0 and V prime equal to 0. This is exactly what we drew and this is the ground state and this is the excited state. So ground electronic state and this will be the excited. [00:21:18.896] electronic state and there are transitions happening all our transitions are starting from v double prime equal to zero so if I say it is a line to zero that means it is starting from v double prime equal to zero and which and it is entering [00:21:36.624] up at V prime equal to 2. So that would represent a [00:21:42.832] two zero line similarly any of these lines are can be seen here these are all electronic transition and yes this requires minimum energy zero zero and then energy gap goes on increasing therefore the lines also will appear towards the higher energy [00:22:00.496] And then we have these lines coming closer together as we go to the higher values of energy, and this shows very clearly that the there is an harmonicity in the excited state. So [00:22:19.904] From this spectrum, we can get parameters. [00:22:26.276] of the excited state very clearly. Now in some uh reactions we know that the excited state is very transient. It appears for a very short while so therefore calculating the various parameters is going to be very difficult. [00:22:44.128] But if you can record this electronic spectrum, then we can get all the information about the bond length, bond strength. If you're doing only the vibrational core structure, you can get the bond strength, and if you can do the rotational fine structure as [00:23:01.776] Well, then we can get the bond strength also. So the electronic uh spectr is going to help us understand the excited state. All lines for the ground state are originating from V double prime equal to zero, but nevertheless they are ending up at different levels of [00:23:19.344] The

excited state and therefore we can get information about the excited state if we can record the spectrum. So we have uh two uh uses of this electronic spectroscopy. One is about this excited state, any transient. [00:23:36.912] Excited state, we can get all the parameters, and for homonuclear diatomic molecules also, we can get all the information. In the next class, we are going to talk about the intensities of these lines. [00:23:54.020] So far what we discussed was about the position of these lines in the vibrational core structure, where the position is determined by the energy gap, but now we have to look at the intensities of these lines, and that will be understood with the help of the Frank Condon principle. [00:24:13.220] Thank you.

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Full transcript: 3489 normalized words, 101 timestamped segments, 1454.933333 seconds duration. Low-confidence segments flagged: 0.

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9 Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential

This lesson is indexed from the source video titled Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000146 · EV0000147

Graphs and diagrams

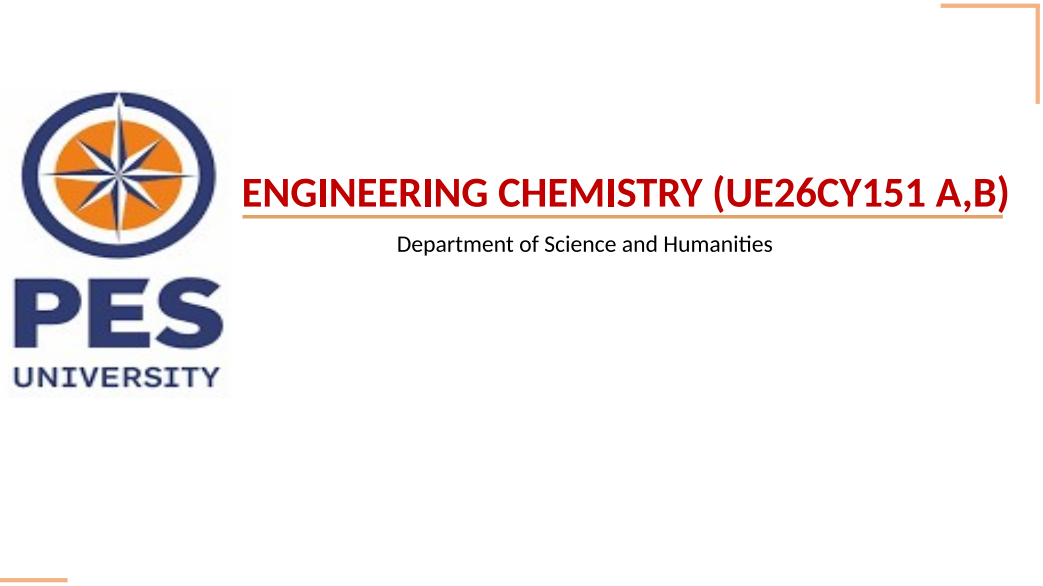

Figure G000076. Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential · source slide page 1.

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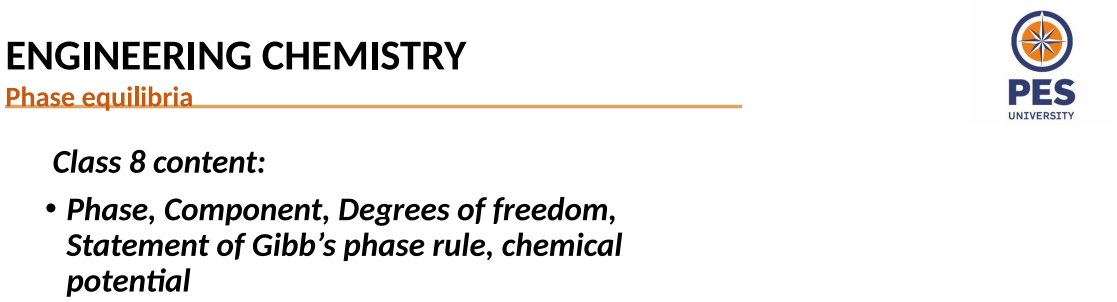

Figure G000077. Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential · source slide page 2.

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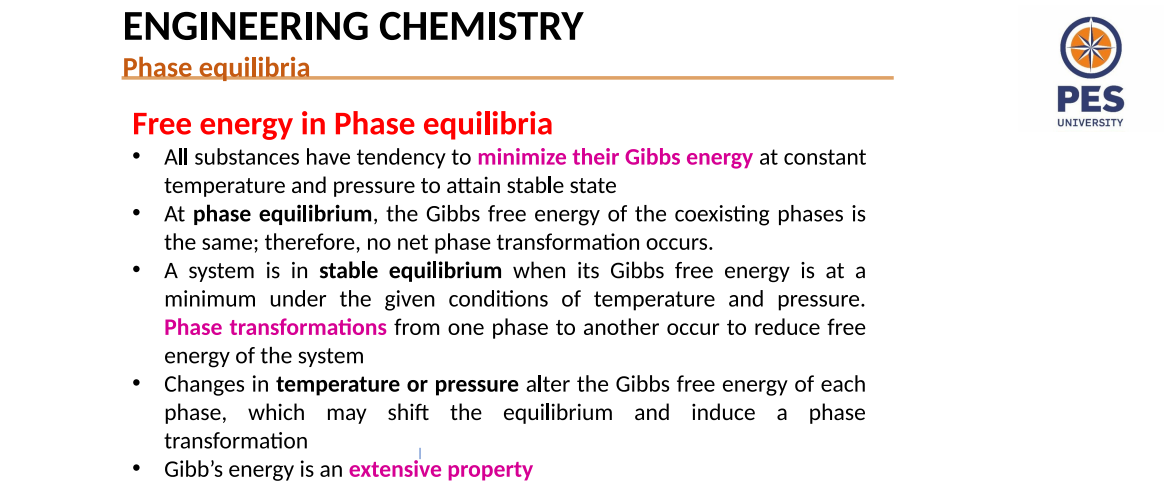

Figure G000078. Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential · source slide page 3.

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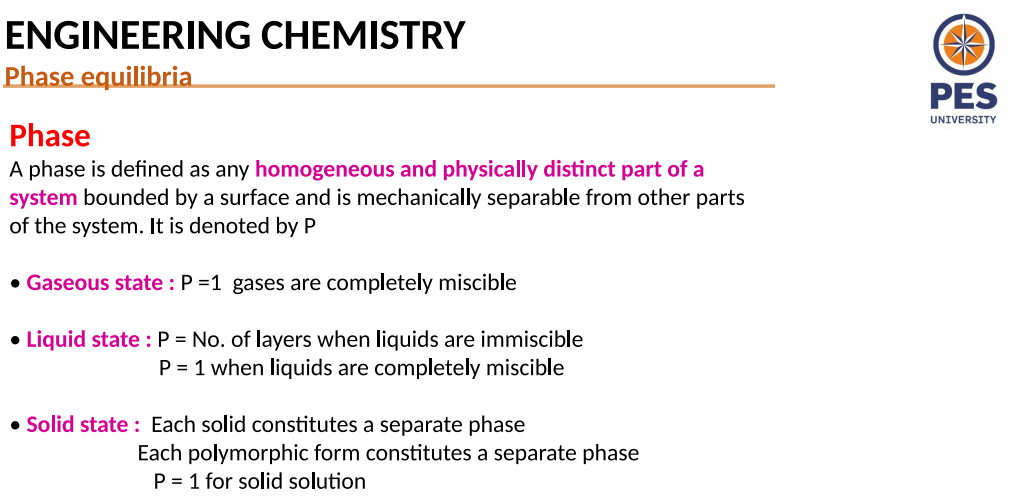

Figure G000079. Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential · source slide page 4.

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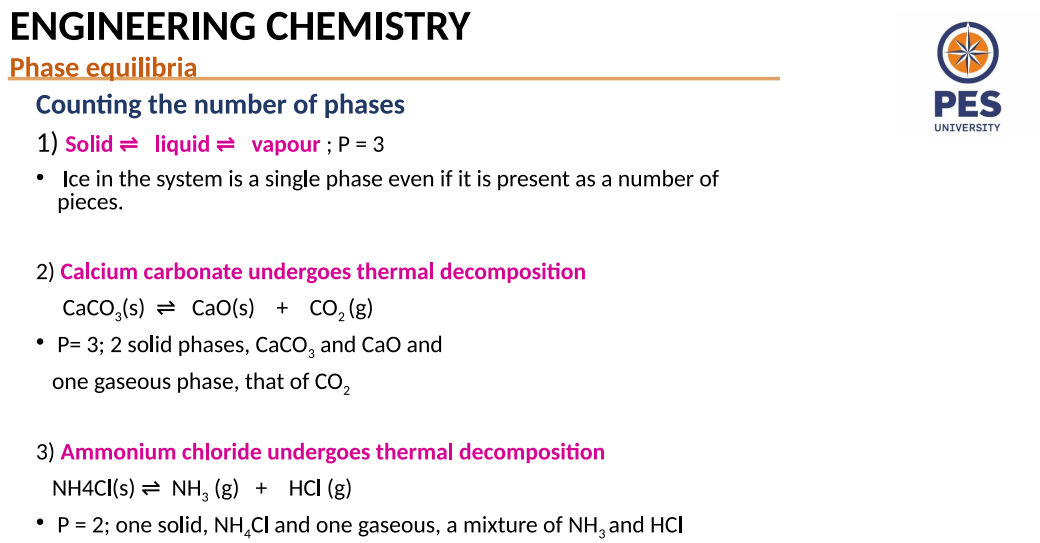

Figure G000080. Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential · source slide page 5.

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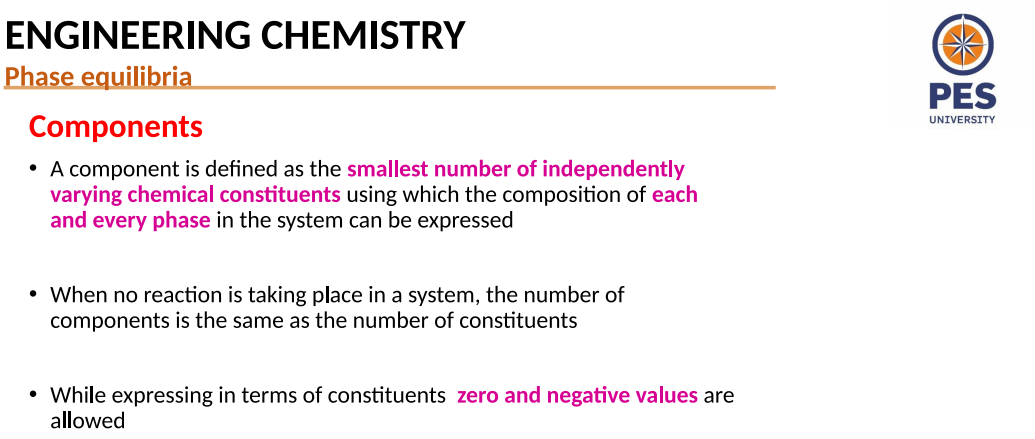

Figure G000081. Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential · source slide page 6.

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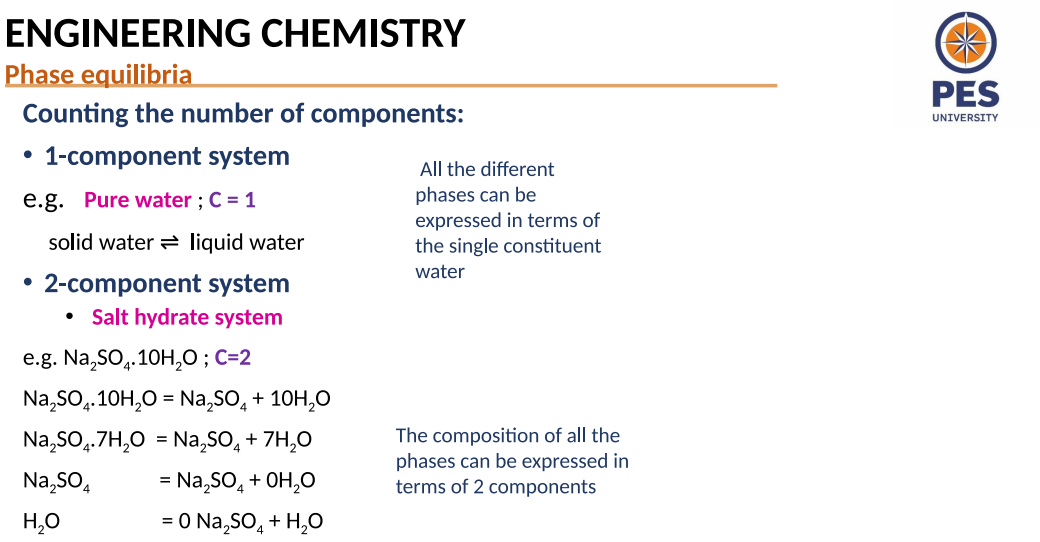

Figure G000082. Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential · source slide page 7.

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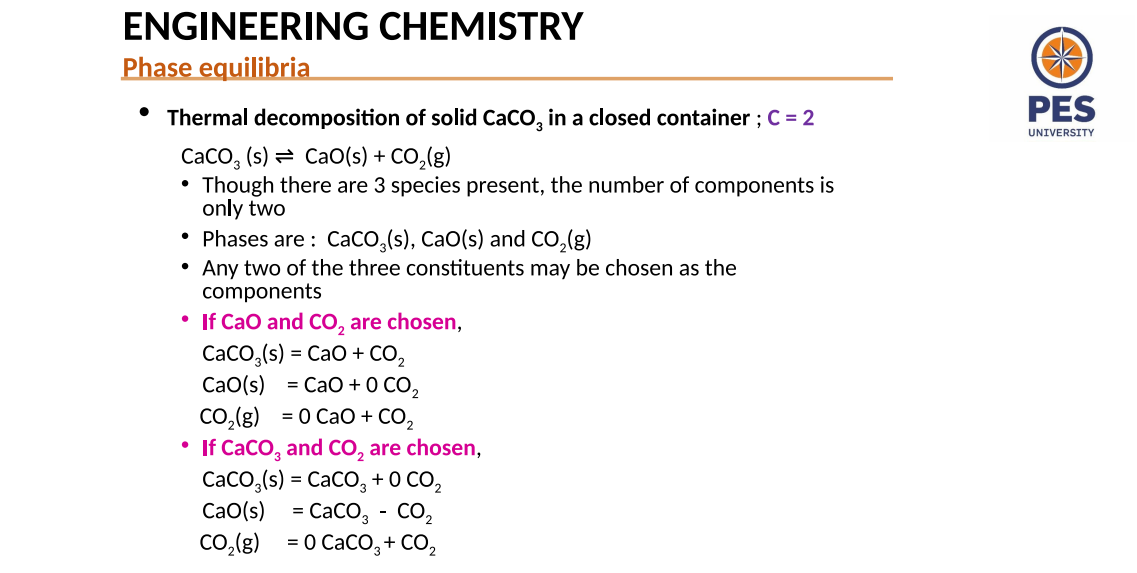

Figure G000083. Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential · source slide page 8.

EV0000505

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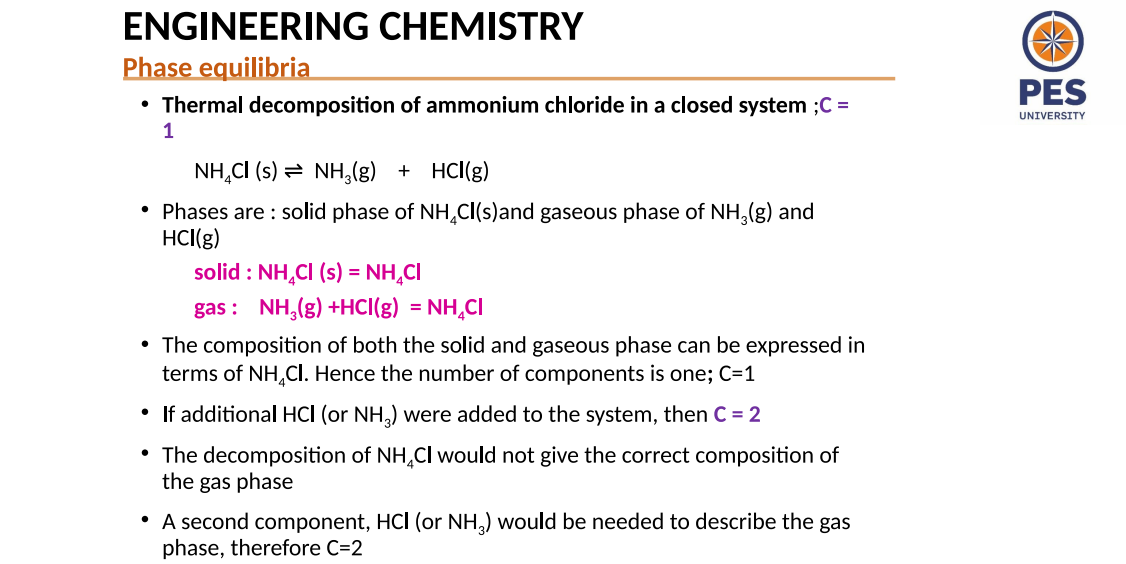

Figure G000084. Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential · source slide page 9.

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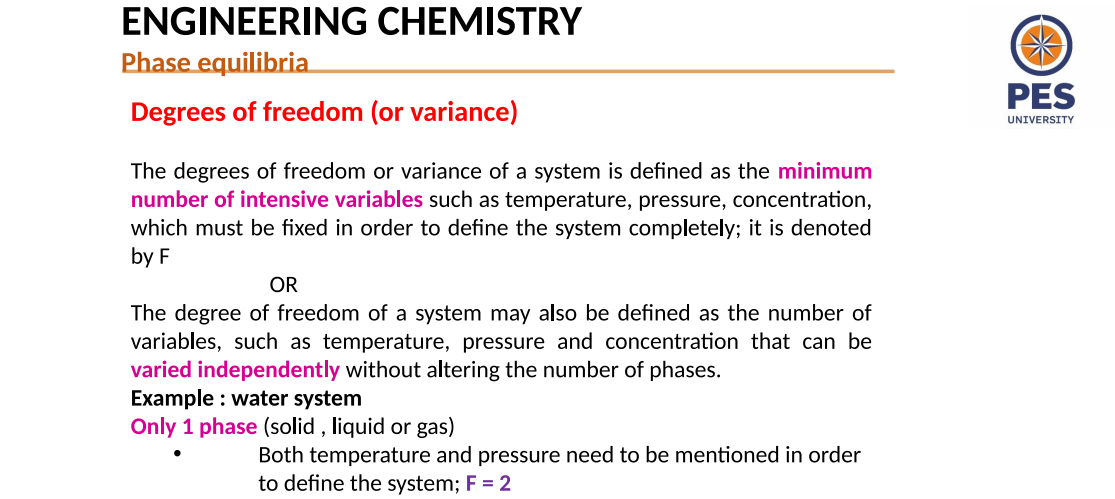

Figure G000085. Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential · source slide page 10.

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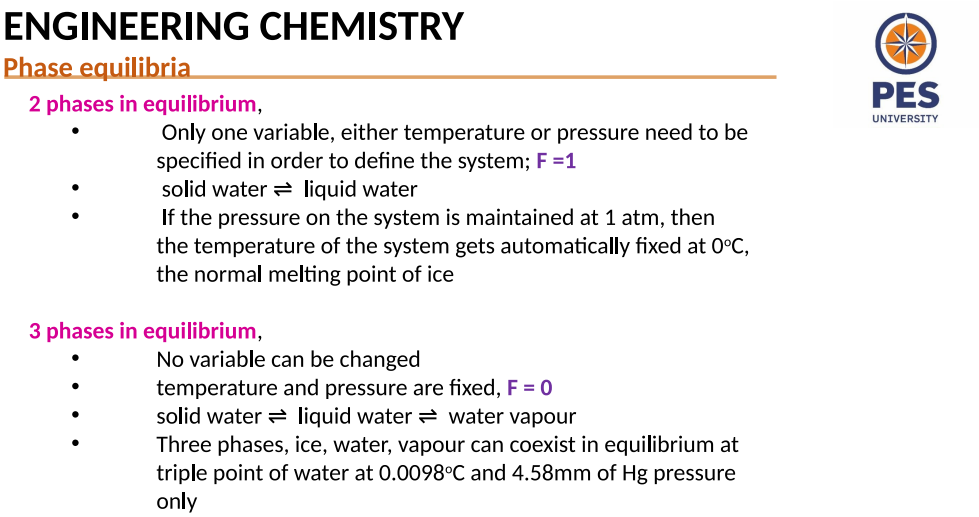

Figure G000086. Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential · source slide page 11.

EV0000514

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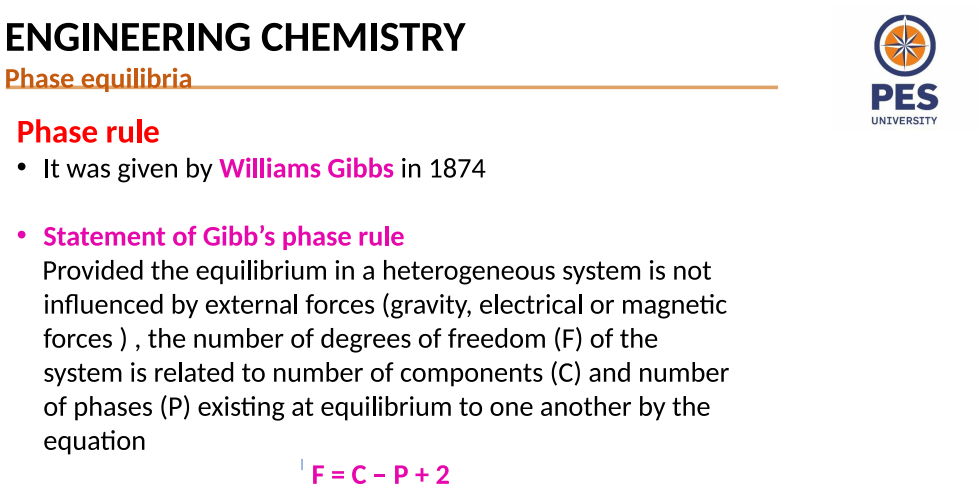

Figure G000087. Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential · source slide page 12.

EV0000517

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Figure G000088. Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential · source slide page 13.

EV0000520

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000483 · EV0000484 · EV0000485
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Class 8 content: Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential EV0000486 · EV0000487 · EV0000488
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Free energy in Phase equilibria All substances have tendency to minimize their Gibbs energy at constant temperature and pressure to attain stable state At phase equilibrium, the Gibbs free energy of the coexisting phases is the same; therefore, no net phase transformation occurs. A system is in stable equilibrium when its Gibbs free energy is at a minimum under the given conditions of temperature and pressure. Phase transformations from one phase to another occur to reduce free energy of the system Changes in temperature or pressure alter the Gibbs free energy of each phase, which may shift the equilibrium and induce a phase transformation Gibb's energy is an extensive property EV0000489 · EV0000490 · EV0000491

4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Phase A phase is defined as any homogeneous and physically distinct part of a system bounded by a surface and is mechanically separable from other parts of the system. It is denoted by P • Gaseous state : P =1 gases are completely miscible • Liquid state : P = No. of layers when liquids are immiscible P = 1 when liquids are completely miscible • Solid state : Each solid constitutes a separate phase Each polymorphic form constitutes a separate phase P = 1 for solid solution EV0000492 · EV0000493 · EV0000494
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Counting the number of phases 1) Solid ⇌ liquid ⇌ vapour ; P = 3 Ice in the system is a single phase even if it is present as a number of pieces. 2) Calcium carbonate undergoes thermal decomposition CaCO3(s) ⇌ CaO(s) + CO2 (g) P= 3; 2 solid phases, CaCO3 and CaO and one gaseous phase, that of CO2 3) Ammonium chloride undergoes thermal decomposition NH4Cl(s) ⇌ NH3 (g) + HCl (g) P = 2; one solid, NH4Cl and one gaseous, a mixture of NH3 and HCl EV0000495 · EV0000496 · EV0000497
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Components A component is defined as the smallest number of independently varying chemical constituents using which the composition of each and every phase in the system can be expressed When no reaction is taking place in a system, the number of components is the same as the number of constituents While expressing in terms of constituents zero and negative values are allowed EV0000498 · EV0000499 · EV0000500
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Counting the number of components: 1-component system e.g. Pure water ; C = 1 solid water ⇌ liquid water 2-component system Salt hydrate system e.g. Na2SO4.10H2O ; C=2 Na2SO4.10H2O = Na2SO4 + 10H2O Na2SO4.7H2O = Na2SO4 + 7H2O Na2SO4 = Na2SO4 + 0H2O H2O = 0 Na2SO4 + H2O All the different phases can be expressed in terms of the single constituent water The composition of all the phases can be expressed in terms of 2 components EV0000501 · EV0000502 · EV0000503
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Thermal decomposition of solid CaCO3 in a closed container ; C = 2 CaCO3 (s) ⇌ CaO(s) + CO2(g) Though there are 3 species present, the number of components is only two Phases are : CaCO3(s), CaO(s) and CO2(g) Any two of the three constituents may be chosen as the components If CaO and CO2 are chosen, CaCO3(s) = CaO + CO2 CaO(s) = CaO + 0 CO2 CO2(g) = 0 CaO + CO2 If CaCO3 and CO2 are chosen, CaCO3(s) = CaCO3 + 0 CO2 CaO(s) = CaCO3 - CO2 CO2(g) = 0 CaCO3 + CO2 EV0000504 · EV0000505 · EV0000506
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Thermal decomposition of ammonium chloride in a closed system ;C = 1 NH4Cl (s) ⇌ NH3(g) + HCl(g) Phases are : solid phase of NH4Cl(s)and gaseous phase of NH3(g) and HCl(g) solid : NH4Cl (s) = NH4Cl gas : NH3(g) +HCl(g) = NH4Cl The composition of both the solid and gaseous phase can be expressed in terms of NH4Cl. Hence the number of components is one; C=1 If additional HCl (or NH3) were added to the system, then C = 2 The decomposition of NH4Cl would not give the correct composition of the gas phase A second component, HCl (or NH3) would be needed to describe the gas phase, therefore C=2 EV0000507 · EV0000508 · EV0000509

10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Degrees of freedom (or variance) The degrees of freedom or variance of a system is defined as the minimum number of intensive variables such as temperature, pressure, concentration, which must be fixed in order to define the system completely; it is denoted by F OR The degree of freedom of a system may also be defined as the number of variables, such as temperature, pressure and concentration that can be varied independently without altering the number of phases. Example : water system Only 1 phase (solid , liquid or gas) Both temperature and pressure need to be mentioned in order to define the system; F = 2 EV0000510 · EV0000511 · EV0000512
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Phase equilibria 2 phases in equilibrium, Only one variable, either temperature or pressure need to be specified in order to define the system; F =1 solid water ⇌ liquid water If the pressure on the system is maintained at 1 atm, then the temperature of the system gets automatically fixed at 0oC, the normal melting point of ice 3 phases in equilibrium, No variable can be changed temperature and pressure are fixed, F = 0 solid water ⇌ liquid water ⇌ water vapour Three phases, ice, water, vapour can coexist in equilibrium at triple point of water at 0.0098oC and 4.58mm of Hg pressure only EV0000513 · EV0000514 · EV0000515
12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Phase rule It was given by Williams Gibbs in 1874 Statement of Gibb's phase rule Provided the equilibrium in a heterogeneous system is not influenced by external forces (gravity, electrical or magnetic forces) , the number of degrees of freedom (F) of the system is related to number of components (C) and number of phases (P) existing at equilibrium to one another by the equation F = C – P + 2 EV0000516 · EV0000517 · EV0000518
13. Source slide page 13 (extracted text): THANK YOU Department of Science and Humanities EV0000519 · EV0000520 · EV0000521

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.572] Hello, everyone. Welcome back to this course on engineering chemistry. We will start with unit two. First, we will look at module three on phase equilibrium. The module content goes like this. We will understand what is phase equilibria. [00:00:21.220] We will derive Gibbs phase rule. Using the Gibbs phase rule, we will construct the phase diagram of a one component system and then go on to apply the Gibbs phase rule to a two component system and construct the phase diagram. And then we'll study the iron carbon phase diagram. [00:00:41.892] So today's class we are going to talk about free energy in phase equilibrium. We will introduce a new term called chemical potential and then we look up certain definitions in phase equilibria that is about the phase component and degree of freedom. [00:01:01.828] Free energy in phase equilibrium. What exactly do we mean by phase equilibrium? We have different phases of matter, you're aware, of solid, liquid, and gas. [00:01:17.028] And we find that in certain conditions the solid phase is more stable [00:01:23.204] In other conditions, liquid phase is more stable, and in some other conditions, gaseous phase becomes more stable. The most common example we can take is that of water. We know that below zero degrees we find solid phase ice is stable. [00:01:38.852] At from 0 to 100, we find it's the liquid phase, and then beyond 100, it is the water vapor. What is the driving force? Why is it that below zero degrees we find only the solid phase and not the liquid phase? So free energy is what defines that. [00:01:57.188] Whichever phase has the lowest free energy, that phase is going to be present. We know that all systems try to reduce their free energy and [00:02:07.872] In this case, we find that solid is having

lower free energy than liquid at zero degrees. When you cross zero and up to 100 degrees, we find that the liquid phase has minimum free energy, and similarly, gaseous phase will have uh less uh lower free energy when it comes to beyond 100 degrees. [00:02:27.588] So every system tries to get minimum free energy, and therefore the phase which has minimum free energy will be most stable phase. Now we also find that phase transformations happen. That is, solid changes into liquid. [00:02:44.592] Now if you talk of water, that happens at exactly zero degrees, which is known as the melting point or the fusion point for uh water or ice. So what happens there? Ice is getting transformed into the liquid phase. [00:02:59.056] beyond the melting point. When you heat the system beyond the melting point, it the solid changes into liquid. Why would such a transformation happen? Because [00:03:08.548] Between 0 to 100, it is a liquid phase which is more stable. So it is moving into the more stable phase. And what would that stable phase mean? Just now I said it will have a lower free energy. So all systems, whenever phase transformation happens, they are moving from one. [00:03:26.464] uh which has a higher free energy to one which has lower free energy. So any such phase transformation would require would would happen when the system wants to reduce the free energy. Reduce free energy of the system. Now [00:03:44.772] That means Gibbs free energy plays a very important role in deciding what kind of phase transformations will happen under a certain set of conditions. But Gibbs energy is an extensive property. [00:03:59.204] So since it depends on the size of the system or the mass or the volume, we cannot depend on Gibbs free energy to compare two different systems. So [00:04:13.732] we have to define a new term which is known as the chemical potential. What is chemical potential? It's basically [00:04:23.344] gives free energy divided by the number of modes. The moment you do that, you have made it into an intensive property, and now you can compare systems, phases, etc, with the help of this new term which is known as [00:04:37.712] Chemical potential. Now we know that in a mixture, it is possible that several components are there. So just saying gives free energy divided by [00:04:49.200] the number of moles that is molar Gibbs free energy would not be sufficient because you have so many components, we say chemical potential, we define it as partial molar Gibbs free energy. It is not just Gibbs free energy, it is partial molar Gibbs free energy. So

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:08.516] How would I mathematically [00:05:11.812] give an expression for this partial molar gives free energy. And remember it is denoted by a symbol mu. So mu i what is i i is the i component in the mixture. So for this component chemical potential is defined as [00:05:28.864] del G by del Ni. NI is the number of moles of that particular component in the system. And since a partial derivative, so pressure, temperature, and number of moles of other components are taken as constant. [00:05:44.452] So let us see how chemical potential is applicable to phase equilibrium. Whenever we say phases are in equilibrium, now instead of Gibbs free energy, we'll refer to chemical potential. [00:05:59.140] So, what would be the condition for phases to be in equilibrium? The chemical potential of a particular component should be the same in all the phases which are present in equilibrium. So if we have [00:06:16.016] Three phases, then in that case we find that suppose it is solid, liquid, and vapor, then we find what is the requirement for a system to be at equilibrium, that the chemical potential [00:06:30.384] the chemical potential of a component is equal in all the phases which are present in equilibrium. So that means at the triple point of water where you have solid, liquid, and the vapor in equilibrium, [00:06:45.872] What is the condition for phase equilibrium? The chemical potential of the solid form is equal to the chemical potential of the liquid form, which is equal to the

chemical potential of the vapor form. Now, if the system is not at equilibrium. [00:07:02.544] what would happen in that case, then at some point in some phase the chemical potential is more as compared to the chemical potential in another phase. So what would happen because of that? There would be a transformation, phase transformation from higher chemical potential to lower chemical potential. [00:07:20.612] So when we say that the system when the phases are in equilibrium with one another. [00:07:27.524] If it is a heterogeneous system, of course, when we say different phases are present, it is a heterogeneous system. There can be no transfer of energy or mass from one phase to another. Only when there is no transfer of mass or energy, we say the system is at equilibrium, or [00:07:45.296] Another way of looking at it is when a system is in equilibrium, the various phases must have the same temperature and pressure, and the composition must remain constant all along. So this is [00:08:01.456] These are the different ways by which we can understand phase equilibrium. Now let us define a few terms that is phase, component, and degrees of freedom. [00:08:18.340] First, let us take up phase. A phase is defined as a homogeneous and physically distinct part of a system. And there will always be a boundary between phases. So how would you define a phase? [00:08:36.032] Whenever you see a boundary, when one phase from the other, there will be a boundary. So these are bounded by a surface and mechanically separable from other parts of the system. And it is denoted by P, and P is a number. [00:08:53.316] Whenever we say phase, we are referring to the number of phases in a particular system. So let us see what are the number of phases. So when we have a gaseous system, we know that gases are completely miscible. So for any gaseous system, we find that P is equal to one. [00:09:11.492] Now, if we have a liquid system, the number of layers, we know that when liquids are immissible, the layers going to be seen in that system, and therefore the number of layers will decide the number of phases. [00:09:26.832] Provided they are invisible. If they are completely miscible, for example, alcohol and water, then P is going to be equal to one. Okay. When we have a solid state, then each state, each of these solid will constitute a separate. [00:09:43.408] Phase we have to count each solid as a separate phase. All solids are not taken as one phase because how did we define phase? There is a boundary. They are mechanically separable. Therefore, each solid is going to constitute one phase. And not only that, each polymorphic form or allotropic form.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.088] Is also going to constitute a separate phase. What do we mean? For example, we have carbon. Carbon exists as graphite as well as diamond. [00:10:09.552] Both of them cannot be treated as one phase, or we cannot say p is equal to one here. Why? Because both of them have boundaries, they can be easily separated. So each of the polymorphic forms is [00:10:22.628] treated as one phase. And if there is a solid solution as in isomorphic alloys, then P is taken as one. So let us take up some cases and see whether we are able to understand or calculate [00:10:40.784] determine the number of phases. So we have a system at triple point where uh where we have solid, liquid, and vapor in equilibrium. Now what are the number of phases here? Here we see P is equal to C three. Why? Because it may be the same system. It may be [00:10:58.500] It may be water system. So here ice will constitute one system, the liquid water will be another another phase, solid ice is one phase, liquid water water is another phase, and water vapor is another phase. So we say P is equal to three. [00:11:13.904] Now let us take a case system where carbon calcium carbonate is undergoing thermal decomposition. [00:11:20.544] Like this is given by this equation, what is the number of phases? Remember that two solids will contribute to two different phases and one gas. So we have two solid

phases, calcium carbonate, calcium oxide, and P is equal to three now because there's also carbon dioxide gas. [00:11:38.192] So in this case we find P is equal to 3. Now if we take the case of ammonium chloride which undergoes thermal decomposition, ammonium chloride solid is giving you ammonia gas and HCl gas. In this case, again, P is equal to 2. Why? [00:11:55.120] Because solid is one, but gases are completely miscible, therefore ammonia and Cl together will constitute one phase. [00:12:06.052] Now let's come to the next definition, and that is of a component. A component is defined as the smallest number. We need to pay attention to these. [00:12:18.448] uh parts which have been underlined, smallest number of independently varying chemical constituents using which the composition of each and every phase [00:12:31.652] These two are to be marked very carefully. Smallest number and each and every phase of the system we should be able to describe with the help of these components. So again, this is a number. This is a number and is denoted. [00:12:49.424] By C. C will be the number of components. And when there is no reaction taking place, obviously as many number of constituents will be equal to the number of components. And since a number, one important uh [00:13:05.488] Fact which we have to remember is that when we are expressing in terms of constituent zero and negative values are also allowed. Let us see what we mean by this by taking up certain examples. Now let us take the case of a one component system. [00:13:25.540] Water. Water is a one component system where C is equal to one. Why? Even if say we have two phases in equilibrium, that is at the melting point, the composition of both the phases will be the same. So if we know [00:13:42.144] what is the composition of water or any any of any of the uh component. If since it's a single constituent, there is both of them will have the same composition. Both of them will be the same. [00:13:56.208] So we need only one constituent water that is enough to describe all the phases. We may have what are the phases here? What are the number of phases? We have two phases here. [00:14:07.824] Solid water as well as liquid water, but both can be expressed in terms of a single constituent. Now let us take up a two-component system. That is a salt hydration. This is one of the systems we will consider. Salt hydrate system. [00:14:25.072] So let us take Na2SO410H2O. Salt hydrate meaning what? It has a water of hydration. So Na2SO410H2O is the composition of the salt. Now here we say C is equal to 2. Why? [00:14:42.020] Because NA2SO410H2O can be expressed as the sum of NA2SO4 plus 10H2O. Only Na2SO4, if you know, that is not sufficient for me to describe this phase. [00:14:56.612] Another phase of the same salt hydrate could be Na2SO47H2. Now what happens? We will write it as Na2SO4 plus 7H2.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:09.060] If we take the anhydrous salt, then what will happen? NE2SO4 is NE2SO4 plus 0H2O. As I mentioned, in the case of components, 0 can also be used. So here, though [00:15:23.952] We are calling it a two-component system. One component is enough to describe this particular phase, but we define component as what? [00:15:32.368] The total number of phases, the minimum number, the smallest number which is described which is used to describe all the phases. You should be able to describe all the phases using that minimum number. Here, though it is one, minimum you need is two to describe. [00:15:50.000] All the phases. So we say C is equal to 2. Now you can also have water itself. Now suppose you take this salt hydrate and evaporate the water completely. Evaporate the water completely and [00:16:02.096] collect it, then you will have only the water phase. Meaning how does or how do all these phases come about here? Because you are changing the conditions, you can have different forms of the same salt hydrate. So this is how we get these different phases.

[00:16:19.600] And how do we describe these phases? In order to describe the phases, now when you come to water, you will say zero of any two so four and water. So that means you need at least two components to describe this uh particular system. So C is equal to two. [00:16:37.136] Now let us take up the third case where we talk of solid calcium carbonate getting decomposed in a closed container. Again here we say C is equal to 2. Why? [00:16:48.880] Because though we have three different phases or three different constituents, there's an equation which is binding them together. So we need not describe each one of them. We need not use all the three constituents. [00:17:06.112] Two components are sufficient to describe all the phases. How? Let us see. [00:17:12.388] What are the phases? We have calcium carbonate solid, calcium oxide solid, and CO2 gas. Let us see if we can describe all the three phases using any two constituents. So let us say we choose CAO and CO2. Then the first phase CACO3 solid will be described as CAO plus CO2. [00:17:30.276] CAO itself will be described as CAO plus 0CO2 and CO2 can be described as CAO plus CO2. So I needed only two of these components to describe all the three phases. Now let us say I take CACO3 and CO2. [00:17:47.584] How will I describe the phases now? We have CACO3 and CO2. So CACO3 becomes CACO3 plus 0 CO2. But if I have to describe CAO now, I don't have CAO with me. How will I use these two? I can say CACO3 minus of CO2. So here [00:18:06.736] we have used a negative quantity of CO2. Now CO2 gas will be described as 0 of CaCO3 plus CO2. So what do we observe? In this case two components are sufficient for me to describe all the three phases. [00:18:25.668] Let us take up another case where we have thermal decomposition of ammonium chloride in a closed system. Here, though the reaction looks similar to the calcium carbonate reaction, C becomes equal to 1. Why? [00:18:41.924] What are the phases? Here we have only two phases. Remember, the one is a solid phase of ammonium chloride, and the other is the gaseous phase where both ammonia and HCl are present in the molar ratio 1 is to 1. So the gaseous phase now is a [00:18:57.968] Mixture of NH3 plus H Cl, but the num the phase is only one because gases are completely miscible. So we have to describe the solid phase as well as the gaseous phase. Now when we talk of the solid phase. [00:19:11.076] In this case, the solid phase is ammonium chloride. So ammonium chloride is sufficient for us to describe the solid phase. But when we come to the gaseous phase, we have ammonia and HCl in equal uh quantities. So therefore, ammonia [00:19:25.792] A plus HCl can also be represented as ammonium chloride. That's so in this case one component is sufficient for me to describe okay. [00:19:36.944] I could use two components also. What ammonia and HCl. If I want to use ammonia and HCl, it will become ammonia plus HCl in this case, and the gaseous form will also be ammonia plus HCl. But the by definition component is the smallest number of [00:19:53.456] constituents which are required. So in this case ammonium chloride itself is sufficient to describe the solid as well as the gas phase. So we say C is equal to one. Now

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:05.860] The composition of both these phases can be described with C is equal to one. Right. But this is if it was a closed system. Now suppose we add a certain amount of HCl or ammonia into the system, when we do that. [00:20:21.008] Now the C becomes equal to two. Why? Because [00:20:26.272] Now the gaseous phase cannot be described by ammonium chloride alone. You have added certain extra amount of HCl or ammonia, so that has to be accounted for. Therefore, C becomes equal to 2 in this case. So [00:20:42.800] Just ammonium chloride is enough for the solid phase, but not enough for the gaseo phase because we added an extra amount of HCl or ammonia. So now you require two components

in order to describe the system or to describe all the phases of the system. [00:21:02.756] Now, let us look at the degrees of freedom. The degrees of freedom or variance of a system is defined as the minimum number of intensive variables, such as temperature, pressure, or concentration, that must be ascertained in order to describe the system. [00:21:22.128] In other words, we can also say what is my degree of freedom to change these variables independently without altering the number of phases, or rather without altering the equilibrium in the system. So [00:21:38.768] Let us look at how to understand this. Let us take up the water system. Now, in the water system, suppose there is only one phase. At a particular time, there is only one phase, so solid, liquid, or the gas. [00:21:54.912] Now, in this case, we know that both temperature as well as pressure can be varied independently. [00:22:02.832] the without affecting the system, I can vary the temperature or pressure. In fact, solid, liquid, or gas gas form of water can be present at different temperature and pressure. Therefore, the degree of freedom or the variance is F that is equal to 2. [00:22:21.156] Suppose we have two phases in equilibrium. Two phases in equilibrium where now temperature or pressure can be [00:22:31.652] They read or can be specified. What does it what do we mean? The degree of freedom now becomes one. Let us take up this system where we have solid water in equilibrium with liquid water. When does this happen? [00:22:43.344] This happens at zero degrees centigrade, but the moment I say zero degrees centigrade, the pressure becomes fixed to one atmosphere. If I change the pressure, then the melting point is also going to change. That zero degrees will not remain zero degrees. [00:22:59.584] Though we have solid water and liquid water in equilibrium, now I can either change the temperature or I can change the pressure. Therefore, when two phases are in equilibrium, f becomes equal to one. [00:23:15.172] Then suppose we come to a time when we have a part of the system where we have three phases in equilibrium. Here, none of the variables can be changed, meaning that neither the temperature nor the pressure can be changed, and therefore we say f is equal to. [00:23:32.368] Zero. Now here where would you find such a uh case? That would be a triple point point of water. Solid water is in equilibrium with liquid water, which is in equilibrium with water vapor. All the three phases are present together, they coexist in equilibrium, and now [00:23:49.840] Neither the temperature nor the pressure can be changed because those are fixed quantities. And what is that value? 0.0098 degrees centigrade and 4.6 millimeter of mercury for water. So when you have a single phase. [00:24:07.332] In a system, F becomes equal to 2. You can change temperature and pressure independent of each other. When you come to two phases in equilibrium, in that case, you can either change the temperature or you can change the [00:24:21.792] pressure so f is equal to one and when you have three phases in equilibrium then f becomes equal to zero. You can neither change the temperature nor you can change the pressure. So we have defined all the three [00:24:35.396] quantities that is phase, component, and degree of freedom. Remember they are numbers. Okay. So now in the next class we are going to look at [00:24:47.108] Gibbs phase rule, which is going to combine all these three components, phase and degree of freedom, and give you a very nice elegant thermodynamic equation. Thank you.

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Full transcript: 3680 normalized words, 100 timestamped segments, 1502.72 seconds duration. Low-confidence segments flagged: 0.

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10 Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part II

This lesson is indexed from the source video titled Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part II. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000166 · EV0000167

Graphs and diagrams

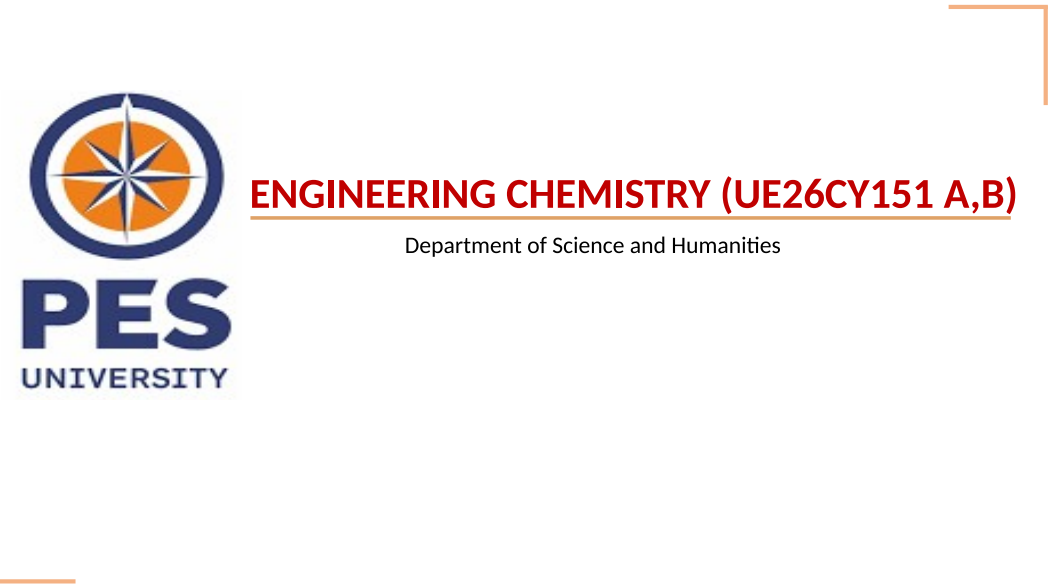

Figure G000089. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 1.

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Figure G000090. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 2.

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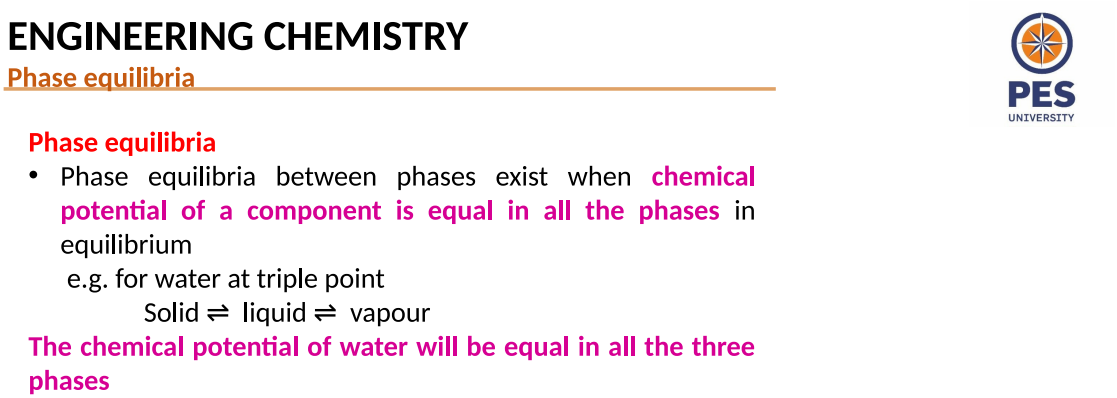

Figure G000091. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 3.

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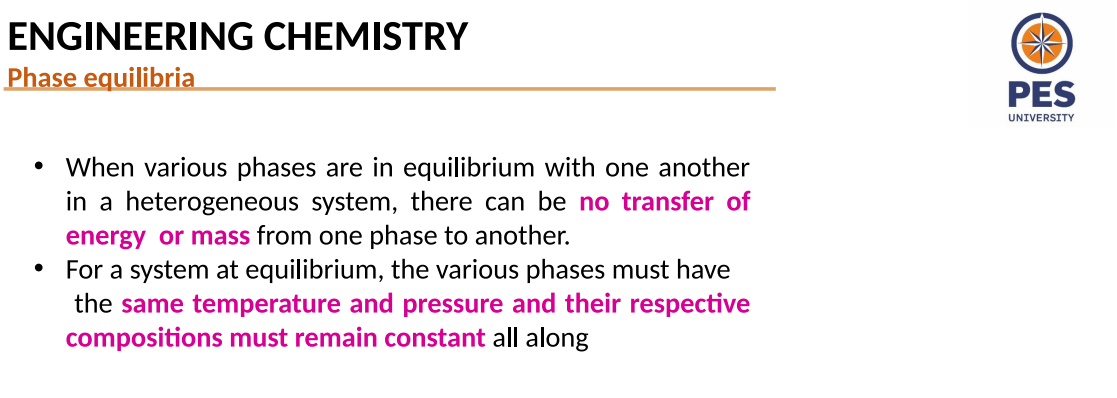

Figure G000092. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 4.

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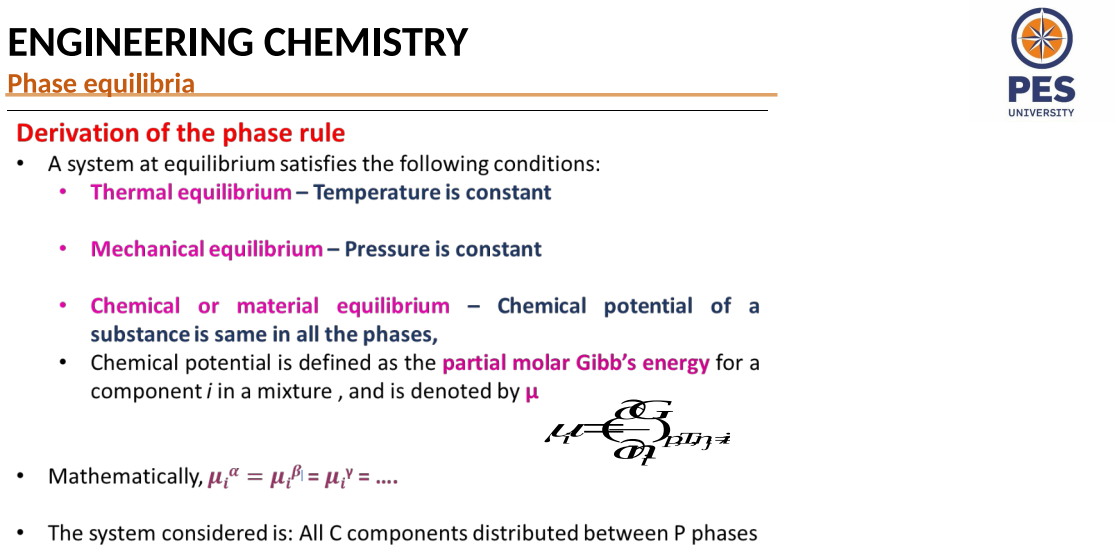

Figure G000093. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 5.

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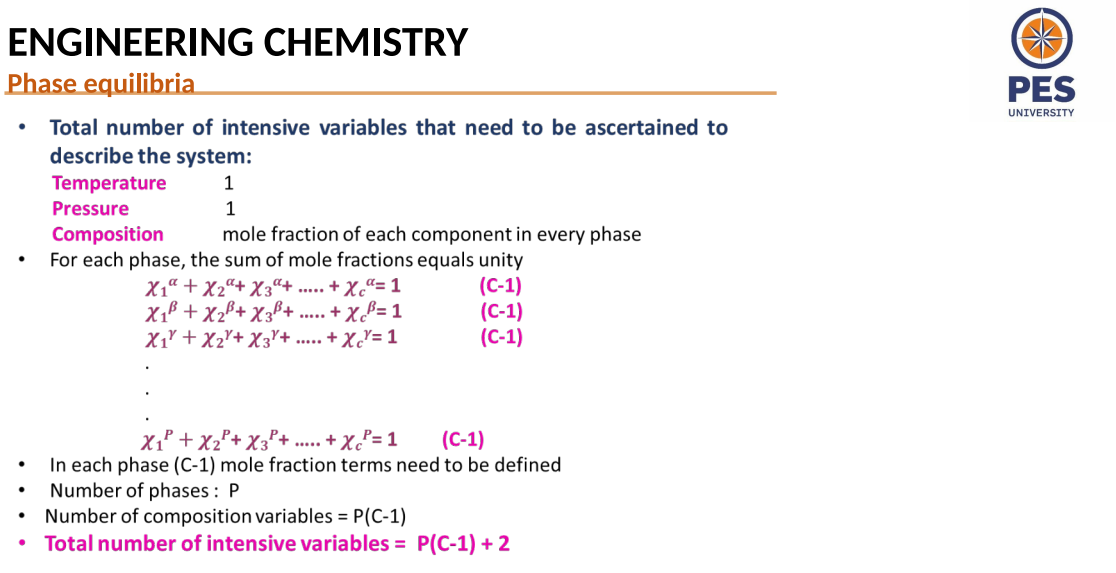

Figure G000094. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 6.

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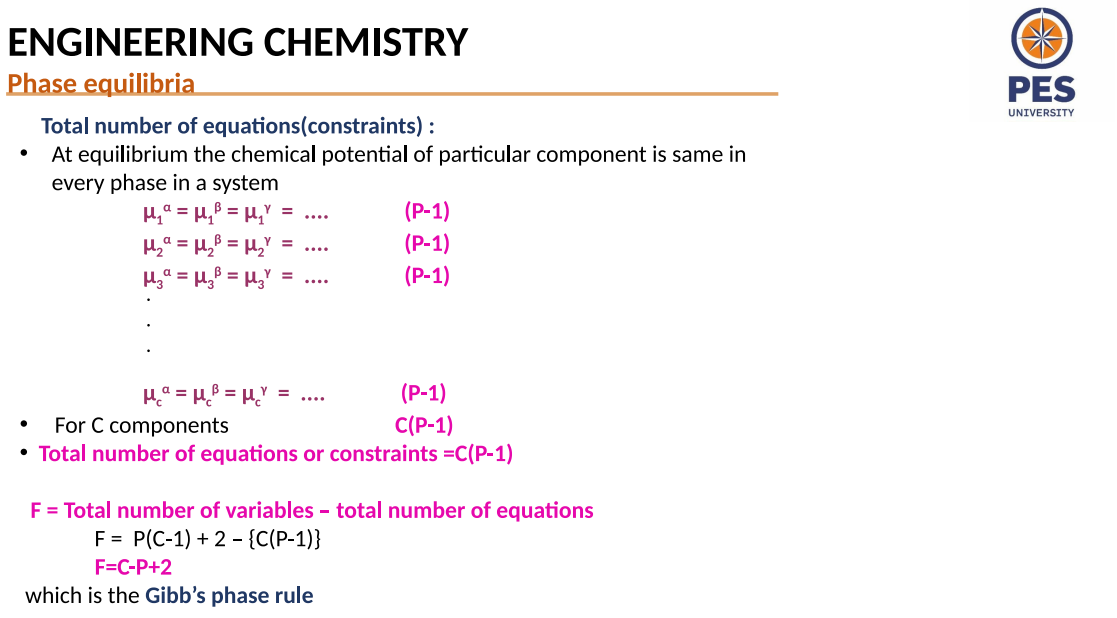

Figure G000095. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 7.

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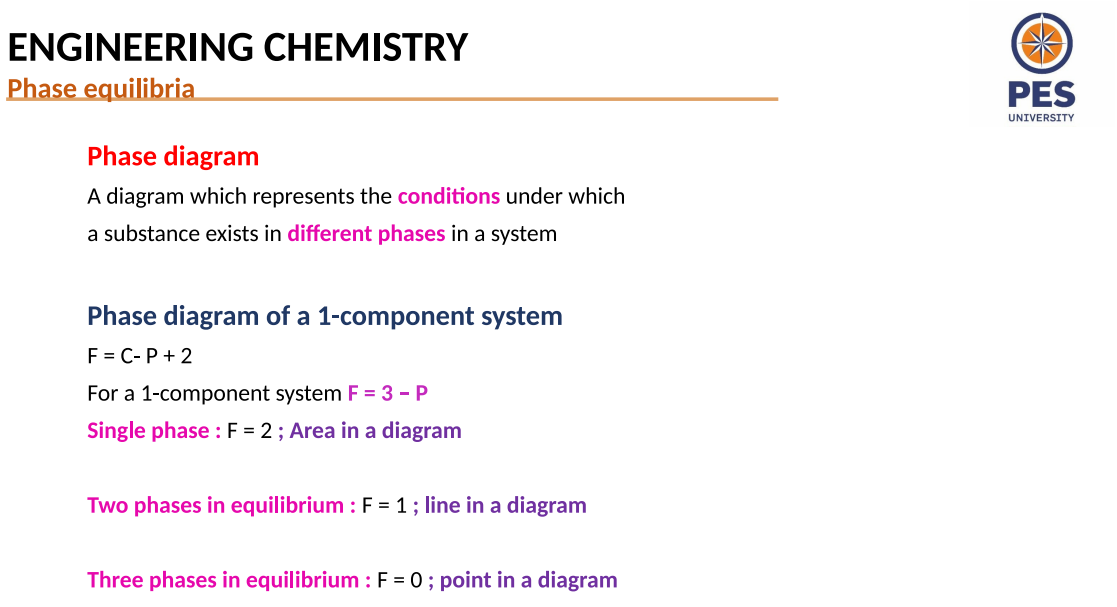

Figure G000096. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 8.

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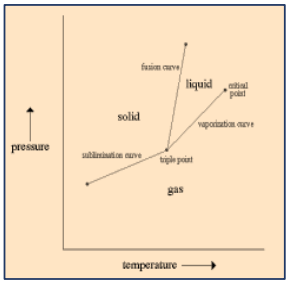
ENGINEERING CHEMISTRY

Phase equilibria



High pressure, low temperature: **solid phase**
High temperature, low pressure : **vapour phase**
In between : **liquid phase**

solid ⇌ liquid
liquid ⇌ vapour
vapour ⇌ solid

Solid ⇌ liquid ⇌ vapour

Source:http://abyss.uoregon.edu/~js/glossary/triple_point.html

Figure G000097. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 9.

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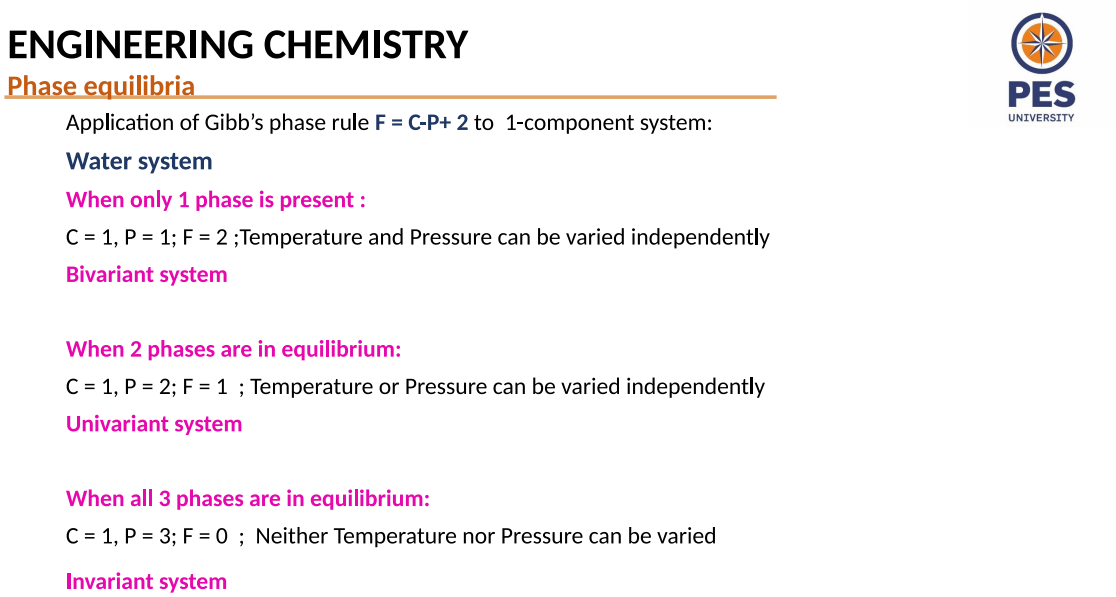

Figure G000098. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 10.

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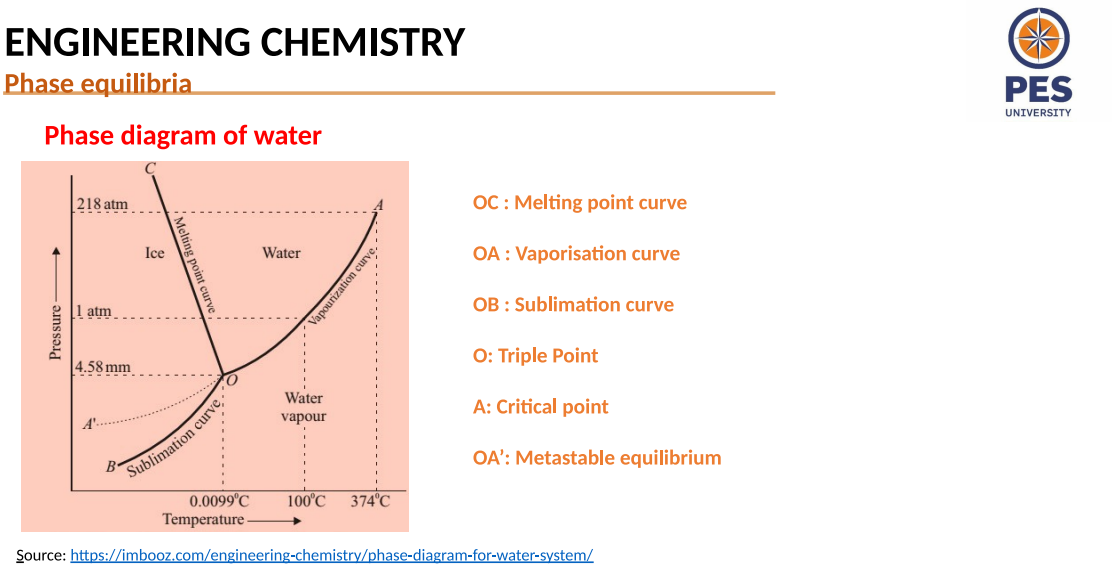

Source: https://imbooz.com/engineering-chemistry/phase-diagram-for-water-system/

Figure G000099. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 11.

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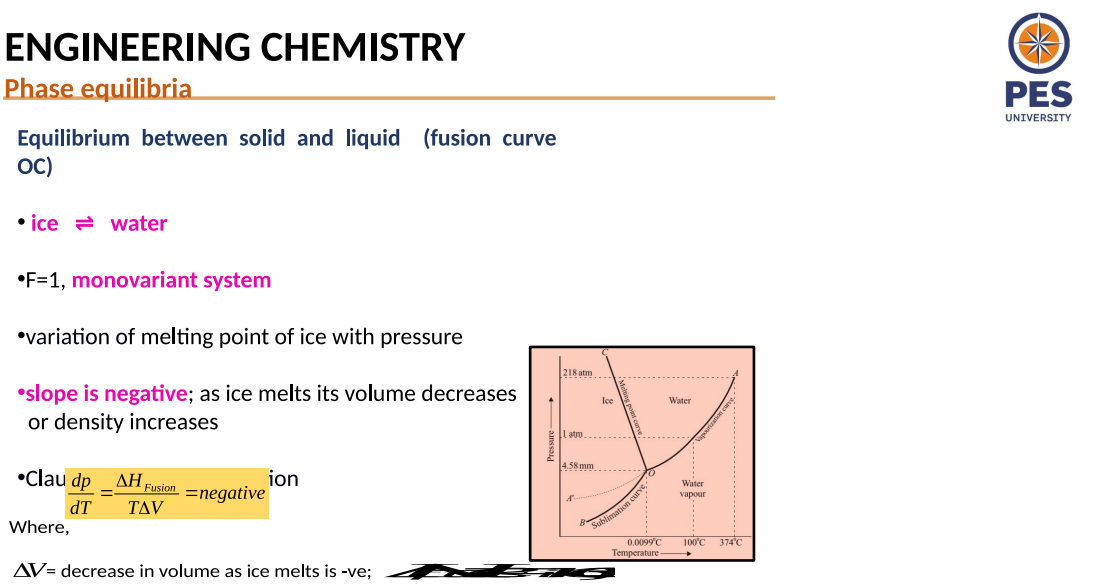

Figure G000100. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 12.

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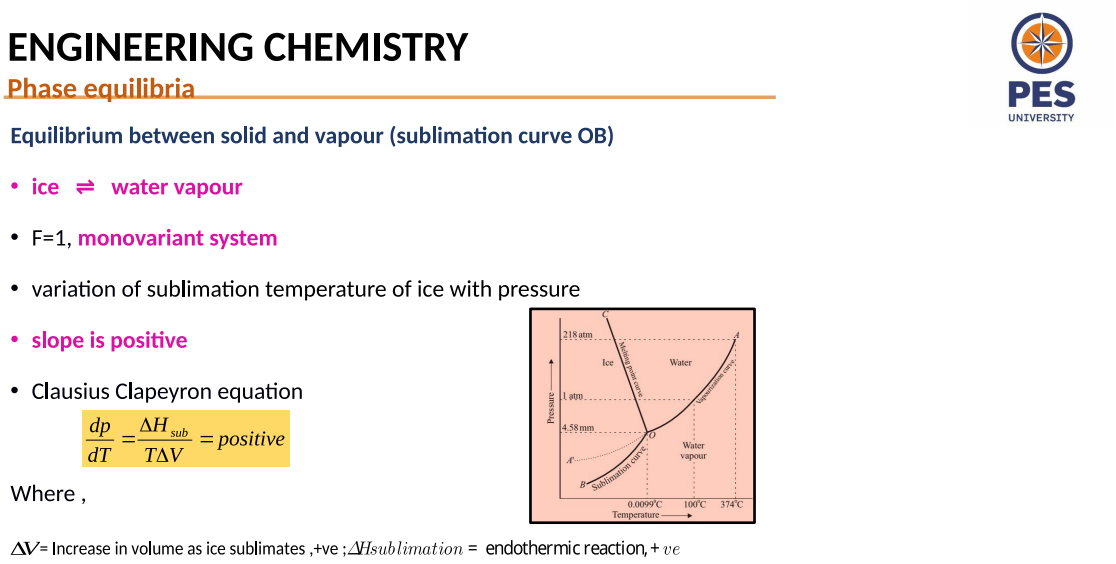

Figure G000101. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 13.

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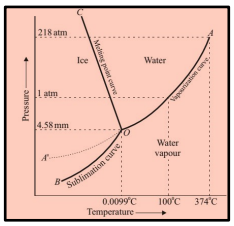
ENGINEERING CHEMISTRY

Phase equilibria

Equilibrium between liquid and vapour (vaporization curve OA)

- liquid water ⇌ vapour
- F=1, monovariant system
- variation of boiling temperature of water with pressure
- slope is positive
- Clausius - Clapeyron equation

$$\frac{dp}{dT} = \frac{\Delta H_{vapourisation}}{T\Delta V} = positive$$



Where,

ΔV= Increase in volume as liquid water vapourises,+ve; $\Delta H_{Vapourisation}$ = endothermic reaction, + ve

Figure G000102. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 14.

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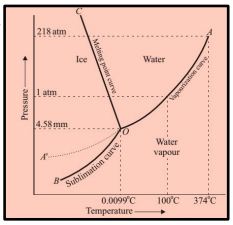
ENGINEERING CHEMISTRY

Phase equilibria

Triple point "O":

- Represents equilibrium between liquid, vapour and solid water (ice)
- All three phases are present together
- F = 0, **invariant system**
- Triple point for water lies at **0.0098 °C and 4.58 mmHg**



Critical point "A":

- the interface between liquid water and water vapour vanishes
- a point above which water does not exist in liquid state
- Critical point lies at **374°C and 218 atm** pressure

Figure G000103. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 15.

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ENGINEERING CHEMISTRY

Phase equilibria

Metastable equilibrium (OA'):

- Ice fails to form at the triple point and water continues to exist in liquid phase
- The vapour pressure of the liquid continues along OA'
- This is called **super cooled water** and represents metastable equilibrium involving liquid and vapour phases.
- Any disturbance will cause the system to go back to stable equlilibrium (OB)
- The vapour pressure of the system in the metastable region is more than that of the stable system ice at the same temperature

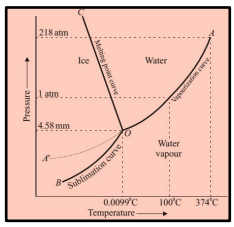

Figure G000104. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 16.

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Figure G000105. Derivation of Gibb s phase rule Phase diagram of one component system water · source slide page 17.

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000522 · EV0000523 · EV0000524
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Class 9 content: Derivation of Gibb's Phase rule EV0000525 · EV0000526 · EV0000527
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Phase equilibria Phase equilibria between phases exist when chemical potential of a component is equal in all the phases in equilibrium e.g. for water at triple point Solid ⇌ liquid ⇌ vapour The chemical potential of water will be equal in all the three phases For systems not at equilibrium, the chemical potential will point to the direction in which the system can move in order to achieve equilibrium viz. , the system moves from higher chemical potential to lower chemical potential EV0000528 · EV0000529 · EV0000530
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Phase equilibria When various phases are in equilibrium with one another in a heterogeneous system, there can be no transfer of energy or mass from one phase to another. For a system at equilibrium, the various phases must have the same temperature and pressure and their respective compositions must remain constant all along EV0000531 · EV0000532 · EV0000533
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Phase equilibria EV0000534 · EV0000535 · EV0000536

6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Phase equilibria EV0000537 · EV0000538 · EV0000539
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Total number of equations(constraints) : At equilibrium the chemical potential of particular component is same in every phase in a system μ1α = μ1β = μ1γ = ... (P-1) μ2α = μ2β = μ2γ = ... (P-1) μ3α = μ3β = μ3γ = ... (P-1) μcα = μcβ = μcγ = ... (P-1) For C components C(P-1) Total number of equations or constraints =C(P-1) F = Total number of variables – total number of equations F = P(C-1) + 2 – {C(P-1)} F=C-P+2 which is the Gibb's phase rule . . . EV0000540 · EV0000541 · EV0000542
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Phase diagram A diagram which represents the conditions under which a substance exists in different phases in a system Phase diagram of a 1-component system F = C- P + 2 For a 1-component system F = 3 – P Single phase : F = 2 ; Area in a diagram Two phases in equilibrium : F = 1 ; line in a diagram Three phases in equilibrium : F = 0 ; point in a diagram EV0000543 · EV0000544 · EV0000545
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Phase equilibria High pressure, low temperature: solid phase High temperature, low pressure : vapour phase In between : liquid phase solid ⇌ liquid liquid ⇌ vapour vapour ⇌ solid Solid ⇌ liquid ⇌ vapour Source:http://abyss.uoregon.edu/~js/glossary/triple_point.html EV0000546 · EV0000547 · EV0000548
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Application of Gibb's phase rule F = C-P+ 2 to 1-component system: Water system When only 1 phase is present : C = 1, P = 1; F = 2 ;Temperature and Pressure can be varied independently Bivariant system When 2 phases are in equilibrium: C = 1, P = 2; F = 1 ; Temperature or Pressure can be varied independently Univariant system When all 3 phases are in equilibrium: C = 1, P = 3; F = 0 ; Neither Temperature nor Pressure can be varied Invariant system EV0000549 · EV0000550 · EV0000551
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Phase diagram of water [Image text, OCR] 218atm Ice Water atm 4.58mm Q Water vapour B 0.0099C 100°C 374°℃ Temperature Fig.Phasediagramforthewatersystem OC : Melting point curve OA : Vaporisation curve OB : Sublimation curve O: Triple Point A: Critical point OA': Metastable equilibrium Source: https://imbooz.com/engineering-chemistry/phase-diagram-for-water-system/ EV0000552 · EV0000553 · EV0000554
12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Equilibrium between solid and liquid (fusion curve OC) ice ⇌ water F=1, monovariant system variation of melting point of ice with pressure slope is negative; as ice melts its volume decreases or density increases Clausius-Clapeyron equation EV0000555 · EV0000556 · EV0000557
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Equilibrium between solid and vapour (sublimation curve OB) ice ⇌ water vapour F=1, monovariant system variation of sublimation temperature of ice with pressure slope is positive Clausius Clapeyron equation Where , EV0000558 · EV0000559 · EV0000560
14. Source slide page 14 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Equilibrium between liquid and vapour (vaporization curve OA) liquid water ⇌ vapour F=1, monovariant system variation of boiling temperature of water with pressure slope is positive Clausius - Clapeyron equation EV0000561 · EV0000562 · EV0000563

15. Source slide page 15 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Triple point “O”: Represents equilibrium between liquid, vapour and solid water (ice) All three phases are present together F = 0, invariant system Triple point for water lies at 0.0098 0C and 4.58 mmHg Critical point “A”: the interface between liquid water and water vapour vanishes a point above which water does not exist in liquid state Critical point lies at 3740C and 218 atm pressure EV0000564 · EV0000565 · EV0000566
16. Source slide page 16 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Metastable equilibrium (OA’): Ice fails to form at the triple point and water continues to exist in liquid phase The vapour pressure of the liquid continues along OA’ This is called super cooled water and represents metastable equilibrium involving liquid and vapour phases. Any disturbance will cause the system to go back to stable equlilibrium (OB) The vapour pressure of the system in the metastable region is more than that of the stable system ice at the same temperature EV0000567 · EV0000568 · EV0000569
17. Source slide page 17 (extracted text): THANK YOU Department of Science and Humanities EV0000570 · EV0000571 · EV0000572

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.580] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module three on phase equilibria and today's class we will talk about phase diagram of water and we will discuss reduced phase rule for a two component system. [00:00:18.884] In the previous class, we derived the Gibbs phase rule, f is equal to C minus P plus 2. And for a one component system, F is equal to 3 minus P. [00:00:32.704] So if we have one phase, then f becomes equal to two and that would be represented by area on the phase diagram. If we have two phases in equilibrium, then f becomes equal to one and that would be represented by a line and when [00:00:50.160] There are three phases in equilibrium. F could be equal to 0 and that would be represented by a point. So now let us apply this phase rule for water and draw the phram of water. [00:01:04.932] So when we have just one phase, say the solid phase that would be represented by this area, and when we have the liquid phase that would be in this area, and when we have water vapor, then that would be represented by this area. [00:01:22.372] We have solid ice and liquid or water in equilibrium that is given by this line OC. So let us see here, OC, we will call it as a melting point curve. And what do we have here? We have the solid in equilibrium with [00:01:40.368] Liquid and since there are two phases in equilibrium, this will be represented by a line, and then OA similarly would be the vaporization curve. What would that be? Water is in equilibrium, liquid water is in equilibrium with the water vapor. And OB stands for the sublimation curve. [00:01:58.000] where we have the solidized in equilibrium. [00:02:01.440] with the water vapor. And an important point which we should note here is the triple point O, which is where all the three phases are in equilibrium, the solid, liquid as well as the water vapor. And then there's another point here A, which is known as [00:02:19.088] As the critical point. We will just discuss this in the next few slides. And then we also have an equilibrium OA prime, which is shown by the dotted lines over here, and this is known as the metastable equilibrium, which we will discuss soon. [00:02:39.268] Let us first discuss the equilibrium between solid and liquid, that is the fusion curve. [00:02:45.392] O C in this diagram, which is also known as the melting point curve. So we have equilibrium between ice and water, and the phase rule being F is equal to C minus P plus 2. It's a one component system and there are

two phases in equilibrium. Therefore [00:03:03.024] O F becomes equal to 1 and the system is a monovariant system. What does it mean? That between pressure and temperature we can change only one variable. If we change the pressure, the temperature is constant. [00:03:17.284] Is defined, or if we change the temperature, then the pressure is defined. What does this line represent here? The variation of melting point of ice with pressure. As we change [00:03:33.520] The pressure as we increase the pressure, we find that the melting point of ice decreases and it's very clear that there is a negative slow. Now this happens only for water. Most of the materials we find that the fusion curve or the melting point. [00:03:51.184] Curve has a positive slope. But in case of water, the slope is negative because of the anomalous behavior of water. Now the reason for that is as ice melts, its volume decreases or density. [00:04:08.272] Increases. Therefore, we know that ice floats on water because the density of ice is less than that of water. So this is only true in case of water. All other substances, the solid is always has a higher density as compared to the liquid. [00:04:28.048] uh melt. Now this can be explained with the help of Glossier's Claperon equation which says dp by dt is equal to delta h fusion by T delta V. [00:04:39.792] Now this will come out to be negative in the case of water. Why did I choose dp by dt? Because here pressure versus temperature is what we have plotted and dp by dt would give me the slope of the [00:04:54.240] Go now in the case where we have ice is equilibrium with water, ice is changing into water. We know that we have to use delta H fusion or delta H melting. Now delta H melting is always going to be a positive value. We know

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:11.888] That heat is going to be absorbed, so therefore it is going to be an endothermic reaction. So delta H fusion is always positive for the conversion ice into water, and whereas delta V is going to be negative. [00:05:27.728] In the case of ice changing into water, because we know that there is going to be a decrease in volume as ice melts. So since delta H is positive, delta V is negative. So DP by Dt is going to be negative. So we find that the slope is [00:05:45.392] Is negative in the case of water. One more important point we need to notice here that when ice changes into water, there is the value of delta V is going to be [00:06:01.632] a very small value and therefore we find that the slope delta h by t delta v that is dp by dt has a very high value. So two points we need to consider when we talk of [00:06:17.424] Equilibrium between ice and water. Here the slope is going to be negative and the slope is going to be very steep. [00:06:29.380] Now coming to the next equilibrium that is the solid and the vapor which is known as the sublimation curve here the curve OB. Here we have ice in equilibrium with water vapor and similarly when we apply the phase rule F is equal to one, so it's again a monovariant system. Now [00:06:47.152] Here, what do we see along this? We understand the variation of the sublimation temperature of ice with pressure. As we change the pressure, what happens to the sublimation temperature? And here we find that the slope is positive. In the case of [00:07:04.720] We found that the slope is negative, whereas here we find the slope is going to be positive. Again using the Clausius Clabron equation we will be able to explain this. [00:07:16.576] DB by D T is equal to delta H sublimation by T delta V. Now again delta H sublimation is going to be a positive value because when ice changes into water vapor, there is the reaction is going to be endothermic, it's going to absorb heat. [00:07:34.020] And delta V is also going to be positive in this case when ice changes to water vapor because there is going to be a big change in the volume. When ice changes to water vapor, there's a large change in the volume. Therefore [00:07:50.800] Both being positive delta H sublimation as well as delta V is positive, and

therefore dB by Dt is coming up is going to be positive. So that is what we see here, a positive slope for the sublimation curve. And one more point to notice here is that delta V being large. [00:08:08.432] This delta V is very large, therefore dp by dt slope is much less as compared to the melting point curve. Now coming to the [00:08:21.536] Vaporization curve the equilibrium between the liquid and the vapor which is represented by this line over here. Again we have liquid water in equilibrium with vapor and f is equal to one, so it's a monovariant system, therefore it's represented by a line. We can either change the temperature or the [00:08:39.216] Pressure pressure. Now what does this line represent? The variate variation of boiling point of water with pressure. As we increase the pressure, we notice that the boiling point increases. And therefore we see that the slope is positive using the [00:08:56.784] Closer's Clapebron equation again we can explain that dp by dt is equal to delta H vaporization by T delta V. Now delta H vaporization is going to be again positive because it's an endothermic reaction, liquid water changing into water vapor requires heat. [00:09:14.352] And then T delta V, so delta V [00:09:18.404] It is again positive here, liquid water changing into vapor, again there is a big change or a large change in the volume, and therefore we'll find that delta V is also positive. So delta H vaporization by delta V both being positive T delta V both being positive. So over [00:09:36.176] all the slope is going to be positive. And since delta V again is a large value we find that the slope is much less as compared to that of the melting point curve. Let us talk about the triple point O. [00:09:53.392] Now at this point we have the solid, liquid and the vapor in equilibrium. Since all the three phases are present together, P is equal to 3. So therefore F is equal to which is equal to C minus P plus 2. So since P is 3, F becomes

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:11.056] equal to zero, so the system becomes invariant. What does it mean? That we can neither change the temperature nor the pressure. Now in the case of water the triple point [00:10:23.748] is at 0.0098 degrees centigrade and 4.58 millimeter of mercury. Another point of importance is A, which is known as the critical point. At the point A [00:10:39.104] the interface between the liquid water and the water vapor [00:10:44.912] vanishes. So if you see here, there is the line stops here, it doesn't extend beyond this. That means we cannot distinguish between the liquid water and the water vapor. So beyond this point we do not find [00:11:02.128] the water in the liquid state. Such a point is known as the critical point. So the critical point for water lies at 374 degrees centigrade and 220 atmosphere pressure. Of course this [00:11:18.436] plot is not made to scale because we are using different uh pressure units. So here we are saying it's 218 atmospheres and here we are saying 4.58 millimeter. [00:11:32.752] of mercury. So this plot is basically indicative it is not to scale. In fact we have even used different uh units for pressure. Let us discuss the metastable equilibrium represented by the line OA prime on this graph. [00:11:50.980] When water is cooled below the triple point, it is supposed to get converted into ice. [00:11:57.328] But under certain conditions, it is found that water continues to be in the liquid phase and does not get converted into ice. This is known as super cooled water and it represents a metastable equilibrium. [00:12:15.056] Which involves the liquid and the vapor phases. It is called metastable equilibrium because even a small disturbance is going to cause the system to go back to the stable equilibrium which is represented by the line OB. [00:12:34.212] The vapor pressure of the system now in the metastable region is higher than that of the stable system of ice at the same temperature. And therefore, we find that the line OA prime is always shown above that of the sublimation curve. [00:12:54.212] Let us talk about the phase

diagram of a two component system. We know that the phase rule is F is equal to C minus P plus 2. But in the case of a two component system, we will not be able to use this ordinary phase rule. The reason being that [00:13:11.984] Since C is equal to two here, F is going to be four minus P. Now if we have [00:13:17.984] One phase then F becomes equal to three. So that means there are three degrees of freedom. So three variables have to be represented. So we need three dimensional space in order to draw the phase diagram. So in order to simplify so that it can [00:13:35.696] We consider one of the three variables to be [00:13:41.760] uh constant. Most of the time it is assumed that pressure is constant because we carry out measurements at atmospheric pressure. So now we have a new rule, Gibbs phase rule, where the degree of freedom has been reduced by one. So [00:13:59.408] So f is equal to C minus P plus two minus one. [00:14:04.292] The new rule, therefore, is F is equal to C minus P plus one. And this is known as the reduced phase rule. The name is derived from the fact that we have reduced the degree of freedom by one because we are assuming [00:14:22.148] That one of the variables is constant. And this is mostly applicable to solid liquid equilibria because in this case the pressure changes. [00:14:34.928] Will not affect the equilibrium to a great extent as compared to when there would be a gas phase present in the system. So systems in which gas phase is absent, they are known as condensed system, and since we are assuming that pressure is [00:14:52.560] Constant. So this phase rule will also be known as the condensed phase rule. So f is equal to C minus P plus 1 is the phase rule which we are going to use for a two component system that is known as the reduced phase rule or

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:10.128] the condensed phase rule. So in the next class we are going to study the construction of the phase diagram for a two component system and specifically we will look at the lead silver system. Thank you.

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Full transcript: 2269 normalized words, 61 timestamped segments, 925.845333 seconds duration. Low-confidence segments flagged: 0.

EV0000166 · EV0000167 · EV0000168 · EV0000169

11 Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I

This lesson is indexed from the source video titled Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000182 · EV0000183

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.452] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module three on phase equilibria and in today's class we are going to talk about Gibbs phase rule. [00:00:14.244] We will

derive the Gibbs phle and we will discuss phase diagrams specifically phase diagram of a one component system. The phase rule was given by Williams Gibbs in eighteen seventy four [00:00:28.356] In the previous class we talked about phases, components, and degrees of freedom. If you remember, P stands for the number of phases, C stands for the number of components, and F stands for the degrees of freedom. [00:00:46.564] The Gibbs phase rule is going to give us a relation between the number of phases, the number of components, and the number of degrees of freedom for a system existing at equilibrium. [00:01:02.340] So the statement of Gibbs phase rule is provided the equilibrium in a heterogeneous system is not influenced by external forces like gravity, electrical or magnetic forces, the number of degrees of freedom of the system is [00:01:18.432] is related to the number of components and the number of phases existing at equilibrium to one another by the equation F is equal to C minus P plus 2. [00:01:32.644] So now let us derive this Gibbs phase rule. F is equal to C minus P plus 2. [00:01:40.004] Before deriving the Gibbs phase rule, let us understand what we mean by a system at equilibrium. What are the conditions a system needs to satisfy in order to be considered to be at equilibrium? [00:01:55.012] Firstly, it should be at thermal equilibrium, meaning that the temperature should be constant throughout the system. If there exists a difference in temperature, then [00:02:07.936] The system will try to equalize the temperature and it will reach equilibrium only when the temperature is same throughout the system. Similarly, for mechanical equilibrium the pressure should be constant. Now in the previous class we [00:02:25.476] Talked about a new term chemical potential. Now there's another equilibrium which needs to be satisfied, and that is chemical or material equilibrium. When we talk of any chemical reaction, we know that gibbs free energy decides. [00:02:44.356] the direction of the reaction. [00:02:47.424] Wherever there is decrease in free energy, we know that the reaction is going to move in that direction because every system tries to go towards minimum free energy. But free energy cannot be used as a criteria for chemical equilibrium because it is [00:03:05.168] An extensive property. Therefore, a new term was defined which is known as the chemical potential. So chemical potential is partial molar Gibbs free energy. So Gibbs free energy divided by the number of moles will give us the [00:03:21.968] Uh molar gives free energy which is chemical potential and since there are so many species in the system we have to consider partial molar gives free energy. So for a system to be at chemical or material equilibrium, the condition is that chemical [00:03:39.600] potential of the substance should be same in all the phases. So suppose we have a component I, then there exists say alpha, beta, gamma, so on [00:03:53.540] the phases then the mathematically if we want to write that would be that would be written as mu i [00:04:01.408] Alpha is equal to mu i beta is equal to mu i gamma. What does it mean? That the chemical potential of the I component in the alpha phase is equal to the chemical potential of the I component in the beta phase, which is again equal to chemical potential of the I component in gamma. [00:04:19.120] phase and so on. So what does it show that the chemical potential has to be same in all the phases for a substance if the system is at chemical equilibrium. [00:04:34.692] In order to derive the Gibbs phase rule, let us take a particular system. Our system is where all the C components are distributed between P phases. So in the system there are P phases and there are C components, but in all the phases we have these C components. [00:04:56.004] Firstly, let us find out what are the total number of intensive variables that need to be ascertained to describe the system. If we want to talk about any system at equilibrium, first and foremost, we should know what are the variables we need to ascertain.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:14.020] Firstly we will have temperature [00:05:17.156] We should know the temperature, so that should be one variable and then we should know the pressure, so that is again another variable. So two variables we need to ascertain and since the system is at equ thermal equilibrium and mechanical equilibrium, we know that there will be just one temperature which we need to. [00:05:34.928] To know and also the pressure of the system which we need to know. So there are two variables corresponding to the temperature and pressure. Now the third variable which is required is composition. Now we said that there are C components and they are present in all the phases. So in order to determine [00:05:52.496] The composition we need to know the mole fraction of each component in every face. [00:05:59.876] For each phase, we know that the sum of the mole fractions is equal to unity, because this is a relative way of expressing composition, mole fraction. So the sum of the mole fractions is always equal to unity. [00:06:16.000] for all the components in that particular uh phase. So how do we express it mathematically? So first let us talk of um [00:06:24.880] Say alpha phase. In the alpha phase, we have so many components. There are C components. So the equation would be given by the mole fraction of the first component in the alpha phase plus the mole fraction of the second component in the alpha phase. [00:06:42.544] Plus mole fraction of the third component of the alpha phase, so on, till we come to the last component that is the mole fraction of the component C in the alpha phase. And all this should sum up to one. [00:06:57.572] We are interested in the number of components. So using this in the alpha phase, we can get C minus 1 components compositions which I need to know. [00:07:11.652] Why C minus 1? Because we know that all these will sum up to 1. So the last component can be determined by subtracting the sum of all these from [00:07:24.608] One. So for each phase we have to know C minus one number of [00:07:32.128] compositions. Now coming to the second phase, say that is the beta phase, again we have an equation like this where we have the first component mode fraction, the second component mode fraction, third component mode fraction, so on adding up to one. So again in the beta phase we will we will require C minus [00:07:49.776] And similarly for the gamma phase, and as we go on, we know that there are p number of phases. So therefore for the last phase it will be the mole fraction of the first component in p phase plus the mole fraction of the second component in p phase, so on. [00:08:07.344] Up to the last component in the P phase, again this all adds up to one. Therefore, we have C minus one compositions again need to be determined. So what do we understand here? For each of the phases, we need C minus 1 more fraction terms which have to be [00:08:24.912] define and since we have p number of phases therefore the number of composition variables will be p times c minus one. Now p times c minus one [00:08:39.408] Plus we have temperature and pressure which we need to ascertain. So therefore the total number of intensive variables will become P times C minus one plus two. If we want to find out what are these variables. [00:08:56.260] We need to know the number of equations or the constraints. So, what are the equations which are available to us in this system? At equilibrium, we know that chemical potential of a particular component is the same in every phase of the system. [00:09:16.516] Just now we define chemical equilibrium and talked about this equation. So let us look at it for the first component. So the first component chemical potential in the alpha phase should be equal to [00:09:30.864] The chemical potential of the first component in the beta phase and also that should be equal to chemical potential of the same component in the gamma phase and so on. So uh if this goes on it will be up to [00:09:44.848] P phases. So the first component chemical potential should

be equal in all the p phases. So what are the number of equations here? So here for each of these components we will get p minus 1.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.116] Now coming to the second component, again the same uh equation or the same constraint is there. So [00:10:10.144] Second component chemical potential in alpha phase should be equal to the chemical potential of the second component in the beta phase, and also that again equal to the chemical potential of the second component in gamma phase. So overall then this also adds up to up to this goes on up to mu to. [00:10:28.964] P because P is the total number of phases, therefore what we get here is again P minus one number of equations. So like that for the third phase, and if we go on like this, we come to the last uh component, which is the which is C. [00:10:46.148] So mu C alpha will be equal to mu C beta, which is equal to mu C gamma, and this goes on up to mu C P. Therefore, again we have P minus one number of equations. So all these equations are available to us. So since C components, [00:11:03.920] equations and each of these components we have p minus 1 equations. So therefore the total number of equints now we have is C times P minus 1. [00:11:16.432] So we got the number of variables and we also got the number of equations. Now we want to find out what is the degree of freedom. [00:11:26.308] Since we know the number of variables and the number of equations, the degrees of freedom will be given by the total number of variables minus the number of equations which we have. Let us understand this. Now suppose we have two variables x and y which we need to be which need to be [00:11:44.080] Determine now since there exists no relation, suppose there are independent variables, then that means I need to ascertain both X as well as Y in order to define the system. But suppose there is an equation which is [00:12:00.964] Relating X and Y. So for example, say X plus Y is equal to 1. Now there is a restriction. [00:12:07.600] I cannot change X and Y independently. If X changes then Y is fixed if Y changes then X is fixed. So what happened now? Because of this one equation the degree of freedom or the [00:12:22.404] My independence to change X and Y is reduced by 1. So we can say the number of variables [00:12:31.844] will be reduced by as many number of equations. So the more the if we have so many number of equations then those restrictions are going to reduce the [00:12:45.520] the degrees of freedom. So total number of variables minus the total number of equations is going to determine the degree of freedom. So the degree of freedom reduces by the number of equations or constraints which we are going to put [00:13:03.184] on the selection of the variables. So f is equal to total number of variables minus total number of equations. So we can write this now we know the total number of f variables is p times c minus 1 plus 2 and the total number of equations is C times P minus 1 [00:13:20.752] one. So what we get is F is equal to C minus P plus 2. And this is Gibbs phase root. [00:13:28.464] So let us apply the Gibbs phase rule, f is equal to C minus P plus two for a one component system. We will take up the water system because that is the easiest for us to uh relate to. So when we have only one phase which is present. [00:13:44.576] In that case, C is one because it is a one component system. P is one again. [00:13:51.696] Uh because we are talking about one phase and when we apply the Gibbs phase rule F is equal to two. So what does it mean? That temperature and pressure can be varied independently. So it is a bivariant system. And when we have two phases present in [00:14:09.328] In equilibrium. So either the solid is in equilibrium with the liquid or the liquid is in equilibrium with the vapor or the solid is in equilibrium with the vapor, then we know that there are two phases, C is again one because it's a one component system. So when we apply the Gibbs phase

rule, F is equal to [00:14:26.896] One. So that means now the system has become a univariant system, and we know that when we have two phases in equilibrium, either the temperature or the pressure can be varied independently, but not both. And when we have all the three phases in equilibrium. [00:14:44.464] And this system now, what happens? F is equal to 0. C is equal to 1, P is equal to 3. So when we apply the Gibbs phase rule, f is equal to 0. So neither temperature nor the pressure can be varied, and hence the system becomes.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:02.032] an invariant system. So we see that the Gibbs phase rule is easily applicable to the water system. Let us now discuss the phase diagram. [00:15:16.548] It is a diagram which represents the conditions under which a substance exists in different phases in a system. So suppose we have the solid, liquid, or the gaseous phase. If I have a map which is going to tell me [00:15:31.428] the conditions under which each of these phases are stable or the conditions under which the phase transformation from one phase to the other will happen or [00:15:43.776] the conditions under which there could be equilibrium between the phases, then such a map would be very useful because I can use it for applications to different situations or in order to understand the [00:16:01.104] uh material or the system under consideration. So let us see phase diagram of a one component system. How do I construct a phase diagram of a one component system? [00:16:13.136] The phase rule we know which we are going to apply is F is equal to C minus P plus 2. Now if it is a one component system then F becomes equal to 3 minus P [00:16:26.020] If we have only a single phase, we want to represent a single phase in the diagram, we know that using this f is equal to 3 minus p, since it's one phase, f becomes equal to 2. And in a diagram, this would be represented by an area. [00:16:43.684] Now, if we have two phases in equilibrium, that would mean that f is equal to 3 minus p, so f becomes equal to 1, and when f is equal to 1, then it would be represented by a line on the [00:16:58.608] diagram and when we have three phases in equilibrium then using the Gibbs phase rule we have f equal to zero. So when f is equal to zero this would be represented by just a point on the [00:17:15.632] diagram. So when we have two phases in equilibrium then we have a line. When we have three phases in equilibrium then it will be represented by a point, but if you have just a single phase that would be shown by an area in the diagram. [00:17:35.492] Let us see what the phase diagram is going to look like. So here we are going to since it's a one component system, so we are going the variables used is pressure versus temperature. [00:17:51.204] Suppose we have just one phase, and just now we said that when we have one phase, phase rule tell us that there is going to it is going to be represented by an area on the diagram. [00:18:03.908] So if we have say high pressure and low temperature, we know by our experience that if we are talking of high pressure and low temperature, then the phase which is uh present is going to be the solid phase. So you can see here that [00:18:19.696] This area would be the solid phase. Similarly, if we have high temperature and low pressure, the phase which is present is going to be the gaseous phase. And in between we have the liquid phase. So we have got areas where we [00:18:38.352] Say this area is going to be the solid phase, this area will be the gaseous phase, and in between we have the liquid phase. So this is the area, the solid phase, the vapor phase, and the liquid phase. That would be represented by an area. [00:18:53.504] Now if we have two phases in equilibrium, say a solid in equilibrium with the liquid. Now since we know this area is the solid and this is the liquid phase, and when we have two phases in equilibrium we know that it is going to be a line. So here this line would represent the equilibrium. [00:19:11.168] Equilibrium between solid and the liquid. So if you want to show an

equilibrium between solid and liquid, then it has to be shown by a line. And similarly, liquid and vapor, so that would be this line here. And then if there is vapor and solid, then this line would represent the [00:19:30.832] So when we have solid and liquid in equilibrium, then we call it as the fusion curve or the melting point curve, and when we have liquid and vapor in equilibrium, we call it as the [00:19:45.168] uh vaporization curve or the boiling point curve and vapor and solid would be the sublimation curve. And then if we have the solid liquid as well as the vapor [00:19:56.484] present in equilibrium together, then that would be represented by a point, and that's exactly what we see here. So this would represent all the solid, the liquid

EV0000194 · EV0000195 · EV0000196 · EV0000197

Transcript block 5 (source interval 001200-001499 seconds): [00:20:08.768] as well as the vapor in equilibrium and that would be a point on the phram. So roughly this is how we are going to draw the phase diagram for a one component system. So in the next class [00:20:23.456] We will take up the water system and see exactly what the phram is going to look like for a water system. Thank you.

EV0000198 · EV0000199 · EV0000200 · EV0000201

Full transcript: 3114 normalized words, 84 timestamped segments, 1234.090667 seconds duration. Low-confidence segments flagged: 0.

EV0000182 · EV0000183 · EV0000184 · EV0000185

12 Reduced phase rule, phase diagram of two-component system - Pb-Ag system

This lesson is indexed from the source video titled Reduced phase rule, phase diagram of two-component system - Pb-Ag system. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence.

EV0000202 · EV0000203

Graphs and diagrams

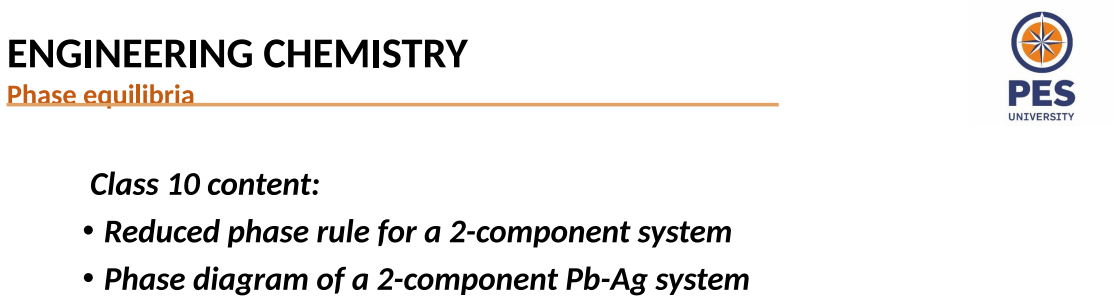

Figure G000106. Reduced phase rule phase diagram of two component system Pb Ag system · source slide page 1.

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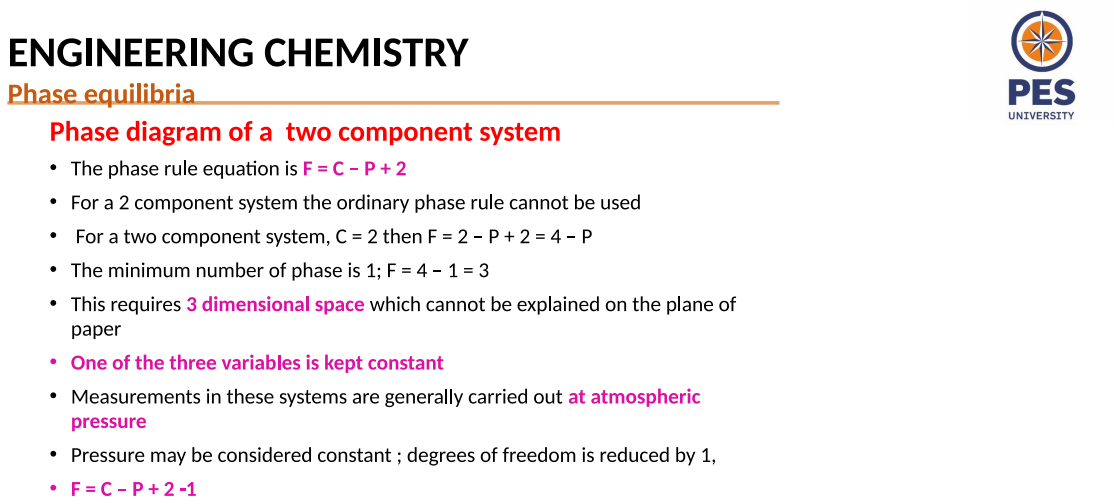

Figure G000107. Reduced phase rule phase diagram of two component system Pb Ag system · source slide page 2.

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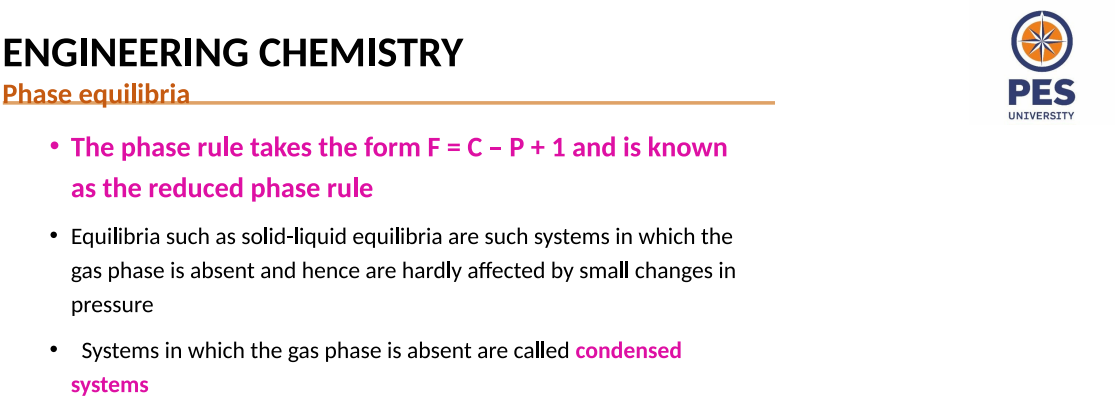

Figure G000108. Reduced phase rule phase diagram of two component system Pb Ag system · source slide page 3.

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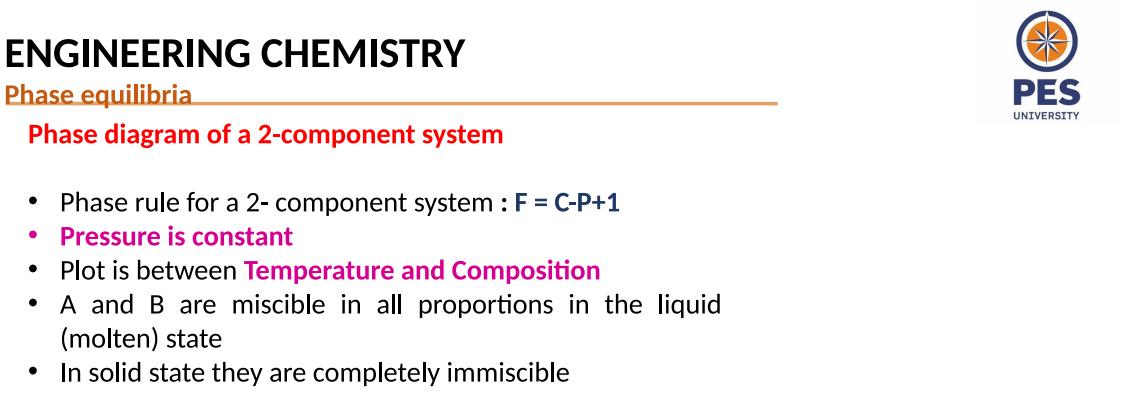

Figure G000109. Reduced phase rule phase diagram of two component system Pb Ag system · source slide page 4.

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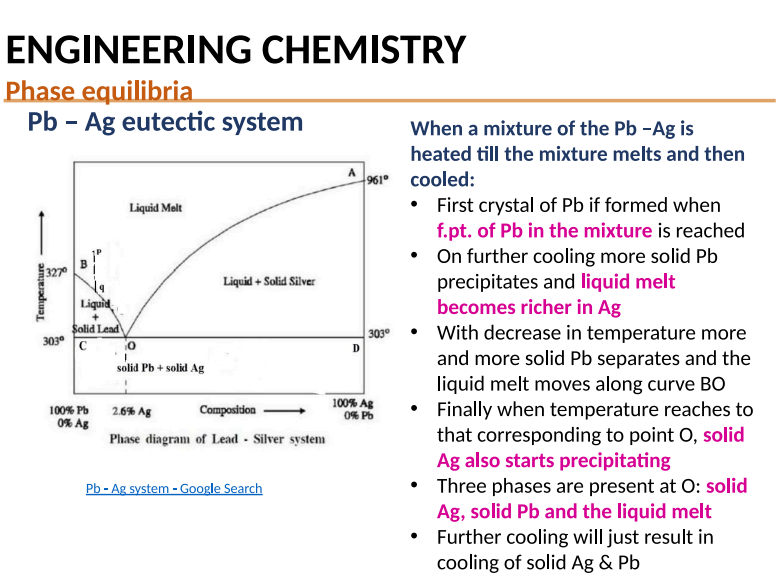

Figure G000110. Reduced phase rule phase diagram of two component system Pb Ag system · source slide page 5.

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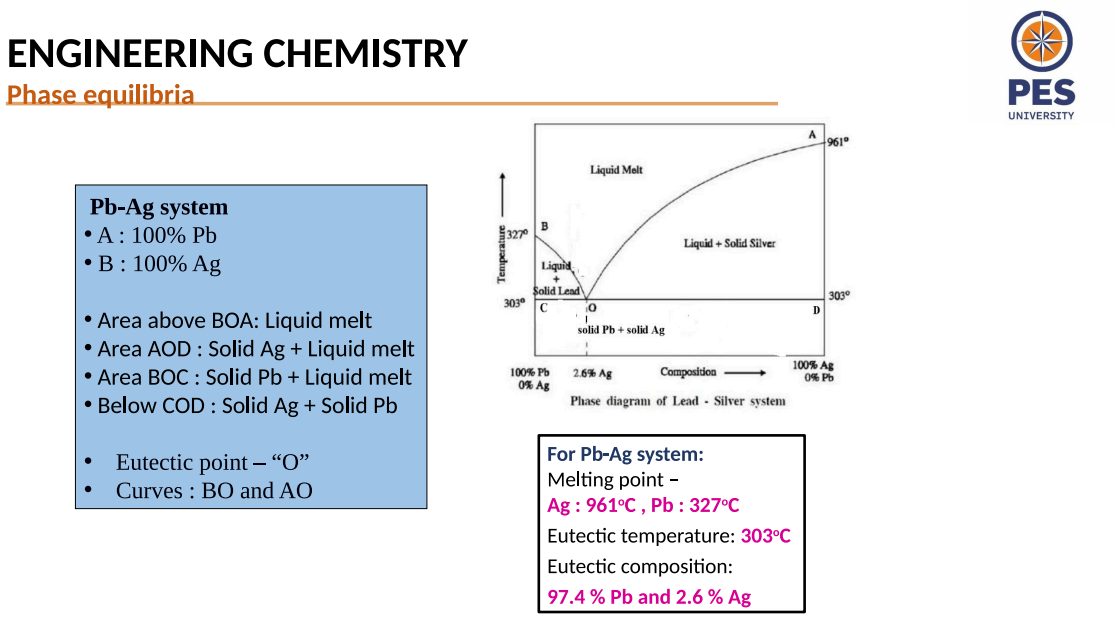

Figure G000111. Reduced phase rule phase diagram of two component system Pb Ag system · source slide page 6.

EV0000589

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ENGINEERING CHEMISTRY

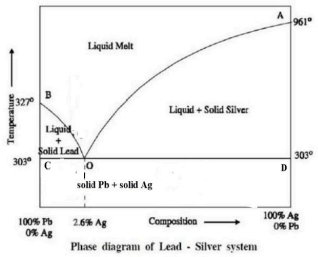
Phase equilibria

CURVE BO : freezing point curve of Pb

- It shows decrease in freezing point / melting point of Pb due to the addition of Ag to Pb
- Solid Pb is in equilibrium with liquid melt of Ag in Pb
- Here C = 2 and P = 2, then the reduced phase rule is F = C – P + 1 = 2 – 2 +1 = 1
- Hence the system is **univariant**

CURVE AO: freezing point curve of Ag

- It shows decrease in freezing point of Ag due to the addition of Pb to Ag
- Solid Ag is in equilibrium with liquid melt of Pb in Ag
- Here C = 2 and P = 2, then the reduced phase rule is F = C – P + 1 = 2 – 2 +1 = 1
- Hence the system is **univariant**



Phase diagram of Lead - Silver system

Figure G000112. Reduced phase rule phase diagram of two component system Pb Ag system · source slide page 7.

EV0000592

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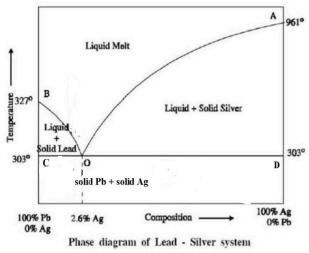
ENGINEERING CHEMISTRY

Phase equilibria

Eutectic point

- **Eutectic mixture:** A mixture of two components which has the lowest freezing point of all the possible mixtures of the components
- Number of phases at eutectic point O = **3**
- F = C-P+1; C =2, P = 3,so **F = 0**; invariant point
- It has a **definite composition** and a **sharp melting point**



Phase diagram of Lead - Silver system

Figure G000113. Reduced phase rule phase diagram of two component system Pb Ag system · source slide page 8.

EV0000595

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Figure G000114. Reduced phase rule phase diagram of two component system Pb Ag system · source slide page 9.

EV0000598

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Class 10 content: Reduced phase rule for a 2-component system Phase diagram of a 2-component Pb-Ag system EV0000573 · EV0000574 · EV0000575
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Phase diagram of a two component system The phase rule equation is F = C – P + 2 For a 2 component system the ordinary phase rule cannot be used For a two component system, C = 2 then F = 2 – P + 2 = 4 – P The minimum number of phase is 1; F = 4 – 1 = 3 This requires 3 dimensional space which cannot be explained on the plane of paper One of the three variables is kept constant Measurements in these systems are generally carried out at atmospheric pressure Pressure may be considered constant ; degrees of freedom is reduced by 1, F = C – P + 2 -1 EV0000576 · EV0000577 · EV0000578
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Phase equilibria The phase rule takes the form F = C – P + 1 and is known as the reduced phase rule Equilibria such as solid-liquid equilibria are such systems in which the gas phase is absent and hence are hardly affected by small changes in pressure Systems in which the gas phase is absent are called condensed systems F = C – P + 1 is also known as condensed phase rule EV0000579 · EV0000580 · EV0000581
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Phase diagram of a 2-component system Phase rule for a 2- component system : F = C-P+1 Pressure is constant Plot is between Temperature and Composition A and B are miscible in all proportions in the liquid (molten) state In solid state they are completely immiscible EV0000582 · EV0000583 · EV0000584

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Pb – Ag eutectic system When a mixture of the Pb –Ag is heated till the mixture melts and then cooled: First crystal of Pb if formed when f.pt. of Pb in the mixture is reached On further cooling more solid Pb precipitates and liquid melt becomes richer in Ag With decrease in temperature more and more solid Pb separates and the liquid melt moves along curve BO Finally when temperature reaches to that corresponding to point O, solid Ag also starts precipitating Three phases are present at O: solid Ag, solid Pb and the liquid melt Further cooling will just result in cooling of solid Ag & Pb [Image text, OCR] A 9610 LiquidMelt 3270 B Liquid+SolidSilver Liquid SolidLead 3030 3030 C D solidPb+solidAg 100%Ag 100%Pb 2.6%Ag Composition 0%Ag 0%Pb Phase diagram ofLead-Silver system Pb - Ag system - Google Search EV0000585 · EV0000586 · EV0000587
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Phase equilibria [Image text, OCR] A 9610 LiquidMelt B 3270 Liquid+SolidSilver Liquid + SolidLead 3030 3030 C D solidPb+solidAg 100%Ag 100%Pb 2.6%Ag Composition 0%Ag 0%Pb Phase diagram ofLead-Silver system Pb-Ag system A : 100% Pb B : 100% Ag Area above BOA: Liquid melt Area AOD : Solid Ag + Liquid melt Area BOC : Solid Pb + Liquid melt Below COD : Solid Ag + Solid Pb Eutectic point – “O” Curves : BO and AO For Pb-Ag system: Melting point – Ag : 961oC , Pb : 327oC Eutectic temperature: 303oC Eutectic composition: 97.4 % Pb and 2.6 % Ag EV0000588 · EV0000589 · EV0000590
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Phase equilibria CURVE BO : freezing point curve of Pb It shows decrease in freezing point / melting point of Pb due to the addition of Ag to Pb Solid Pb is in equilibrium with liquid melt of Ag in Pb Here C = 2 and P = 2, then the reduced phase rule is F = C – P + 1 = 2 – 2 +1 = 1 Hence the system is univariant CURVE AO: freezing point curve of Ag It shows decrease in freezing point of Ag due to the addition of Pb to Ag Solid Ag is in equilibrium with liquid melt of Pb in Ag Here C = 2 and P = 2, then the reduced phase rule is F = C – P + 1 = 2 – 2 +1 = 1 Hence the system is univariant [Image text, OCR] A 9610 LiquidMelt B 3270 Liquid+SolidSilver Liquid + SolidLead 3030 3030 C D solidPb+solidAg 100%Ag 100%Pb 2.6%Ag Composition 0%Ag 0%Pb Phase diagram ofLead-Silver system EV0000591 · EV0000592 · EV0000593
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Phase equilibria Eutectic point Eutectic mixture: A mixture of two components which has the lowest freezing point of all the possible mixtures of the components Number of phases at eutectic point O = 3 F = C-P+1; C =2, P = 3,so F = 0; invariant point It has a definite composition and a sharp melting point [Image text, OCR] A 9610 LiquidMelt B 3270 Liquid+SolidSilver Liquid + SolidLead 3030 3030 C D solidPb+solidAg 100%Ag 100%Pb 2.6%Ag Composition 0%Ag 0%Pb Phase diagram ofLead-Silver system EV0000594 · EV0000595 · EV0000596
9. Source slide page 9 (extracted text): THANK YOU Department of Science and Humanities EV0000597 · EV0000598 · EV0000599

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.484] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module three on phase equilibria and in today's class we are going to learn how to construct a phase diagram of a two component system in general and specifically the lead silver system. [00:00:19.300] Then we study

the determination of solid liquid equilibria and finally the Patterson's process. [00:00:27.076] When we talk of a two component system, we know that we need to use the phase rule F is equal to C minus P plus 1, which is known as the reduced phase rule. [00:00:39.588] It is known as the reduced phase rule because we have reduced the degree of freedom by one. In the previous class we learnt that the Gibbs phase rule f is equal to C minus P plus 2 cannot be applied for a two component system for the simple reason that if P equals [00:00:57.360] Is equal to one, then the degree of freedom becomes equal to three, and we need to draw a three dimensional phram. So in order to simplify and make it into a two dimensional phase diagram, we need to [00:01:11.632] keep one of the variables as constant. Most often we use pressure as constant variable. So therefore because we reduce the degree of freedom by one, the reduced phase rule is now applicable. F is equal to c minus p plus one. [00:01:31.140] And now out of pressure, temperature and composition, since pressure is being maintained constant, and especially for solid liquid equilibria, pressure is chosen constant variable. So now your graph is going to be between the temperature and composition. [00:01:50.596] When we are drawing this phase diagram of a two component system, this particular phase diagram will be known as a simple eutectic system. And for that, the criteria is that suppose we have two components A and B, then A and B must be miscible in [00:02:10.384] all proportions in the molten state or the liquid state and in the solid state they are completely immissible. So this is the criteria for getting a simple eutect diagram for [00:02:26.976] the components A and B. Now if you look at this plot we have used two temperature scales. One temperature scale stands for pure A and the other temperature scale stands for pure B. And on the x axis we have the composition scale. Now here [00:02:44.032] What is happening is here we have 100% A and as we move along this axis we are adding B into A. So we get different mixtures. [00:02:55.120] of various compositions which are represented by A and B, the percentage of A and B. So here we have 100% A and 0% B, but as we move along the amount of B goes on increasing and when we come here we have 100% B and 0% B [00:03:12.752] Now the first thing we need to know is the melting point of pure A. [00:03:19.984] Now suppose say it is at some value which is a high value as compared to that of pure B. Now that melting point because this axis represents pure A and the temperature scale. So this would be the melting point of pure A. So let us say that is some. [00:03:37.712] the alphabet M. Now as we go on adding B into A, the we are actually adding an impurity into the pure substance A, and therefore we know that there is going to be a depression in freezing point. So [00:03:56.496] When we add a certain amount of B, say a small amount of B, we find that it reduces the freezing point of pure A. And as we go on adding more and more of A, we find that this goes on decreasing like this. [00:04:13.392] So therefore if I have to join these points I get a plot like this [00:04:21.124] Now similarly, if pure B has a melting point somewhere here, then let us say it is N. And now if I add A into the pure B again I will notice that there is a decrease in the [00:04:36.672] melting point and if I join these again I get a line like this. If I join these two they happen to meet at a point which is known as the utectic point. So if I draw a line [00:04:52.708] like this joining the two axis then this point is known as the utect point. So for the pure comp for the components A and B

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:04.580] If we draw a phram, then this is what the phram is going to look like. Now let us understand this diagram in detail. [00:05:16.036] So as I mentioned just now, this system when you get a diagram like this phase diagram looks like this, it is known as a simple eutect system. [00:05:25.988] Now let us see if we

are able to label all these areas which we can see here. For that, what we do is the simplest way is to take any composition of A and B, mixture of A and V, and we heat it. [00:05:43.140] beyond this point. So this is where what does this point represent just now when I told you when I drew the diagram we saw that this would represent the melting point of A in this mixture. [00:05:59.908] So that means if I go beyond this point, then the liquid is what we are going to get is the liquid melt. So therefore we have the liquid melt now. What I do now is I start cooling it. When I come to this point here, [00:06:19.268] I find that we have that I have reached the melting point or the freezing point of A in the mixture. So what will I observe? The first crystal of A will start forming here. [00:06:35.556] Remember that this is not pure A, it is a mixture, and in a mixture now the melting point has reduced or the freezing point has reduced. So at this temperature, which is much lower than this, we find that the crystal A will start forming. As we go further down. [00:06:55.076] We find that more and more of this uh [00:06:58.832] A is going to uh freeze out of the mixture. Now when A is coming out of the mixture, obviously the liquid milk which is remaining back is becoming richer in pure B. That's exactly what happens here, and how do we know what is the [00:07:16.464] Liquid melt now it is at this point say if [00:07:20.752] More of A has come out, we find that the liquid melt can be given by this point. So what happened to the composition now? See, it has become richer in B. So as we go down further, we go on cooling it, more and more of A will precipitate out and therefore [00:07:38.384] So the liquid melt now becomes suppose we are at this point what will happen more of liquid A and what happened to the liquid composition? Liquid melt has now reached this composition. So you can see here that it is becoming richer in B. [00:07:53.520] So now we come to a very important point. When we reach this point of the graph, what we notice is that the liquid melt has now reached this point. Now the composition is such that [00:08:08.740] We have reached even the freezing point of B. Therefore, we will find that B also starts precipitating at this point. So basically what we have here, we have the liquid melt, we have the [00:08:26.180] the pure A and the pure B. So solid A, solid B as well as the liquid melt is what we get at E and this E point point E is known as the U tactic point. Now if we are going to [00:08:42.528] uh further cool it then we don't see much happening just the solids A and B are going to uh just cool down. So here what happens? The liquid is cooling initially and then we get the first crystal of A [00:08:59.792] As we cool down further, more and more A separates out and the liquid melt becomes richer and richer in B. So basically as we go on cooling it we find that this [00:09:11.748] The liquid melt now is along this curve, it will give you the composition along this curve will be the uh composition of liquid uh of the liquid melt, which is now we can see getting richer in B, but moment we reach the point E, pure the B [00:09:28.240] uh solid also will start respirating out and below this point which is known as a eutactic point we find that there is further uh just cooling of the solid and the solid A and solid B. [00:09:45.348] Now A B system. Let us now label each of these areas. Now what would be at here we have A that is 100% of A is there and at B we have 100% of B.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:03.972] Now but we know that we have drawn this space diagram and similarly what we did for this side we can also get it from this side also. And both of them will meet at the same utactic point E. [00:10:16.356] Now beyond this, this area above MEN would be the liquid melt. Why? Because all these points represent the melting points of the mixtures. Therefore, this whole area will be the liquid melt. [00:10:33.668] Now let us look at this area MEF that is this area. We find here that we have the liquidment as

well as the solid A because in this area what is happening? Solid A is being [00:10:48.400] uh frozen out and precipitating out and we have the liquid melt. So liquid melt now contains both A as well as B. So this area would represent liquid melt [00:11:01.456] plus the solid A and similarly this area would represent the liquid melt plus solid B because if you are cooling on this side we will find that B starts respirating out. So we have the [00:11:13.872] Liquid mill which again is uh has both A and B and here we have the solid B in this area plus the liquid mill. Now below this line E F G [00:11:26.948] F E G. So what will happen here? We have both solid A as well as solid B because below this point what do we observe? The liquid melt now completely converted into the solids A and B and here we see there is a [00:11:44.196] So this whole area would be only solid A and solid B. Apart from this, the the another point of interest which I have been mentioning is the [00:11:55.536] point E which is known as the eutectic point and also these curves are of interest to us the curve ME and the curve NE. Let us first talk about the eutectic point which is shown by the point E here on the graph. [00:12:13.056] found to be a eutectic mixture. It's a mixture of the two components A and B, which has the lowest freezing point of all the possible mixtures of the components. Now looking at the graph here we know that this composition has the minimum [00:12:30.208] temperature the minimum melting point whereas all these compositions have melting points higher than this point and even if you go to this side again the melting points are higher than this point. So this [00:12:43.056] point which has the lowest melting point is known as the eutectic point and the mixture is known as the eutectic mixture. [00:12:53.828] Here the name eutectic is from a Greek word which means easily melting. And since has the lowest melting point amongst all mixtures of A and B, this is referred to as the eutectic mixture. [00:13:09.952] Now at this point we have three phases in equilibrium. One is the liquid phase, one is solid A, and the other is solid B. Now using the phase rule, the reduced phase rule, F is equal to C minus P plus 1, we will find out the degrees of [00:13:27.632] Freedom at this point. Now C is equal to two and P is equal to three. Therefore F comes out to be zero. So therefore this point is invariant. Now we know that when we have an invariant point, it means that we cannot change [00:13:45.200] the temperature or the composition. So in this case for this particular system A and B, the temperature and the composition are going to be fixed. So it has a definite composition and also a definite melting point [00:14:04.160] point. So the composition is referred to as the eutectic composition and the melting point the temperature or the freezing point is known as the eutectic temperature. And one more important point which we need to understand here is that the melting point is very very sharp. [00:14:22.980] unlike other mixtures where the melting point is over a range of temperatures, in the case of a eutectic mixture it is found that it it has a very sharp melting point. [00:14:34.960] In fact, earlier it was assumed that it is actually a compound, because compounds usually have a characteristic melting point, but later on microscopic examinations and other evidence showed that this [00:14:48.932] Temperature, this composition is actually that of a mixture. It is not a compound, it is a mixture, hence the name eutectic mixture, but it has a very sharp melting point. So

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:02.660] The important features of the utected point is that the mixture has the lowest melting point amongst all other compositions, then the it is an invariant point and also it has a very sharp melting point. [00:15:19.600] Now a very common example for a simple eutectic system would be the lead silver system. [00:15:26.276] The melting point of pure lead is 961 degrees centigrade that would be

represented here, and that of pure lead is 327 degrees centigrade. And the eutectic temperature and composition become fixed and the eutectic temperature is found to be [00:15:44.048] To be 303 degrees centigrade, and the eutectic composition is 97.4% of lead and 2.6% of silver. Let us understand these curves ME as well as NE. [00:16:02.992] Now this ME curve first is known as the freezing point curve of A. Why? I have already explained that as we go on adding more and more of B, the melting point or the freezing point of A is decreasing. So this would represent the freezing point curve of A [00:16:20.720] A and it also shows that as we go on adding B into A, the freezing point or the melting point is decreasing. So what are the phases which are in equilibrium on this curve? Here we have solid A [00:16:38.000] which is in equilibrium with the liquid melt which has which has B in A. So all along this line here this curve B have solid A which is in equilibrium with the liquid melt of B in A. [00:16:56.132] And now the number of components are 2, the degree the number of phases is also 2, therefore the reduced phase rule is applicable. F is equal to C minus P plus 1 is equal to so 2 minus 2 plus 1 and therefore now it is [00:17:13.264] 1. F is equal to 1 along this curve and therefore it is a univariant system. Similarly, we will have we will look at this curve N E which will be the freezing point curve of B and it shows the decrease in freezing point of B due to the addition of A. [00:17:30.896] A into B again here we have on the curve solid B in equilibrium with the liquid melt which is now A added to B. And again we will use the reduced phase rule we find that [00:17:44.708] F is equal to 1, so the system is univariant along this curve. So at the U Tecty point it is invariant along the these curves ME and NE, the system is univariant. [00:18:02.116] Now let us study how do we determine the solid-liquid equilibria. So we have to understand through what is called the thermal analysis. [00:18:16.420] Thermal analysis is the study of what we call as the cooling curves of various compositions of a system during solidification. So if we want to [00:18:29.892] do thermal analysis we need to plot a cooling curve and that would be temperature versus time. [00:18:39.364] This curve, whatever curve we get here when we study the temperature versus time. So say something like this, we get a curve like this. So this curve would be uh [00:18:50.736] uh it will be called as a cooling curve. So basically what we are doing here is we are reducing we are cooling the system al and plotting the graph with time and using these cooling curves [00:19:07.008] We can determine the freezing point of the component in the mixture and also the euvectic point. So let us see the phase diagram. How to construct the phase diagram using cooling curves. [00:19:22.180] We have studied the simple eutectic system, so here we can see that the phase diagram is live like this. [00:19:28.976] We will take up different compositions of A and B and try to plot the cooling curve for each of these compositions. So if we take pure A here, we find that [00:19:42.724] We the initially if we uh when we when it is in the liquid form and we start cooling it, there is just [00:19:50.212] cooling will happen and we don't see anything specific, but when it reaches the melting point or the freezing point, then solid A starts to form. And we know that at the freezing point the temperature gets arrested till

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:07.268] all the liquid has got converted into solid and that is what we see because this is the this is the time axis. So this is temperature and this is the time. So we find that at this at this temperature there is an arrest [00:20:24.144] And then again when all this uh liquid has got converted into solid, we find that again cooling is going to happen. So this now temperature can be [00:20:35.168] plotted in this graph over here and we find that this is the point here. Now let us take up the next composition. Say this is the composition and what are we doing here for cooling curve? We take this

composition heat it to a high temperature beyond the uh [00:20:52.880] A high temperature when we see that the liquid now the the mixture is in the liquid form, it's in the melt state. Then we start cooling it. When we cool it initially, we don't see anything, there is just a reduction in temperature. But the moment it reaches this point. [00:21:10.884] We find that the first crystal of A will start to form. And we know whenever there is crystallization, then there is a latent heat of fusion involved. So heat is going to be liberated. Now we are trying to hold the system and there is a liberation of heat. [00:21:28.656] Definitely there is going to be a [00:21:31.184] change in the slope and that is where we find a change here and that can be taken as the freezing point of A in the mixture B. Then further it keeps on cooling down till we come to the eutectic point. So when we come when we are cooling this we reach the eutectic [00:21:48.848] protective point we know that now the temperature [00:21:53.296] again gets arrested because now B is also uh being uh precipitated. We have three phases in equilibrium, F becomes equal to zero as we learned just now, and therefore the temperature becomes constant. So it starts behaving as if it was a pure compound and has a very sharp [00:22:10.960] melting point. So which temperature are we at now? This is known as the eutectic temperature. So here we see an arrest in the temperature and then further when all the u the eutectic melt the uh has got converted into the solid [00:22:28.324] Which contains A and B of the same composition as the eutectic mill. Then after that, there is just a reduction in the temperature, that is there is just cooling. Now, so from this graph, the second graph, we can get [00:22:44.064] This this break will tell us what is the freezing point of A in when B is added to it and this point also tells us the eutectic temperature. [00:22:56.800] Similarly, if we take another composition, say this composition here, we find that when we melt it and start cooling it, now the freezing will happen at a lower temperature because we are we are here we have added more of B. If you look here, this is we have added. [00:23:14.448] Added more of B into this, and therefore the freezing point will be much lower, and that's exactly what we see here. The first break now when the [00:23:23.716] A uh pure A will now start coming out of the system, starts precipitating out, we will get the freezing point of A in the mixture, and now this will happen at a much lower temperature because the freezing point has decreased further due to more of B in the mixture. [00:23:41.488] Then after that, because of this freezing of A, we find a change in the slope, and that is what helps us to determine the freezing point. And then further, when we go down again, we see an arrest in the temperature, and that is the eutectic temperature. So we notice that whether it is this. [00:23:59.056] Composition or discomposition, the eutectic temperature is always constant. So that is what we meant that for a system A and B the temperature as well as the composition is going to be fixed at the eutectic point. Further, there is just cooling. Now [00:24:16.624] Now if we look at this side, now we come to this side, which has uh which is pure B, pure B will show show a similar uh graph as pure A, and there is cooling, then there is arrest in the temperature. What is happening here? The compound. [00:24:32.484] The A now is changing is crystallizing, it is changing from the liquid to the solid, so there is a [00:24:39.824] Arrest in temperatures till all of the liquid has got converted to solid and then there is cooling. So we get this point A, uh this point over here. And then coming to the next point, we have added more of A into B, and therefore we find there is a decrease in the freezing point that [00:24:57.456] Can be noted from the cooling curve where the first crystal of B starts appearing, there is a change in the slope. The slope was like this, now suddenly there's a change in the slope. Wherever that inflection point, that change in the slope happens, that will be the freezing point of B.

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Transcript block 6 (source interval 001500-001799 seconds): [00:25:14.788] when A is added into the mixture and then further cooling comes when it reaches the eutectic point. So you are cooling this mixture when it reaches the eutectic point. Again, so whether you approach from this side or you approach from this side the eutectic temperature [00:25:32.560] You are going to get the same value of utectic temperature. So here we reach the eutectic temperature. Now A also starts prespiring out, and that is how we get a arrest in the temperature, then further cooling happens. So again from this graph we got the utectic temperature and also the freezing point and [00:25:50.128] And similarly, if we take this now the eutectic point itself, that would be very interesting because what would happen now when you take this eutectic mixture and you heat it, we find that [00:26:04.548] Initially nothing happens, there's just cooling and the the temperature just keeps falling down till we reach this point. Now [00:26:13.312] The because we are at the eutectic point A and B both will start precipitating out and therefore we have three phases and this behaves so if you behave like a pure compound and we get the temperature arrest over here till the eutectic melt has completely converted into the [00:26:32.944] two solids and the composition remains the same. The eutectic melt and the uh the solids which are formed have the same composition. Therefore we find this arrest in temperature and then there is for the cooling. So if you see the eutectic mixture has a similar cooling curve [00:26:52.352] like the as if it were or the pure compounds A and B. But here we find that there is you can uh know the utectic temperature. So now using all these points which we got the freezing points of A and we got the freezing points of B for various compositions. [00:27:10.064] And we already know what is the eutectic temperature because that remains constant throughout. So now when we plot the graph. [00:27:17.568] we can find out wherever these two lines curves meet that would be the eutectic composition. So this is how we study the solid liquid equilibria and we construct the phase diagram with the help of cooling curves. [00:27:33.508] Let us now talk about an application of the phase diagram of a two component system, the lead silver system. The lead silver system shows a simple eutectic diagram and in metallurgy there is a process known as the Pattenson's process. [00:27:51.280] Which is used for desilverization of Argentiferous lead. The process of heating Argentiferous lead containing a very small quantity of silver, about 0.1%, and cooling it to get pure lead and a liquid melt which is richer in silver. [00:28:11.044] Now Argentiferus lead is available in nature and it contains a very very small quantity of silver 0.1%. So that would be in this area, this region of the phase diagram. So we have to heat this mixture to [00:28:27.856] this ore and when it goes beyond this point we know that it is going to melt. When we start cooling it, when we reach this point here, we find that [00:28:38.880] pure lead will start precipitating out and as we cool it further more and more of lead will precipate out and the liquid is going to get richer in silver. [00:28:52.708] We go further cool it and then we see that more and more of the lead will precipitate out and the liquid melt composition now is moving towards the area where there is more of silver. The liquid melt is getting richer in silver. [00:29:13.924] The lead now that is separating out floats and it is continuously removed by layers. Now when the temperature of the liquid melt reaches the utectic [00:29:26.320] temperature, then what is going to happen is that we have now reached the eutectic composition. So now the composition is [00:29:35.760] 2.6% of silver. The liquid melt now contains 2.6% of silver. So basically we have increased the percentage of silver from 0.1%, which was in the ore. [00:29:51.312] to a composition where the liqu of the liquid weld which has which is much richer in silver and it has now the composition is two point six percent of

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Transcript block 7 (source interval 001800-002099 seconds): [00:30:01.808] uh silver. So the solid mixture now of lead and silver is subjected to other processes for the recovery of silver. So this is known as the Pattenson's process and using the phase diagram we can understand that how you are naturally [00:30:19.440] Natural ore which contains such a small quantity of silver can be made richer in the percentage of silver. [00:30:29.444] So, in the next class we are going to talk about a more complicated system, that is the iron carbon system. Thank you.

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Full transcript: 4580 normalized words, 124 timestamped segments, 1840.341333 seconds duration. Low-confidence segments flagged: 0.

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13 Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part I

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Graphs and diagrams

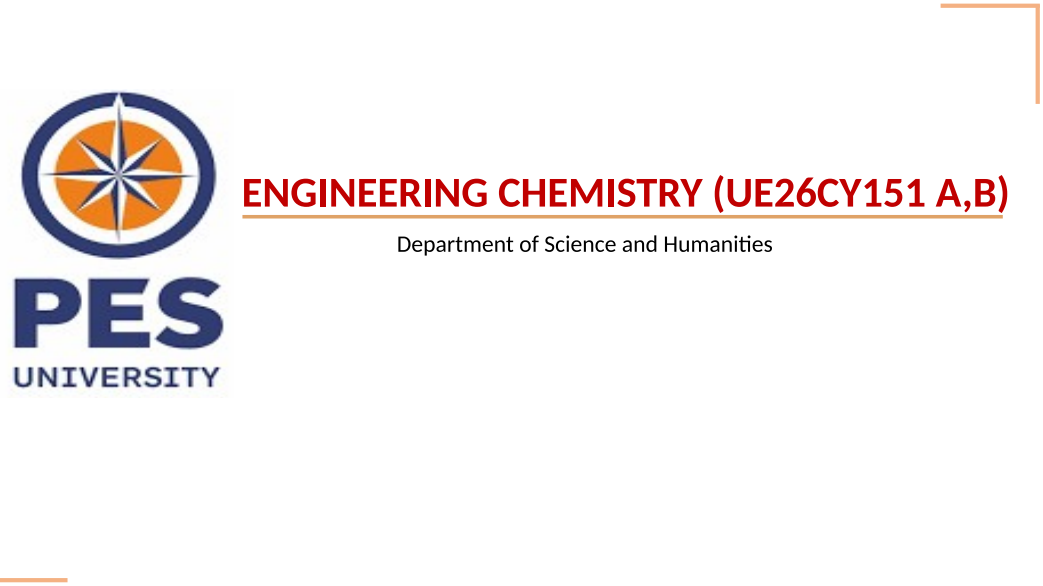

Figure G000115. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 1.

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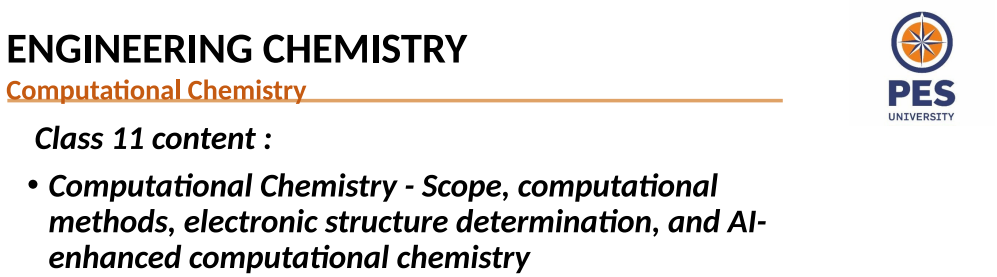

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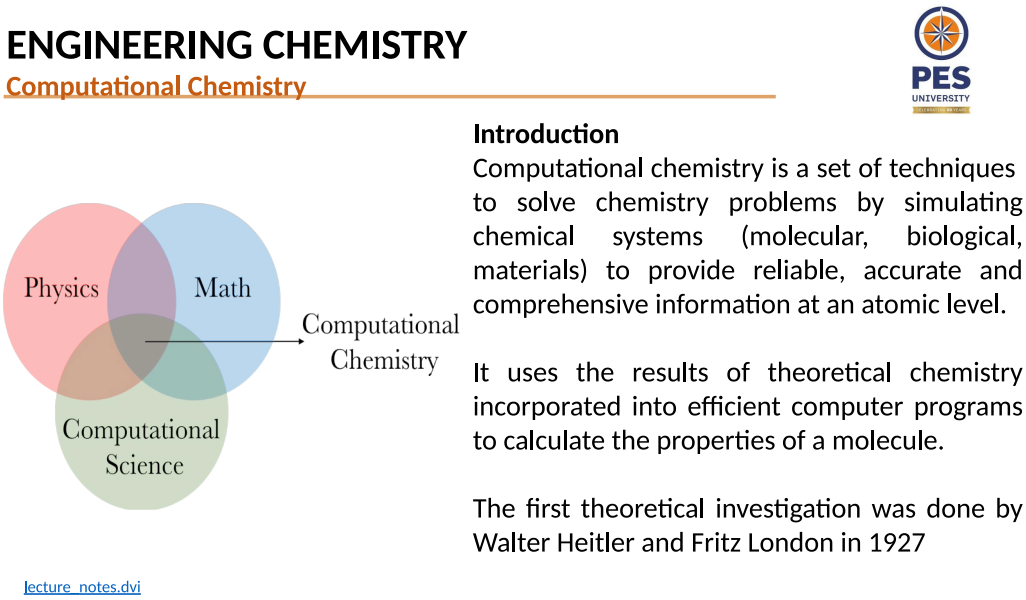

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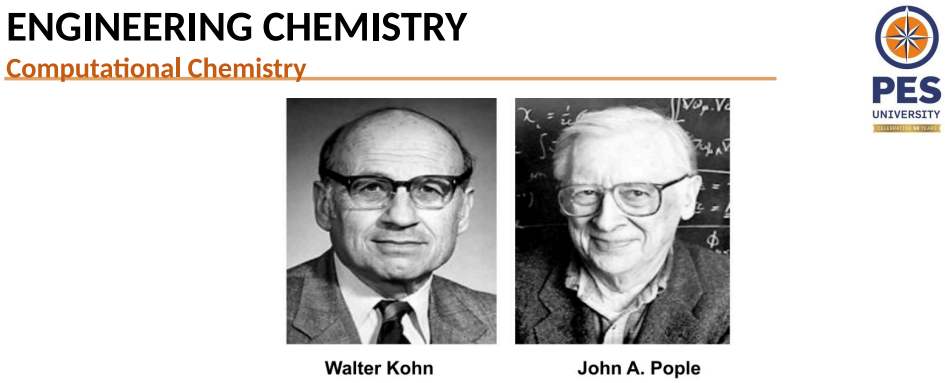

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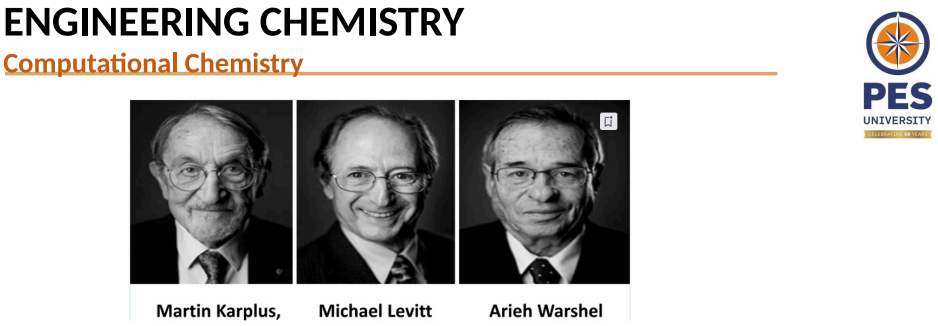

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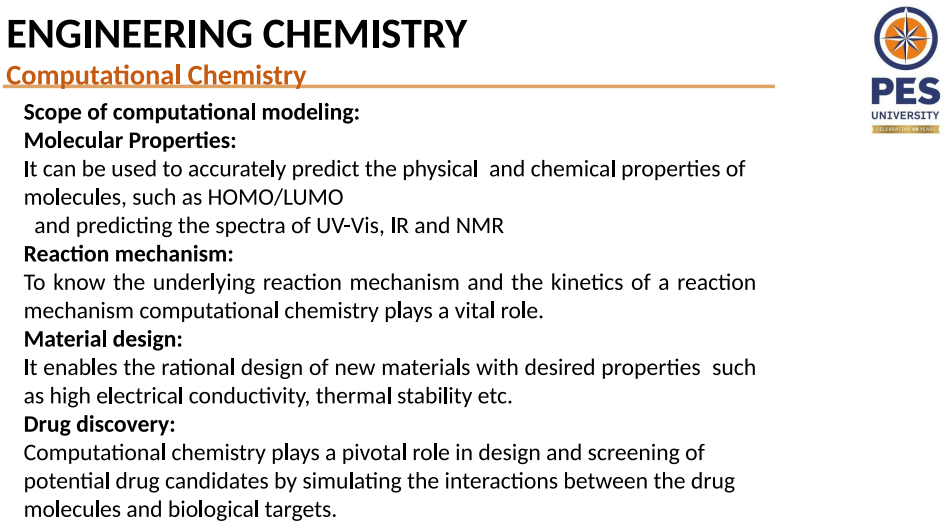

Figure G000120. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 6.

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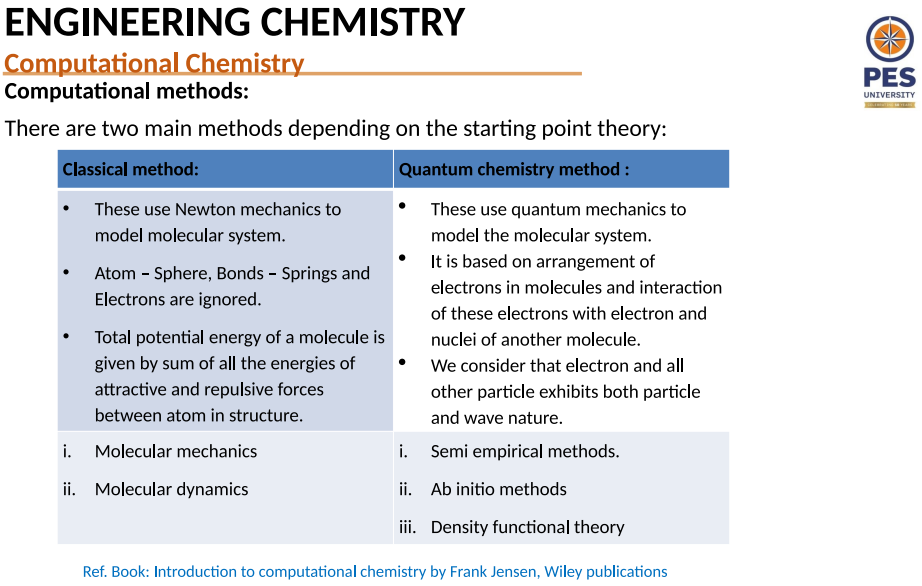

Figure G000121. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 7.

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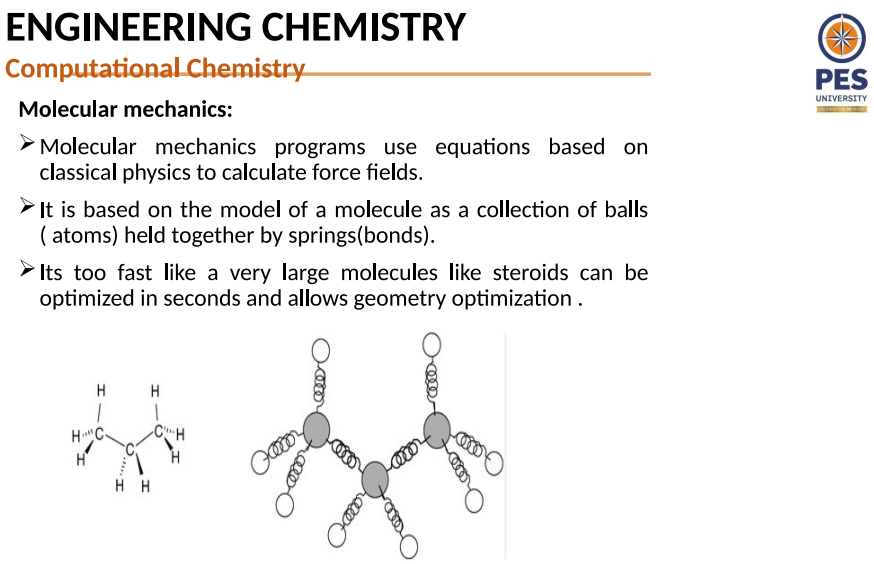


Figure G000122. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 8.

EV0000622

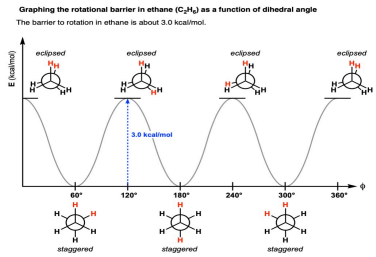
How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Computational Chemistry

Molecular dynamics:

- Molecular dynamics is a molecular mechanics program designed to mimic the movement of atoms within a molecule. The laws of motion to molecule is the base for it.
- Molecular dynamics can be carried out on a molecule to generate different conformation which on energy minimization.



Staggered vs Eclipsed Conformations of Ethane - Master Organic Chem

Figure G000123. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 9.

EV0000625

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

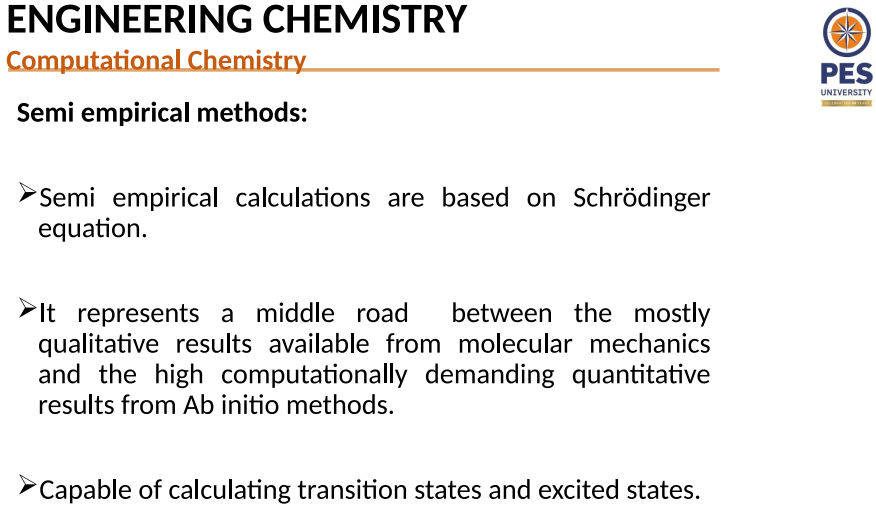


Figure G000124. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 10.

EV0000628

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

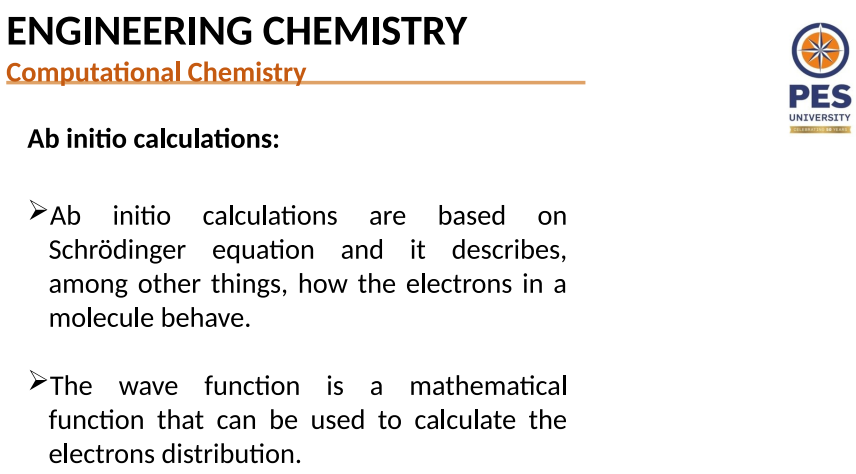


Figure G000125. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 11.

EV0000631

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

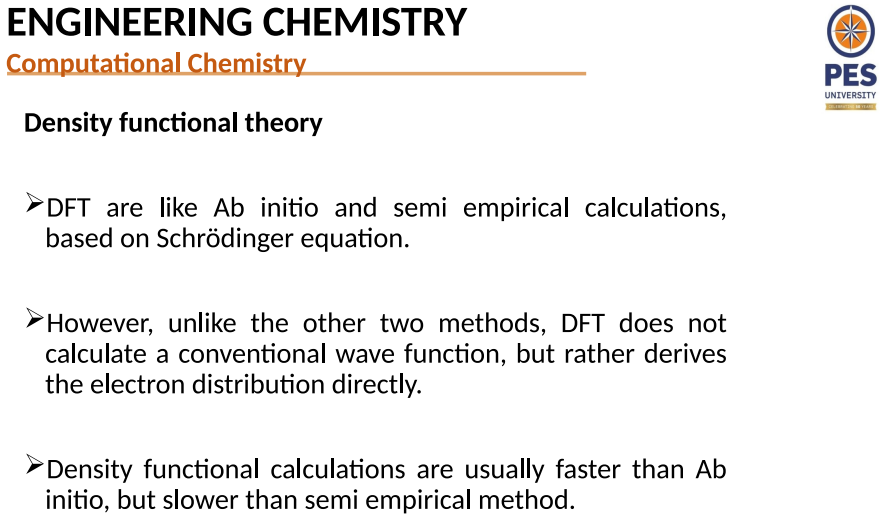


Figure G000126. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 12.

EV0000634

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

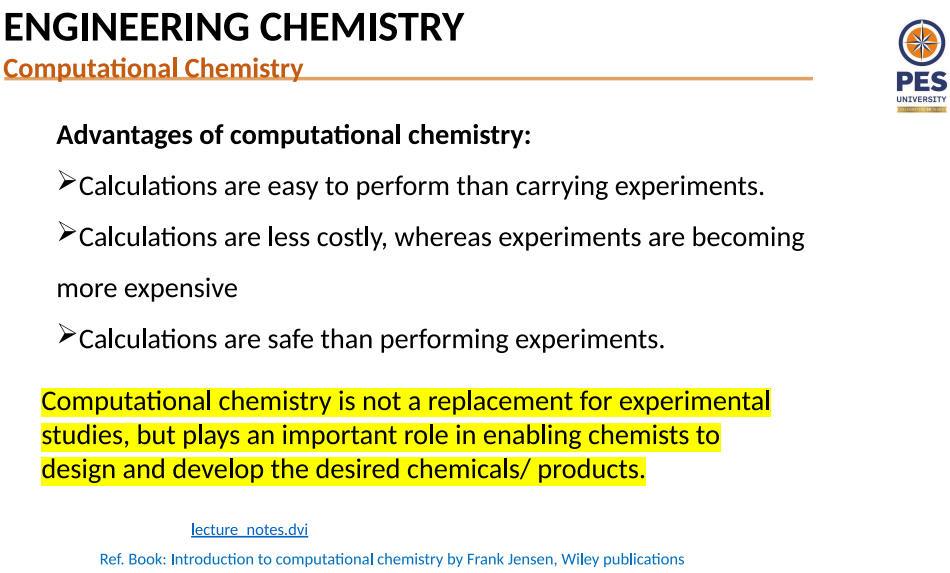


Figure G000127. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 13.

EV0000637

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

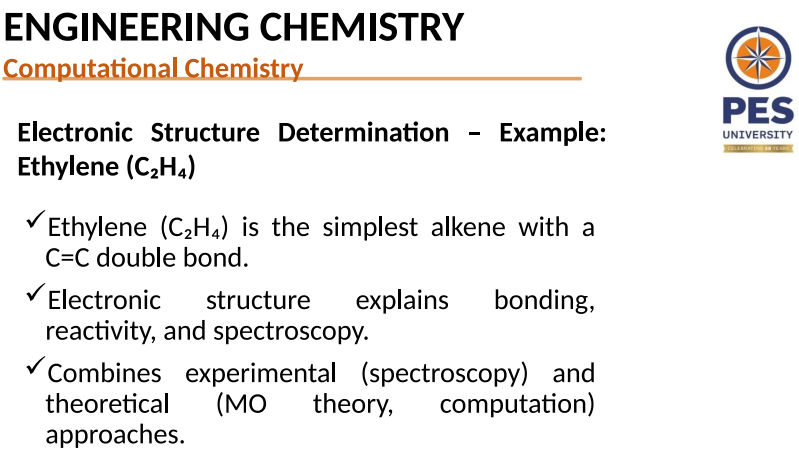


Figure G000128. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 14.

EV0000640

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

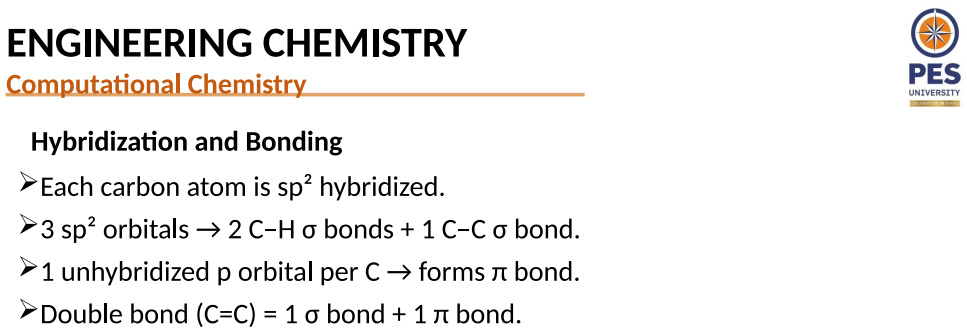


Figure G000129. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 15.

EV0000643

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

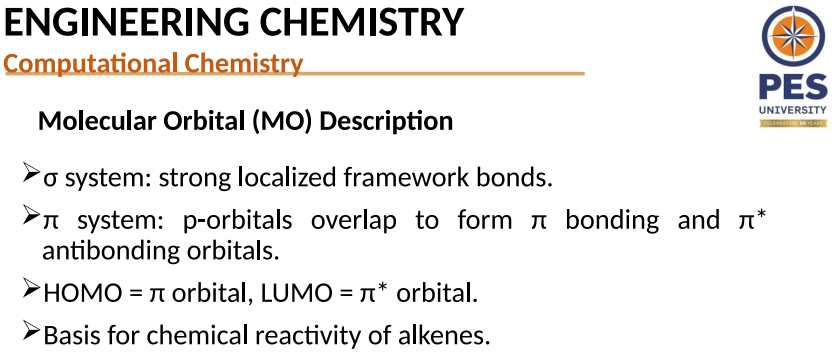


Figure G000130. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 16.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

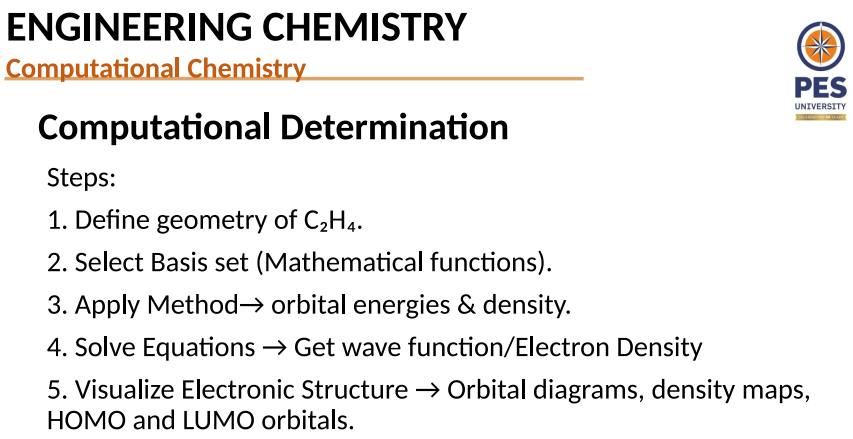

Figure G000131. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 17.

EV0000649

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

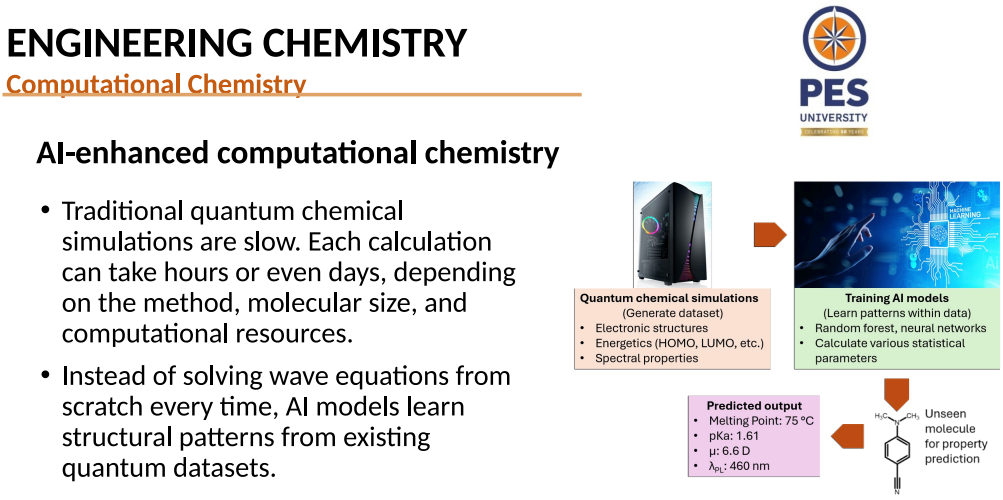

Figure G000132. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 18.

EV0000652

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

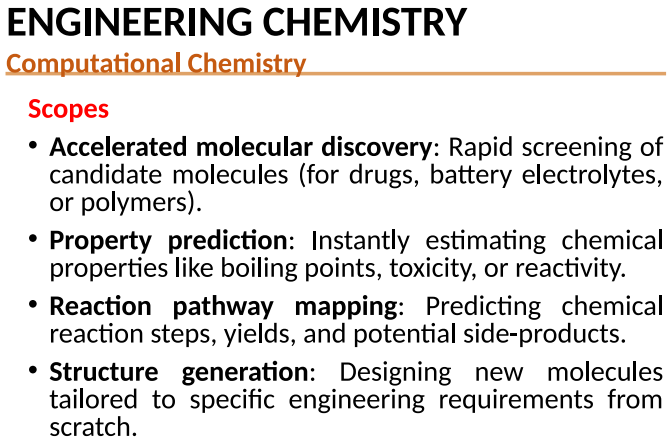

Figure G000133. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 19.

EV0000655

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

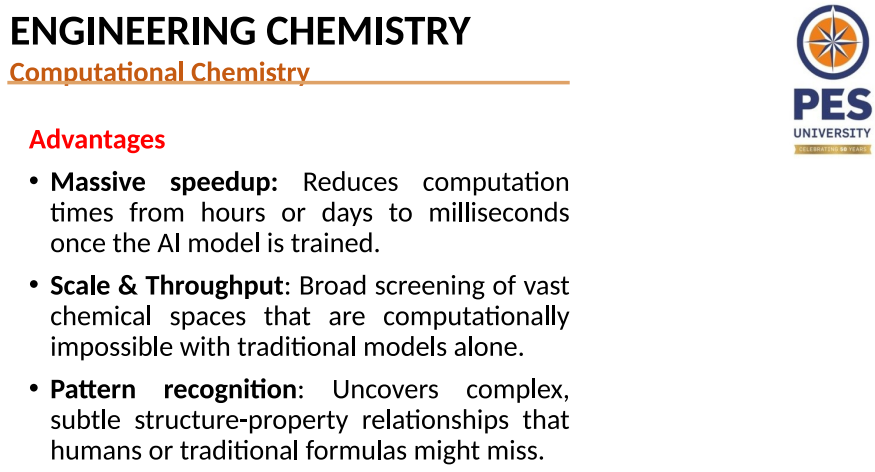

Figure G000134. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 20.

EV0000658

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000135. Computational chemistry Scope computational methods electronic structure determination AI enhanced computation · source slide page 21.

EV0000661

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000600 · EV0000601 · EV0000602
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Class 11 content : Computational Chemistry - Scope, computational methods, electronic structure determination, and AI-enhanced computational chemistry EV0000603 · EV0000604 · EV0000605
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Introduction Computational chemistry is a set of techniques to solve chemistry problems by simulating chemical systems (molecular, biological, materials) to provide reliable, accurate and comprehensive information at an atomic level. It uses the results of theoretical chemistry incorporated into efficient computer programs to calculate the properties of a molecule. The first theoretical investigation was done by Walter Heitler and Fritz London in 1927 [Image text, OCR] Physics Math Computational Science lecture_notes.dvi EV0000606 · EV0000607 · EV0000608
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry [Image text, OCR] Walter Kohn John A. Pople 1998 Walter Kohn : For the development of density functional theory. John A. Pople : For the development of computational methods in quantum chemistry. EV0000609 · EV0000610 · EV0000611
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry [Image text, OCR] Martin Karplus, Michael Levitt AriehWarshel 2013 For the development of computer-based methods to model complex systems. EV0000612 · EV0000613 · EV0000614

6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Scope of computational modeling: Molecular Properties: It can be used to accurately predict the physical and chemical properties of molecules, such as HOMO/LUMO and predicting the spectra of UV-Vis, IR and NMR Reaction mechanism: To know the underlying reaction mechanism and the kinetics of a reaction mechanism computational chemistry plays a vital role. Material design: It enables the rational design of new materials with desired properties such as high electrical conductivity, thermal stability etc. Drug discovery: Computational chemistry plays a pivotal role in design and screening of potential drug candidates by simulating the interactions between the drug molecules and biological targets. EV0000615 · EV0000616 · EV0000617
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Computational methods: There are two main methods depending on the starting point theory: Classical method: | Quantum chemistry method : These use Newton mechanics to model molecular system. Atom – Sphere, Bonds – Springs and Electrons are ignored. Total potential energy of a molecule is given by sum of all the energies of attractive and repulsive forces between atom in structure. | These use quantum mechanics to model the molecular system. It is based on arrangement of electrons in molecules and interaction of these electrons with electron and nuclei of another molecule. We consider that electron and all other particle exhibits both particle and wave nature. Molecular mechanics Molecular dynamics | Semi empirical methods. Ab initio methods Density functional theory Ref. Book: Introduction to computational chemistry by Frank Jensen, Wiley publications EV0000618 · EV0000619 · EV0000620
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Molecular mechanics: Molecular mechanics programs use equations based on classical physics to calculate force fields. It is based on the model of a molecule as a collection of balls (atoms) held together by springs(bonds). Its too fast like a very large molecules like steroids can be optimized in seconds and allows geometry optimization . [Image text, OCR] H H — H H H H H H EV0000621 · EV0000622 · EV0000623
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Molecular dynamics: Molecular dynamics is a molecular mechanics program designed to mimic the movement of atoms within a molecule. The laws of motion to molecule is the base for it. Molecular dynamics can be carried out on a molecule to generate different conformation which on energy minimization. [Image text, OCR] Graphing the rotational barrier in ethane (C2Hs) as a function of dihedral angle Thebarriertorotationinethaneisabout3.0kcal/mol. (o/o)) eclipsed eclipsed eclipsed eclipsed E 3.0kcal/mol 60° 120° 180° 240° 300° 360° H H staggered staggered staggered Staggered vs Eclipsed Conformations of Ethane - Master Organic Chem EV0000624 · EV0000625 · EV0000626
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Semi empirical methods: Semi empirical calculations are based on Schrödinger equation. It represents a middle road between the mostly qualitative results available from molecular mechanics and the high computationally demanding quantitative results from Ab initio methods. Capable of calculating transition states and excited states. EV0000627 · EV0000628 · EV0000629
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Ab initio calculations: Ab initio calculations are based on Schrödinger equation and it describes, among other things, how the electrons in a molecule behave. The wave function is a mathematical function that can be used to calculate the electrons distribution. EV0000630 · EV0000631 · EV0000632

12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Density functional theory DFT are like Ab initio and semi empirical calculations, based on Schrödinger equation. However, unlike the other two methods, DFT does not calculate a conventional wave function, but rather derives the electron distribution directly. Density functional calculations are usually faster than Ab initio, but slower than semi empirical method. EV0000633 · EV0000634 · EV0000635

13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Advantages of computational chemistry: Calculations are easy to perform than carrying experiments. Calculations are less costly, whereas experiments are becoming more expensive Calculations are safe than performing experiments. Computational chemistry is not a replacement for experimental studies, but plays an important role in enabling chemists to design and develop the desired chemicals/ products. lecture_notes.dvi Ref. Book: Introduction to computational chemistry by Frank Jensen, Wiley publications EV0000636 · EV0000637 · EV0000638

14. Source slide page 14 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Electronic Structure Determination – Example: Ethylene (C_2H_4) Ethylene (C_2H_4) is the simplest alkene with a C=C double bond. Electronic structure explains bonding, reactivity, and spectroscopy. Combines experimental (spectroscopy) and theoretical (MO theory, computation) approaches. EV0000639 · EV0000640 · EV0000641

15. Source slide page 15 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Hybridization and Bonding Each carbon atom is sp^2 hybridized. 3 sp^2 orbitals → 2 C–H σ bonds + 1 C–C σ bond. 1 unhybridized p orbital per C → forms π bond. Double bond (C=C) = 1 σ bond + 1 π bond. EV0000642 · EV0000643 · EV0000644

16. Source slide page 16 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Molecular Orbital (MO) Description σ system: strong localized framework bonds. π system: p-orbitals overlap to form π bonding and π* antibonding orbitals. HOMO = π orbital, LUMO = π* orbital. Basis for chemical reactivity of alkenes. EV0000645 · EV0000646 · EV0000647

17. Source slide page 17 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Computational Determination Steps: 1. Define geometry of C_2H_4. 2. Select Basis set (Mathematical functions). 3. Apply Method→ orbital energies & density. 4. Solve Equations → Get wave function/Electron Density 5. Visualize Electronic Structure → Orbital diagrams, density maps, HOMO and LUMO orbitals. EV0000648 · EV0000649 · EV0000650

18. Source slide page 18 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry AI-enhanced computational chemistry [Image text, OCR] MACHINE LEARNING Quantum chemical simulations Training Al models (Generate dataset) (Learn patterns within data) Electronic structures Random forest, neural networks Energetics (HOMO, LUMO, etc.) Calculate various statistical Spectral properties parameters Predicted output Melting Point: 75 °C pKa: 1.61 μ: 6.6 D 入pL: 460 nm Traditional quantum chemical simulations are slow. Each calculation can take hours or even days, depending on the method, molecular size, and computational resources. Instead of solving wave equations from scratch every time, AI models learn structural patterns from existing quantum datasets. EV0000651 · EV0000652 · EV0000653

19. Source slide page 19 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Scopes Accelerated molecular discovery: Rapid screening of candidate molecules (for drugs, battery electrolytes, or polymers). Property prediction: Instantly estimating chemical properties like boiling points, toxicity, or reactivity. Reaction pathway mapping: Predicting chemical reaction steps, yields, and potential side-products. Structure generation: Designing new molecules tailored to specific engineering requirements from scratch. EV0000654 · EV0000655 · EV0000656
20. Source slide page 20 (extracted text): ENGINEERING CHEMISTRY Computational Chemistry Advantages Massive speedup: Reduces computation times from hours or days to milliseconds once the AI model is trained. Scale & Throughput: Broad screening of vast chemical spaces that are computationally impossible with traditional models alone. Pattern recognition: Uncovers complex, subtle structure-property relationships that humans or traditional formulas might miss. EV0000657 · EV0000658 · EV0000659
21. Source slide page 21 (extracted text): THANK YOU Department of Science and Humanities EV0000660 · EV0000661 · EV0000662

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.708] Hi, welcome to PS University Online Teaching and Learning Practices. In this session, let us learn Computational Chemistry Introduction, Computational Methods and Applications. [00:00:17.636] Computational chemistry is a multidisciplinary area. It involves maths, physics, and also chemistry basic information. So it can be defined as [00:00:34.820] Computational chemistry is a branch of chemistry that uses mathematical algorithms, statistics, and computer simulations to study and solve chemical problems. [00:00:49.476] It does not directly replace experimental chemistry, but it complements predicting molecular structure, properties and reactions, explaining experimental observation. [00:01:06.756] reducing the cost and time of the experiment. Computational chemistry refers to a collection of techniques used to investigate chemical problems with the aid of computers. [00:01:24.164] Rather than relying on laboratory experiments with chemicals, it applies the principles of theoretical chemistry through efficient computer program to predict and calculate molecular properties. [00:01:42.660] The earliest theoretical study was done in this field. The experiment was carried out by Walter Hitler and Fritz Linden in 1927. [00:02:00.740] In the year nineteen ninety eight, Nobel Prize is awarded in chemistry to Walter Conn and John A Popel. [00:02:12.292] For the development of density functional theory, as well as for the contributions to the development of computational methods in quantum chemistry, which greatly advanced under practical applications of theoretical models. [00:02:33.156] In the year 2013, the Nobel Prize was awarded in chemistry for the development of computer based methods and simulate. [00:02:45.892] Complex chemical system enabling a deeper understanding of molecular process. [00:02:56.260] Objectives of computational chemistry. It predicts and estimates the structure, stability, reactivity, and properties of molecules. [00:03:10.180] To provide reliable, accurate, and compressive information at anatomic level. The advantages of computational chemistry are calculations are easy to perform than carrying out the experiments in the laboratory. [00:03:30.148] Calculations are less costly whereas experiments are becoming more expensive because not using any chemical molecule in the calculation process, so it is less costly. [00:03:46.564] Calculations are safer than performing the experiment in the lab. These are the advantages of computational chemistry. Scope of computational modeling. [00:04:03.620] The computational modeling can be used to accurately

predict physical and chemical properties of molecule such as bond length, bond angles, humor and lumo interactions. [00:04:19.972] It can help in predicting the spectr of ultraviolet visible and IR spectroscopy, as well as proton nuclear magnetic resonance spectroscopy. [00:04:33.316] And also able to understand thermodynamic and kinetic properties. According to thermodynamic properties are stable and unstable. [00:04:46.212] Whether the system is stable or unstable with respect to the chemistry molecule, whether the molecule is stable or highly unstable. This can be determined thermodynamically. [00:04:58.948] Similarly, kinetic properties it is nothing but leby and inertness. Labelity means immediately it can undergo reaction. Inertness means certain extent it is stable, higher the activation energy of the molecule.

EV0000230 · EV0000231 · EV0000232 · EV0000233

Transcript block 2 (source interval 000300-000599 seconds): [00:05:17.444] The electronic structure and all unmechanical properties can be determined by using computational modelling. [00:05:27.364] So reaction mechanism. What we need to understand in the reaction, one can easily understand the activation energy. After understanding the activation energy, [00:05:39.856] At what temperature a reaction can be carried out? What is the pressure? Whether it requires to use the catalyst or not. All these can be easily determined by using computational tool. [00:05:54.852] To know the underlying reaction mechanism and the kinetics of a reaction mechanism, computational chemistry plays a vital role in understanding. [00:06:06.276] What is the temperature, pressure, or catalyst is required or not to carry out the experiment in the laboratory. And also it helps in material design. [00:06:19.780] In the by using computational modeling enables the rational design of new material with desired properties such as electrical conductivity, thermal stability, optical properties, magnetic properties. [00:06:40.036] and other properties can be designed easily with the help of computational tool. So [00:06:47.748] This leads to the development of advanced material for energy production, storage application, especially in the batteries, capacitor, supercapacitor, electronics, and much more applications. [00:07:07.844] This is very, very important one in the pharmaceutical industry in the determination of drug. Drug discovery. [00:07:18.468] Computational chemistry plays a pivotal role in design and the screening of potential drug candidates by simulating the interaction between the drug molecule and biological target, particularly in case of [00:07:36.644] The cancer cells in a particular part of the DNA or an RNA molecule affected by cancer cells, where by using this computational model, one can easily discover a target molecule for the cancer cell. [00:07:56.100] develop in the human system. That is why in the drug discovery computational model is very much useful, especially structure activity relationship of the drug molecule can be easily determined. [00:08:16.676] Thank you.

EV0000234 · EV0000235 · EV0000236 · EV0000237

Full transcript: 742 normalized words, 33 timestamped segments, 499.265433 seconds duration. Low-confidence segments flagged: 0.

EV0000230 · EV0000231 · EV0000232 · EV0000233

14 Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part III

This lesson is indexed from the source video titled Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part III. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000238 · EV0000239

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.420] Hi, welcome back to the course on engineering chemistry. In this session we learn electronic structural determination by taking an example of ethylene molecule. Ethylene molecule is the simplest algae having carbon double bound carbon. [00:00:19.844] In the determination of electronic structure, the process of finding the arrangement of electrons and atoms and molecules, it is important because chemical bonding reactivity and properties depending on electronic configuration. [00:00:38.244] It can be achieved by using theoretical models, computational methods, and spectroscopic techniques. You want understood about the iodization and bonding. An ethylene molecule is simplest molecule. [00:00:56.836] Each carbon atom is sp2 iperized. Three sp2 orbitals, two Cs sigma bonds, and one carbon single bond carbon. [00:01:09.988] One unibrodized p orbital per carbon. Per carbon forms pi bond. Double bonds. One sigma bond and one pi bond. Molecular orbital. [00:01:25.444] Description Sigma system means where there is mutual sharing of electrons between two atoms leads to the formation of sigma bond. Hence it is called a strong localized framework bonds, call it as sigma system. [00:01:44.004] Pi bond is formed where there is sideways overlapping between p orbitals leads to the formation of pi bonding. It can be represented as pi bonding and pi star antebonding orbitals. Totally it is called as pi system. [00:02:02.884] Similarly, highest occupied molecular orbitals can be represented pi orbital, lowest unoccupied molecular orbitals can be represented as pi star orbital. This is the basis for chemical reactivity. [00:02:20.708] In the computational determination, some steps are involved. So there are five steps are involved in the feeding the information about molecule into the computational tool. [00:02:35.876] The first step is to define the system with respect to the molecule, atom, and geometry. Second one, select basis set. It is nothing but mathematical functions. Applying method. We have to apply method either at in Gio. [00:02:55.460] or density functional theory or semi empirical method. Solve the equity get wave function or electron density. [00:03:05.700] Visualize electronic structure, orbital diagram, density maps, frontier orbitals like Homo and Lomo. Applications in spectroscopy. [00:03:20.132] Photoelectron spectroscopy measures sigma and pi energies. Usually in ultraviolet visible spectroscopy, sigma to sigma star transition, pi to pi star, pi to n. [00:03:38.532] And SIMO N transition takes place in particular. Pi to pi star transition takes place between six one sixty five to

one eighty nanometer. Similarly, in IL spectroscopy [00:03:55.876] Characteristic stretching frequency is observed for carbon double bond carbon at 1650 reciprocome centimeter. In the same way, proton NMR spectroscopy, we will measure proton change. In general application, [00:04:15.588] of computational tool. It helps in predicting the molecular structure and it helps in understanding electrophilic addition reactions, molecular geometry, bonding nature, [00:04:31.460] And spectroscopic transitions. It provides Homo Lomo insights for the reactivity and study reaction mechanism. Here one cell explanatory video is playing. [00:04:50.372] All of you observe carefully. [00:04:59.876] We have to use Goggian software is a powerful tool to understand reactivity and humor interactions. All of you observe carefully all these things in the self-explanatory video.

EV0000238 · EV0000239 · EV0000240 · EV0000241

Transcript block 2 (source interval 000300-000599 seconds): [00:07:36.644] [ASR low confidence; check audio] Yeah.

EV0000242 · EV0000243 · EV0000244 · EV0000245

Full transcript: 492 normalized words, 20 timestamped segments, 642.581 seconds duration. Low-confidence segments flagged: 1. Review locators: 00:07:36.644

EV0000238 · EV0000239 · EV0000240 · EV0000241

15 Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part II

This lesson is indexed from the source video titled Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part II. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000246 · EV0000247

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.964] Hi, welcome back to the course on engineering chemistry. In this session, we learn methods of computational chemistry. There are two main methods depending on starting point theory. [00:00:17.732] One is called as a classical method, and another one is called as quantum chemistry method. Let us understand what is classical method. In this method, we are using Newton mechanics to model molecular system. At the same time, [00:00:36.580] We need to consider atom as a spare or ball, and bonds is nothing but a spring, and electrons are ignored here. Total potential energy of a molecule is given by [00:00:50.212] The sum of all the energies of attractive and repulsive forces between atoms in the structure. Namely molecular dynamics and molecular mechanics. In case of quantum chemistry methods, [00:01:07.172] This uses quantum mechanics to model the molecular system. It is based

on the arrangement of electrons in molecule and the interaction of these electrons with electron and the nuclei of another molecule. [00:01:22.276] Here we will consider that electron and all other particles exhibit both particle and wave nature. And three important methods are there in the quantum chemistry based on this semi empirical methods, Abin show method and density functional theory. [00:01:41.028] Let us understand in detail about these three methods. First, particularly the molecular mechanics as well as molecular dynamics, followed by this, and we have to learn another three methods, namely semi-empirical methods eventually methods, density functional theory in detail. [00:02:01.636] What is molecular mechanics? Molecular mechanics is a computational method used to study and predict the structure, stability, and properties of molecules using classical physics rather than quantum mechanics. [00:02:20.324] Here also we do consider atoms are treated as space or bone and bond is treated as spring and connecting to the atom. [00:02:33.220] So, molecular mechanics based on the programs use equations based on the classical physics to calculate force field. It is also based on the ball and spring model. [00:02:47.876] It's too fast like a very large molecules like steroids can be optimized in seconds and almost geometry optimization by using molecular mechanics in the classical methods. [00:03:01.956] The another technique is called molecular dynamics. What is this molecular dynamics? Molecular dynamics is a [00:03:10.596] computer simulation technique in which time dependent behavior of a molecular system is studied by solving Newton's equation of motion of atoms and molecules. [00:03:24.612] Molecular dynamics is a molecular mechanism program and designed to mimic the movement of atoms within a molecule, again based on the law of motion of molecule, based on the molecular structure. [00:03:38.340] Molecular dynamics can be carried out on a molecule to generate different conformations which are energy minimizing minimization. I hope you people have understood in the lower standards. Take butane molecule. Butane molecule is having different conformations. [00:03:57.732] What are these conformation? The spati arrangement of methyl group in the plane. So based on this, we do measure stabilization energy, and this stabilization energy can be calculated by using molecular dynamics. [00:04:20.100] In the semi-empirical form methods, particularly in the quantum mechanics, we do use semi-empirical method. What is the semi-empirical method? What is the definition? [00:04:34.212] Semi-empirical methods are quantum chemistry approaches that simplify the Schrödinger wave equation by making approximation and incorporating experimental data, particularly empirical parameters are used. [00:04:52.132] Usually semi-empirical methods values lie between av and show methods, completely quantum, very accurate, but expensive. And molecular mechanics, fast but no electron detail is given. So

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:09.924] In semi empirical calculations are based on completely Schrödinger equations. [00:05:16.580] It represents a middle road between the mostly qualitative results available from molecular mechanics and the high computationally demanding quantitative results from adventure methods, capable of calculating transition state and excited states. [00:05:42.052] So let us understand ab in show method. What is abinshoe method? First and foremost thing abinshoe is the Latin word. It means from the beginning. [00:05:54.724] What is the definition of events show calculation or events method? In computational chemistry, abinshow methods are first principles, quantum, mechanical calculation, and do not use empirical parameters. [00:06:11.236] Everything is derived from solving the Schrödinger equation using fundamental constants like electron charge, Planck's constant, etc. [00:06:23.428] In the F and Shoe method as already discussed based on the Schrödinger equation, it

describes among other things how the electrons in a molecule behave. The wave function is a mathematical function that can be used to calculate electron distribution. [00:06:46.852] Most accurate quantum chemical techniques, but computationally computationally demanding, they serve as gold standard for validating other computational methods. [00:07:04.036] Let us understand what is density functional theory. What is the definition for this? Density functional theory is a quantum mechanical method used to investigate the electronic structure, especially ground state properties of atoms. [00:07:23.204] molecules and solids instead of solving for the mini electron wave function because it becomes very complex. DFT works with electron density as the central variable. [00:07:39.172] So, DFT are like abinshium and semi-empirical calculation based on the Schrodinger equation. However, [00:07:48.420] Unlike the other two methods, DFT does not calculate conventional wave function, but rather derives the electron distribution directly. Density functional calculations are usually faster than adio. [00:08:06.564] but slower than empirical methods. In a simpler manner, DFD is a quantum method using electron density instead of wave functions. Balances, accuracy, and computational efficiency. [00:08:24.804] So now we need to understand what are the important applications of computational chemistry. Computational chemistry is very much essential nowadays. We are going to use in the determination of electronic structure. [00:08:41.572] And predictions, energy minimization. And we can find out the transition structure and reaction parts, molecular docking, protein protein coupling, protein protein interaction, protein ligand interactions. All these are useful in [00:09:01.124] developing particular molecule especially in the cancer drug anti-cancer drug molecule. So already we understood in the previous slide by using conformations, especially energy stabilization. [00:09:17.700] uh stagger, anti-gauche, gauche, and all these confirmation what is the stabilization energy that can be calculated by using computational chemistry. [00:09:30.148] Thermochemistry Usually in thermochemistry two reaction takes place one is exothermic and endothermic reactions and the heat of reactions can be easily calculated, energy of activation of the molecule can be calculated using computational chemistry. [00:09:47.492] Calculation of many other molecular physical and chemical properties can be explored by using computational chemistry methods. Computational chemistry plays a vital role in progenitic world.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:04.420] particularly in advancing our understanding of chemical system and accelerating scientific discovery. [00:10:11.876] It has one of the important advantages is that cost effectiveness, efficiency, safety, accessibility, versatility, and accuracy make it indispensable tool in modern research and development. [00:10:29.072] Computational chemistry is not a replacement from the lab. You know? Why? Because we need to [00:10:38.244] Perform the experiment to get a desired molecule. But this computational chemistry will help us whether we are going to get desired molecule or not before conducting experiment in the lab. [00:10:51.968] That is what computational chemistry is not a replacement for experimental studies, but plays an important role in enabling chemists to design and develop the design chemicals and the product molecule. Thank you very much for patient watching.

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Full transcript: 1167 normalized words, 42 timestamped segments, 670.741 seconds duration. Low-confidence segments flagged: 0.

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16 Source slide deck: Born Oppenheimer approximation

This entry preserves the complete slide deck titled Born Oppenheimer approximation as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Chemistry, Unit_01, to avoid an unsupported match. EV0000312 · EV0000313

Graphs and diagrams

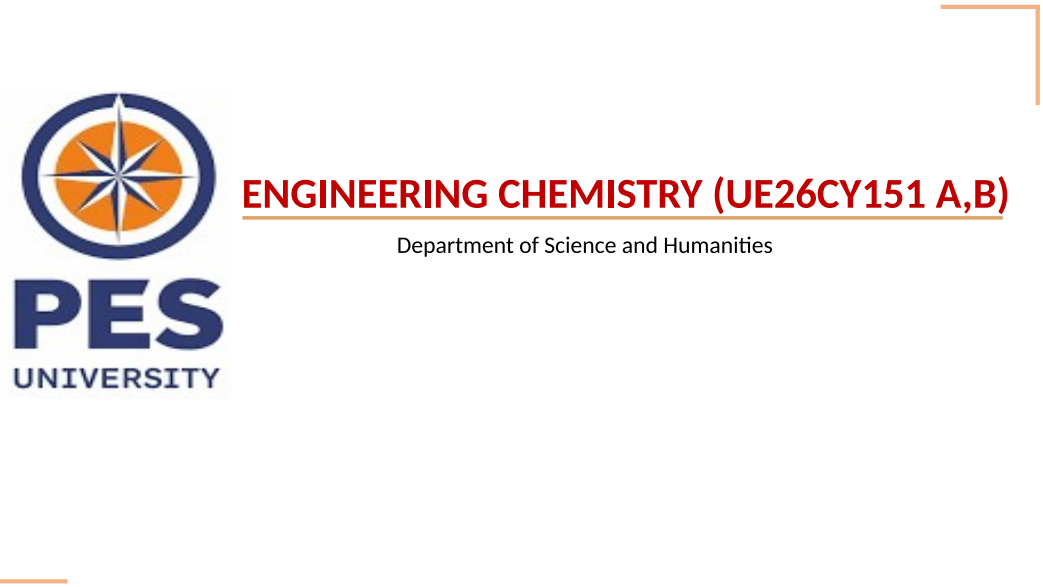

Figure G000019. Born Oppenheimer approximation · source slide page 1.

EV0000313

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000020. Born Oppenheimer approximation · source slide page 2.

EV0000316

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

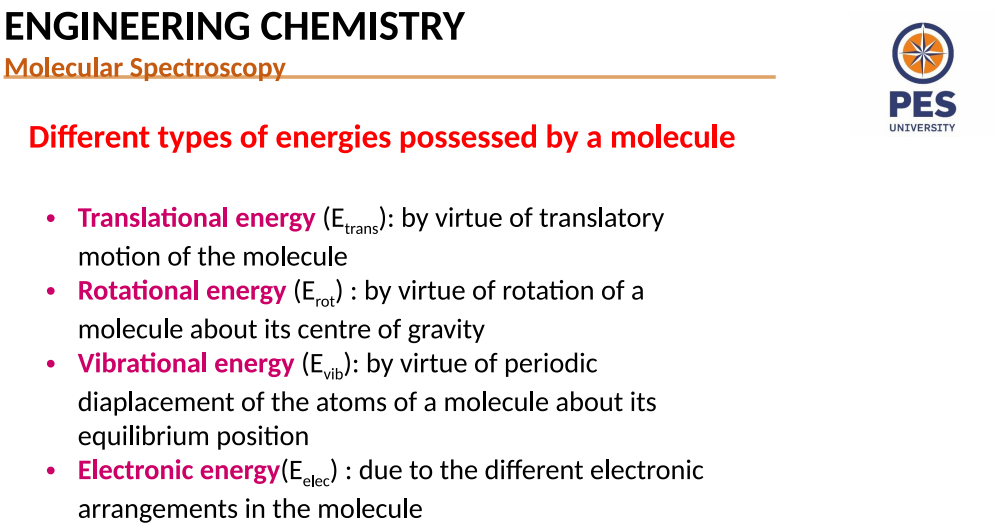

Figure G000021. Born Oppenheimer approximation · source slide page 3.

EV0000319

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

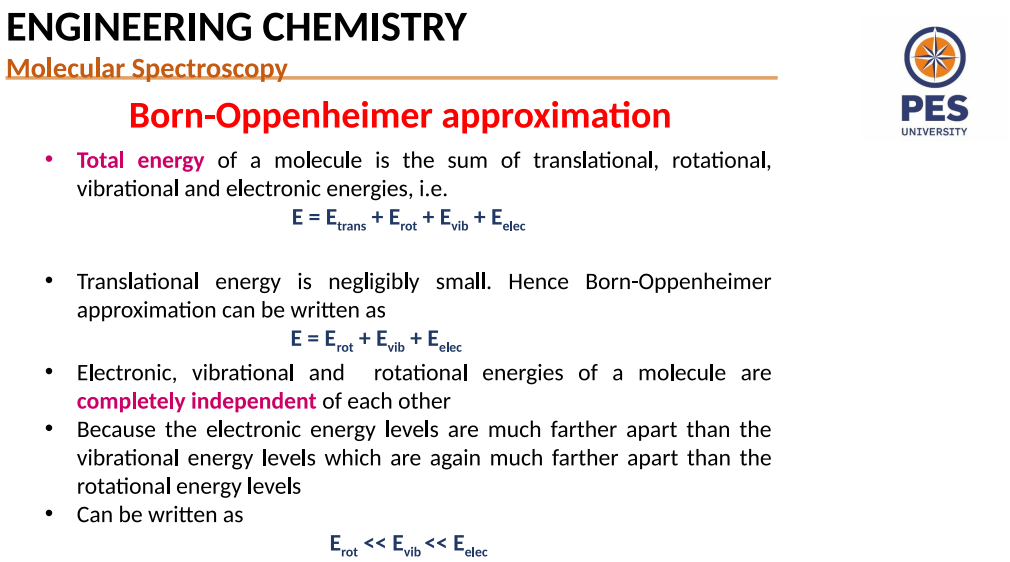

Figure G000022. Born Oppenheimer approximation · source slide page 4.

EV0000322

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

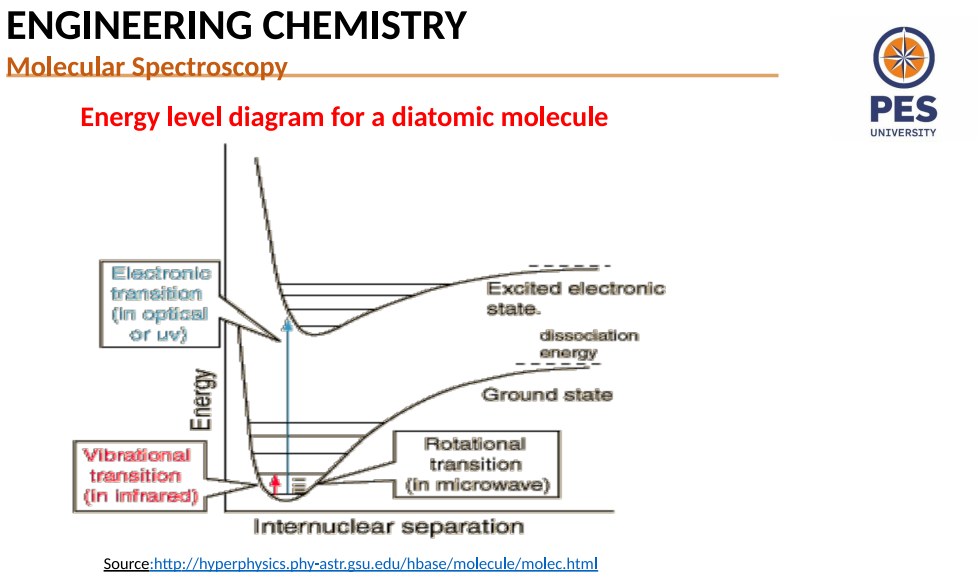

Figure G000023. Born Oppenheimer approximation · source slide page 5.

EV0000325

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000024. Born Oppenheimer approximation · source slide page 6.

EV0000328

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000312 · EV0000313 · EV0000314
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Class 3 content: Born - Oppenheimer approximation EV0000315 · EV0000316 · EV0000317
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Different types of energies possessed by a molecule Translational energy (Etrans): by virtue of translatory motion of the molecule Rotational energy (Erot) : by virtue of rotation of a molecule about its centre of gravity Vibrational energy (Evib): by virtue of periodic diaplacement of the atoms of a molecule about its equilibrium position Electronic energy(Eelec) : due to the different electronic arrangements in the molecule EV0000318 · EV0000319 · EV0000320
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Born-Oppenheimer approximation Total energy of a molecule is the sum of translational, rotational, vibrational and electronic energies, i.e. E = Etrans + Erot + Evib + Eelec Translational energy is negligibly small. Hence Born-Oppenheimer approximation can be written as E = Erot + Evib + Eelec Electronic, vibrational and rotational energies of a molecule are completely independent of each other Because the electronic energy levels are much farther apart than the vibrational energy levels which are again much farther apart than the rotational energy levels Can be written as Erot << Evib << Eelec EV0000321 · EV0000322 · EV0000323

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Molecular Spectroscopy Energy level diagram for a diatomic molecule [Image text, OCR] Electronic transition Excitedelectronic (in optical state. or uv) dissoclation energy Groundstate Vibrational Rotational transition transition (in infrared) (inmicrowave) Internuclearseparation Source:http://hyperphysics.phy-astr.gsu.edu/hbase/molecule/molec.html EV0000324 · EV0000325 · EV0000326
6. Source slide page 6 (extracted text): THANK YOU Department of Science and Humanities EV0000327 · EV0000328 · EV0000329

SHORT-NOTE RECAP

Source deck preserved: Born Oppenheimer approximation. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000312

17 Source slide deck: Experiential learning Based on Phase Diagram

This entry preserves the complete slide deck titled Experiential learning Based on Phase Diagram as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Chemistry, Unit_01, to avoid an unsupported match. EV0000663 · EV0000664

Graphs and diagrams

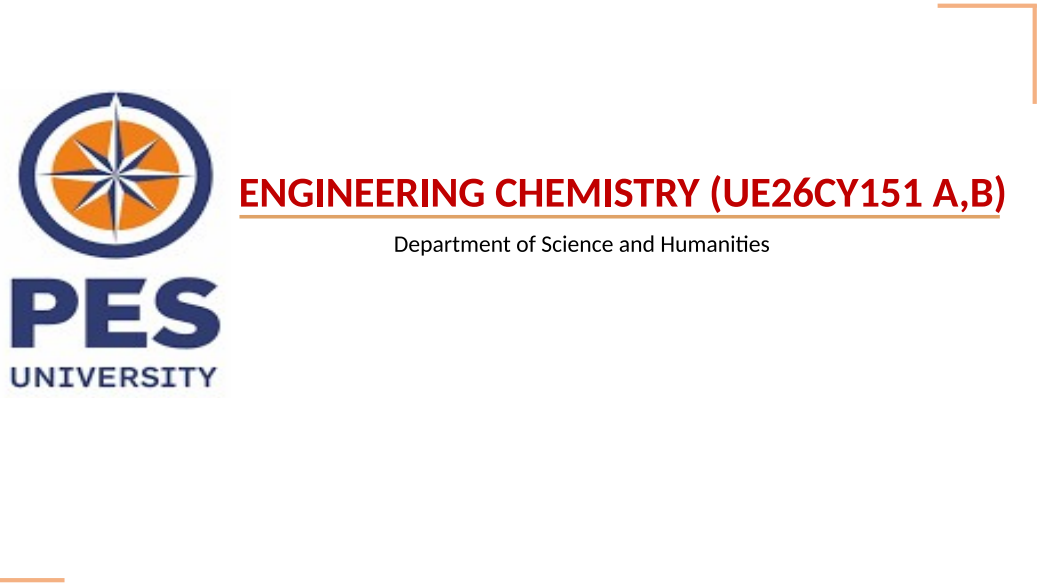

Figure G000136. Experiential learning Based on Phase Diagram · source slide page 1.

EV0000664

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

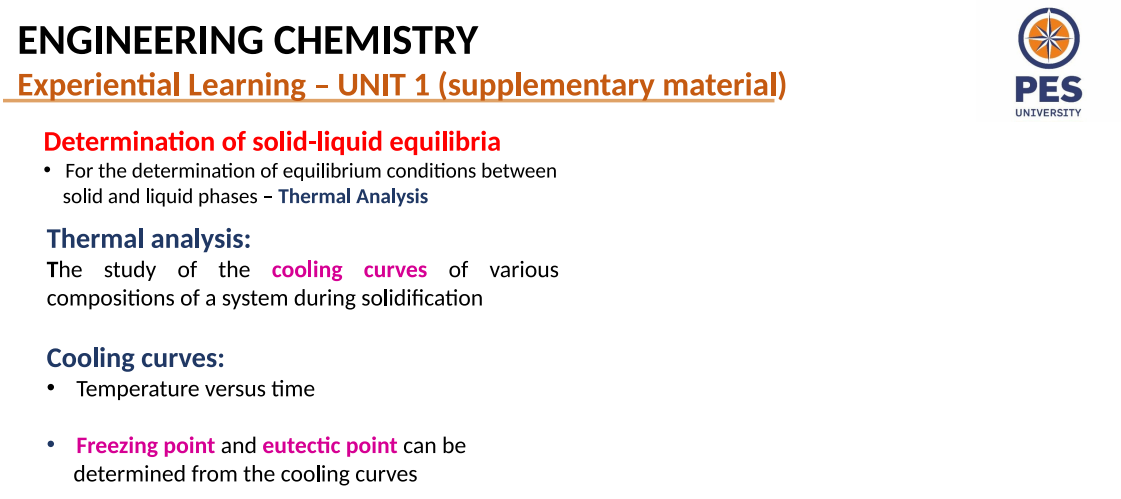

Figure G000137. Experiential learning Based on Phase Diagram · source slide page 2.

EV0000667

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

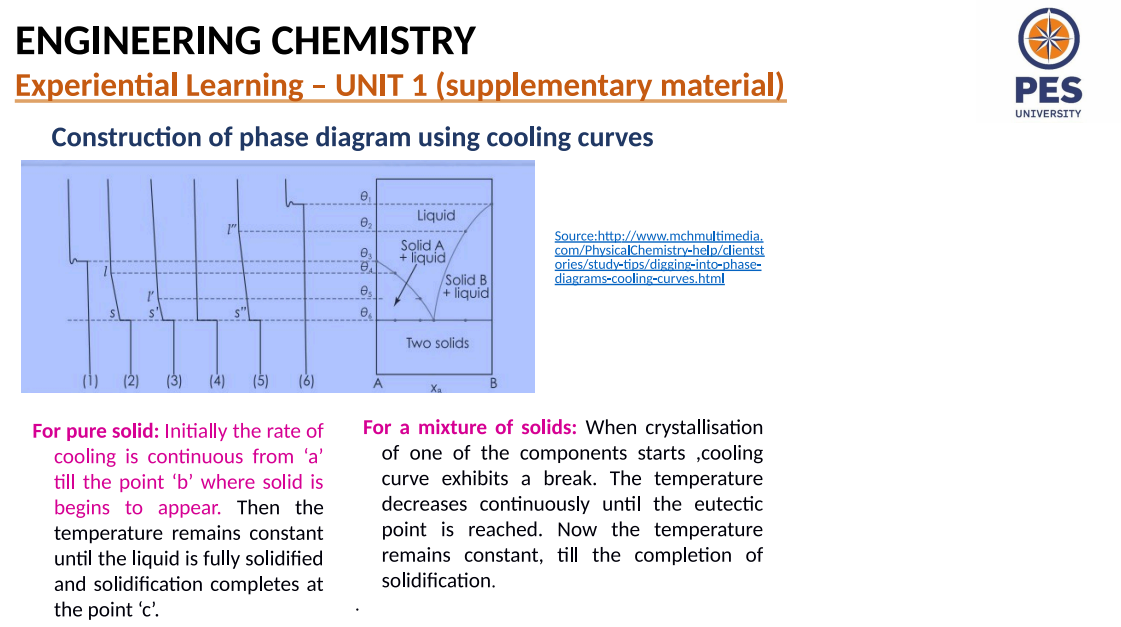

Figure G000138. Experiential learning Based on Phase Diagram · source slide page 3.

EV0000670

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000139. Experiential learning Based on Phase Diagram · source slide page 4.

EV0000673

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000663 · EV0000664 · EV0000665
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Experiential Learning – UNIT 1 (supplementary material) Determination of solid-liquid equilibria For the determination of equilibrium conditions between solid and liquid phases – Thermal Analysis Thermal analysis: The study of the cooling curves of various compositions of a system during solidification Cooling curves: Temperature versus time Freezing point and eutectic point can be determined from the cooling curves EV0000666 · EV0000667 · EV0000668

3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Experiential Learning – UNIT 1 (supplementary material) Construction of phase diagram using cooling curves [Image text, OCR] ϑ2 Liquid SolidA ϑ3 +liquid SolidB ϑ +liquid Twosolids 1) (2) (3) (4) (5) (6) XB B Figure6.17 Useofthermalanalysis.Demonstrationofhowthermalanalysiscanbeusedto determineaphasediagram.Coolingcurves(1)and(6)representbehaviorforpure AandB,respectively. PhysicalChemistry(e-book);Laidler,Meiser,Sanctuary|www.mchmultimedia.com Source:http://www.mchmultimedia.com/PhysicalChemistry-help/clientstories/study-tips/digging-into-phase-diagrams-cooling-curves.html For a mixture of solids: When crystallisation of one of the components starts ,cooling curve exhibits a break. The temperature decreases continuously until the eutectic point is reached. Now the temperature remains constant, till the completion of solidification. . For pure solid: Initially the rate of cooling is continuous from 'a' till the point 'b' where solid is begins to appear. Then the temperature remains constant until the liquid is fully solidified and solidification completes at the point 'c'. EV0000669 · EV0000670 · EV0000671
4. Source slide page 4 (extracted text): [Image text, OCR] PES UNIVERSITY THANK YOU Department of Science and Humanities EV0000672 · EV0000673 · EV0000674

SHORT-NOTE RECAP

Source deck preserved: Experiential learning Based on Phase Diagram. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000663

18 Historical question-source pages · unit mapping candidates

This section preserves historical paper/question-bank pages whose extracted terms matched Engineering Chemistry, Unit_01 (Molecular spectroscopy, phase equilibria, and computational chemistry) above the dataset's candidate threshold. The association has not been manually confirmed, source text/OCR may be incomplete, and the pages are not rewritten or presented as solved questions. EV0000675 · EV0000676

Important points and why

1. Archive page 12 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.8333; source text/OCR may be partial. ENGINEERING CHEMISTRY Module I- COMPUTATIONAL CHEMISTRY Computational methods: There are two main methods depending on the starting point theory: Classical method: Quantum chemistry method : • These use Newton mechanics to model molecular system. • Atom – Sphere, Bonds – Springs and Electrons are ignored. • Total potential energy of a molecule is given by sum of all the energies of attractive and repulsive forces between atom in structure. ☒ These use quantum mechanics to model the molecular system. ☒ It is based on arrangement of electrons in molecules and interaction of these electrons with electron and nuclei of another molecule. ☒ We consider that electron and all other particle exhibits both particle and wave nature. i. Molecular mechanics ii. Molecular dynamics i. Semi empirical methods. ii. Ab initio methods iii. Density functional theory Ref. Book: Introduction to computational chemistry by Frank Jensen, Wiley publications Classical Method Quantum Chem Method EV0000675
2. Archive page 2 from 10__1vFS6E1bKV.pdf; unit match candidate score 1.0; source text/OCR may be partial. (4.0 Marks) (2.0 Marks) (5.0 Marks) 1.d. For HBr molecule: (4M) (i) The rotational spectrum shows the first line at 17.19 cm . Calculate the bond length of the HBr molecule (use rigid rotor model). (ii) The vibrational spectrum shows fundamental and the first overtone at 2559.08 cm and 5027.54 cm respectively. Calculate the anharmonicity constant for the molecule (use the anharmonic oscillator model). (1 amu = 1.66 x 10 kg, h = 6.627 x 10 J s, c = 3 x 10 cm s , N = 6.023 x 10 mol , gram molar mass of H = 1.0 and Br = 79.9). 1.e. Give the two step synthesis of alpha Aluminium oxide in solution combustion method (2M) 2.a. i.In the given phase diagram of water; in fusion curve OC, identify the number of phase, variance and reason out the negative slop with variation of melting point of ice with pressure. (3M) ii. In the following phase equilibrium, identify the number of phases and components (2M) a. Ammonium chloride undergoes thermal decomposition NH4Cl(s) ⇌ NH3 (g) + HCl (g) (closed system) b. CaCO (s) ↔CaO(s) + CO (g) (Closed System) –1 –1 –1 –27 –34 10 –1 A 23 –1 3 2 EV0000676
3. Archive page 2 from 11__1zfeA5C-J7.pdf; unit match candidate score 0.6667; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (6.0 Marks) 1.d. What is the reducing agent used during the' synthesis of α - Al O nano metal oxide by solution combustion method' experiment? Why should the reaction be carried out in open containers with minimum quantity of redox mixtures in the lab? 2.a. (i) What is the maximum % of carbon taken along the composition axis in the phase diagram of two component Fe-C system? Why? (2marks) (ii) Why is reduced/condensed phase rule used for two component Pb-Ag and Fe-C systems? (iii) Which are the three phases in equilibrium at the eutectic point in the two component Pb - Ag system? What is the degree of freedom of the system at this point? 2.b. (i) For the cell represented below, calculate the E . Pt/H (10 atm)/H (0.3M) / H (3atm)/Pt (2 marks) (ii) A glass electrode is coupled with saturated calomel electrode to measure unknown pH. The cell potentials measured are 0.215V and 0.385V in contact with a solution of pH = 7 and with solution of unknown pH respectively. Calculate the pH of unknown solution. Given E = 0.244V 2.c. (i) For the cell represented below- Ag/AgCl/Cl (0.1M) // Fe (0.02M), Fe (0.1)/ Pt (i) Write the overall cell reaction. (2 marks) (ii)Calculate E and E at 298 K (4 marks) (Given: The standard reduction potential of E Fe /Fe = 0.77 V and that of E = 0.222 V, R = 8.314 J/K/mol, F =96500 C/mol) 2 3 cell 2 + 2 SCE -1 2+ 3+ oCell Cell 0 3+ 2+ 0Ag/AgCl EV0000677

4. Archive page 1 from 1__1rBrYYqw3s.pdf; unit match candidate score 1.0; source text/OCR may be partial. (6.0 Marks) (6.0 Marks) (4.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan - June 2024 UE23CY151B - Engineering Chemistry Total Marks : 100.0 1.a. For a rigid rotor diatomic molecule: (i) Draw the energy level diagram up to J=5. (ii) Draw the rotational spectrum. (iii) Calculate the moment of inertia and the distance between two spectral line in the spectra if B = 10.56 cm . (Given : c = 3 x 10 m/s, π =3.14, h = 6.625x10 Js respectively) 1.b. (i) The force constant of HF molecule showing simple harmonic oscillation at 970 N/m. Calculate the oscillation frequency (cm) and zero point energy. (ii) Why does the spectrum shows only one line in simple harmonic oscillator? (Given: c =3 x 10 m/sec, π=3.14, N = 6.02 x 10 , 1 amu =1.66x10 kg, molar mass of H =1.008 amu, F = 18.99 amu). 1.c. Derive phase rule for a system of C componens and P phases which are in equilibrium with each other. -1 8 -34 -1 8 23 -27 EV0000678
5. Archive page 1 from 2__11pj9jsL0V.pdf; unit match candidate score 1.25; source text/OCR may be partial. (6.0 Marks) (6.0 Marks) (4.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan - June 2024 UE23CY151B - Engineering Chemistry Total Marks : 100.0 1.a. For a rigid rotor hetero diatomic molecule (AX), The first line in the pure rotational spectrum corresponds to 21.18 cm . Derive an expression for the moment of inertia and Calculate the moment of inertia and bond length of the AX molecule. (Given: Atomic masses of A = 1.008 amu and X = 35.45 amu, Avagadro's number = 6.023 x 10 , Speed of light = 3 x 10 cm/s, π = 3.14, Planck's constant = 6.627 x 10 Js, 1 amu = 1.66 x 10 kg). 1.b. i). The radiation absorbed by CO during a vibrational transition is 2168 cm . Calculate ground state vibrational energy and force constant. (Simple harmonic oscillator model Given c = 3×10 cm/s, Planck's constant = 6.625×10 Js, N = 6.02×10 ,1 amu = 1.66 ×10 kg, atomic masses: C=12, O=15.999). ii). Write a note on the failure of a simple harmonic oscillator model. 1.c. Justify the Franck-Condon principle by drawing the vibrational-electronic transitions in the following cases and draw the spectra arising from them: 1. When the internuclear distances in the ground and excited states are the same. 2. When the internuclear distance in the excited state is much larger than that in the ground state. -1 23 10 - 34 -27 -1 10 -34 23 -27 EV0000679
6. Archive page 1 from 3__1MwY1CTl_j.pdf; unit match candidate score 1.0; source text/OCR may be partial. 3/25/25, 3:51 PM ESA - Dec 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 1/7 (5.0 Marks) (5.0 Marks) (4.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Dec 2024 UE23CY151B - Engineering Chemistry Total Marks : 100.0 1.a. Derive the expression for the moment of inertia for a diatomic rigid rotor 1.b. Briefly describe the Born-Oppenheimer aproximation and draw the ground and excited electronic states of a diatomic molecule showing the rotational and vibrational energy levels (using anharmonic oscillator model). 1.c. The force constant of CO is 1840 Nm . Calculate the oscillation frequency and wave number in cm . (1 amu = 1.66 x 10 kg, c = 3 x 10 ms , Gram molar mass of C = 12.000 and O = 15.9994) -1 -1 -27 8 -1 EV0000680

7. Archive page 2 from 3__1MwY1CTl_j.pdf; unit match candidate score 0.5833; source text/OCR may be partial. 3/25/25, 3:51 PM ESA - Dec 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 2/7 (8.0 Marks) (3.0 Marks) (5.0 Marks) 1.d. (i) Draw the phase diagram of water, label the areas and the phase boundaries, and calculate degrees of freedom in the areas, phase boundaries, and the triple point. (4M) (ii) With the help of Clausius-Clapeyron equation, discuss why the melting point curve of the water phase diagram has a negative slope (2M) (iii) What is reduced phase rule and under which situation it is used? (2M) 1.e. State the Beer-Lambert law. Why is ammonia added during colorimetric estimation of copper? (2M + 1M) 2.a. Derive the Nernst equation for a single electrode undergoing the following electrode reaction: M + ne → M n+ EV0000681

8. Archive page 1 from 4__1eKMFaJ2lx.pdf; unit match candidate score 1.0; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 1/7 (6.0 Marks) (6.0 Marks) (4.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan - June 2024 UE23CY151B - Engineering Chemistry Total Marks : 100.0 1.a. For a rigid rotor diatomic molecule: (i) Draw the energy level diagram up to J=5. (ii) Draw the rotational spectrum. (iii) Calculate the moment of inertia and the distance between two spectral line in the spectra if B = 10.56 cm . (Given : c = 3 x 10 m/s, π =3.14, h = 6.625x10 Js respectively) 1.b. (i) The force constant of HF molecule showing simple harmonic oscillation at 970 N/m. Calculate the oscillation frequency (cm) and zero point energy. (ii) Why does the spectrum shows only one line in simple harmonic oscillator? (Given: c =3 x 10 m/sec, π=3.14, N = 6.02 x 10 , 1 amu =1.66x10 kg, molar mass of H =1.008 amu, F = 18.99 amu). 1.c. Derive phase rule for a system of C componens and P phases which are in equilibrium with each other. -1 8 -34 -1 8 23 -27 EV0000682

9. Archive page 1 from 6__1H6QH_PVNf.pdf; unit match candidate score 1.0; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 1/7 (6.0 Marks) (6.0 Marks) (4.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan - June 2024 UE23CY151B - Engineering Chemistry Total Marks : 100.0 1.a. For a rigid rotor diatomic molecule: (i) Draw the energy level diagram up to J=5. (ii) Draw the rotational spectrum. (iii) Calculate the moment of inertia and the distance between two spectral line in the spectra if B = 10.56 cm . (Given : c = 3 x 10 m/s, π =3.14, h = 6.625x10 Js respectively) 1.b. (i) The force constant of HF molecule showing simple harmonic oscillation at 970 N/m. Calculate the oscillation frequency (cm) and zero point energy. (ii) Why does the spectrum shows only one line in simple harmonic oscillator? (Given: c =3 x 10 m/sec, π=3.14, N = 6.02 x 10 , 1 amu =1.66x10 kg, molar mass of H =1.008 amu, F = 18.99 amu). 1.c. Derive phase rule for a system of C componens and P phases which are in equilibrium with each other. -1 8 -34 -1 8 23 -27 EV0000683

10. Archive page 1 from 7__16pEjcSTVO.pdf; unit match candidate score 1.0; source text/OCR may be partial. (6.0 Marks) (6.0 Marks) (6.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan - May 2024 UE22CY151B - Engineering Chemistry Total Marks : 100.0 1.a. (i) State Born Openhemier approximation? (ii) What does an appearance of continnum in an electronic spectra lead us to conclude. 1.b. (i) Derive an expression for the moment of inertia for a diatomic molecule (rigid rotar model). (ii) Why is carbon dioxide infrared active but microwave inactive? 1.c. (I) For a HBr molecule undergoing simple harmonic motion the vibrational spectrum shows a fundamental frequency at 2649 cm . determine force constant of HBr Given the reduced mass of HBr is 1.639x10−27 Kg and c= 3x10 m/s. (ii) What are three dimensional nanomaterials? Give one example. -1 8 EV0000684
11. Archive page 2 from 7__16pEjcSTVO.pdf; unit match candidate score 0.5; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (6.0 Marks) 1.d. Why should the synthesis of α Al O be done in open container with a minimum quantity of reactants and in a well ventilated set up? 2.a. (i) Give Gibbs Phase rule and explain the tems involved. (ii) At triple point of one component water system, what are the phases in equilibrium and the the degree of freedom. 2.b. (i) For the cell represented below, calculate the E Au/Au (0.05M) // Au (0.12 M)/ Au (ii) When a glass electrode is combined with saturated calomel electrode and immersed in a solution of pH 7 the E recorded was 0.215V, when the same electrode assembly was immersed in a solution of unknown pH the E was recorded as 0.385V. Using the data determine the pH of unknown solution. Given E is 0.24V. 2.c. For the below mentioned cell representation- Fe/Fe (0.05M) //Ag (0.1M)/Ag Given : E = −0.44V ; E = 0.8 V (i) write the anodic, cathodic and net cell reaction. (ii) Calculate the E and E 2 3 cell 3+ 3+ cell cell SCE 2+ + 0Fe / Fe +2 0Ag /Ag + 0cell cell EV0000685
12. Archive page 2 from 9__14fBQb1-hv.pdf; unit match candidate score 0.5; source text/OCR may be partial. 11/8/24, 1:02 PM ESA - DEC 2023 - UE22CY151B/UE22CY151A (set- 1) about:blank 2/9 (4.0 Marks) (2.0 Marks) (5.0 Marks) 1.d. The first line in the pure rotational spectrum of HCl appears at 21.18 cm-1 . Calculate the bond length of the molecule. Given atomic masses of H and Cl are 1.008 amu and 35.45 amu respectively. (1 amu = 1.66 x 10-27 kg, c = 3 x 108 ms-1, h = 6.6x 10-34 Js) 1.e. Mention the precautions to be taken while synthesis of alfa-Al2O3 nano metal oxide via compustion method 2.a. The study of phase diagram of a 2-component system has formed the basis for Pattinson's process for the desilverisation of argentiferous lead, illustrate with the phase diagram of lead-silver systerm. EV0000686

13. Archive page 1 from Chem-RR[2025]__1xN30tqzNj.pdf; unit match candidate score 1.3333; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 1) about:blank 1/7 (6.0 Marks) (7.0 Marks) (6.0 Marks) PES University, Bengaluru (Established under Karnataka Act 16 of 2013) END SEMESTER ASSESSMENT (ESA) - Jan-Feb UE24CY151A - Engineering Chemistry Total Marks : 100.0 1.a. i) Why HI molecule is IR active but H and I are IR inactive? (1M) ii) The first spectral line in the rotational spectrum of HI molecule appears at 13.022 cm . Calculate the bond length. (3M) (Given: c = 3 x 10 m/s, 1amu=1.66 x 10 kg, gram molar mass of H =1.0 amu and I=126.9 amu, π=3.14, Planck's constant = 6.627x10 Js) iii) Draw an energy level diagram of ground and excited electronic energy states of a diatomic molecule and show vibrational and rotational energy levels within them. (2M) 1.b. i) Using harmonic oscillator model calculate zero point vibrational energy of NO molecule given that the force constant of NO is 1530 N/m. (4M) ii) Draw the vibrational energy level diagram for the above model and draw the corresponding spectra obtained from the transition. (2M) iii) Calculate the energy of v=3 and v=4 level. (1M) (Given: c = 3x10 m/s, 1amu=1.66x10 kg, gram molar mass of N =14.0 amu and O=16 amu, π=3.14, Planck's constant = 6.627x10 Js) 1.c. i) Draw a neat labeled phase diagram of water system (mention the areas, lines, triple point, critical point, pressure, temperature) and calculate the degree of freedom along the freezing point curve. (4M) ii) What is condensed phase rule and why is it applied to the two component lead-silver system? (2M) 2 2 -1 8 -27 -34 8 -27 -34 EV0000687

SHORT-NOTE RECAP

13 historical archive pages are included as question-source references. Inspect the original paper before relying on OCR or unit placement; no answers are newly authored here.

EV0000675

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Interaction of radiation with matter transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/001_Interaction_of_radiation_with_matter__1eb05f6c3f230eec.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Interaction of radiation with matter transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/001_Interaction_of_radiation_with_matter__1eb05f6c3f230eec.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Interaction of radiation with matter transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/001_Interaction_of_radiation_with_matter__1eb05f6c3f230eec.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Interaction of radiation with matter transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/001_Interaction_of_radiation_with_matter__1eb05f6c3f230eec.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electromagnetic spectrum transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/002_Electromagnetic_spectrum__d84acfbd19ed2af4.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electromagnetic spectrum transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/002_Electromagnetic_spectrum__d84acfbd19ed2af4.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electromagnetic spectrum transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/002_Electromagnetic_spectrum__d84acfbd19ed2af4.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9997; only the JSON segmentation supplies the displayed timestamps.
Electromagnetic spectrum transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/002_Electromagnetic_spectrum__d84acfbd19ed2af4.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Rotational spectroscopy transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/004_Rotational_spectroscopy__cda73f2030eb1a62.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Rotational spectroscopy transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/004_Rotational_spectroscopy__cda73f2030eb1a62.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Rotational spectroscopy transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/004_Rotational_spectroscopy__cda73f2030eb1a62.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9997; only the JSON segmentation supplies the displayed timestamps.
Rotational spectroscopy transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/004_Rotational_spectroscopy__cda73f2030eb1a62.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(harmonic oscillator) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/005_Vibrational_spectroscopy_harmonic_oscillator__bd1d3b496879b403.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(harmonic oscillator) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/005_Vibrational_spectroscopy_harmonic_oscillator__bd1d3b496879b403.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(harmonic oscillator) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/005_Vibrational_spectroscopy_harmonic_oscillator__bd1d3b496879b403.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Vibrational spectroscopy(harmonic oscillator) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/005_Vibrational_spectroscopy_harmonic_oscillator__bd1d3b496879b403.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(anharmonic oscillator) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/006_Vibrational_spectroscopy_anharmonic_oscillator__4fe4d43ab2b66d24.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(anharmonic oscillator) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/006_Vibrational_spectroscopy_anharmonic_oscillator__4fe4d43ab2b66d24.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(anharmonic oscillator) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/006_Vibrational_spectroscopy_anharmonic_oscillator__4fe4d43ab2b66d24.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(anharmonic oscillator) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/006_Vibrational_spectroscopy_anharmonic_oscillator__4fe4d43ab2b66d24.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy - Franck-Condon principle (Part II) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_-_Franck-Condon_principle_Part_II__70e8e6f497126f7d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Electronic spectroscopy - Franck-Condon principle (Part II) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_-_Franck-Condon_principle_Part_II__70e8e6f497126f7d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy - Franck-Condon principle (Part II) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_-_Franck-Condon_principle_Part_II__70e8e6f497126f7d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9987; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy - Franck-Condon principle (Part II) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_-_Franck-Condon_principle_Part_II__70e8e6f497126f7d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy(Progression) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_Progression__6dd27636f794caa1.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy(Progression) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_Progression__6dd27636f794caa1.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electronic spectroscopy(Progression) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_Progression__6dd27636f794caa1.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Electronic spectroscopy(Progression) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/007_Electronic_spectroscopy_Progression__6dd27636f794caa1.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__a7da1e88f96b163b.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__a7da1e88f96b163b.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__a7da1e88f96b163b.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__a7da1e88f96b163b.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part II transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_II__2a5eb65b0585fe28.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part II transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_II__2a5eb65b0585fe28.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part II transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_II__2a5eb65b0585fe28.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9967; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part II transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_II__2a5eb65b0585fe28.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_I__f611c716baed093a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_I__f611c716baed093a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_I__f611c716baed093a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9976; only the JSON segmentation supplies the displayed timestamps.
Derivation of Gibb's phase rule, Phase diagram of one-component system - water - Part I transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water_-_Part_I__f611c716baed093a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Reduced phase rule, phase diagram of two-component system - Pb-Ag system transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__8e25eaa5432159a1.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Reduced phase rule, phase diagram of two-component system - Pb-Ag system transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__8e25eaa5432159a1.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Reduced phase rule, phase diagram of two-component system - Pb-Ag system transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__8e25eaa5432159a1.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9987; only the JSON segmentation supplies the displayed timestamps.
Reduced phase rule, phase diagram of two-component system - Pb-Ag system transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__8e25eaa5432159a1.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part I transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__0671dbe2a6d868d4.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part I transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__0671dbe2a6d868d4.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part I transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__0671dbe2a6d868d4.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9907; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part I transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__0671dbe2a6d868d4.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part III transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__5a0b9f520a3c63b3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part III transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__5a0b9f520a3c63b3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part III transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__5a0b9f520a3c63b3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9860; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part III transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__5a0b9f520a3c63b3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part II transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__74b8d88c6bf2ef3c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part II transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__74b8d88c6bf2ef3c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part II transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__74b8d88c6bf2ef3c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9940; only the JSON segmentation supplies the displayed timestamps.
Computational chemistry: Scope, computational methods, electronic structure determination, AI-enhanced computational chemistry - Part II transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__74b8d88c6bf2ef3c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Interaction of electromagnetic radiation with matter	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Interaction of electromagnetic radiation with matter	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Interaction of electromagnetic radiation with matter (converted PDF; original hash 4c9c45c71266e6a0e44f1edeeaa2f0ff7a2976a7d76a178093c7c5b191877db5)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Electromagnetic spectrum	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Electromagnetic spectrum	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Electromagnetic spectrum (converted PDF; original hash fae8af7f3abd0f8e1640189a707e1813a92e86f60f6178090157f5495ecbd9ed)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Born Oppenheimer approximation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Born Oppenheimer approximation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Born Oppenheimer approximation (converted PDF; original hash 01f7b22262b35a5b19fe37907dd488fade2c6b0d34195e39f50256e30e05b682)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Microwave spectroscopy diatomic rigid rotor model and the rotational spectrum (converted PDF; original hash fe26fc505240d4a63cbfd4a4728a57f0fe4e56fda0e7888be1d7ed342a585ed1)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: IR spectroscopy diatomic harmonic oscillator model	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: IR spectroscopy diatomic harmonic oscillator model	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for IR spectroscopy diatomic harmonic oscillator model (converted PDF; original hash 449aef5af8ed12ee270a3dd50b67af77db157bd304cf802b9d0a7db1012c76f8)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: IR spectroscopy anharmonic oscillator model	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: IR spectroscopy anharmonic oscillator model	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for IR spectroscopy anharmonic oscillator model (converted PDF; original hash d8de97dd12513d33d5f25e51c96df34737da95e7c8da27403471afd7c29b952b)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Electronic spectroscopy vibrational coarse structure progressions Franck Condon principle (converted PDF; original hash ab6e8532f77273da103d72afdcbb94b3f20bda0bf1afe42fc4d49febb73bf784)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Phase Component Degrees of freedom Statement of Gibb s phase rule chemical potential (converted PDF; original hash e528631d79770489e2f8b6b485441c06503be6f69b0dc9fa6db4c18735f96c20)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Derivation of Gibb s phase rule Phase diagram of one component system water	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Derivation of Gibb s phase rule Phase diagram of one component system water	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Derivation of Gibb s phase rule Phase diagram of one component system water (converted PDF; original hash 72fc1f0d9f3152dc670b650a3653f9702ffcf97fd1d13ef290d1f8547dd43d5e)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Reduced phase rule phase diagram of two component system Pb Ag system	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Reduced phase rule phase diagram of two component system Pb Ag system	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Reduced phase rule phase diagram of two component system Pb Ag system (converted PDF; original hash a52eedc2c4ce354e84cf7c3619435ef387196e6a3aa19475ccc063cd5cb89bec)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Computational chemistry Scope computational methods electronic structure determination AI enhanced computation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Computational chemistry Scope computational methods electronic structure determination AI enhanced computation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Computational chemistry Scope computational methods electronic structure determination AI enhanced computation (converted PDF; original hash bbd5a5fef818c52e5f7de4b010ff65e9086c2fe3947500de1f80f4fbc2e1b37a)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Experiential learning Based on Phase Diagram	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/012_Experiential_learning_Based_on_Phase_Diagram__42cc75d52ba7f0e4.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Experiential learning Based on Phase Diagram	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/012_Experiential_learning_Based_on_Phase_Diagram__42cc75d52ba7f0e4.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Experiential learning Based on Phase Diagram (converted PDF; original hash 4f25bcdae7b386b7085c93346836685287dafdaced1ee62de23f9a750f12b5bc)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/012_Experiential_learning_Based_on_Phase_Diagram__42cc75d52ba7f0e4.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Extracted historical pages: Chem_Extra_Topics__1-HfOeZNwo.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Extra/Chem_Extra_Topics__1-HfOeZNwo.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 10__1vFS6E1bKV.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/10__1vFS6E1bKV.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 11__1zfeA5C-J7.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/11__1zfeA5C-J7.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 1__1rBrYYqw3s.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/1__1rBrYYqw3s.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 2__11pj9jsL0V.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/2__11pj9jsL0V.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Source	Path	Use
Extracted historical pages: 3__1MwY1CTl_j.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/3__1MwY1CTl_j.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 4__1eKMFaJ2lx.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/4__1eKMFaJ2lx.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 6__1H6QH_PVNf.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/6__1H6QH_PVNf.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 7__16pEjcSTVO.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/7__16pEjcSTVO.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 9__14fBQb1-hv.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/9__14fBQb1-hv.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: Chem-RR[2025]__1xN30tqzNj.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2025/Chem-RR[2025]__1xN30tqzNj.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
EV0000001	SRC00001/catalogue	covered	Unit title and index copied from the official units.json catalogue; human content review is pending.
EV0000002	SRC00002/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000003	SRC00003/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000004	SRC00004/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000005	SRC00005/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000006	SRC00002/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000007	SRC00003/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000008	SRC00004/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000009	SRC00005/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000010	SRC00002/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000011	SRC00003/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000012	SRC00004/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000013	SRC00005/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000014	SRC00002/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0083. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000015	SRC00003/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0083. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000016	SRC00004/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0083. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000017	SRC00005/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0083. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000018	SRC00002/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0084 to s0103. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000019	SRC00003/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0084 to s0103. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000020	SRC00004/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0084 to s0103. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000021	SRC00005/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0084 to s0103. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000022	SRC00006/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000023	SRC00007/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000024	SRC00008/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000025	SRC00009/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000026	SRC00006/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000027	SRC00007/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000028	SRC00008/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000029	SRC00009/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000030	SRC00006/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000031	SRC00007/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000032	SRC00008/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000033	SRC00009/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000034	SRC00006/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000035	SRC00007/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000036	SRC00008/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000037	SRC00009/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000038	SRC00006/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0102. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000039	SRC00007/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0102. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000040	SRC00008/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0102. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000041	SRC00009/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0102. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000042	SRC00006/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0103 to s0118. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000043	SRC00007/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0103 to s0118. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000044	SRC00008/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0103 to s0118. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000045	SRC00009/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0103 to s0118. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000046	SRC00010/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000047	SRC00011/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000048	SRC00012/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000049	SRC00013/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000050	SRC00010/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000051	SRC00011/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000052	SRC00012/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000053	SRC00013/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000054	SRC00010/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000055	SRC00011/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000056	SRC00012/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000057	SRC00013/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000058	SRC00010/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000059	SRC00011/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000060	SRC00012/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0082. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000061	SRC00013/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000062	SRC00010/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0090. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000063	SRC00011/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0090. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000064	SRC00012/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0090. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000065	SRC00013/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0090. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000066	SRC00014/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000067	SRC00015/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000068	SRC00016/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000069	SRC00017/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000070	SRC00014/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000071	SRC00015/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000072	SRC00016/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000073	SRC00017/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000074	SRC00014/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000075	SRC00015/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000076	SRC00016/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000077	SRC00017/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000078	SRC00014/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000079	SRC00015/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000080	SRC00016/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000081	SRC00017/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000082	SRC00014/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0100. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000083	SRC00015/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0100. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000084	SRC00016/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0100. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000085	SRC00017/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0100. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000086	SRC00014/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0109. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000087	SRC00015/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0109. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000088	SRC00016/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0109. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000089	SRC00017/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0109. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000090	SRC00018/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000091	SRC00019/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000092	SRC00020/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000093	SRC00021/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000094	SRC00018/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000095	SRC00019/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000096	SRC00020/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000097	SRC00021/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000098	SRC00018/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000099	SRC00019/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000100	SRC00020/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000101	SRC00021/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000102	SRC00018/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000103	SRC00019/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000104	SRC00020/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000105	SRC00021/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001106	SRC00018/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0089. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001107	SRC00019/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0089. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001108	SRC00020/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0089. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001109	SRC00021/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0089. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001110	SRC00022/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001111	SRC00023/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001112	SRC00024/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000113	SRC00025/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000114	SRC00022/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000115	SRC00023/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000116	SRC00024/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000117	SRC00025/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000118	SRC00022/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000119	SRC00023/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000120	SRC00024/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000121	SRC00025/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000122	SRC00022/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0065 to s0075. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000123	SRC00023/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0065 to s0075. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000124	SRC00024/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0065 to s0075. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000125	SRC00025/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0065 to s0075. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000126	SRC00026/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000127	SRC00027/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000128	SRC00028/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000129	SRC00029/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000130	SRC00026/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000131	SRC00027/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000132	SRC00028/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000133	SRC00029/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000134	SRC00026/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000135	SRC00027/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000136	SRC00028/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000137	SRC00029/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000138	SRC00026/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000139	SRC00027/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000140	SRC00028/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0084. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000141	SRC00029/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000142	SRC00026/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0085 to s0101. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000143	SRC00027/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0085 to s0101. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000144	SRC00028/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0085 to s0101. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000145	SRC00029/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0085 to s0101. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000146	SRC00030/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000147	SRC00031/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000148	SRC00032/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000149	SRC00033/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000150	SRC00030/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000151	SRC00031/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000152	SRC00032/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000153	SRC00033/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000154	SRC00030/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000155	SRC00031/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000156	SRC00032/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000157	SRC00033/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000158	SRC00030/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000159	SRC00031/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000160	SRC00032/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000161	SRC00033/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000162	SRC00030/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0100. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000163	SRC00031/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0100. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000164	SRC00032/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0100. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000165	SRC00033/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0100. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000166	SRC00034/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000167	SRC00035/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000168	SRC00036/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9967; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000169	SRC00037/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000170	SRC00034/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000171	SRC00035/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000172	SRC00036/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 0.9967; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000173	SRC00037/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000174	SRC00034/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000175	SRC00035/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000176	SRC00036/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 0.9967; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000177	SRC00037/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000178	SRC00034/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000179	SRC00035/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000180	SRC00036/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0061. Sidecar variants compare at similarity 0.9967; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000181	SRC00037/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000182	SRC00038/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000183	SRC00039/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000184	SRC00040/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000185	SRC00041/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000186	SRC00038/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000187	SRC00039/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000188	SRC00040/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000189	SRC00041/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000190	SRC00038/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000191	SRC00039/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000192	SRC00040/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000193	SRC00041/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000194	SRC00038/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000195	SRC00039/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000196	SRC00040/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 97	SRC00041/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 98	SRC00038/t00120 0-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 99	SRC00039/t00120 0-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 00	SRC00040/t00120 0-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0084. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 01	SRC00041/t00120 0-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 02	SRC00042/t00000 0-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 03	SRC00043/t00000 0-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000204	SRC00044/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000205	SRC00045/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000206	SRC00042/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000207	SRC00043/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000208	SRC00044/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000209	SRC00045/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000210	SRC00042/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00002 11	SRC00043/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 12	SRC00044/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 13	SRC00045/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 14	SRC00042/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 15	SRC00043/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 16	SRC00044/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 17	SRC00045/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000218	SRC00042/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0102. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000219	SRC00043/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0102. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000220	SRC00044/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0102. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000221	SRC00045/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0102. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000222	SRC00042/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0103 to s0121. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000223	SRC00043/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0103 to s0121. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000224	SRC00044/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0103 to s0121. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000225	SRC00045/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0103 to s0121. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000226	SRC00042/t001800-002099	covered	Timestamp interval t001800-002099; canonical segment IDs s0122 to s0124. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000227	SRC00043/t001800-002099	covered	Timestamp interval t001800-002099; canonical segment IDs s0122 to s0124. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000228	SRC00044/t001800-002099	covered	Timestamp interval t001800-002099; canonical segment IDs s0122 to s0124. Sidecar variants compare at similarity 0.9987; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000229	SRC00045/t001800-002099	covered	Timestamp interval t001800-002099; canonical segment IDs s0122 to s0124. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000230	SRC00046/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000231	SRC00047/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000232	SRC00048/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9907; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000233	SRC00049/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000234	SRC00046/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0033. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000235	SRC00047/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0033. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000236	SRC00048/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0033. Sidecar variants compare at similarity 0.9907; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000237	SRC00049/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0033. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000238	SRC00050/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000239	SRC00051/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000240	SRC00052/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9860; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000241	SRC00053/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000242	SRC00050/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000243	SRC00051/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000244	SRC00052/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0020. Sidecar variants compare at similarity 0.9860; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000245	SRC00053/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000246	SRC00054/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000247	SRC00055/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000248	SRC00056/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9940; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000249	SRC00057/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000250	SRC00054/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000251	SRC00055/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000252	SRC00056/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 0.9940; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000253	SRC00057/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000254	SRC00054/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000255	SRC00055/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000256	SRC00056/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0042. Sidecar variants compare at similarity 0.9940; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000257	SRC00057/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000258	SRC00058/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000259	SRC00060/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000260	SRC00059/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000261	SRC00058/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000262	SRC00060/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000263	SRC00059/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000264	SRC00058/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000265	SRC00060/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000266	SRC00059/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 67	SRC00058/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 68	SRC00060/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 69	SRC00059/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 70	SRC00058/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 71	SRC00060/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 72	SRC00059/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000273	SRC00058/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000274	SRC00060/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000275	SRC00059/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000276	SRC00058/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000277	SRC00060/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000278	SRC00059/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000279	SRC00058/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000280	SRC00060/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000281	SRC00059/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000282	SRC00058/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000283	SRC00060/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000284	SRC00059/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000285	SRC00058/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000286	SRC00060/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000287	SRC00059/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000288	SRC00058/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000289	SRC00060/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/001_Interaction_of_electromagnetic_radiation_with_matter__9d3205fc29225eeb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000290	SRC00059/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000291	SRC00061/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000292	SRC00063/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000293	SRC00062/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000294	SRC00061/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000295	SRC00063/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000296	SRC00062/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000297	SRC00061/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002 98	SRC00063/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 99	SRC00062/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 00	SRC00061/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 01	SRC00063/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 02	SRC00062/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 03	SRC00061/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 04	SRC00063/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000305	SRC00062/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000306	SRC00061/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000307	SRC00063/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000308	SRC00062/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000309	SRC00061/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000310	SRC00063/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/002_Electromagnetic_spectrum__ef06d432886fbdb5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000311	SRC00062/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000312	SRC00064/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000313	SRC00066/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000314	SRC00065/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000315	SRC00064/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000316	SRC00066/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000317	SRC00065/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000318	SRC00064/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000319	SRC00066/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000320	SRC00065/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000321	SRC00064/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000322	SRC00066/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000323	SRC00065/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000324	SRC00064/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000325	SRC00066/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00003326	SRC00065/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00003327	SRC00064/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003328	SRC00066/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/003_Born_Oppenheimer_approximation__342acd62ba6e1f57.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003329	SRC00065/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00003330	SRC00067/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003331	SRC00069/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003332	SRC00068/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000333	SRC00067/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000334	SRC00069/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000335	SRC00068/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000336	SRC00067/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000337	SRC00069/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000338	SRC00068/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000339	SRC00067/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000340	SRC00069/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000341	SRC00068/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000342	SRC00067/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000343	SRC00069/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000344	SRC00068/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000345	SRC00067/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000346	SRC00069/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000347	SRC00068/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000348	SRC00067/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000349	SRC00069/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000350	SRC00068/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003351	SRC00067/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003352	SRC00069/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003353	SRC00068/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00003354	SRC00067/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003355	SRC00069/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003356	SRC00068/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000357	SRC00067/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000358	SRC00069/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000359	SRC00068/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000360	SRC00067/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000361	SRC00069/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000362	SRC00068/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003363	SRC00067/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003364	SRC00069/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003365	SRC00068/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV00003366	SRC00067/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003367	SRC00069/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/004_Microwave_spectroscopy_-_diatomic_rigid_rotor_model_and_the_rotational_spectrum__416fab5550d2cc51.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003368	SRC00068/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000369	SRC00070/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000370	SRC00072/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000371	SRC00071/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000372	SRC00070/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000373	SRC00072/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000374	SRC00071/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003375	SRC00070/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003376	SRC00072/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003377	SRC00071/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003378	SRC00070/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003379	SRC00072/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003380	SRC00071/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003381	SRC00070/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003382	SRC00072/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003383	SRC00071/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00003384	SRC00070/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003385	SRC00072/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003386	SRC00071/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000387	SRC00070/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000388	SRC00072/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000389	SRC00071/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000390	SRC00070/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000391	SRC00072/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000392	SRC00071/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000393	SRC00070/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000394	SRC00072/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000395	SRC00071/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000396	SRC00070/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000397	SRC00072/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000398	SRC00071/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 99	SRC00070/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 00	SRC00072/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/005_IR_spectroscopy_-_diatomic_harmonic_oscillator_model__044502d9a73710b7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 01	SRC00071/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 02	SRC00073/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 03	SRC00075/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 04	SRC00074/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000405	SRC00073/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000406	SRC00075/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000407	SRC00074/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000408	SRC00073/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000409	SRC00075/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000410	SRC00074/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000411	SRC00073/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000412	SRC00075/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000413	SRC00074/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000414	SRC00073/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000415	SRC00075/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000416	SRC00074/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000417	SRC00073/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000418	SRC00075/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000419	SRC00074/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000420	SRC00073/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000421	SRC00075/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000422	SRC00074/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000423	SRC00073/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000424	SRC00075/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000425	SRC00074/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000426	SRC00073/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000427	SRC00075/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000428	SRC00074/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000429	SRC00073/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000430	SRC00075/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/006_IR_spectroscopy_-_anharmonic_oscillator_model__ac64a226f45a8e06.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000431	SRC00074/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000432	SRC00076/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000433	SRC00078/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000434	SRC00077/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000435	SRC00076/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000436	SRC00078/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000437	SRC00077/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000438	SRC00076/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000439	SRC00078/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000440	SRC00077/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000441	SRC00076/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000442	SRC00078/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000443	SRC00077/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000444	SRC00076/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000445	SRC00078/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000446	SRC00077/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000447	SRC00076/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000448	SRC00078/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000449	SRC00077/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000450	SRC00076/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000451	SRC00078/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000452	SRC00077/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000453	SRC00076/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000454	SRC00078/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000455	SRC00077/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000456	SRC00076/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000457	SRC00078/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000458	SRC00077/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000459	SRC00076/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000460	SRC00078/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000461	SRC00077/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000462	SRC00076/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000463	SRC00078/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000464	SRC00077/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 65	SRC00076/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 66	SRC00078/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 67	SRC00077/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 68	SRC00076/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 69	SRC00078/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 70	SRC00077/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000471	SRC00076/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000472	SRC00078/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000473	SRC00077/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000474	SRC00076/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000475	SRC00078/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000476	SRC00077/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 77	SRC00076/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 78	SRC00078/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 79	SRC00077/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 80	SRC00076/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 81	SRC00078/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/007_Electronic_spectroscopy_-_vibrational_coarse_structure_progressions_Franck-Condon_principle__51ffd9a09d4ec41e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 82	SRC00077/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000483	SRC00079/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000484	SRC00081/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000485	SRC00080/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000486	SRC00079/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000487	SRC00081/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000488	SRC00080/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000489	SRC00079/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000490	SRC00081/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000491	SRC00080/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000492	SRC00079/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000493	SRC00081/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000494	SRC00080/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 95	SRC00079/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 96	SRC00081/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 97	SRC00080/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 98	SRC00079/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 99	SRC00081/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 00	SRC00080/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005 01	SRC00079/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 02	SRC00081/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 03	SRC00080/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 04	SRC00079/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 05	SRC00081/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 06	SRC00080/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000507	SRC00079/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000508	SRC00081/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000509	SRC00080/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000510	SRC00079/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000511	SRC00081/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000512	SRC00080/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000513	SRC00079/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000514	SRC00081/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000515	SRC00080/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000516	SRC00079/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000517	SRC00081/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000518	SRC00080/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000519	SRC00079/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000520	SRC00081/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__eb00184a944a5a23.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000521	SRC00080/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000522	SRC00082/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000523	SRC00084/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000524	SRC00083/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000525	SRC00082/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000526	SRC00084/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000527	SRC00083/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000528	SRC00082/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000529	SRC00084/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000530	SRC00083/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000531	SRC00082/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000532	SRC00084/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000533	SRC00083/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000534	SRC00082/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000535	SRC00084/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000536	SRC00083/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000537	SRC00082/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000538	SRC00084/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000539	SRC00083/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000540	SRC00082/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000541	SRC00084/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000542	SRC00083/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000543	SRC00082/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000544	SRC00084/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000545	SRC00083/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000546	SRC00082/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000547	SRC00084/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000548	SRC00083/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000549	SRC00082/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000550	SRC00084/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000551	SRC00083/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000552	SRC00082/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000553	SRC00084/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000554	SRC00083/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000555	SRC00082/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000556	SRC00084/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000557	SRC00083/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000558	SRC00082/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000559	SRC00084/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000560	SRC00083/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000561	SRC00082/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000562	SRC00084/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000563	SRC00083/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000564	SRC00082/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000565	SRC00084/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000566	SRC00083/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000567	SRC00082/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000568	SRC00084/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000569	SRC00083/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000570	SRC00082/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000571	SRC00084/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/009_Derivation_of_Gibb_s_phase_rule_Phase_diagram_of_one-component_system_-_water__15b0d3b4d5281d6f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000572	SRC00083/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000573	SRC00085/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000574	SRC00087/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000575	SRC00086/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000576	SRC00085/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000577	SRC00087/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000578	SRC00086/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005 79	SRC00085/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 80	SRC00087/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 81	SRC00086/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 82	SRC00085/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 83	SRC00087/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 84	SRC00086/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000585	SRC00085/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000586	SRC00087/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000587	SRC00086/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000588	SRC00085/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000589	SRC00087/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000590	SRC00086/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000591	SRC00085/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000592	SRC00087/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000593	SRC00086/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000594	SRC00085/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000595	SRC00087/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000596	SRC00086/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000597	SRC00085/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000598	SRC00087/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/010_Reduced_phase_rule_phase_diagram_of_two-component_system_-_Pb-Ag_system__da2fe68080b8e9cb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000599	SRC00086/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000600	SRC00088/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000601	SRC00090/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000602	SRC00089/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000603	SRC00088/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000604	SRC00090/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000605	SRC00089/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000606	SRC00088/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000607	SRC00090/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000608	SRC00089/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000609	SRC00088/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000610	SRC00090/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000611	SRC00089/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000612	SRC00088/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000613	SRC00090/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000614	SRC00089/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000615	SRC00088/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000616	SRC00090/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000617	SRC00089/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000618	SRC00088/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000619	SRC00090/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000620	SRC00089/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000621	SRC00088/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000622	SRC00090/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000623	SRC00089/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000624	SRC00088/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000625	SRC00090/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000626	SRC00089/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000627	SRC00088/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000628	SRC00090/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000629	SRC00089/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000630	SRC00088/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000631	SRC00090/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000632	SRC00089/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000633	SRC00088/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000634	SRC00090/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000635	SRC00089/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000636	SRC00088/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000637	SRC00090/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000638	SRC00089/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000639	SRC00088/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000640	SRC00090/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000641	SRC00089/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000642	SRC00088/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000643	SRC00090/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000644	SRC00089/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000645	SRC00088/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000646	SRC00090/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000647	SRC00089/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000648	SRC00088/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000649	SRC00090/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000650	SRC00089/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000651	SRC00088/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000652	SRC00090/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000653	SRC00089/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000654	SRC00088/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000655	SRC00090/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000656	SRC00089/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.
EV0000657	SRC00088/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000658	SRC00090/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000659	SRC00089/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.
EV0000660	SRC00088/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000661	SRC00090/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/011_Computational_chemistry_Scope_computational_methods_electronic_structure_determination_AI-enhanced_computation__07fe702917862787.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000662	SRC00089/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.
EV0000663	SRC00091/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/012_Experiential_learning_Based_on_Phase_Diagram__42cc75d52ba7f0e4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000664	SRC00093/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/012_Experiential_learning_Based_on_Phase_Diagram__42cc75d52ba7f0e4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000665	SRC00092/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000666	SRC00091/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/012_Experiential_learning_Based_on_Phase_Diagram__42cc75d52ba7f0e4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000667	SRC00093/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/012_Experiential_learning_Based_on_Phase_Diagram__42cc75d52ba7f0e4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000668	SRC00092/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000669	SRC00091/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/012_Experiential_learning_Based_on_Phase_Diagram__42cc75d52ba7f0e4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000670	SRC00093/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/012_Experiential_learning_Based_on_Phase_Diagram__42cc75d52ba7f0e4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000671	SRC00092/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000672	SRC00091/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/012_Experiential_learning_Based_on_Phase_Diagram__42cc75d52ba7f0e4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000673	SRC00093/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/Slides/012_Experiential_learning_Based_on_Phase_Diagram__42cc75d52ba7f0e4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000674	SRC00092/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000675	SRC00094/q12	covered	Original archive PDF page 12; candidate unit association is text-based and unconfirmed.
EV0000676	SRC00095/q2	covered	Original archive PDF page 2; candidate unit association is text-based and unconfirmed.
EV0000677	SRC00096/q2	covered	Original archive PDF page 2; candidate unit association is text-based and unconfirmed.
EV0000678	SRC00097/q1	covered	Original archive PDF page 1; candidate unit association is text-based and unconfirmed.
EV0000679	SRC00098/q1	covered	Original archive PDF page 1; candidate unit association is text-based and unconfirmed.

Evidence	Source unit	Status	Decision
EV0000680	SRC00099/q1	covered	Original archive PDF page 1; candidate unit association is text-based and unconfirmed.
EV0000681	SRC00099/q2	covered	Original archive PDF page 2; candidate unit association is text-based and unconfirmed.
EV0000682	SRC00100/q1	covered	Original archive PDF page 1; candidate unit association is text-based and unconfirmed.
EV0000683	SRC00101/q1	covered	Original archive PDF page 1; candidate unit association is text-based and unconfirmed.
EV0000684	SRC00102/q1	covered	Original archive PDF page 1; candidate unit association is text-based and unconfirmed.
EV0000685	SRC00102/q2	covered	Original archive PDF page 2; candidate unit association is text-based and unconfirmed.
EV0000686	SRC00103/q2	covered	Original archive PDF page 2; candidate unit association is text-based and unconfirmed.
EV0000687	SRC00104/q1	covered	Original archive PDF page 1; candidate unit association is text-based and unconfirmed.

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